Reductive groups over a local field: II. Group schemes. Existence of a valued root datum

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JACQUES TITS

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REDUCTIVE GROUPS OVER A LOCAL FIELD

by F. BRUHAT and J. TITS

CHAPTER II

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In memory of Claude Chevalley

INTRODUCTION

In a first chapter of this work (denoted I, see bibliography), we introduced and studied from the point of view of "abstract" groups the notion of "valued root datum". One of the results of this second chapter — a result which *a posteriori* motivates this study — is that, under fairly general conditions to which we shall return in a moment, the group $G(K)$ of rational points of a connected reductive group G over a henselian valued field K possesses a valued root datum "compatible with the valuation of K" (5.1.20, 5.1.23); this datum is unique up to conjugation and equipollence (I, 6.2.5). Thus, everything that was said in I applies to $G(K)$: description and classification of maximal bounded subgroups, Iwasawa, Cartan, and Bruhat decompositions, existence and properties of the subgroups U_f and other related subgroups (I, § 6), existence and properties of the building of the group, etc.

The group schemes in question in the title are both an essential tool for achieving this result and the main object of study of the chapter. They are group schemes over the ring of integers $\mathcal{O}$ of K, with generic fiber G, most often smooth and affine. Suppose for example, to avoid entering immediately into technical complications, that G is semi-simple and the valuation of K is discrete; then we associate to every bounded subset Ω of an apartment of the building $\mathcal{I}$ of G such a group scheme $\mathcal{G}$ (denoted $\mathcal{G}^\dagger_\Omega$ in 5.1.9) whose group of integer points $\mathcal{G}(\mathcal{O})$ is the stabilizer of Ω in $G(K)$ (which operates on $\mathcal{I}$, let us recall). If moreover Ω is a facet of the building $\mathcal{I}$ (I, 2.1.1), the set of facets whose closure contains Ω identifies with the "spherical" building of the identity component of the closed fiber of $\mathcal{G}$, that is, of the algebraic group $\overline{\mathcal{G}}_k$ obtained from $\mathcal{G}$ by reduction modulo the maximal ideal $\mathfrak{m}$ of $\mathcal{O}$ (5.1.32); when $\overline{k}$ is perfect, the spherical building in question is the building of $\overline{k}$-parabolic subgroups of $\overline{\mathcal{G}}_k$ ([37], § 5). Thus, the reduction mod $\mathfrak{m}$ corresponds geometrically to a "localization" in the building, and the study of the closed fibers of the schemes associated with facets provides information of a combinatorial nature about the building. The stabilizers of bounded subsets of apartments are only some of the subgroups introduced and studied in §§ 6 and 7 of I (groups U_f, P_f, $P^\dagger_f$, etc.) and many of which are also interpreted as groups of integer points of smooth schemes with generic fiber G (4.6.2, 4.6.21, 4.6.28).

The proof of the existence of the valued root datum (hence of the building) of $G(K)$ is obtained by the succession of two descent processes, a *quasi-descent*

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...split, transition from a field on which G splits to a subfield on which it is quasi-split, and an *étale descent*, transition from a field of "quasi-splitting" to a subfield of which it is an étale extension (in the Henselian case, where we place ourselves for the moment, an algebraic extension is said to be étale if it is unramified and if the residual extension is separable). The quasi-split descent is the easiest. It could be done by explicit calculations (4.2.11), but instead of this "down to earth" method, we prefer another, more geometric one (it makes use of the building); a bit less elementary than the first, it requires on the other hand no case-by-case distinction and has above all, for us, the advantage of having been prepared in Chapter I by a descent theorem (I, 9.2.10) of which quasi-split descent is an immediate application (4.2.3). Let us recall incidentally that quasi-split groups had been studied (at least over a p-adic field) by H. Hijikata [18], after N. Iwahori and H. Matsumoto had studied the split case in a fundamental memoir [21] which is at the origin of our research.

The same Theorem 9.2.10 of I applies also but less directly to étale descent. The difficulty consists in showing that the conditions for using the theorem are satisfied. The key lemma here is the existence (always under certain conditions: see further below) of a torus of G defined over the base field and which becomes maximal split over an étale extension on which G is quasi-split. It is to prove this lemma that we need the group schemes mentioned above. In an earlier state of our work (sketched in [8], [11], [13], [40]) we only had these schemes available in the split case. To perform the étale descent, and notably to prove the lemma in question, we had to be content with providing the facet stabilizers — for a quasi-split group — with proalgebraic structures, deduced by descent (quasi-split and totally ramified) from analogous structures in the split case, which were obtained from the schemes using the Greenberg functor. The method we use here has several advantages: it gives more precise results (a scheme over $\mathcal{O}$ has "more structure" than the corresponding proalgebraic group), it is more general (we no longer need to assume the discrete valuation nor the residual field perfect), and schemes are a more powerful and convenient tool than proalgebraic groups (thus, cumbersome processes of successive approximations are replaced by simple applications of Hensel's lemma). On the other hand, the schemes we need for quasi-split groups are no longer obtained by descent from the split case, and must be constructed directly; this requires the machinery set up in §§ 2 and 3, which we will discuss in more detail in the analytical part of this introduction.

Let us note in passing that since the publication of [11], [13], the existence of valued root data and buildings for non-quasi-split groups has also been obtained by other methods. Recall first of all that the case of classical groups is treated in § 10 of I. In [20], H. Hijikata (who had already previously studied [19] certain particular classical groups) shows by direct calculations the existence of a

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Tits system of affine type (hence of a building) for every semi-simple group over a locally compact field; however, he needs for this to know the vanishing of H^1 for simply connected groups (a result of M. Kneser [22] and Bruhat-Tits [13]). A geometric descent method which should be applicable to tamely ramified descent was developed by G. Rousseau [27], but he pointed out to us the inaccuracy of a lemma, crucial for the case of dense valuations. Finally, using the classification [34], one of us [38] verified, case by case, by a reduction process to rank one, the existence of a valued root datum for "most" groups of relative rank ≥ 2 over any Henselian field.

This leads us to speak again about the conditions in which we place ourselves in this work. Our existence theorem of the valued root datum (5.1.20) is proven under the following hypotheses: K is Henselian, G is quasi-split over the strict Henselization (maximal étale extension) $\widehat{K}$ of K and when the valuation is dense, the splitting field of G above $\widehat{K}$ is unramified ("admissible" extension: cf. 1.6.1) and G satisfies a condition (5.1.5 (DE)) which is perhaps always true and is in any case as soon as the completion of K is maximally complete. These hypotheses are satisfied for every reductive group G over a Henselian discrete valuation field with a perfect residue field, since in this case, a well-known result of R. Steinberg (cf. [32], [6] and also [29], II, § 3) implies that G is quasi-split over $\widehat{K}$.

When the valuation is dense, the scheme method that we use probably cannot extend to the case where the group G does not split over any unramified extension: the fact that the ring of integers of a ramified extension is not a module of finite type over the ring of integers of the base field seems to be an insurmountable obstacle. On the other hand, several intermediate results are established under more general conditions. Thus, the treatment of quasi-split groups (valued root datum, building, schemes associated with subsets of apartments, etc.) does not require K to be Henselian, and some of the more general constructions of §§ 2 and 3 are valid for any Prüfer ring, or even for any integral domains.

Let us now review the content of the various paragraphs.

Reminders and generalities on algebraic groups and group schemes have been gathered in § 1. These are most often known concepts and results but for which we have not always found an adequate reference. It even sometimes seemed useful to us to develop these generalities a little more than was strictly necessary for our needs. This is notably the case for what concerns distributions (1.3) and the restriction of scalars (1.5).

Let G be a reductive group over any field K, S a K-split torus of G, Z the centralizer of S, X the character group of S and Φ the set of *root rays*, that is to say open half-lines of $X \otimes \mathbf{R}$ which contain at least one weight of S in the Lie algebra of G. For $a \in \Phi$, there exists a largest subgroup U_a of G normalized by S and such that the weights of S in U_a belong to a.

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It is known that if the elements of Φ are arranged in a suitable order, the product map from $\prod U_a \times Z$ into G is an open immersion whose image C traditionally bears the name of *big cell*. Now let A be a subring of K having K as its field of fractions, suppose that G is a linear group, $G \subset GL(V)$, and let us be given in the vector space V a finitely generated projective A-module M such that $M \otimes K = V$ and that M is the direct sum of its intersections with the isotypic components of the S-module V. The giving of M provides G, Z and the U_a with structures of A-schemes $\mathfrak{G}$, $\mathfrak{Z}$, $\mathfrak{U}_a$ (schematic closures of G, Z, U_a in the group scheme $\mathfrak{GL}(M)$). In § 2, we are mainly interested in the case where $\mathfrak{G}, \mathfrak{Z}, \mathfrak{U}_a$ are flat (this is the case, for example, if A is a valuation ring or a Dedekind ring), hence they are group schemes. We show in particular that the product map defines an isomorphism of $\prod \mathfrak{U}_a \times \mathfrak{Z}$ (for the same order on Φ as above) onto an open set $\mathfrak{C}$ ("big cell") of $\mathfrak{G}$; it is deduced from this that $\mathfrak{G}$ (assumed flat) is smooth as soon as $\mathfrak{Z}$ and the $\mathfrak{U}_a$ are.

The problem studied in § 3 is in some sense inverse. Here, we are given A, as above, and flat group schemes $\mathfrak{U}_a$ ($a \in \Phi$), $\mathfrak{Z}$ "extending the U_a and Z" (that is, having the U_a and Z as generic fibers), whence a big cell $\mathfrak{C} = \prod \mathfrak{U}_a \times \mathfrak{Z}$, which we propose to embed as an open set in a group scheme $\mathfrak{G}$ extending G. The method we use consists in reducing back to the situation of § 2, that is, in looking for a linear representation $\rho : G \to GL(V)$ and a sub-A-module M of V possessing the properties stated above and such that the $\mathfrak{U}_a$ and $\mathfrak{Z}$ "are" the schematic closures of the U_a and of Z in $\mathfrak{GL}(M)$. This is *grosso modo* the path followed by C. Chevalley to show the existence of the schemes that bear his name [16]. In 3.1.1,

we state four conditions (on the $\mathfrak{U}_a$ and $\mathfrak{Z}$) manifestly necessary for the existence of ρ and M. These consist, for the most part, of conditions expressing that certain morphisms involving the $\mathfrak{U}_a$ and $\mathfrak{Z}$ and expressing commutation or conjugation relations extend into morphisms of schemes this time involving the $\mathfrak{U}_a$ and $\mathfrak{Z}$; when these conditions, denoted (DRS 0) to (DRS 3), are satisfied, we say that the system $((\mathfrak{U}_a)_{a\in\Phi}, \mathfrak{Z})$ is a *schematic root datum.* The main result of § 3 is that, in the case that interests us most, that of a quasi-split group G and a valuation ring A (or, more generally, a Prüfer ring), the necessary conditions (DRS 0) to (DRS 3) are also sufficient for the existence of ρ and M, and thus for the group scheme $\mathfrak{G}$. Certain intermediate results apply, moreover, to non-quasi-split groups or to non-Prüferian rings.

Another method for constructing $\mathfrak{G}$ from $\mathfrak{C}$, a method used by M. Demazure in the case of Chevalley schemes, would consist in providing $\mathfrak{C}$ with a "group kernel" structure, the existence of which follows again from the properties (DRS 0) to (DRS 3), then to "integrate" this kernel using the Weil-Artin theorem ([SGAD], exposé XVIII). Although this procedure leads perhaps to more general results than ours, we preferred the method of linear representations because

because of its more "concrete" character — it moreover provides more precise information about the schemes obtained — and especially to avoid having to align cumbersome sorites on the not very attractive notion of kernels of group schemes (cf. 3.1.7).

§ 4 is devoted to the study of quasi-split groups over a valued field K. Here, S denotes a maximal K-split torus of the group G under consideration and its centralizer Z is therefore a maximal torus T of G. We begin (no. 4.2) by showing the existence of a valued root datum in $G(K)$. Recall that a real-valued function $f : \Phi \to \mathbf{R}$ on the set Φ of roots of G with respect to S is said to be concave if, for $a, b \in \Phi$ and $p, q \in \mathbf{Z}$ such that $pa+qb \in \Phi$, one has $f(pa+qb) \leq pf(a)+qf(b)$. In no. 4.5, we associate to every concave (or even quasi-concave: cf. I, 6.4.8 and no. 4.5.3 below) function f satisfying, in the case of dense valuations, an additional technical condition (condition (Sch) of no. 4.5.1), a schematic root datum $((U_{a,f}), \mathfrak{T})$: the $U_{a,f}$ are selected from among group schemes with generic fiber U_a constructed in 4.3, and the possible choices of the scheme $\mathfrak{T}$, with generic fiber T, are discussed in no. 4.4. Thus, every function f having the indicated properties gives rise to a "big cell" $\mathfrak{C}_f = \prod U_{a,f} \times \mathfrak{T}$, possessing, according to § 3, an open immersion in a group scheme with generic fiber G. This group scheme is not unique but there exists a "smallest" one, which we denote $\mathfrak{G}_f$, and by passing to the identity component and gluing of translates one obtains others (schemes $\mathfrak{Y}\mathfrak{G}_F^\circ$ of no. 4.6.21), also useful: thus, the schemes attached to bounded parts of apartments, which were mentioned at the beginning of this introduction, are special cases of these $\mathfrak{Y}\mathfrak{G}_F^\circ$. In no. 4.6, we establish the existence of $\mathfrak{G}_f$ and the $\mathfrak{Y}\mathfrak{G}_F^\circ$ (the problem, for these, being to show that they are separated and affine) and we briefly describe the structure (*group of components, split unipotent radical, root system...*) of the closed fibers of the $\mathfrak{Y}\mathfrak{G}_F^\circ$, that is to say the algebraic groups over the residue field deduced from these by reduction mod $\mathfrak{m}$. In the case of a discrete valuation, these last results allow us to establish the isomorphism mentioned above between the star of a facet of the building of G and the spherical building of the algebraic group $\overline{\mathfrak{G}}_F$ deduced by reduction mod $\mathfrak{m}$ of the scheme $\mathfrak{G}_F$ attached to F (4.6.33). When the valuation is dense, a facet F is no longer a set but a germ of a set; one can no longer naturally attach a group scheme to it but we show (4.6.41) that one can still canonically attach to it an algebraic group $\overline{\mathfrak{G}}_F$ over the residue field, whose spherical building is identified with the star of F.

Finally, § 5 is devoted to *étale descent.* Let us retain the preceding notation, but suppose that K is strictly Henselian (*i.e.* Henselian with a separably closed residue field) and that the group G is defined over a subfield $K^\natural$ of K such that K is an étale extension. The main results of no. 5.1 are the existence of a valued root datum in $G(K^\natural)$ (subject to (DE) in the case of a dense valuation), hence the existence of a building of G over $K^\natural$, and the identification of this building with the set of fixed points of $\mathrm{Gal}(K/K^\natural)$ in the building of G over K. These results

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are easily deduced from the descent theorem of I, 9.2.10, once the existence of a $K^\natural$-torus which becomes maximal split over K (5.1.12) is established with the help of no. 4.6.

The end of § 5 is devoted to the generalization by descent of the other results of § 4 and to a more thorough study of the case of discrete valuation (definition of parahoric and Iwahori subgroups, Tits system, conjugation of parahoric subgroups).

Finally, we give in an Appendix a description of the "commutation relations" in quasi-split groups which may be useful and which allows in particular the demonstration of the existence of a valued root datum in a quasi-split group without using the descent theorem of Chapter I (cf. 4.2.11).

We would not like to end this introduction without thanking Guy Rousseau, who kindly read our manuscript attentively, and whose remarks and corrections have been valuable to us, nor without pointing out that the reader will find in this Chapter the proof of all the results stated in [8], [13] and [40], with the exception of those relating to "Dynkin diagrams" and related questions (indices, classification), to applications to cohomology and to certain explicit examples. We hope to return to these questions one day.

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§ 1. NOTATIONS AND RECALLS

Throughout this work, the letter K denotes an infinite field. In §§ 1, 2 and 3, the letter A denotes a subring of K, having K as its field of fractions. Starting from No. 4.2, the field K will be assumed to be equipped with a non-trivial valuation (by valuation of a field, we mean a valuation with values in $\mathbf{R}$; on the other hand, we make no restrictive hypothesis on the valuation rings considered in §§ 2 and 3), and we will take A to be the ring $\mathcal{O}$ of this valuation. Unless explicitly stated otherwise, the A-algebras considered are assumed to be commutative and to possess a unit element.

The word *scheme* means, unless explicitly stated otherwise, *affine scheme*.

1.1. Algebraic groups.

Regarding algebraic groups (which are always assumed to be *affine*), we take the definitions and notations of [2], [4] and [7], with the exception that our field K is denoted by the letter k in these references, to which we refer for the results stated in this section.

1.1.1. — Throughout this work, the letter G denotes an (affine) algebraic group defined over K and connected. The letter S denotes a torus of G, defined and split over K. We denote by $Z = Z(S)$ the centralizer and by $N = N(S)$ the normalizer of S in G. It is known that Z and N are defined over K and that Z is the identity component of N.

1.1.2. — The *roots* of G relative to S are the non-zero weights of S in the adjoint representation of G in its Lie algebra Lie G. They form a finite subset of the group of characters $X^*(S)$ of S, a group which is identified with its canonical image in the real vector space $E = \mathbf{R} \otimes X^*(S)$. A *root ray* of G relative to S is an open half-line with origin o in E containing at least one root; the set of root rays of G relative to S is denoted by $\Phi(S, G)$ or simply Φ.

1.1.3. — For every open half-line a with origin o in E, there exists a largest connected closed subgroup U_a of G, normalized by S and such that the characters of S occurring in the adjoint representation of S in Lie U_a belong to a; it is defined over K, split unipotent over K and it is called the *root subgroup* of G associated to a. We have $U_a \neq \{1\}$ if and only if $a \in \Phi$.

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1.1.4. — If a and b are two open rays from the origin o of E, the set of open rays contained in the cone $a + b$ is also denoted by $a + b$. A subset Ψ of Φ is said to be *closed* if for all $a, b \in \Psi$, we have $(a + b) \cap \Phi \subset \Psi$. If, moreover, the union of the $a \in \Psi$ is contained in an open half-space of E, Ψ is said to be *positively closed*.

1.1.5. — A subset Φ^+ of Φ is called a *system of positive root rays* if there exists a hyperplane H of E not meeting any element of Φ such that Φ^+ is the set of elements of Φ contained in one of the two open half-spaces with boundary H. We then set $\Phi^- = \Phi - \Phi^+$ and the elements of Φ^- are called *negative*. It is clear that Φ^+ and Φ^- are positively closed and that conversely, every positively closed subset is contained in a suitable system of positive root rays.

Si G is *reductive*, we have $\Phi = -\Phi$ and $\Phi^- = -\Phi^+$.

1.1.6. — For $\Psi \subset \Phi$, we denote by U_Ψ the closed subgroup generated by the U_a for $a \in \Psi$. It is a connected subgroup of G, defined over K and normalized by Z. We denote by G_Ψ the closed subgroup generated by Z and U_Ψ; it is defined over K.

We say that Ψ is *quasi-closed* if every weight of S in Lie U_Ψ is an element of an $a \in \Psi$ and *positively quasi-closed* if, in addition, Ψ is contained in an open half-space. If Ψ is *closed* (resp. *positively closed*), then Ψ is *quasi-closed* (resp. *positively quasi-closed*).

1.1.7. — If Ψ is *positively quasi-closed*, then U_Ψ is unipotent and the product map $\prod_{a \in \Psi} U_a \to U_\Psi$ is an isomorphism of K-varieties, regardless of the order put on Ψ. The product map is also an isomorphism of K-varieties from $Z \times U_\Psi$ onto G_Ψ, which is the semi-direct product of Z by U_Ψ.

1.1.8. — Let $a, b \in \Phi$ with $b \neq -a$; then $(a + b) \cap \Phi$ is positively closed and the commutator map is a morphism of K-varieties from $U_a \times U_b$ into $U_{(a+b)\cap\Phi}$, whence, by 1.1.7, a morphism of K-varieties

$$\gamma_{a,b} : U_a \times U_b \to \prod_{c \in (a+b)\cap\Phi} U_c$$

(regardless of the order put on $(a+b)\cap\Phi$; of course, $\gamma_{a,b}$ depends on this order!).

1.1.9. — Let Ψ be a closed subset of Φ and let Φ^+ be a system of positive root rays. Then $\Psi^\pm = \Psi\cap\Phi^\pm$ is positively closed and the product map $U_{\Psi^-}\times Z\times U_{\Psi^+}\to G_\Psi$ is an open immersion. In particular, $U_{\Phi^-}\times Z\times U_{\Phi^+}$ is identified via the product map with an open neighborhood of the identity element in G, which is all of G when G is solvable.

1.1.10. — Let $a\in\Phi$ such that $-a\in\Phi$ (which is always true if G is reductive). Let W_a be the inverse image of the open set $U_{-a}ZU_a$ under the product map from $U_a\times U_{-a}$ into $G_{\{a,-a\}}$ (note that $\{a,-a\}$ is obviously closed!). Then, W_a is

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an open neighborhood of the identity element of $U_a\times U_{-a}$ and the composite of the product map $W_a\to U_{-a}ZU_a$ and the inverse of the product map $U_{-a}ZU_a\to U_{-a}\times Z\times U_a$ is a morphism of K-varieties

$$\beta_a : W_a\to U_{-a}\times Z\times U_a$$

such that prod $=$ prod $\circ\,\beta_a : W_a\to G$.

1.1.11. — The split unipotent radical $R_{ud}(G)$ is the largest unipotent subgroup defined and split over K contained in the radical of G. If K is perfect, R_{ud} is nothing other than the unipotent radical of G and the results that follow are proven in [4]; in the general case, an outline of the proof will be found in [7] pending something better. The subgroup $R_{ud}(G)$ is a normal subgroup of G, stable under separable extension of K, and the product map

$$\prod_{a\in\Phi^+}(R_{ud}(G)\cap U_a)\times R_{ud}(Z)\times\prod_{a\in\Phi^-}(R_{ud}(G)\cap U_a)\to R_{ud}(G)$$

is an isomorphism of K-varieties. The subgroups $R_{ud}(G)\cap U_a$ are split unipotent over K. If Ψ is a closed subset of Φ such that $(-\Psi)\cap\Phi\subset\Psi$, then $R_{ud}(G_\Psi)=R_{ud}(G)\cap G_\Psi$.

1.1.12. — We say that G is quasi-reductive (over K) if $R_{ud}(G)=\{1\}$. In the general case, we will set ${}^qG = G/R_{ud}(G)$ and we will denote by q the canonical map from G to ${}^qG$; the group ${}^qG$ is quasi-reductive. It is known ([2], p. 363) that if H is a normal unipotent subgroup of G, defined and split over K, then the canonical map from $G(K)$ to $(G/H)(K)$ is surjective; in particular $q(G(K))={}^qG(K)$.

1.1.13. — Suppose until the end of 1.1 that S is a maximal K-split torus. Then, the image $q(S)$ of S is a maximal K-split torus of ${}^qG$ and $q:S\to q(S)$ is an isomorphism, which allows us to identify henceforth $X^*(S)$ and $X^*(q(S))$. The image $q(Z)$ (resp. $q(N)$) is the centralizer (resp. normalizer) of $q(S)$ in ${}^qG$. For all $a\in\Phi$, the restriction of q to U_a defines an isomorphism from $U_a/(R_{ud}(G)\cap U_a)$ onto the root subgroup U'_a of ${}^qG$ associated to a.

The set R of roots of $q(S)$ in ${}^qG$ is a root system whose Weyl group is naturally identified with the groups $N(K)/Z(K)$ and $q(N)(K)/q(Z)(K)$ acting canonically on $X^*(S)=X^*(q(S))$. There exists a generating root datum $(q(Z)(K),(V_\alpha)_{\alpha\in R})$ in ${}^qG(K)$, of type R (in the sense of I, 6.1.1), such that $U'_\alpha(K)$ is equal to V_α or to $V_\alpha\cdot V_{2\alpha}$ depending on whether the root ray $a\in\Phi(q(S),{}^qG)$ contains a single root α or two roots α and 2α.

1.1.14. — The set R^+ of elements of R belonging to the elements of Φ^+ is a system of positive roots of R. Let B be the corresponding base of R; for any

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subset J of B, let us denote by Ψ_J^+ the set of root rays containing either a positive root or a linear combination of the elements of J: it is a closed subset of $\Phi(q(S),{}^gG)$. We set

$$P_J = q^{-1}(({}^gG)_{\Psi_J^+})$$

and we say that P_J is the *standard K-pseudo-parabolic subgroup* associated with S, R^+, and J. More generally, the K-pseudo-parabolic subgroups of G are by definition the conjugates of the P_J by elements of $G(K)$. Their groups of rational points over K are exactly the parabolic subgroups of the Tits system which is the inverse image in $G(K)$ of the Tits system of ${}^gG(K)$ associated with the root datum of 1.1.13 (cf. I, 6.1.12 and 1.2.10).

If K is *perfect* or if G is *reductive*, the K-pseudo-parabolic subgroups are none other than the K-parabolic subgroups of G in the sense of [4].

For the properties of K-pseudo-parabolic subgroups (which are roughly the same as those of K-parabolic subgroups in the reductive case), see [7]. Let us mention a few of them:

- two K-pseudo-parabolic subgroups are equal (resp. conjugate in G) if and only if their groups of rational points are equal (resp. conjugate in $G(K)$);
- two minimal K-pseudo-parabolic subgroups are conjugate by an element of $G(K)$;
- every K-pseudo-parabolic subgroup is its own normalizer;
- if K' is a *separable* extension of K, the K'-pseudo-parabolic subgroups are the K-closed K'-pseudo-parabolic subgroups.

1.1.15. — Every coset of N according to Z contains an element rational over K. Indeed, let $n \in N$; the restriction to $q(S)$ of the inner automorphism of ${}^gG$ defined by $q(n)$ provides an automorphism of the root system R. As $N(K)$ induces on R its entire Weyl group (1.1.13), there exists $n' \in n \cdot N(K)$ preserving R^+, thus normalizing the minimal K-pseudo-parabolic subgroup $P_\emptyset$ of G, and thus belonging to $P_\emptyset$ (1.1.14). But as S is contained in the radical of $P_\emptyset$, its normalizer in $P_\emptyset$ coincides with its centralizer Z. Thus $n' \in Z$ and $n \in Z \cdot N(K)$.

1.2. Schemes.

1.2.1. — Let us recall that the word *scheme* means, unless expressly mentioned to the contrary, *affine scheme*.

If $\mathfrak{X}$ is an A-scheme, we denote by $A[\mathfrak{X}]$ its affine algebra, by $\mathfrak{X}_K$ its generic fiber, that is to say the K-scheme $\operatorname{Spec} K \times_{\operatorname{Spec} A} \mathfrak{X}$, with affine algebra $K[\mathfrak{X}] = K[\mathfrak{X}_K] = K \otimes_A A[\mathfrak{X}]$, and we identify $\mathfrak{X}_K$ and its canonical image in $\mathfrak{X}$ (via the second projection).

If $u : \mathfrak{X} \to \mathfrak{Y}$ is a morphism of A-schemes, we denote by u^* the corresponding homomorphism...

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...sponding of A-algebras from $A[\mathfrak{Y}]$ into $A[\mathfrak{X}]$ (defined by $f \mapsto f \circ u$), and $u_K : \mathfrak{X}_K \to \mathfrak{Y}_K$ the restriction of u to $\mathfrak{X}_K$; we have $u_K^* = \mathrm{id} \otimes u^* : K \otimes A[\mathfrak{Y}] \to K \otimes A[\mathfrak{X}]$.

1.2.2. — To a linear algebraic group G defined over K there corresponds canonically a smooth (affine) K-group scheme, whose affine algebra is the affine algebra of G, and vice-versa. We will allow ourselves to denote both by the same letter G. The product (resp. the inverse) in the algebraic group G or in the K-group scheme G is given by the same homomorphism from $K[G]$ into $K[G] \otimes K[G]$ (resp. $K[G]$). The K-closed subsets (resp. closed subsets defined over K) of the algebraic group G correspond bijectively to the reduced (resp. geometrically reduced) closed subschemes of the K-scheme G: both correspond bijectively to the ideals I of $K[G]$ such that $K[G]/I$ is reduced, *i.e.* without nilpotent elements $\neq 0$ (resp. such that $\tilde{K} \otimes K[G]/I$ is reduced for every algebraic closure $\tilde{K}$ of K) (cf. [2], p. 45). To a closed subgroup of G defined over K there corresponds thus a smooth group subscheme and vice-versa (cf. [DG], p. 238).

1.2.3. — Let $\mathfrak{X}$ be an A-scheme. For the generic fiber $\mathfrak{X}_K$ to be dense in $\mathfrak{X}$, it is necessary and sufficient that $\mathfrak{X}$ be "torsion-free", that is to say that the A-module $A[\mathfrak{X}]$ be torsion-free: each of these conditions means that the canonical map $j_{\mathfrak{X}} : A[\mathfrak{X}] \to K[\mathfrak{X}]$ is injective. We will then identify $A[\mathfrak{X}]$ and its image in $K[\mathfrak{X}]$. This applies in particular if $\mathfrak{X}$ is flat, that is to say if the A-module $A[\mathfrak{X}]$ is flat.

1.2.4. — Let X be a K-scheme. An extension of X is a pair $(\mathfrak{X}, i)$ formed of an A-scheme $\mathfrak{X}$ and an isomorphism i of K-schemes from $\mathfrak{X}_K$ to X. We will then identify X and $\mathfrak{X}_K$ and we will say that $\mathfrak{X}$ extends X.

De façon analogue, if $\mathfrak{X}$ and $\mathfrak{Y}$ are A-schemes and u a morphism of K-schemes from $\mathfrak{X}_K$ into $\mathfrak{Y}_K$, an extension of u is a morphism $\tilde{u}$ of A-schemes from $\mathfrak{X}$ into $\mathfrak{Y}$ such that $\tilde{u}_K = u$. If $\mathfrak{X}$ is torsion-free (e.g. flat), then u extends to $\mathfrak{X}$ if and only if $\mathrm{Im}(u^* \circ j_{\mathfrak{Y}}) \subset A[\mathfrak{X}]$ and in this case the extension $\tilde{u}$ is unique and defined by $\tilde{u}^* = u^* \circ j_{\mathfrak{Y}} : A[\mathfrak{Y}] \to A[\mathfrak{X}]$. In other words if u and v are two morphisms from $\mathfrak{X}$ into $\mathfrak{Y}$ such that $u_K = v_K$, then $u = v$: we will say that the equality $u = v$ is obtained from $u_K = v_K$ by extension by continuity.

1.2.5. — Let $f : \mathfrak{X} \to \mathfrak{Y}$ be a morphism of A-schemes and let $\mathfrak{Z}$ be a closed subscheme of $\mathfrak{Y}$. Let us say that f maps $\mathfrak{X}$ into $\mathfrak{Z}$ if f factors as $\mathfrak{X} \to \mathfrak{Z} \xrightarrow{i} \mathfrak{Y}$, where i is the canonical immersion of $\mathfrak{Z}$ into $\mathfrak{Y}$. If $\mathfrak{X}$ is torsion-free and if f_K maps $\mathfrak{X}_K$ into $\mathfrak{Z}_K$, then f maps $\mathfrak{X}$ into $\mathfrak{Z}$: indeed, if z belongs to the ideal defining $\mathfrak{Z}$, we have $j_{\mathfrak{X}} \circ f^*(z) = f_K^* \circ j_{\mathfrak{Y}}(z) = 0$, whence $f^*(z) = 0$ since $j_{\mathfrak{X}}$ is injective.

1.2.6. — Let $f : \mathfrak{Y} \to \mathfrak{X}$ be a morphism of A-schemes. The closed image of f (or by abuse of notation of $\mathfrak{Y}$) in $\mathfrak{X}$ is the smallest closed subscheme $\mathfrak{Y}'$ of $\mathfrak{X}$ such that f maps $\mathfrak{Y}$ into $\mathfrak{Y}'$. Its ideal is the kernel of f^* and its support is the closure of...

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the image $f(\mathfrak{Y})$ (cf. [EGA] I, pp. 176-177). If f is the canonical immersion of an (affine) sub-scheme $\mathfrak{Y}$ of $\mathfrak{X}$, we also say that $\mathfrak{Y}'$ is the *schematic closure* of $\mathfrak{Y}$ in $\mathfrak{X}$. The scheme $\mathfrak{Y}$ is then an open sub-scheme of its schematic closure $\mathfrak{Y}'$ (loc. cit., 9.5.10).

Let further B be an integral A-algebra, flat over A. Then, the closed image of $f_B : \mathfrak{X}_B \to \mathfrak{Y}_B$ identifies with $\mathfrak{Y}'_B$ ([DG], p. 56).

Suppose $\mathfrak{X}$ is torsion-free. Let Y be a closed sub-K-scheme of $\mathfrak{X}_K$ and let f be the composition of the immersion of Y into $\mathfrak{X}_K$ and the canonical injection of $\mathfrak{X}_K$ into $\mathfrak{X}$. Let $\mathfrak{Y}$ be the closed image of f (which we will also call the schematic closure of Y in $\mathfrak{X}$) and let $I(\mathfrak{Y})$ (resp. $I(Y)$) be the ideal of $A[\mathfrak{X}]$ (resp. $K[\mathfrak{X}]$) defining $\mathfrak{Y}$ (resp. Y). We have

$$I(\mathfrak{Y}) = j_{\mathfrak{X}}^{-1}(I(Y)), \tag{1}$$

from which it immediately follows that $\mathfrak{Y}$ is torsion-free and if Y is reduced (resp. reduced and irreducible) (that is to say if $I(Y)$ is equal to its root (resp. is prime)), then the same holds for $\mathfrak{Y}$. Furthermore, from the exact sequence $I(\mathfrak{Y}) \to A[\mathfrak{X}] \to A[\mathfrak{Y}] \to 0$ we obtain the exact sequence $K \otimes I(\mathfrak{Y}) \to K[\mathfrak{X}] \to K[\mathfrak{Y}] \to 0$, whence $I(\mathfrak{Y}_K) = Kj_{\mathfrak{X}}(I(\mathfrak{Y})) = I(Y)$, and

$$\mathfrak{Y}_K = Y. \tag{2}$$

Conversely, if $\mathfrak{Y}'$ is a torsion-free closed sub-A-scheme of $\mathfrak{X}$ such that $\mathfrak{Y}'_K = Y$, then $\mathfrak{Y}' = \mathfrak{Y}$: indeed, on the one hand, the injection of Y into $\mathfrak{X}$ factors through $\mathfrak{Y}'$ and on the other hand $\mathfrak{Y}' \subset \mathfrak{Y}$ according to (2) and 1.2.5. The operation of schematic closure thus establishes a *bijection between the set of closed sub-K-schemes of $\mathfrak{X}_K$ and the set of torsion-free closed sub-A-schemes of $\mathfrak{X}$.*

1.2.7. — Let $\mathfrak{X}$ be an A-group scheme, let $p : \mathfrak{X} \times \mathfrak{X} \to \mathfrak{X}$ [1] be the product morphism and $s : \mathfrak{X} \to \mathfrak{X}$ be the inversion morphism of $\mathfrak{X}$ (which we will call inversion). If $\mathfrak{Y}$ is a *flat* closed sub-scheme of $\mathfrak{X}$ such that $\mathfrak{Y}_K$ is a group sub-scheme of $\mathfrak{X}_K$, then $\mathfrak{Y}$ is a group sub-scheme of $\mathfrak{X}$. Indeed, $\mathfrak{Y} \times \mathfrak{Y}$ is flat (note, on the other hand, that $\mathfrak{Y}$ torsion-free does not imply $\mathfrak{Y} \times \mathfrak{Y}$ torsion-free!) and p_K (resp. s_K) sends $\mathfrak{Y}_K \times \mathfrak{Y}_K = (\mathfrak{Y} \times \mathfrak{Y})_K$ (resp. $\mathfrak{Y}_K$) into $\mathfrak{Y}_K$. Consequently p (resp. s) sends $\mathfrak{Y} \times \mathfrak{Y}$ (resp. $\mathfrak{Y}$) into $\mathfrak{Y}$ according to 1.2.5.

In particular, if Y is a closed group sub-K-scheme of $\mathfrak{X}_K$ and if the schematic closure $\mathfrak{Y}$ of Y is a flat A-scheme, then $\mathfrak{Y}$ is a flat closed group sub-A-scheme of $\mathfrak{X}$.

1.2.8. — Let $\mathfrak{X}$ be an A-scheme, $p : \mathfrak{X} \times \mathfrak{X} \to \mathfrak{X}$, $e : \operatorname{Spec} A \to \mathfrak{X}$ and $s : \mathfrak{X} \to \mathfrak{X}$ be morphisms. Suppose that $\mathfrak{X}$ is *flat* and that p_K is a group law on the K-scheme $\mathfrak{X}_K$, with unit section e_K and inversion s_K. Then, p is a group law on $\mathfrak{X}$, with unit section e and inversion s. Indeed, this assertion translates into a certain number of equalities $u_i = v_i$ of morphisms of schemes of $\mathfrak{X} \times \mathfrak{X} \times \mathfrak{X}$ or of $\mathfrak{X}$ into $\mathfrak{X}$, expressing associativity, etc. (cf. [DG], p. 140). The hypothesis implies $(u_i)_K = (v_i)_K$, whence $u_i = v_i$ by extension by continuity (1.2.4).

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Similarly, let $\mathfrak{X}$ and $\mathfrak{Y}$ be two A-group schemes and let $u : \mathfrak{X} \to \mathfrak{Y}$ be a morphism of A-schemes; if $\mathfrak{X}$ is flat and if u_K is a morphism of K-group schemes, then u is a morphism of A-group schemes.

1.2.9. — An A-scheme $\mathfrak{X}$ is smooth ([EGA] IV, § 17, notably 17.5.1) if it is flat, of finite presentation (i.e. $A[\mathfrak{X}]$ is an A-algebra of finite presentation) and if its fibers are smooth, that is to say "are" (cf. 1.2.2) varieties of which all geometric points are simple (cf. [2], p. 72). Suppose $\mathfrak{X}$ is smooth and let $s : \operatorname{Spec} A \to \mathfrak{X}$ be a section of $\mathfrak{X}$; there corresponds to it a homomorphism $s^* : A[\mathfrak{X}] \to A$. Let I be the kernel ideal of s^*, so that $A/I = A$. The A-module I/I^2 is projective of finite type and the obvious canonical homomorphism from the symmetric algebra $\operatorname{Sym}(I/I^2)$ into the graded A-algebra

[1] Of course, the notation $\mathfrak{X} \times \mathfrak{Y}$ (resp. $\mathfrak{X}_K \times \mathfrak{Y}_K$) designates the fiber product of the A-schemes $\mathfrak{X}$ and $\mathfrak{Y}$ over Spec A (resp. of the K-schemes $\mathfrak{X}_K$ and $\mathfrak{Y}_K$ over Spec K).

Gr $A[\mathfrak{X}] = \coprod I^n/I^{n+1}$ is bijective (see [EGA] IV, § 17 and 0_{IV}, § 19). In particular, A/I^n is a projective A-module of finite type for every integer n.

1.2.10. — We denote by $\mathfrak{Add}$ (resp. $\mathfrak{Mult}$) the smooth A-group scheme which to every A-algebra R associates the additive group of R (resp. the multiplicative group of invertible elements of R) (cf. [DG], pp. 148 and 149, where these schemes are noted α_A and μ^A respectively). We have $A[\mathfrak{Add}] = A[T]$ (resp. $A[\mathfrak{Mult}] = A[T, T^{-1}]$), where T is the identity function, which is algebraically free over A, and $\mathfrak{Mult}$ is an open subscheme of $\mathfrak{Add}$.

Let us remark that $\mathfrak{Add}$ can be considered as an A-module scheme, by endowing each $\mathfrak{Add}(R) = R$ with its canonical R-module structure. We will then denote it by $\mathfrak{D}$ or $\mathfrak{D}_A$. A line over A is an A-module scheme isomorphic to $\mathfrak{D}_A$.

1.2.11. — Let S be a torus defined and split over K. The characters of S form a basis of the K-module $K[S]$ and the K-algebra $K[S]$ is none other than the group algebra of $X^*(S)$, the group of characters of S with coefficients in K. If $p : S \times S \to S$ is the group law of S, we have

$$p^*(\alpha) = \alpha \otimes \alpha \quad \text{for all } \alpha \in X^*(S). \tag{1}$$

Consider then the group algebra of $X^*(S)$ with coefficients in A: it is the affine algebra of a torsion-free A-scheme $\mathfrak{S}$, with generic fiber S, and it is immediate that p extends to a group law on $\mathfrak{S}$, still given by (1). We say that $\mathfrak{S}$ is the canonical A-group scheme associated with S. A K-isomorphism from S to $(\mathfrak{Mult}_K)^n$ corresponds to the choice of a basis of $X^*(S)$ and extends in one and only one way to an isomorphism of A-group schemes from $\mathfrak{S}$ to $(\mathfrak{Mult})^n$. In particular, $\mathfrak{S}$ is smooth.

1.2.12. — Let $\mathfrak{G}$ be an A-group scheme. We will say that $\mathfrak{G}$ is connected if, for every $\mathfrak{p} \in \operatorname{Spec} A$, its fiber $\mathfrak{G}_\mathfrak{p}$ above $\mathfrak{p}$ is connected. For any A-algebra B, the B-group scheme $\mathfrak{G}_B$ is then connected ([SGAD], Exp. VI_A, 2.1.1).

Suppose that $\mathfrak{G}$ is smooth at the points of its identity section. Then, the union $\mathfrak{G}^0$

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of the identity components $\mathfrak{G}_p^0$ of the fibers $\mathfrak{G}_p$ for $p \in \operatorname{Spec} A$ is a smooth connected open A-subgroup scheme of $\mathfrak{G}$, normal and even characteristic in $\mathfrak{G}$ ([SGAD], Exp. VI_B, th. 3.10), which we call the *identity component* of $\mathfrak{G}$, and $\mathfrak{G}$ is connected if and only if $\mathfrak{G} = \mathfrak{G}^0$.

1.2.13. — Let $\mathfrak{G}$ and $\mathfrak{H}$ be two group schemes, $j_K : \mathfrak{G}_K \to \mathfrak{H}_K$ a morphism of K-group schemes. Suppose that $\mathfrak{G}$ is smooth and connected and that there exists an open neighborhood $\mathfrak{U}$ of the identity section of $\mathfrak{G}$ such that the restriction of j_K to $\mathfrak{U}_K$ extends to a morphism j of A-schemes from $\mathfrak{U}$ into $\mathfrak{H}$. Then, j_K extends (in one and only one way) to a morphism of A-group schemes from $\mathfrak{G}$ into $\mathfrak{H}$. In fact, it suffices to show that j_K extends to $\mathfrak{G}$ (1.2.8), or even that for all $x \in \mathfrak{G}$, of projection $p \in \operatorname{Spec} A$, of local ring $\mathcal{O}_x$, and all $f \in A[\mathfrak{H}]$, one has $j_K^*(f) \in \mathcal{O}_x$. For this, one can replace A by A_p and suppose A local with maximal ideal $\mathfrak{p}$. Let then A' be the strict henselization of A: it is a local A-algebra, of maximal ideal $\mathfrak{p}'$, of residue field k' separably closed, faithfully flat over A ([EGA] IV, 18.8.8). The canonical morphism $\mathfrak{G}_{A'} \to \mathfrak{G}_A$ is faithfully flat ([DG], p. 42, prop. 2.5) hence surjective. Let $x' \in \mathfrak{G}_{p'}$, with image x. Since $\mathfrak{G}_{p'}$ is connected, we have $\mathfrak{G}_{p'} = \mathfrak{U}(k').\mathfrak{U}_{p'}$. Since $\mathfrak{G}$ is smooth and A' is henselian, a section of $\mathfrak{U}$ over A' passes through every point of $\mathfrak{U}(k')$; there exists then $y \in \mathfrak{U}(A')$ such that $x' \in y.\mathfrak{U}_{A'}$ and the morphism $y.u \mapsto j_{A'}(y).j_{A'}(u)$ of the open $y.\mathfrak{U}_{A'}$ into $\mathfrak{H}_{A'}$ extends $j_{K\otimes A'}$. It follows that $j_{K\otimes A'}^*(f)$ belongs to the local ring $\mathcal{O}'_{x'}$ of x' for all $f \in A'[\mathfrak{H}] \subset K \otimes A'[\mathfrak{H}]$ and that if $f \in A[\mathfrak{H}]$, one has $j_K^*(f) \in \mathcal{O}'_{x'} \cap (K \otimes \mathcal{O}_x)$. Now $\mathcal{O}'_{x'} \cap (K \otimes \mathcal{O}_x) = \mathcal{O}_x$ since $\mathcal{O}'_{x'}$ is faithfully flat over $\mathcal{O}_x$ ([NB] AC I, § 3, prop. 10 (ii)), q.e.d.

More generally, let us replace the hypothesis $\mathfrak{G}$ connected by that of the existence of a smooth subgroup scheme $\mathfrak{Z}$ of $\mathfrak{G}$ such that the restriction of j_K to $\mathfrak{Z}_K$ extends to a morphism $j_\mathfrak{Z}$ from $\mathfrak{Z}$ into $\mathfrak{H}$ and that $\mathfrak{G} = \mathfrak{G}(A).\mathfrak{Z}.\mathfrak{G}^0$. Then, j_K extends to $\mathfrak{G}$: to see this, it suffices to replace in the above proof the relation $\mathfrak{G}_{p'} = \mathfrak{U}(k').\mathfrak{U}_{p'}$ by $\mathfrak{G}_{p'} = \mathfrak{G}(A).\mathfrak{Z}(k').\mathfrak{U}(k').\mathfrak{U}_{p'}$ and the morphism $yu \mapsto j_K(x)j_\mathfrak{Z}(z)j(v)j(u)$ of $y.\mathfrak{U}$ into $\mathfrak{H}$, for $y = xzv$ with $x \in \mathfrak{G}(A)$, $z \in \mathfrak{Z}(A')$ and $v \in \mathfrak{U}(A')$.

1.2.14. — Let $\mathfrak{G}$ and $\mathfrak{H}$ be two smooth A-group schemes and let $j : \mathfrak{G} \to \mathfrak{H}$ be a morphism. Suppose that $j_K : \mathfrak{G}_K \to \mathfrak{H}_K$ is an isomorphism and that there exists an open neighborhood $\mathfrak{U}$ (resp. $\mathfrak{V}$) of the identity section of $\mathfrak{G}$ (resp. $\mathfrak{H}$) such that $j(\mathfrak{U}) = \mathfrak{V}$ and that the restriction of j to $\mathfrak{U}$ is an isomorphism of schemes from $\mathfrak{U}$ onto $\mathfrak{V}$. Then j is an isomorphism from $\mathfrak{G}$ onto an open subgroup scheme of $\mathfrak{H}$. In fact, j is of finite presentation ([DG], p. 69). For all $p \in \operatorname{Spec} A$, the morphism $j_p : \mathfrak{G}_p \to \mathfrak{H}_p$ is smooth

at e, thus is smooth ([SGAD], Exp. VI_B, prop. 1.3) and therefore flat. Consequently j is flat ([DG], p. 327, Lemma 2.2) hence smooth. It follows that $j(\mathfrak{G})$ is open ([DG], p. 79, cor. 3.11) and that Ker j is a flat group scheme ([DG], p. 42, prop. 2.5). Since $(\mathrm{Ker}\ j)_K = \{1\}$, Ker $j = \{1\}$, hence the result.

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Note that if there exists a subset Y of $\mathfrak{G}$ or of $\mathfrak{G}(A)$ such that $\mathfrak{H} = j(Y) \cdot \mathfrak{H}^0$, then j is surjective and is therefore an *isomorphism*.

1.3. Distributions.

1.3.1. — Let $\mathfrak{X}$ be an A-scheme, let $s : \mathrm{Spec}\ A \to \mathfrak{X}$ be a section of $\mathfrak{X}$ and let $I = I(s) = \mathrm{Ker}\ s^*$ be the ideal of $A[\mathfrak{X}]$ associated with s (cf. 1.2.9). A distribution of order $\leq m$ on $\mathfrak{X}$ along s is a linear form δ on $A[\mathfrak{X}]$ that vanishes on I^m, in other words an element of the dual ${}^t(A[\mathfrak{X}]/I^m)$. We say that δ is without constant term if $\delta(a) = 0$ for all $a \in A$. We will denote by ${}^s\mathrm{Dist}_m\mathfrak{X}$ (resp. ${}^s\mathrm{Dist}^+_m\mathfrak{X}$) the A-module of distributions of order $\leq m$ along s (resp. and without constant term) and we set ${}^s\mathrm{Dist}\ \mathfrak{X} = \bigcup_{m\geq 0} {}^s\mathrm{Dist}_m\mathfrak{X}$, ${}^s\mathrm{Dist}^+\mathfrak{X} = \bigcup_{m\geq 0} {}^s\mathrm{Dist}^+_m\mathfrak{X}$. Since $A[\mathfrak{X}]$ is the direct sum of A and I, the A-module ${}^s\mathrm{Dist}_m\mathfrak{X}$ is the direct sum of As^* and ${}^s\mathrm{Dist}^+_m\mathfrak{X} = {}^t(I/I^m)$; in particular, ${}^s\mathrm{Dist}^+_1\mathfrak{X}$ is the dual of I/I^2, that is to say the tangent space to $\mathfrak{X}$ along s.

1.3.2. — If $f : \mathfrak{X} \to \mathfrak{Y}$ is a morphism of A-schemes and s a section of $\mathfrak{X}$, the map f^* sends ${}^s\mathrm{Dist}\ \mathfrak{X}$ into ${}^{f(s)}\mathrm{Dist}\ \mathfrak{Y}$: we set $f(\delta) = {}^tf^*(\delta)$ for $\delta \in {}^s\mathrm{Dist}\ \mathfrak{X}$. If f is an immersion (resp. an open immersion), the map $\delta \to f(\delta)$ is injective (resp. bijective).

1.3.3. — Let B be an A-algebra. Let $\mathfrak{X}'$ be the B-scheme deduced from $\mathfrak{X}$ by the base change $A \to B$ and let s' be the section of $\mathfrak{X}'$ deduced from s. The ideal $I(s')^m$ is the canonical image of $B \otimes I(s)^m$ in $B[\mathfrak{X}']$ and $B[\mathfrak{X}']/I(s')^m$ identifies with $B \otimes A[\mathfrak{X}]/I(s)^m$. By duality, one deduces a B-linear map, called canonical, from $B \otimes {}^s\mathrm{Dist}_m\mathfrak{X}$ into ${}^{s'}\mathrm{Dist}_m\mathfrak{X}'$, then from $B \otimes {}^s\mathrm{Dist}\ \mathfrak{X}$ into ${}^{s'}\mathrm{Dist}\ \mathfrak{X}'$.

Let us take $B = K$. The canonical map from ${}^s\mathrm{Dist}\ \mathfrak{X}$ into $K \otimes {}^s\mathrm{Dist}\ \mathfrak{X}$ and that from $K \otimes {}^s\mathrm{Dist}\ \mathfrak{X}$ into ${}^s\mathrm{Dist}\ \mathfrak{X}_K$ are then injective ([NB] A II, p. 117, cor. 1, and p. 197, exer. 29) and allow the identification of ${}^s\mathrm{Dist}\ \mathfrak{X}$ with its image in ${}^s\mathrm{Dist}\ \mathfrak{X}_K$. If furthermore $\mathfrak{X}$ is of finite type (which implies that $A[\mathfrak{X}]/I(s)^m$ is an A-module of finite type), the canonical map from $K \otimes {}^s\mathrm{Dist}\ \mathfrak{X}$ into ${}^s\mathrm{Dist}\ \mathfrak{X}_K$ is bijective (*loc. cit.*).

1.3.4. — Let $\mathfrak{X}$ and $\mathfrak{Y}$ be two A-schemes, s (resp. t) a section of $\mathfrak{X}$ (resp. $\mathfrak{Y}$); then (s,t) is a section of $\mathfrak{X} \times \mathfrak{Y}$ and $I(s,t)$ is the sum of the canonical images of $I(s) \otimes A[\mathfrak{Y}]$ and of $A[\mathfrak{X}] \otimes I(t)$ in $A[\mathfrak{X} \times \mathfrak{Y}] = A[\mathfrak{X}] \otimes A[\mathfrak{Y}]$. Consequently, $I(s,t)^m$ is the sum of the canonical images of the $I(s)^p \otimes I(t)^q$ for $p+q = m$ and the linear form $\delta \otimes \delta'$ on $A[\mathfrak{X} \times \mathfrak{Y}]$ belongs to ${}^{(s,t)}\mathrm{Dist}_{p+q}(\mathfrak{X} \times \mathfrak{Y})$ as soon as $\delta \in {}^s\mathrm{Dist}_p\mathfrak{X}$ and $\delta' \in {}^t\mathrm{Dist}_q\mathfrak{Y}$. Hence a linear map, called canonical, π from ${}^s\mathrm{Dist}\ \mathfrak{X} \otimes {}^t\mathrm{Dist}\ \mathfrak{Y}$ into ${}^{(s,t)}\mathrm{Dist}(\mathfrak{X} \times \mathfrak{Y})$.

1.3.5. — Suppose $\mathfrak{X}$ is smooth. Then the A-modules $A[\mathfrak{X}]/I^m$, I^m/I^{m+1} and ${}^s\mathrm{Dist}_m\mathfrak{X}$ are projective of finite type and the graded A-module Gr ${}^s\mathrm{Dist}\ \mathfrak{X} = \coprod {}^s\mathrm{Dist}_{m+1}\mathfrak{X}/{}^s\mathrm{Dist}_m\mathfrak{X}$

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identifies with the graded dual of Gr $A[\mathfrak{X}] = \coprod I^m/I^{m+1}$ (cf. 1.2.9), or even with the graded dual of the symmetric algebra $\mathrm{Sym}(I/I^2)$ ("divided powers algebra" $\Gamma({}^t(I/I^2))$).

Let us resume the notation of 1.3.3: the canonical map from $B \otimes {}^s\mathrm{Dist}\ \mathfrak{X}$ to ${}^s\mathrm{Dist}\ \mathfrak{X}'$ is then *bijective*: we have indeed ${}^t(B \otimes (A[\mathfrak{X}]/I^m)) = B \otimes {}^t(A[\mathfrak{X}]/I^m)$ since $A[\mathfrak{X}]/I^m$ is projective of finite type. If moreover $A \subset B$, then the canonical map ${}^s\mathrm{Dist}\ \mathfrak{X} \to {}^s\mathrm{Dist}\ \mathfrak{X}'$ is *injective* since ${}^s\mathrm{Dist}\ \mathfrak{X}$ is flat over A.

Similarly, let us resume the notation of 1.3.4 by assuming that $\mathfrak{X}$ is smooth. For every $m \geq 1$, we have the inclusions

$$I(s)^{2m} \otimes A[\mathfrak{Y}] + A[\mathfrak{X}] \otimes I(t)^{2m} \subset I(s,t)^{2m} \subset I(s)^m \otimes A[\mathfrak{Y}] + A[\mathfrak{X}] \otimes I(t)^m \subset I(s,t)^m,$$

whence, by passing to the quotients of $A[\mathfrak{X}\times\mathfrak{Y}] = A[\mathfrak{X}]\otimes A[\mathfrak{Y}]$ by the terms of this sequence, surjections

$$A[\mathfrak{X}]/I(s)^{2m}\otimes A[\mathfrak{Y}]/I(t)^{2m} \xrightarrow{\alpha} A[\mathfrak{X}\times\mathfrak{Y}]/I(s,t)^{2m}$$
$$\xrightarrow{\beta} A[\mathfrak{X}]/I(s)^{m}\otimes A[\mathfrak{Y}]/I(t)^{m} \xrightarrow{\gamma} A[\mathfrak{X}\times\mathfrak{Y}]/I(s,t)^{m},$$

where $\beta\circ\alpha$ (resp. $\gamma\circ\beta$) is the canonical projection. By duality, taking into account that the $A[\mathfrak{X}]/I(s)^m$ are projective of finite type, we deduce injections

$${}^{(s,t)}\mathrm{Dist}_m(\mathfrak{X}\times\mathfrak{Y}) \xrightarrow{{}^t\gamma} {}^s\mathrm{Dist}_m\mathfrak{X}\otimes{}^t\mathrm{Dist}_m\mathfrak{Y}$$
$$\xrightarrow{{}^t\beta} {}^{(s,t)}\mathrm{Dist}_{2m}(\mathfrak{X}\times\mathfrak{Y}) \xrightarrow{{}^t\alpha} {}^s\mathrm{Dist}_{2m}\mathfrak{X}\otimes{}^t\mathrm{Dist}_{2m}\mathfrak{Y},$$

where ${}^t\alpha\circ{}^t\beta$ (resp. ${}^t\beta\circ{}^t\gamma$) is the canonical injection (recall that ${}^s\mathrm{Dist}_m\mathfrak{X}$ is projective of finite type) and where ${}^t\beta$ is the restriction of the canonical map

$$\pi : {}^s\mathrm{Dist}\ \mathfrak{X}\otimes{}^t\mathrm{Dist}\ \mathfrak{Y}\to{}^{(s,t)}\mathrm{Dist}(\mathfrak{X}\times\mathfrak{Y}).$$

By passing to the inductive limit over m, we see that π is *bijective*.

Let us take $\mathfrak{Y}=\mathfrak{X}$ and consider the diagonal map $d:\mathfrak{X}\to\mathfrak{X}\times\mathfrak{X}$. The map $c=\pi^{-1}\circ{}^td^* : {}^s\mathrm{Dist}\ \mathfrak{X}\to{}^s\mathrm{Dist}\ \mathfrak{X}\otimes{}^s\mathrm{Dist}\ \mathfrak{X}$ equips ${}^s\mathrm{Dist}\ \mathfrak{X}$ with an associative and commutative coalgebra structure, with the map $\delta\mapsto\delta(1)$ as counit. For $\delta\in{}^s\mathrm{Dist}\ \mathfrak{X}$, $c(\delta)=\sum\delta_i\otimes\delta_i'$ and $f,g\in A[\mathfrak{X}]$, we have

$$\langle\delta, fg\rangle = \sum\langle\delta_i, f\rangle\cdot\langle\delta_i', g\rangle. \tag{1}$$

1.3.6. — Let $\mathfrak{X}$ be an A-group scheme and let e be the identity section of $\mathfrak{X}$; we set $\mathrm{Dist}\ \mathfrak{X}={}^e\mathrm{Dist}\ \mathfrak{X}$, etc. In view of 1.3.2 and 1.3.4, the product morphism $\pi:\mathfrak{X}\times\mathfrak{X}\to\mathfrak{X}$ defines a linear map from $\mathrm{Dist}\ \mathfrak{X}\otimes\mathrm{Dist}\ \mathfrak{X}$ into $\mathrm{Dist}\ \mathfrak{X}$, and one immediately verifies that $\mathrm{Dist}\ \mathfrak{X}$ is thus equipped with an associative A-algebra structure, not necessarily commutative, admitting e^* as identity element, compatible with the filtration of $\mathrm{Dist}\ \mathfrak{X}$ by the $\mathrm{Dist}_m\mathfrak{X}$. For $\delta,\delta'\in\mathrm{Dist}\ \mathfrak{X}$, we have

$$\delta\delta' = (\delta\otimes\delta')\circ\pi^*.$$

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We set $\mathrm{Lie}\ \mathfrak{X}=\mathrm{Dist}_1^+\mathfrak{X}$: it is known that $[\delta,\delta']=\delta\delta'-\delta'\delta\in\mathrm{Lie}\ \mathfrak{X}$ for all $\delta,\delta'\in\mathrm{Lie}\ \mathfrak{X}$ and that $\mathrm{Lie}\ \mathfrak{X}$ is thus equipped with a structure of A-Lie algebra ([DG], prop. 6.9, p. 230).

If $\mathfrak{X}$ is smooth, the canonical map from $\Gamma(\mathrm{Lie}\ \mathfrak{X})$ to $\mathrm{Gr}\ \mathrm{Dist}\ \mathfrak{X}$ (1.3.5) is an isomorphism of algebras. In particular, if $\mathrm{Lie}\ \mathfrak{X}$ is a free A-module and if $(\delta_1,\dots,\delta_n)$ is a basis for it, there exists a basis $(\delta_\alpha)_{\alpha\in\mathbb{N}^n}$ of $\mathrm{Dist}\ \mathfrak{X}$ such that the δ_α for $|\alpha|=m$ form a basis of $\mathrm{Dist}_m\mathfrak{X}$ modulo $\mathrm{Dist}_{m-1}\mathfrak{X}$ and that $\delta_\alpha\delta_\beta\equiv(\alpha,\beta)\delta_{\alpha+\beta}$ modulo $\mathrm{Dist}_{|\alpha|+|\beta|-1}\mathfrak{X}$ (with $(\alpha,\beta)=(\alpha+\beta)!/\alpha!\beta!$ and $|\alpha|=\alpha_1+\dots+\alpha_n$ (cf. [NB] A III, p. 155)). If moreover A is "of characteristic zero", that is to say, is a $\mathbb{Q}$-algebra, one can take $\delta_\alpha=(1/\alpha!)\delta_1^{\alpha_1}\dots\delta_n^{\alpha_n}$ and $\mathrm{Dist}\ \mathfrak{X}$ identifies with the enveloping algebra of $\mathrm{Lie}\ \mathfrak{X}$.

1.3.7. — Suppose that the generic fiber $\mathfrak{X}_K$ of $\mathfrak{X}$ is connected, reduced, and of finite type. Let I_K denote the ideal of $K[\mathfrak{X}]$ defining the identity element of $\mathfrak{X}_K$, i.e., the canonical image of $K\otimes I(e)$ in $K[\mathfrak{X}]$. Since $K[\mathfrak{X}]$ is Noetherian and integral, we then have $\bigcap_{n\geq 0}I_K^n=\{0\}$ ([NB] AC III, cor. to Krull's thm., p. 65). It follows immediately that if $f\in K[\mathfrak{X}]$ is such that $\delta(f)=0$ for all $\delta\in\mathrm{Dist}\ \mathfrak{X}_K$, then $f=0$.

If moreover $\mathfrak{X}$ is smooth, then $A[\mathfrak{X}]\subset K[\mathfrak{X}]$ and $\mathrm{Dist}\ \mathfrak{X}_K=K\otimes\mathrm{Dist}\ \mathfrak{X}$. Consequently, if $f\in A[\mathfrak{X}]$ is such that $\delta(f)=0$ for all $\delta\in\mathrm{Dist}\ \mathfrak{X}$, then $f=0$.

1.4. Linear Representations.

In this section, we denote by $\mathfrak{X}$ an A-group scheme with product π and identity section e, and we denote by M a projective A-module of finite type.

1.4.1. — The functor which to an A-algebra R associates the additive group $R \otimes M$ is representable by a smooth A-group scheme $\mathfrak{M}$, whose affine algebra is identified (with its bialgebra structure) with the symmetric algebra of the dual of M ([DG], p. 148). If N is a projective A-submodule of finite type of M, then $\mathfrak{N}$ is a closed subgroup scheme of $\mathfrak{M}$ if and only if the canonical map ${}^tM \to {}^tN$ is surjective, that is to say if and only if N is a direct summand in M.

In particular, to an invertible fractional ideal $\mathfrak{a}$ of A (i.e., a projective A-submodule of finite type of K) corresponds an A-group scheme denoted by $\mathfrak{Add}_{\mathfrak{a}}$, with generic fiber the additive group $\mathfrak{Add}_K$.

1.4.2. — Let N be a second projective A-module of finite type. The functor which to an A-algebra R associates the additive group $\mathrm{Hom}_R(R \otimes M, R \otimes N)$ is representable by a smooth A-group scheme $\mathfrak{L}(M, N)$ whose affine algebra identifies (with its bialgebra structure) with the symmetric algebra $\mathrm{Sym}({}^tN \otimes M)$ ([DG], p. 150). If $N = M$, then $\mathfrak{L}(M, M) = \mathfrak{L}(M)$ is an A-algebra scheme.

The linear group $\mathfrak{GL}(M)$ is the open subscheme of $\mathfrak{L}(M)$ induced on the open

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special $\mathfrak{L}(M)_{\det}$ of the points of $\mathfrak{L}(M)$ where the regular function $\det : g \to \det g$ on $\mathfrak{L}(M)$ does not vanish: if R is an A-algebra, we have $\mathfrak{GL}(M)(R) = GL(R \otimes M)$ ([DG], p. 150). It is a smooth group sub-A-scheme of the A-monoid scheme (under multiplication) $\mathfrak{L}(M)$. Transposition is an isomorphism of schemes from $\mathfrak{GL}(M)$ onto $\mathfrak{GL}({}^tM)$ which is an anti-isomorphism of groups.

It is clear that $\mathfrak{L}(M)$ acts on $\mathfrak{M}$ via the morphism φ from $\mathfrak{L}(M) \times \mathfrak{M}$ into $\mathfrak{M}$ defined by $\varphi(u, m) = u(m)$ for $u \in \mathrm{End}(R \otimes M)$ and $m \in R \otimes M$. We will also write $\varphi(u, m) = u.m$. We leave it to the reader to explicate the properties of associativity and bilinearity of this operation ($u.(v.m) = (uv).m$, $u.(m + m') = u.m + u.m'$, etc.). By restriction, we obtain the law of operation of $\mathfrak{GL}(M)$ on $\mathfrak{M}$.

1.4.3. — A linear representation ρ of $\mathfrak{X}$ in M is a homomorphism of group A-schemes from $\mathfrak{X}$ into $\mathfrak{GL}(M)$ (cf. [DG], p. 169). To it corresponds an "action of $\mathfrak{X}$ on $\mathfrak{M}$", i.e., a morphism of schemes $\tilde{\rho} : (x, m) \to \rho(x).m$ from $\mathfrak{X} \times \mathfrak{M}$ into $\mathfrak{M}$ such that the two morphisms $\tilde{\rho} \circ (\pi \times \mathrm{id})$ and $\tilde{\rho} \circ (\mathrm{id} \times \tilde{\rho})$ from $\mathfrak{X} \times \mathfrak{X} \times \mathfrak{M}$ into $\mathfrak{M}$ are equal: we have $\rho(xy).m = \rho(x).(\rho(y).m)$.

For every $m \in M = \mathfrak{M}(A)$, the map $x \to \rho(x).m$ is then a morphism of A-schemes from $\mathfrak{X}$ into $\mathfrak{M}$, and if $m' \in {}^tM \subset A[\mathfrak{M}]$, the map $x \to \langle m', \rho(x).m \rangle$ belongs to $A[\mathfrak{X}]$: we shall denote it by $c^{\rho}_{m',m}$ and we shall say that it is the *coefficient* with indices (m', m) of the representation ρ. It is clear that the map $(m', m) \to c^{\rho}_{m',m}$ from ${}^tM \times M$ into $A[\mathfrak{X}]$ is bilinear. To it therefore corresponds a linear map $c^{\rho} : M \to M \otimes A[\mathfrak{X}] = \mathrm{Hom}({}^tM, A[\mathfrak{X}])$ given by

$$(m' \otimes \mathrm{id})(c^{\rho}(m)) = c^{\rho}_{m',m} \quad (m' \in {}^tM, m \in M) \tag{1}$$

(and also a linear map ${}^tc^{\rho} : {}^tM \to {}^tM \otimes A[\mathfrak{X}]$ given by $(m \otimes \mathrm{id})({}^tc^{\rho}(m')) = c^{\rho}_{m',m}$).

We say that c^{ρ} (resp. ${}^tc^{\rho}$) is the coaction of $\mathfrak{X}$ on M (resp. ${}^tM$). One immediately verifies that c^{ρ} (resp. ${}^tc^{\rho}$) endows M (resp. ${}^tM$) with a structure of comodule over the coalgebra $A[\mathfrak{X}]$ (cf. [30] or [DG], p. 173 sq.). Conversely, if $\gamma : M \to M \otimes A[\mathfrak{X}]$ is a structure of comodule, there exists a unique linear representation ρ of $\mathfrak{X}$ in M such that $\gamma = c^{\rho}$ ([DG], p. 173 sq.).

We denote by $\det \rho$ the image $\rho^*(\det)$ of the element $\det \in A[\mathfrak{GL}(M)]$; it is an invertible element of $A[\mathfrak{X}]$.

1.4.4. — Let ρ be a linear representation of $\mathfrak{X}$ in M and let N be a direct factor submodule of M. We say that N is *invariant* under ρ if for every $n \in N$, the morphism $x \to \rho(x).n$ from $\mathfrak{X}$ into $\mathfrak{M}$ factors through a morphism from $\mathfrak{X}$ into $\mathfrak{N}$ followed by the canonical injection of $\mathfrak{N}$ into $\mathfrak{M}$. One easily verifies that N is invariant under ρ if and only if N is a sub-comodule of M, i.e., if and only if $c^{\rho}(N) \subset N \otimes A[\mathfrak{X}]$ (note that $N \otimes A[\mathfrak{X}] \subset M \otimes A[\mathfrak{X}]$ since N is a direct factor in M), or even if and only if $c^{\rho}_{m',n} = 0$ for all $n \in N$ and m' belonging to the orthogonal of N.

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in ${}^tM$. The restriction of ρ to N is then the representation of $\mathfrak{X}$ in N corresponding to the coaction $c^{\rho} : N \to N \otimes A[\mathfrak{X}]$.

1.4.5. — We say that a representation ρ of $\mathfrak{X}$ in M is *faithful* if the morphism $\rho : \mathfrak{X} \to \mathcal{GL}(M)$ is a closed immersion. For this to be so, it is necessary and sufficient that the coefficients $c^{\rho}_{m',m}$ and $(\det \rho)^{-1}$ generate the A-algebra $A[\mathfrak{X}]$. Indeed, the $m' \otimes m$ and $\det^{-1}$ generate the A-algebra $A[\mathcal{GL}(M)]$ and the condition means that $\rho^* : A[\mathcal{GL}(M)] \to A[\mathfrak{X}]$ is surjective.

If A is a Dedekind ring, *every flat A-group scheme (affine!) of finite type possesses a faithful linear representation.* Indeed, let $\mathfrak{X}$ be such a scheme, with product π and identity section e. There exists then a sub-module M of finite type of $A[\mathfrak{X}]$ (thus torsion-free, thus projective) generating the A-algebra $A[\mathfrak{X}]$ and such that $\pi^*(M) \subset M \otimes A[\mathfrak{X}]$ ([30], §1, prop. 2). The restriction of π^* to M defines an $A[\mathfrak{X}]$-comodule structure on M, thus a linear representation ρ of $\mathfrak{X}$ in M. As $(e^* \otimes \mathrm{id}) \circ \pi^* = \mathrm{id}$ on $A[\mathfrak{X}]$, we have, for every $m \in M$,

$$m = (e^* \otimes \mathrm{id}) \circ \pi^*(m) = (e^* \otimes \mathrm{id}) \circ c^{\rho}(m) = c^{\rho}_{e^*,m}$$

(denoting again by e^* the restriction of e^* to M). Consequently, the coefficients of ρ generate $A[\mathfrak{X}]$ and ρ is faithful.

1.4.6. — Let B be an A-algebra. From a representation ρ of $\mathfrak{X}$ in M, we deduce by base change a representation ρ_B of $\mathfrak{X}_B$ in $B \otimes M$: we will say that ρ_B *comes from* ρ or "descends to A" as ρ.

In particular, let us take $B = K$ and let $\tilde{\rho}$ be a representation of $\mathfrak{X}_K$ in a K-vector space of finite dimension V. Let M be a projective A-sub-module of V (automatically of finite type). We will say that $\mathfrak{X}$ *operates* (resp. *operates faithfully*) on M via $\tilde{\rho}$ if the vector subspace $W = KM$ of V is invariant under $\tilde{\rho}$ (1.4.4) and if the restriction $\tilde{\rho}_W$ of $\tilde{\rho}$ to W comes from a representation (resp. a faithful representation) ρ of $\mathfrak{X}$ in M.

Suppose that the scheme $\mathfrak{X}$ is *flat*. Then $M \otimes A[\mathfrak{X}]$ is identified with an A-sub-module of $W \otimes K[\mathfrak{X}]$ and we see immediately that $\mathfrak{X}$ operates on M if and only if

$$c^{\tilde{\rho}}(M) \subset M \otimes A[\mathfrak{X}]. \tag{1}$$

The representation ρ of $\mathfrak{X}$ in M from which $\tilde{\rho}_W$ comes is then *unique* and its coaction c^{ρ} is the restriction of $c^{\tilde{\rho}}$ to M.

Condition (1) can be translated in terms of coefficients: let us call the *polar* of M in the dual ${}^tV$ of the vector space V and denote by M^0 the set of $m' \in {}^tV$ such that $\langle m', m\rangle \in A$ for all $m \in M$ (note that if M generates V, then M^0 is identified with the dual ${}^tM$ of M); then M is the polar of M^0 in ${}^t({}^tV) = V$ and condition (1) is equivalent to

$$c^{\tilde{\rho}}_{m',m} \in A[\mathfrak{X}] \subset K[\mathfrak{X}] \quad \text{for all } m \in M \text{ and } m' \in M^0. \tag{2}$$

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1.4.7. — Let ρ be a representation of $\mathfrak{X}$ in M. For $\delta \in \operatorname{Dist} \mathfrak{X}$, we denote by $\rho(\delta)$ the endomorphism of M defined by

$$\langle m', \rho(\delta).m\rangle = \langle \delta, c^\circ_{m',m}\rangle \quad \text{for } m \in M \text{ and } m' \in {}^tM. \tag{1}$$

We also have

$$\langle m', \rho(\delta).m\rangle = \langle m' \otimes \delta, c^\circ(m)\rangle = \langle m \otimes \delta, {}^tc^\circ(m')\rangle, \tag{2}$$

or again

$$\rho(\delta) = (\mathrm{id} \otimes \delta) \circ c^\circ. \tag{3}$$

The map ρ is then a homomorphism of A-algebras from $\operatorname{Dist} \mathfrak{X}$ into $\operatorname{End} M$.

1.4.8. — Let ρ be a representation of $\mathfrak{X}$ in M; let $V = K \otimes M$ and suppose that the generic fiber $\mathfrak{X}_K$ is connected reduced, of finite type. For a vector subspace W of V to be invariant under ρ_K (1.4.4), it is necessary and sufficient that $\rho_K(\delta).w \in W$ for all $\delta \in \operatorname{Dist} \mathfrak{X}_K$ and all $w \in W$: the condition is trivially necessary and its sufficiency follows immediately from 1.3.7 and 1.4.4. We will see further on (3.5) generalizations of this result.

1.4.9. — An automorphism of the A-group scheme $\mathfrak{X}$ clearly defines an automorphism of the algebra $\operatorname{Dist} \mathfrak{X}$ (resp. of the A-module $\operatorname{Dist}_m\mathfrak{X}$ for $m \geq 0$). For example, if $\mathfrak{Z}$ is an A-group scheme and $(z, x) \mapsto z.x$ an operation of $\mathfrak{Z}$ on $\mathfrak{X}$ respecting the group structure of $\mathfrak{X}$ (cf. [DG], pp. 160 sq.), then the group $\mathfrak{Z}(A)$ of points of $\mathfrak{Z}$ with values in A acts by automorphisms on $\mathfrak{X}$, hence on the algebra $\operatorname{Dist} \mathfrak{X}$ and on each of the A-modules $\operatorname{Dist}_m\mathfrak{X}$. If moreover $\mathfrak{X}$ is smooth, then for every A-algebra R one has $\operatorname{Dist}_m\mathfrak{X}_R = R \otimes \operatorname{Dist}_m\mathfrak{X}$ (1.3.5) and what precedes provides a functorial homomorphism of $\mathfrak{Z}(R)$ into $\mathrm{GL}(R \otimes \operatorname{Dist}_m\mathfrak{X})$, that is to say a linear representation of $\mathfrak{Z}$ in the A-module (projective of finite type) $\operatorname{Dist}_m\mathfrak{X}$.

1.5. Restriction of scalars.

1.5.1. — Let B be an A-algebra, projective of finite type as an A-module[2]. It is known that A identifies via the map $a \mapsto a1$ with a free submodule direct summand in B (cf. [NB] AC II, §5, exer. 4, p. 176). We can therefore choose elements $e_1, \ldots, e_d \in B$ and $e_1^*, \ldots, e_d^* \in {}^tB$ such that $e_1 = 1$, $e_j^*(e_1) = 0$ for $j > 1$ and $\sum e_i \otimes e_i^* = \mathrm{id}$ in $B \otimes {}^tB$ identified with $\operatorname{End}_A B$. If M is an A-module, one then has

$$x = \sum e_i \otimes (e_i^* \otimes \mathrm{id})(x) \quad \text{for every } x \in B \otimes M. \tag{1}$$

1.5.2. — Let $\mathfrak{X}$ be a B-scheme (affine). It is known that there exists an A-scheme $\mathfrak{X}^\#$ and only one up to unique isomorphism such that the functors $R \mapsto \mathfrak{X}^\#(R)$ and $R \mapsto \mathfrak{X}(B \otimes R)$

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on the category of A-algebras are isomorphic (or also such that the functors $Y \mapsto \operatorname{Mor}(\mathfrak{Y}, \mathfrak{X}^\#)$ and $\mathfrak{Y} \mapsto \operatorname{Mor}(\mathfrak{Y}_B, \mathfrak{X})$ on the category of A-schemes are isomorphic): it is called the A-scheme obtained from $\mathfrak{X}$ by *restriction of scalars* from B to A and is denoted $\Pi_{B/A}\mathfrak{X}$ (or also $\mathfrak{X}^\#$ when A and B are obvious) ([DG], I, § 1, n° 6.6, p. 30 sq.). The uniqueness of $\mathfrak{X}^\#$ is obvious; an explicit construction of it will be given later (1.5.7).

1.5.3. — Let A' be an A-algebra and let $B' = A' \otimes B$. If $\mathfrak{X}$ is a B-scheme, the A'-scheme $\Pi_{B'/A'}\mathfrak{X}_{B'}$ is canonically identified with $(\Pi_{B/A}\mathfrak{X})_{A'}$.

If C is a projective B-algebra of finite type and $\mathfrak{X}$ a C-scheme, one has

$$\Pi_{C/A}\mathfrak{X} = \Pi_{B/A}(\Pi_{C/B}\mathfrak{X}).$$

1.5.4. — It is clear that the correspondence $\mathfrak{X} \mapsto \mathfrak{X}^\#$ is functorial and commutes with the product of schemes. If $f : \mathfrak{X} \to \mathfrak{Y}$ is a morphism of B-schemes, one denotes by $\Pi_{B/A}f$ or $f^\#$ the morphism corresponding to $\mathfrak{X}^\#$ in $\mathfrak{Y}^\#$. If $\mathfrak{X}$ is a B-group scheme, then $\mathfrak{X}^\#$ is an A-group scheme, etc.

[2] In what follows up to the 1st paragraph of 1.5.12 inclusive, it is not necessary to assume A is integral.

Let $f, g : \mathfrak{X} \to \mathfrak{Y}$ be two morphisms of B-schemes. If $f^\# = g^\#$, then $f = g$. Indeed, let R be a B-algebra; since B/A is projective, hence flat, the canonical map from R into $B \otimes_A R$ is injective ([NB] AC I, § 2, prop. 4). The hypothesis $f^\# = g^\#$ implies $\xi \circ f^* = \xi \circ g^*$ for all $\xi \in \mathfrak{X}(B \otimes_A R)$, hence *a fortiori* for all $\xi \in \mathfrak{X}(R)$, whence $f^* = g^*$ by taking $R = B[\mathfrak{X}]$ and $\xi = \mathrm{id}$.

1.5.5. — Let N be a projective B-module of finite type and let $N_{[A]}$ be the underlying A-module (which is also projective of finite type). If $\mathfrak{X} = \mathfrak{N}$ is the B-group scheme canonically associated to N (1.4.1), then $\mathfrak{X}^\#$ is the A-group scheme $\mathfrak{N}_{[A]}$ canonically associated to $N_{[A]}$.

One knows that $B[\mathfrak{X}] = \mathrm{Sym}_B(\mathrm{Hom}_B(N, B))$ and $A[\mathfrak{X}^\#] = \mathrm{Sym}_A({}^tN)$ (where ${}^tN$ denotes the dual of N *as an A-module*) (1.4.1). In particular, if $N = B \otimes M$, where M is a projective A-module of finite type, one has $\mathrm{Hom}_B(N, B) = B \otimes {}^tM$ and ${}^tN = {}^tB \otimes {}^tM$, whence

$$B[\mathfrak{X}] = \mathrm{Sym}_B(B \otimes {}^tM) = B \otimes \mathrm{Sym}_A{}^tM,$$
$$A[\mathfrak{X}^\#] = \mathrm{Sym}_A({}^tB \otimes {}^tM).$$

1.5.6. — More generally, let E be an (arbitrary) A-module. Let $\mathfrak{G}$ be the B-scheme $\mathrm{Spec}(B \otimes \mathrm{Sym}_A E)$ and let $\mathfrak{G}'$ be the A-scheme $\mathrm{Spec}\ \mathrm{Sym}_A({}^tB \otimes E)$. One then has $\mathfrak{G}^\# = \mathfrak{G}'$ ([DG], *loc. cit.*). Indeed, if R is an A-algebra,

$$\mathfrak{G}(B \otimes R) = \mathrm{Hom}_{B\text{-alg}}(\mathrm{Sym}_B(B \otimes E), B \otimes R)$$

is identified with $\mathrm{Hom}_B(B \otimes E, B \otimes R)$ and $\mathfrak{G}'(R) = \mathrm{Hom}_{A\text{-alg}}(\mathrm{Sym}_A({}^tB \otimes E), R)$ is identified with $\mathrm{Hom}_A({}^tB \otimes E, R)$. Since B is a projective A-module of finite type, these two

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A-modules are functorially isomorphic by the map which to $\varphi \in \mathrm{Hom}_A({}^tB \otimes E, R)$ associates the element $\widetilde{\varphi} \in \mathrm{Hom}_B(B \otimes E, B \otimes R)$ given by

$$\widetilde{\varphi}(b \otimes x) = \sum be_i \otimes \varphi(e_i^* \otimes x) \quad (b \in B, x \in E) \tag{1}$$

(cf. [NB] A II, p. 77 (12)[3]).

One can also say this: let us denote by

$$j : \mathrm{Sym}_B(B \otimes E) \to B \otimes \mathrm{Sym}_A({}^tB \otimes E)$$

the unique B-algebra homomorphism which extends the linear map from $B \otimes E$ into $B \otimes {}^tB \otimes E$ given by

$$j(b \otimes x) = \sum be_i \otimes e_i^* \otimes x \quad (b \in B, x \in E).$$

Then the bijection $\varphi \mapsto \widetilde{\varphi}$ from $\mathfrak{G}^\#(R) = \mathrm{Hom}_{A\text{-alg}}(\mathrm{Sym}_A({}^tB \otimes E), R)$ to $\mathfrak{G}(B \otimes R) = \mathrm{Hom}_{B\text{-alg}}(\mathrm{Sym}_B(B \otimes E), B \otimes R)$ is given by

$$\widetilde{\varphi} = (\mathrm{id} \otimes \varphi) \circ j, \tag{2}$$

since the two sides of (2) coincide on $B \otimes E$, in view of (1).

1.5.7. — Let $\mathfrak{X}$ be a B-scheme. One can always write $B[\mathfrak{X}]$ as a quotient of a free commutative algebra, that is to say of the form $\mathrm{Sym}_B(B \otimes E)$ where E is a free A-module, by an ideal I. Then, the algebra $A[\mathfrak{X}^\#]$ is identified with the quotient of $\mathrm{Sym}_A({}^tB \otimes E)$ by the ideal $I^\#$ generated by the elements $(u \otimes \mathrm{id}) \circ j(x)$ for $u \in {}^tB$ and $x \in I$ (which, moreover, demonstrates the existence of $\mathfrak{X}^\#$). In view of 1.5.6, whose notation we adopt, it suffices to show that if $\varphi \in \mathfrak{G}'(R)$, then $\widetilde{\varphi}$ is zero on I if and only if φ is zero on $I^\#$. Now $\widetilde{\varphi}$ zero on I is equivalent to $\mathrm{Ker}(\mathrm{id} \otimes \varphi) = B \otimes \mathrm{Ker}\ \varphi \supset j(I)$, and it then follows from 1.5.1 (1) that $I^\#$ is the smallest ideal of $\mathrm{Sym}_A({}^tB \otimes E)$ such that $B \otimes I^\# \supset j(I)$, hence our assertion.

1.5.8. — If $\mathfrak{X}$ is of finite type (resp. of finite presentation), one can take E to be free of finite type (resp. and I of finite type), which immediately shows that $\mathfrak{X}^\#$ is of finite type (resp. of finite presentation).

[3] N. Bourbaki only treats the case of free modules of finite type here; the extension to the projective case is immediate. The same remark applies to the references to N. Bourbaki's algebra book given further on.

On the other hand, note that $\mathfrak{X}$ flat does not necessarily imply $\mathfrak{X}^{\#}$ flat, even if A is a complete discrete valuation ring. However, we know that $\mathfrak{X}$ smooth implies $\mathfrak{X}^{\#}$ smooth ([DG], I, § 4, cor. 4.8, p. 112).

1.5.9. — The map $I \mapsto I^{\#}$ from 1.5.7 is obviously increasing. It follows that if $\mathfrak{Y}$ is a closed B-subscheme of $\mathfrak{X}$, then $\mathfrak{Y}^{\#}$ is identified with a closed A-subscheme of $\mathfrak{X}^{\#}$.

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1.5.10. — Suppose for example that B is a free A-module with basis (e_i) ((e_i^*) is then its dual basis) and that $B[\mathfrak{X}] = B[X_1, \ldots, X_n]/I$, where I is the ideal generated by the polynomials $P_1, \ldots, P_q$. Let us take for E the free A-module with basis $X_1, \ldots, X_n$: we have $\mathrm{Sym}_B(B \otimes E) = B[X_1, \ldots, X_n]$ and $\mathrm{Sym}_A({}^tB \otimes E)$ is the polynomial algebra in the indeterminates $Y_{i,j} = e_i^* \otimes X_j$ for $1 \leq i \leq d$ and $1 \leq j \leq n$. One checks immediately that if $P \in B[X_1, \ldots, X_n]$, then $j(P)$ is the polynomial obtained by substituting $\sum_i e_i Y_{i,j}$ for X_j in P for $1 \leq j \leq n$ and is written in one and only one way in the form $\sum_k e_k j_k(P)$, with $j_k(P) \in A[(Y_{i,j})]$. It is then clear that the ideal $I^{\#}$ is generated by the polynomials $j_k(P_l)$ for $1 \leq k \leq d$ and $1 \leq l \leq q$:

$$A[\mathfrak{X}^{\#}] = A[(Y_{i,j})]/((j_k(P_l))).$$

1.5.11. — Let $\mathfrak{Y}$ be an A-scheme. The identity map of $\mathfrak{Y}_B$ gives a morphism i, called canonical, from $\mathfrak{Y}$ to $(\mathfrak{Y}_B)^{\#}$ (cf. 1.5.2); let us show that it is a closed immersion. Let us first take the case $\mathfrak{Y} = \mathrm{Spec}\ \mathrm{Sym}_A E$ (with the notation of 1.5.6). Let ε be the A-algebra homomorphism from $\mathrm{Sym}_A({}^tB \otimes E)$ to $\mathrm{Sym}_A E$ defined by $\varepsilon(u \otimes x) = u(1)x$ for $u \in {}^tB$ and $x \in E$; it is surjective since $e_1^*(1) = 1$. It suffices then to show that $i = \mathrm{Spec}\ \varepsilon$, or even that $\tilde{j} = (\mathrm{id} \otimes \varepsilon) \circ j$ is the identity of $B \otimes \mathrm{Sym}_A E$, which is immediate:

$$\tilde{j}(b \otimes x) = (\mathrm{id} \otimes \varepsilon)\left(\sum be_i \otimes e_i^* \otimes x\right) = \sum be_i \otimes e_i^*(1)x = b \otimes x.$$

In the general case, one can write $\mathfrak{Y}$ as a closed subscheme of an A-scheme $\mathfrak{Z}$ of the form $\mathrm{Spec}\ \mathrm{Sym}_A E$ and our assertion follows immediately from the commutative diagram

$$\begin{array}{ccc} \mathfrak{Y} & \longrightarrow & \mathfrak{Z} \\ \downarrow & & \downarrow \\ (\mathfrak{Y}_B)^{\#} & \longrightarrow & (\mathfrak{Z}_B)^{\#} \end{array}$$

Let for example M be a projective A-module of finite type and take $\mathfrak{Y} = \mathfrak{M}$: we saw in 1.5.5 that $(\mathfrak{Y}_B)^{\#}$ is identified with the A-scheme associated with the A-module $B \otimes M$ and it is immediate that i is the morphism associated with the canonical injection $m \to 1 \otimes m$ of M into $B \otimes M$.

1.5.12. — If $\mathfrak{U}$ is an open subscheme of $\mathfrak{X}$, then $\mathfrak{U}^{\#}$ is identified with an open subscheme of $\mathfrak{X}^{\#}$: if I is an ideal of $B[\mathfrak{X}]$ defining $\mathfrak{U}$ (i.e. such that $\mathfrak{U}$ is the set of points of $\mathfrak{X}$ where at least one element $f \in I$ does not vanish), then $\mathfrak{U}^{\#}$ is the open subset of $\mathfrak{X}^{\#}$ defined by the ideal J formed of $a \in A[\mathfrak{X}^{\#}]$ such that $1 \otimes a$ belongs to the ideal of $B \otimes A[\mathfrak{X}^{\#}]$ generated by $j(I)$ ([DG], pp. 31-32). We note that $J \subset I^{\#}$, but the closed subschemes of $\mathfrak{X}^{\#}$ defined by J and $I^{\#}$ are in general distinct and do not even have the same underlying space.

Suppose I is principal: in other words, $\mathfrak{U}$ is the special open subset of $\mathfrak{X}$ defined by a

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function $f \in B[\mathfrak{X}]$. Let $\varphi \in \mathfrak{X}^{\#}(R)$ and let $\widetilde{\varphi}$ be the corresponding element of $\mathfrak{X}(B \otimes R)$; then φ comes from a point of $\mathfrak{U}^{\#}(R)$ if and only if $\widetilde{\varphi}(f)$ is an invertible element of $B \otimes R$, or also if $\mathrm{Norm}_{B \otimes R/R}\, \widetilde{\varphi}(f)$[4] is invertible in R. It follows that $\mathfrak{U}^{\#}$ is the special open subset of $\mathfrak{X}^{\#}$ defined by the norm in $A[\mathfrak{X}^{\#}]$ of the element $j(f) \in B \otimes A[\mathfrak{X}^{\#}]$.

[4]Since A is an integral domain, B has a rank r ([NB] AC II, p. 141) and $B \otimes R$ is a projective R-module of rank r. One can therefore consider the determinant of an R-endomorphism of $B \otimes R$ ([NB] AC II, § 5, exerc. 9, p. 177) and the norm $\mathrm{Norm}_{B \otimes R/R}\, x$ of an element $x \in B \otimes R$ is the determinant of the multiplication by x. It is given by the value at x of the map from $B \otimes R$ to R defined by a homogeneous element $\mathrm{Norm}_{B \otimes R/R}$ of degree r of $\mathrm{Sym}_R(R \otimes {}^tB) = R \otimes \mathrm{Sym}_A\, {}^tB$, depending functorially on R: we have $\mathrm{Norm}_{B \otimes R/R} = 1 \otimes \mathrm{Norm}_{B/A}$.

1.5.13. — **Examples:** 1) Let $\mathfrak{X} = \mathfrak{Add}$ and $\mathfrak{U} = \mathfrak{Mult}$; taking for f the identity function, we see that

$$A[\mathfrak{Mult}^{\#}] = (\mathrm{Sym}_A {}^t B)[1/\mathrm{Norm}_{B/A}].$$

Let $i : \mathfrak{Mult}_A \to \Pi_{B/A}\mathfrak{Mult}_B$ be the canonical closed immersion (1.5.11). It is a morphism of group schemes and one verifies immediately (e.g., by considering $\mathfrak{Mult}$ as an open subset of $\mathfrak{Add}$ and applying the last lines of 1.5.11) that for every A-algebra R, the homomorphism $i(R)$ is nothing other than the canonical injection of $R^\times$ into $(B \otimes R)^\times$.

2) Let N be a projective B-module of rank r. Let us take $\mathfrak{X} = \mathfrak{L}(N)$ (1.4.2): in view of 1.5.5, $\mathfrak{X}^{\#}$ is the A-scheme canonically associated with the A-module $\mathrm{End}_B N$. Now the latter is a direct factor submodule in $\mathrm{End}_A N$[5]. Consequently, $\mathfrak{L}(N)^{\#}$ is identified with a closed group subscheme (and even a sub-algebra) of $\mathfrak{L}(N_{[A]})$.

Now let us take $\mathfrak{U} = \mathfrak{GL}(N)$: it is the special open subset of $\mathfrak{L}(N)$ defined by the function $\det \in B[\mathfrak{X}]$. Consequently, $\mathfrak{U}^{\#}$ is the special open subset of $\mathfrak{L}(N)^{\#}$ defined by the function Norm(det), which is none other than the restriction to $\mathfrak{L}(N)^{\#}$ of the function det on $\mathfrak{L}(N_{[A]})$ (cf. [NB] A III, p. 112). In other words, $\mathfrak{GL}(N)^{\#} = \mathfrak{L}(N)^{\#} \cap \mathfrak{GL}(N_{[A]})$ and $\mathfrak{GL}(N)^{\#}$ is identified with a closed group subscheme of $\mathfrak{GL}(N_{[A]})$.

In particular, if $\mathfrak{G}$ is a B-group scheme and if ρ is a linear representation of $\mathfrak{G}$ in N, then $\rho^{\#}$ is a linear representation of $\mathfrak{G}^{\#}$ in $N_{[A]}$; if ρ is faithful, the same holds for $\rho^{\#}$.

1.5.14. — Let L be a finite extension of degree d of the field K. Everything that precedes applies of course to the restriction of scalars from L to K of an L-scheme.

Suppose L is separable over K and let K' be an extension of K. It is known ([NB] A VIII, § 7, cor. 1 to th. 1, p. 86) that $L' = K' \otimes_K L$ is a product $\prod_{i \in I} L_i$, where the L_i form a system of representatives of the isomorphism classes of the extensions composed of L and K'. If Ω is an algebraically closed extension of K, of transcendence degree...

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[5] $\mathrm{End}_A N/\mathrm{End}_B N$ is an A-module of finite presentation as a quotient of a module of finite presentation by a finitely generated submodule. Taking into account ([NB] AC II, § 3, cor. 1 to prop. 12, p. 114), we can thus assume that A is local. Then B is semi-local ([NB] AC V, § 2, prop. 3, p. 40) and N, having a rank, is a free B-module ([NB] AC II, § 5, prop. 5, p. 143). If (n_k) is a basis of N over B, the set of A-endomorphisms of N that vanish on the n_k is a complement of $\mathrm{End}_B N$ in $\mathrm{End}_A N$.

greater than that of K', the L_i correspond bijectively to pairs of K-isomorphisms of L and K' into Ω taken up to a K-automorphism of Ω. If one chooses such a pair and if one identifies L and K' with their images in Ω via this pair, one easily sees that one can take for I the set of double classes $\mathrm{Gal}(\Omega/K')\backslash\mathrm{Gal}(\Omega/K)/\mathrm{Gal}(\Omega/L)$ and take $L_i = K'(\sigma_i(L))$, where σ_i is an element of the double class i. The isomorphism $j : K' \otimes L \to \prod L_i$ is then given by

$$j(x \otimes y) = (x\sigma_i(y))_{i\in I} \quad (x \in K', y \in L).$$

If K' is a quasi-Galois (also called "normal") extension of K containing L, then $L_i = K'$ for all $i \in I$ and $I = \mathrm{Gal}(\Omega/K)/\mathrm{Gal}(\Omega/L)$ is identified with $\mathrm{Gal}(K'/K)/\mathrm{Gal}(K'/L)$, hence $\mathrm{Card}\, I = d$ and j is an isomorphism of $K'\otimes L$ onto K'^d. The Galois group $\mathrm{Gal}(K'/K)$ operates on $K'\otimes L$ by $(\gamma, x\otimes y) \mapsto \gamma(x)\otimes y$ and if one transfers this operation to K'^d via j, one easily verifies that

$$\gamma \cdot (\xi_i)_{i\in I} = (\eta_i)_{i\in I} \quad \text{with } \eta_i = \gamma(\xi_{\gamma^{-1}\cdot i}). \tag{1}$$

1.5.15. — Let now $\mathfrak{X}$ be an L-scheme and $\mathfrak{X}^{\#} = \Pi_{L/K}\mathfrak{X}$. We have

$$(\Pi_{L/K}\mathfrak{X})_{K'} = \prod_{i\in I}(\Pi_{L_i/K'}\mathfrak{X}_{L_i}),$$

where, of course, L_i is considered as an extension of L by means of $\sigma_i : L \to L_i$.

If K' is quasi-Galois and contains L, we have $L_i = K'$ and $\Pi_{L_i/K'}\mathfrak{X}_{L_i} = \mathfrak{X}_{L_i}$ for all $i \in I$. Every $\gamma \in \mathrm{Gal}(K'/K)$ provides an isomorphism of $\mathfrak{X}_{L_i}$ onto $\mathfrak{X}_{L_{\gamma\cdot i}}$, still denoted by γ, and the natural operation of γ on $(\Pi_{L/K}\mathfrak{X})_{K'}$ transferred to $\prod_{i\in I}\mathfrak{X}_{L_i}$ is again given by 1.5.14 (1).

1.5.16. — Conversely, let $\mathfrak{Y}$ be a K-scheme and let K' be a Galois extension of K such that $\mathfrak{Y}_{K'}$ decomposes into a product $\mathfrak{Y}_0 \dots \mathfrak{Y}_{d-1}$ of K'-schemes permuted transitively by the natural action of $\mathrm{Gal}(K'/K)$. Let L be the field of invariants of the stabilizer of $\mathfrak{Y}_0$ in $\mathrm{Gal}(K'/K)$. One sees by Galois descent that $\mathfrak{Y}_0$ arises by base extension from L to K' of an L-scheme $\mathfrak{X}$ and one easily verifies that $\mathfrak{Y}_{K'}$ equipped with the action of $\mathrm{Gal}(K'/K)$ is identified with $(\Pi_{L/K}\mathfrak{X})_{K'}$ and consequently that $\mathfrak{Y}$ is isomorphic to $\Pi_{L/K}\mathfrak{X}$.

1.5.17. — Let us take for $\mathfrak{X}$ the L-group scheme $\mathfrak{Mult}$ and set $T = \Pi_{L/K}\mathfrak{Mult}$. If K' is a Galois extension of K containing L, then 1.5.15 shows that $T_{K'}$ is isomorphic to $\mathfrak{Mult}^d$, hence T is a torus, which splits over K'. The group $X(T)$ of characters of T is isomorphic to $\mathbf{Z}^d$ and admits a basis (χ_i), indexed by $I = \mathrm{Gal}(K'/K)/\mathrm{Gal}(K'/L)$, stable under the natural action of $\mathrm{Gal}(K'/K)$ in $X(T)$ and such that $\gamma \cdot \chi_i = \chi_{\gamma\cdot i}$. One sees that L_i is the field of definition of the character χ_i, that the smallest extension of K splitting T is the Galois extension of K generated by L and that T is split over K if and only if $L = K$. Finally, the characters of T rational over K are the integer multiples of the character $\sum \chi_i$, which is none other than the func-

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tion $N = \mathrm{Norm}_{L/K}$ on T (cf. 1.5.13): indeed, if R' is a K'-algebra, and if $x = (x_i) \in T(R') = \prod_{i\in I}(L_i \otimes_{K'} R')^{\times}$, we have

$$N(x) = \mathrm{Norm}_{L\otimes_{K'}R'/R'}x = \prod_{i\in I}\mathrm{Norm}_{L_i\otimes_{K'}R'/R'}x_i = \prod_{i\in I}x_i = \left(\sum \chi_i\right)(x).$$

It follows that the maximal K-split subtorus S of T is of dimension 1 and therefore coincides with the canonical image of $\mathfrak{Mult}$ (1.5.11).

Conversely, if T is a torus defined over K such that the Galois group $\mathrm{Gal}(\Omega/K)$ transitively permutes a basis $\chi_0, \dots, \chi_{d-1}$ of the character group $X(T)$ of T, and if L is the field of definition of χ_0, then T is K-isomorphic to $\prod_{L/K}\mathfrak{Mult}$: this follows from 1.5.16 or from the fact that the character groups $X^*(T)$ and $X^*(\prod_{L/K}\mathfrak{Mult})$ are isomorphic $\mathrm{Gal}(\Omega/K)$-modules.

More generally, we shall say that a torus T defined over K is an *induced torus* if there exists a basis Ξ of the character group $X(T)$ which is stable under the natural action of $\mathrm{Gal}(\Omega/K)$. Then $X(T)$ decomposes into a direct product of subgroups generated by the different orbits of $\mathrm{Gal}(\Omega/K)$ in Ξ, and T is isomorphic to a direct product of tori of the form $\prod_{L/K}\mathfrak{Mult}$ for finite-degree separable extensions of K. The converse is obvious.

1.6. Admissible and étale extensions.

All valuations considered in the following are real-valued. If L is a field equipped with a non-trivial[6] valuation, we denote by $\mathcal{O}_L$ the ring of this valuation, $\mathfrak{m}_L$ its maximal ideal, and $\bar{L}$ its residue field. In this no. 1.6, we assume K is equipped with a non-trivial valuation ω and we set $\mathcal{O} = \mathcal{O}_K$, $\mathfrak{m} = \mathfrak{m}_K$.

If L is an extension of K, and if ω_L is a valuation of L extending ω, we denote by $f = f(\omega_L/\omega)$ the residue degree of ω_L, i.e., the degree of $\bar{L}$ over $\bar{K}$, and by $e = e(\omega_L/\omega)$ the ramification index of ω_L, i.e., the index of the subgroup $\omega(K^\times)$ in $\omega_L(L^\times)$. We always have $ef \leq [L : K]$. If $e = 1$, we say that ω_L (or L) is *unramified*.

Definition 0.1 (1.6.1)**.** Let L be an extension of K. We say that L is *univalent* if there exists a unique valuation ω_L of L extending ω. We say that L is *admissible* if it is finite-degree separable, univalent, and if the ring $\mathcal{O}_L$ of ω_L is a free $\mathcal{O}$-module of rank $[L : K]$.

Suppose L is separable of degree n; the following facts are easy consequences of well-known theorems (cf. [NB] AC VI, § 8, *passim*, notably Cor. 1 to Prop. 1, Thm. 2 and its corollaries, Exer. 6):

a) L is univalent if and only if the integral closure of $\mathcal{O}$ in L is a local ring, then equal to $\mathcal{O}_L$. If K is Henselian, in particular if K is complete, then every algebraic extension of K is univalent.

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b) Suppose L is univalent; then L is admissible if and only if $\mathcal{O}_L$ is a free $\mathcal{O}$-module (resp. of finite type). If ω is discrete, every separable extension of finite degree that is univalent is admissible.

c) If ω is discrete, L is univalent, hence admissible, if and only if $ef = n$; if L is Galois and ω_L is invariant by $\mathrm{Gal}(L/K)$, then L is admissible.

If ω is not discrete, then L is admissible if and only if $f = n$, which implies $e = 1$.

d) Suppose L is admissible. Let π be a uniformizer of L if ω is discrete, and $\pi = 1$ if ω is not discrete. Let $(x_i)_{1\leq i\leq f}$ be elements of $\mathcal{O}_L$ such that their images $\bar{x}_i$ form a basis of $\bar{L}$ over $\bar{K}$. Then the elements $x_i\pi^j$ for $1 \leq i \leq f$ and $0 \leq j < e$ form a basis of $\mathcal{O}_L$ over $\mathcal{O}$.

e) Let $x \in L$ such that $L = K(x)$; if the minimal polynomial of x over K has coefficients in $\mathcal{O}$ and if its image in $\bar{K}[X]$ remains irreducible, then L is admissible unramified: indeed $f \geq n$ for every valuation ω_L extending ω.

f) If ω is discrete and K is Henselian, every separable extension of finite degree of K is admissible.

Definition 0.2 (1.6.2)**.** An extension K' of finite degree of K is said to be *étale* if it is admissible, unramified and if its residue field $\bar{K}'$ is a separable extension of $\bar{K}$. An arbitrary extension of K is said to be étale if it is a union of étale extensions of finite degree.

An étale extension is separable, univalent, unramified, with a separable residue field over $\bar{K}$. If K' is a Galois étale extension of K, then $\bar{K}'$ is Galois and the natural map is an isomorphism from $\mathrm{Gal}(K'/K)$ onto $\mathrm{Gal}(\bar{K}'/\bar{K})$. On the other hand, $\bar{K}'$ can be Galois without K' being so (see however 1.6.7 *a*)).

1.6.3. — Let L be an extension of finite degree of K and let L' be a sub-extension of L. For L to be univalent (resp. univalent unramified, resp. admissible, resp. étale), it is necessary and sufficient that L' be so and that L be a univalent (resp. ...) extension of L' (equipped with the unique valuation extending ω).

Proposition 0.1 (1.6.4)**.** *An extension K' of finite degree n of K is étale if and only if there exists $x \in K'$ such that $K' = K(x)$, the minimal polynomial of x over K has coefficients in $\mathcal{O}$ and the image of this polynomial in $\bar{K}[X]$ is irreducible and separable.*

[6] [[UNCERTAIN: non-trivial — CONFIDENCE: 95% — REASON: In the context of valuations, "impropre" refers to the trivial valuation.]]

Proof. That the condition is sufficient follows immediately from 1.6.1 *e*). Conversely, if K' is étale, one can choose $x \in \mathcal{O}_{K'}$ whose canonical image $\bar{x}$ in $\bar{K}'$ generates $\bar{K}'$ over $\bar{K}$. Let P be the minimal polynomial of x over K: we have $\deg P \leq n$, $P \in \mathcal{O}[X]$ since x is integral over $\mathcal{O}$, $\bar{P}(\bar{x}) = 0$ hence $\deg \bar{P} \geq n$. Consequently $\deg P = n$, $K' = K(x)$ and P is separable because $\bar{P}$ is the minimal polynomial of $\bar{x}$ over $\bar{K}$ and is thus separable. □

Corollary 0.2 (1.6.5)**.** *Let k' be a separable algebraic extension of $\bar{K}$. There exists an étale extension K' of K whose residual extension is isomorphic to k'.*

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One immediately reduces to the case where k' is of finite degree (1.6.3). It then suffices to take $K' = K[X]/(P)$, where P is a unitary polynomial with coefficients in $\mathcal{O}$ whose image in $\overline{K}[X]$ is the minimal polynomial of a generator of k' over $\overline{K}$.

1.6.6. Corollary. — *Every étale extension is contained in a maximal étale extension. The residue field of a maximal étale extension is a separable closure of $\overline{K}$.*

1.6.7. — Suppose K is Henselian. Then every algebraic extension of K is univalent and Henselian.

a) Let L be an étale extension of K: it is known that the canonical map from the group of K-automorphisms of L to the group of $\overline{K}$-automorphisms of $\overline{L}$ is bijective. In particular, L is Galois if and only if $\overline{L}$ is Galois and one then has $\mathrm{Gal}(L/K) = \mathrm{Gal}(\overline{L}/\overline{K})$.

b) Let L be an admissible extension of K and let $\overline{L}_0$ be the separable closure of $\overline{K}$ in $\overline{L}$. There exists a largest étale sub-extension $L_{\text{ét}}$ of L and its residue field is $\overline{L}_0$: one obtains it for example by considering a generator $\overline{x}$ of $\overline{L}_0$ over $\overline{K}$, its minimal polynomial $\overline{P}$, and by choosing a unitary polynomial $P \in \mathcal{O}[X]$ with canonical image $\overline{P}$ in $\overline{K}[X]$. Since $\overline{P}$ is separable, P has a root x in $\mathcal{O}_L$ with image $\overline{x}$ and one has $L_{\text{ét}} = K(x)$.

If L is Galois, then $L_{\text{ét}}$ is also Galois, $\overline{L}_0$ is Galois, $\overline{L}$ is quasi-Galois and the canonical map from $\mathrm{Gal}(L/K)$ to $\mathrm{Gal}(\overline{L}/\overline{K}) = \mathrm{Gal}(\overline{L}_0/\overline{K})$ is surjective.

1.6.8. Proposition. — *Let Ω be an algebraically closed extension of K, let L be an admissible sub-extension and let K' be a sub-extension equipped with a valuation ω' extending ω, unramified.*

(i) Suppose that L is Galois, that $L \cap K'$ is an étale Galois extension of K and that the residue fields $\overline{L}$ and $\overline{K}'$ are linearly disjoint extensions of $\overline{L \cap K'}$ [7]*. Let $I = Gal(L \cap K'/K)$ and, for every $i \in I$, let $\sigma_i \in Gal(L/K)$ be an extension of i. Then*

$$K' \otimes_K L = \prod_{i \in I} L'_i$$

where $L'_i = L' = K'(L)$ and where the projection $p_i : K' \otimes L \to L'_i$ is given by $p_i(x \otimes y) = x\sigma_i(y)$ for $x \in K'$ and $y \in L$. Moreover, L' is an admissible extension of K' and $p = (p_i)$ induces an isomorphism

$$\mathcal{O}_{K'} \otimes_{\mathcal{O}} \mathcal{O}_L \xrightarrow{\sim} \prod_{i \in I} \mathcal{O}_{L'_i}.$$

(ii) Suppose $\overline{L}$ and $\overline{K}'$ are linearly disjoint over $\overline{K}$. Then L and K' are linearly disjoint over K, $L' = K' \otimes_K L$ is an admissible extension of K', one has $\mathcal{O}_{L'} = \mathcal{O}_{K'} \otimes_{\mathcal{O}} \mathcal{O}_L$ and $\omega(L') = \omega(L)$.

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[7] We say that two extensions M and N of a field E, at least one of which is of finite degree, are linearly disjoint over E if $M \otimes_E N$ is a field. This is equivalent to saying that $u(M)$ and $v(N)$ are linearly disjoint sub-extensions of F (in the classical sense) for every triple or, which amounts to the same thing, for at least one triple (F, u, v), where F is an extension of E and $u : M \to F$ and $v : N \to F$ are isomorphisms onto sub-extensions of F (cf. [NB] A V, §2, no 5).

Let us prove (i). The assertion relative to the decomposition of $K' \otimes L$ has already been seen (1.5.14), taking into account that L' is a Galois extension of K' and that

$$\begin{aligned}\mathrm{Gal}(\Omega/L)\backslash\mathrm{Gal}(\Omega/K)/\mathrm{Gal}(\Omega/K') &= \mathrm{Gal}(\Omega/K)/(\mathrm{Gal}(\Omega/L)\cdot\mathrm{Gal}(\Omega/K')) \\ &= \mathrm{Gal}(L/K)/\mathrm{Gal}(L/L\cap K') = \mathrm{Gal}(L\cap K'/K) = I\end{aligned}$$

(the second equality resulting from the fact that the image of $\mathrm{Gal}(\Omega/K')$ in $\mathrm{Gal}(L/K)$ is equal to $\mathrm{Gal}(L/L\cap K')$ (cf. [NB] A V, p. 68)). We have

$$\mathrm{Card}\ I = [L\cap K' : K]. \tag{1}$$

Let us choose a valuation ω'' of $L' = K'(L)$ extending ω'. The restriction of ω'' to L is ω_L. Consequently, the residue field of L' contains $\overline{L}$ and $\overline{K}'$, and the hypothesis of linear disjointness implies

$$f(\omega''/\omega') \geq [\overline{L} : \overline{L\cap K'}]. \tag{2}$$

Since K' is unramified, we also have

$$e(\omega''/\omega') \geq e(\omega_L/\omega) = e(\omega_L/\omega_{L\cap K'}). \tag{3}$$

On the other hand, L is an admissible extension of $L\cap K'$, itself an admissible extension of K (whence $\omega' = \omega_L$ on $L\cap K'$) (1.6.3), thus

$$[L : L\cap K'] = [\overline{L} : \overline{L\cap K'}]e(\omega_L/\omega). \tag{4}$$

Finally, we have

$$[L' : K'] \geq e(\omega''/\omega')f(\omega''/\omega'). \tag{5}$$

Using all these inequalities, we deduce:

$$\begin{aligned}[K'\otimes L : K'] &= \sum_{i\in I}[L'_i : K'] \geq (\mathrm{Card}\ I)e(\omega''/\omega')f(\omega''/\omega') \\ &\geq [L\cap K' : K]e(\omega_L/\omega)[\overline{L} : \overline{L\cap K'}] \\ &\geq [L\cap K' : K][L : L\cap K'] = [L : K] = [K'\otimes L : K'].\end{aligned} \tag{6}$$

Consequently, the inequalities (2), (3), (5), (6) are equalities, and it then follows from equality (5) that L' is an admissible extension, taking into account that if ω is not discrete, equality (3) implies that $e(\omega''/\omega') = e(\omega_L/\omega) = 1$. Furthermore, equality (2) shows that $[\overline{L}' : \overline{K}'] = [\overline{L} : \overline{L\cap K'}]$, whence

$$\overline{L}' = \overline{K}' \otimes_{\overline{L\cap K'}} \overline{L}. \tag{7}$$

Finally, let $(x_j)_{j\in I}$ (resp. $(y_k)_{1\leq k\leq f'}$) be elements of $\mathcal{O}_{L\cap K'}$ (resp. $\mathcal{O}_L$) whose images form a basis of $\overline{L\cap K'}$ over $\overline{K}$ (resp. of $\overline{L}$ over $\overline{L\cap K'}$), let π be a uniformizer of L if ω is discrete and set $\pi = 1$ if ω is not discrete. The $x_j y_k \pi^l$ for $j\in I$, $1\leq k\leq f'$ and $0\leq l < e = e(\omega_L/\omega)$ form a basis of $\mathcal{O}_L$ over $\mathcal{O}_K$ and, in view of (7), the $\sigma_i(y_k\pi^l)$ for $1\leq k\leq f'$ and $0\leq l<e$ form, for each $i\in I$, a basis

of $\mathcal{O}_{L'}$ over $\mathcal{O}_{K'}$ (cf. 1.6.1 d)). Let $u = (u_i)_{i\in I}$ in $\prod \mathcal{O}_{L'_i}$; there exist unique $u_{i,k,l}\in\mathcal{O}_{K'}$ such that

$$u_i = \sum_{k,l} u_{i,k,l}\sigma_i(y_k\pi^l) \quad \text{for } i\in I.$$

But an element v of $\mathcal{O}_{K'}\otimes_{\mathcal{O}}\mathcal{O}_L$ is uniquely written in the form $\sum_{j,k,l} v_{j,k,l}x_jy_k\pi^l$ with $v_{j,k,l}\in\mathcal{O}_{K'}$ and the relation $u = \rho(v)$ is expressed by the system of equations

$$u_{i,k,l} = \sum_{j\in I} v_{j,k,l}\sigma_i(x_j) \tag{8}$$

for $i \in I$, $1 \leq k \leq f'$ and $0 \leq l < e$. Now, since $L \cap K'$ is Galois étale, we have $\det(\sigma_i(x_j)) \in \mathcal{O}_{L\cap K'}^{\times} \subset \mathcal{O}_{K'}^{\times}$ and system (8) has, for any given u, one and only one solution, which shows clearly that $\rho : \mathcal{O}_{K'} \otimes_{\mathcal{O}} \mathcal{O}_L \to \prod \mathcal{O}_{L'_i}$ is bijective.

The proof of (ii) is analogous and simpler: let E be a composite extension of L and K' and let ω'' be a valuation of E extending ω'. We have

$$f(\omega''/\omega') \geq f(\omega_L/\omega), \tag{2 bis}$$

$$e(\omega''/\omega') \geq e(\omega_L/\omega), \tag{3 bis}$$

hence

$$[E : K'] \geq e(\omega''/\omega')f(\omega''/\omega') \geq e(\omega_L/\omega)f(\omega_L/\omega) \geq [L : K] \geq [E : K'], \tag{6 bis}$$

and these inequalities are all equalities. The last equality of (6 bis) gives the linear disjointness of L and K', the first equality of (6 bis) shows that E is an admissible extension of K', considering that, if ω is not discrete, equality (3 bis) gives $e(\omega''/\omega') = 1$. Finally, the last assertion follows immediately from 1.6.1 d), taking into account equality (2 bis).

Remark. — In the proof of (i), the hypothesis that L is Galois served only to show that the restriction operation defines a bijection from $\mathrm{Gal}(\Omega/L)\backslash\mathrm{Gal}(\Omega/K)/\mathrm{Gal}(\Omega/K')$ onto $\mathrm{Gal}(L \cap K'/K)$. It can be replaced by any other condition leading to this conclusion, for example that K' is Galois.

1.6.9. — We use 1.6.8 mainly in the following cases:

a) The residue field of K is finite. Let L be an admissible extension of K; let $q = \mathrm{Card}\ \bar{K}$ and $q^f = \mathrm{Card}\ \bar{L}$. There then exists an étale extension K' of K such that $\bar{K}'$ is infinite and linearly disjoint from $\bar{L}$ over $\bar{K}$, which allows for the application of 1.6.8 (ii). It suffices to use 1.6.5 by taking as the extension k' of $\bar{K}$ the union of the fields of roots of unity of order $q^{p^n} - 1$, where p is a prime number not dividing f and where n runs through the set of integers ≥ 1.

b) One takes for K' the henselization ${}^hK$ of K (i.e., the field of fractions of the henselization ${}^h\mathcal{O}$ of $\mathcal{O}$, equipped with the valuation ${}^h\omega$; for the notion of henselization, see [EGA] IV, § 18 or [NB] AC VI, § 8, exer. 14, p. 193), or the completion $\hat{K}$ of K: it is an extension

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unramified over K (for its canonical valuation), with residue field $\overline{K}$ and, for every admissible extension L of K, we have $L \cap K' = K$, since $L \cap K'$ is an admissible extension of K with residual degree and ramification index equal to 1. The conditions of 1.6.8 (ii) are satisfied.

c) Suppose K is Henselian (one can reduce to this case thanks to *b)*). Let L be an admissible Galois extension of K and $L_{\text{ét}}$ the maximal étale sub-extension (1.6.7 *b)*). Let K' be a strict henselization ${}^{hs}K$ of K containing $L_{\text{ét}}$, i.e., a maximal étale extension of $L_{\text{ét}}$, an extension which is unique up to isomorphism. We then have $K' \cap L = L_{\text{ét}}$, the residue field of K' is a separable closure of $\overline{L}_{\text{ét}}$ and is therefore linearly disjoint from $\overline{L}$ over $\overline{L \cap K'}$, since $\overline{L}$ is purely inseparable over $\overline{L}_{\text{ét}}$. One can therefore apply 1.6.8 (i).

1.7. Stuffed schemes.

We maintain the notation of 1.6. Furthermore, if $\mathfrak{X}$ is an $\mathcal{O}$-scheme, we denote by $\overline{\mathfrak{X}}$ its closed fiber $\mathfrak{X}_{\overline{k}}$.

Definition 0.3 (1.7.1)**.** An $\mathcal{O}$-scheme $\mathfrak{X}$ is said to be **stuffed** if it is flat and if, for every $\mathcal{O}$-scheme $\mathfrak{Y}$, every K-morphism u from $\mathfrak{X}_K$ to $\mathfrak{Y}_K$ such that $u(\mathfrak{X}(\mathcal{O})) \subset \mathfrak{Y}(\mathcal{O})$ extends to an $\mathcal{O}$-morphism from $\mathfrak{X}$ to $\mathfrak{Y}$.

Recall that this extension is then unique since $\mathfrak{X}$ is flat (1.2.4) and that it exists if and only if $u^*(\mathcal{O}[\mathfrak{Y}]) \subset \mathcal{O}[\mathfrak{X}]$. Consequently, $\mathfrak{X}$ is stuffed if and only if it satisfies the following condition:

$$\mathcal{O}[\mathfrak{X}] = \{f \in K[\mathfrak{X}] \mid f(\mathfrak{X}(\mathcal{O})) \subset \mathcal{O}\}. \tag{ET}$$

1.7.2. — In practice, the following two conditions will be satisfied:

(ET 1) For every $f \in K[\mathfrak{X}]$, $f \notin \mathcal{O}[\mathfrak{X}]$, there exists $a \in \mathcal{O}$ such that $af \in \mathcal{O}[\mathfrak{X}]$ and $bf \notin \mathcal{O}[\mathfrak{X}]$ for all $b \in K$ with $\omega(b) < \omega(a)$ (otherwise said $\inf\{t \in \omega(K^\times) \mid af \in \mathcal{O}[\mathfrak{X}] \text{ for } \omega(a) = t\} \in \omega(K^\times)$).

(ET 2) The canonical image of $\mathfrak{X}(\mathcal{O})$ in $\mathfrak{X}(\overline{K})$ is dense in $\overline{\mathfrak{X}}$ [(1)].

Let us show that conditions (ET 1) and (ET 2) imply (ET). Let $f \in K[\mathfrak{X}]$ such that $f(\mathfrak{X}(\mathcal{O})) \subset \mathcal{O}$ and suppose that $f \notin \mathcal{O}[\mathfrak{X}]$. For $a \in \mathcal{O}$ as in (ET 1) we have $a \in \mathfrak{m}$, $af(\mathfrak{X}(\mathcal{O})) \subset \mathfrak{m}$ but $af \notin \mathfrak{m}\mathcal{O}[\mathfrak{X}]$; consequently, the canonical image of af in $\overline{K}[\overline{\mathfrak{X}}]$ is not zero, but vanishes on the canonical image of $\mathfrak{X}(\mathcal{O})$ in $\mathfrak{X}(\overline{K})$, which contradicts (ET 2).

1.7.3. — *a)* (ET 1) is satisfied in each of the following cases:

a_1) ω is discrete;

a_2) $\mathcal{O}[\mathfrak{X}]$ is a free $\mathcal{O}$-module;

---

(1) In the schematic sense: $\overline{\mathfrak{X}}$ is the smallest closed sub-scheme of $\overline{\mathfrak{X}}$ whose underlying space contains the image of $\mathfrak{X}(\mathcal{O})$; this implies that $\overline{\mathfrak{X}}$ is reduced.

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a_3) there exists a finite covering of $\mathfrak{X}$ by affine open sets $\mathfrak{X}_i$ satisfying (ET 1). Indeed, if $f \in K[\mathfrak{X}]$, one has

$$\begin{aligned} &\inf\{k \in \omega(K^\times) \mid af \in \mathcal{O}[\mathfrak{X}] \text{ for } \omega(a) = k\} \\ &= \inf_i \inf\{k \in \omega(K^\times) \mid af \in \mathcal{O}[\mathfrak{X}_i] \text{ for } \omega(a) = k\}. \end{aligned}$$

b) One knows that (ET 2) is satisfied if $\mathfrak{X}$ is smooth and $\mathcal{O}$ is strictly henselian (i.e., henselian with separably closed residue field).

c) Consequently, if ω is discrete and K is strictly henselian, every smooth $\mathcal{O}$-scheme is thick.

d) If $\mathfrak{X}$ is a union of thick affine open sets, then $\mathfrak{X}$ is thick.

e) Suppose $\overline{K}$ is infinite and let M be a free $\mathcal{O}$-module of finite type. Any open $\mathfrak{X}$ of the $\mathcal{O}$-scheme $\mathfrak{M}$ associated to M is a thick $\mathcal{O}$-scheme. In view of *d*), it suffices to show that every special affine open $\mathfrak{X}$ satisfies (ET 1) and (ET 2). Now, one has $\mathcal{O}[\mathfrak{X}] = \mathcal{O}[X_1, \ldots, X_n][d^{-1}]$, where the X_i are indeterminates and where $d \in \mathcal{O}[X_1, \ldots, X_n]$. If $d \in \mathfrak{m}[X_1, \ldots, X_n]$, one has $\mathcal{O}[\mathfrak{X}] = K[\mathfrak{X}]$ and there is nothing to prove. Suppose $d \notin \mathfrak{m}[X_1, \ldots, X_n]$. The canonical image $\overline{d}$ of d in $\overline{K}[X_1, \ldots, X_n]$ is not zero and

$$\overline{K}[\overline{\mathfrak{X}}] = \overline{K}[X_1, \ldots, X_n][\overline{d}^{-1}].$$

For $f \in K[X_1, \ldots, X_n]$ and $k \in \mathbf{N}$, the relation $af/d^k \in \mathcal{O}[\mathfrak{X}]$ is equivalent to $af \in \mathcal{O}[X_1, \ldots, X_n]$; indeed, if $\tilde{\omega}$ is the valuation of $K(X_1, \ldots, X_n)$ extending ω and such that $\tilde{\omega}(X_i) = 0$ for all i (cf. [NB] AC VI, § 10, prop. 2, p. 161), one has $\tilde{\omega}(d) = 0$ and each of the two previous relations is equivalent to $\tilde{\omega}(af) \geq 0$. The condition (ET 1) follows immediately (even if $\overline{K}$ is finite, for that matter).

On the other hand, $\overline{\mathfrak{X}}$ is the special open subset of $\overline{K}^n$ defined by $\overline{d}$. Since $\overline{K}$ is infinite, $\overline{\mathfrak{X}}(\overline{K})$ is dense in $\overline{\mathfrak{X}}$; but the canonical map $\mathfrak{X}(\mathcal{O}) \to \overline{\mathfrak{X}}(\overline{K})$ is surjective, hence (ET 2).

f) Let X be a K-scheme and let P be a subset of $X(K)$, "bounded" in the sense that, for all $f \in K[X]$, one has $\inf_{z \in P} \omega(f(z)) > -\infty$. Let $\mathscr{A}^P = \{f \in K[X] \mid f(P) \subset \mathcal{O}\}$ and $\mathfrak{X}^P = \operatorname{Spec} \mathscr{A}^P$. Then $P \subset \mathfrak{X}^P(\mathcal{O})$ and $\mathfrak{X}^P$ is a thick $\mathcal{O}$-scheme of generic fiber X.

Conversely, if $\mathfrak{X}$ is a thick $\mathcal{O}$-scheme of generic fiber X, then $P = \mathfrak{X}(\mathcal{O})$ is bounded and one has $\mathfrak{X} = \mathfrak{X}^P$.

1.7.4. — Let $\mathfrak{X}$ be a flat $\mathcal{O}$-scheme satisfying (ET 1) and let $\mathfrak{U}$ be an affine open subset of $\mathfrak{X}$ such that $\overline{\mathfrak{U}}$ is dense in $\overline{\mathfrak{X}}$ (e.g., an affine open neighborhood of the unit section when $\mathfrak{X}$ is a connected group $\mathcal{O}$-scheme). Then

$$\mathcal{O}[\mathfrak{X}] = K[\mathfrak{X}] \cap \mathcal{O}[\mathfrak{U}].$$

Indeed, let $f \in K[\mathfrak{X}]$, $f \notin \mathcal{O}[\mathfrak{X}]$ and $f \in \mathcal{O}[\mathfrak{U}]$ (otherwise said, the restriction of f to $\mathfrak{U}_K$ extends to $\mathfrak{U}$), and let $a \in \mathcal{O}$ satisfying (ET 1). Then $\omega(a) > 0$, af is zero on $\overline{\mathfrak{U}}$ hence on $\overline{\mathfrak{X}}$; consequently $af \in \mathfrak{m}\mathcal{O}[\mathfrak{X}]$, which contradicts the minimality of $\omega(a)$.

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1.7.5. — Let K' be an extension of K, equipped with a valuation extending ω, unramified. If $\mathfrak{X}'$ is a flat $\mathcal{O}'$-scheme satisfying (ET 1) for K', then $\mathfrak{X}'$ considered as an $\mathcal{O}$-scheme also satisfies (ET 1). If moreover K' is étale (1.6.2) and if $\mathfrak{X}$ is a flat $\mathcal{O}$-scheme satisfying (ET 1), then $\mathfrak{X}_{\mathcal{O}'}$ satisfies (ET 1) for K': to see this, it suffices to note that there exists a family (e_i) of elements of $\mathcal{O}'$ forming a basis of $\mathcal{O}'$ over $\mathcal{O}$ and of K' over K, and whose canonical images form a basis of $\overline{K}'$ over $\overline{K}$.

1.7.6. Proposition. — *Let ${}^{hs}\mathcal{O}$ be a strict henselization of $\mathcal{O}$ and let $\mathfrak{X}$ be a smooth $\mathcal{O}$-scheme satisfying (ET 1). Then*

$$\mathcal{O}[\mathfrak{X}] = \{f \in K[\mathfrak{X}] \mid f(\mathfrak{X}({}^{hs}\mathcal{O})) \subset {}^{hs}\mathcal{O}\}.$$

If $\mathfrak{Y}$ is an $\mathcal{O}$-scheme and u is a K-morphism from $\mathfrak{X}_K$ to $\mathfrak{Y}_K$, then u extends to an $\mathcal{O}$-morphism from $\mathfrak{X}$ to $\mathfrak{Y}$ if and only if $u(\mathfrak{X}({}^{hs}\mathcal{O})) \subset \mathfrak{Y}({}^{hs}\mathcal{O})$.

One reasons as in 1.7.2.

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§ 2. DECOMPOSITION OF A LINEAR GROUP

2.1. Decomposition of the general linear group.

2.1.1. — Let $(M_i)_{1\leq i\leq n}$ be projective A-modules of finite type and let M be the direct sum of the M_i. We set $\mathfrak{L}_{ij} = \mathfrak{L}(M_j, M_i)$ (for $1 \leq i, j \leq n$): it is a smooth group A-scheme, of affine algebra $\mathrm{Sym}({}^tM_i \otimes M_j)$. The algebra scheme $\mathfrak{L} = \mathfrak{L}(M, M)$ is identified in an obvious way with $\prod_{i,j} \mathfrak{L}_{ij}$, the product in $\mathfrak{L}$ being translated by the 'matrix' product $(g_{ij}) \cdot (h_{ij}) = (\sum_k g_{ik}h_{kj})$. By this identification, the group scheme $\prod_i \mathrm{GL}(M_i) \subset \prod_i \mathfrak{L}_{ii}$ is identified with a closed group sub-scheme $\mathfrak{D}$ of $\mathrm{GL}(M)$ ("group of diagonal matrices").

2.1.2. — Let us set $\Theta = \{(i, j) \mid 1 \leq i \neq j \leq n\}$, $\Theta^+ = \{(i, j) \in \Theta \mid i < j\}$ and $\Theta^- = \{(i, j) \in \Theta \mid i > j\}$. A subset Ψ of Θ is said to be *closed* if $(i, j) \in \Psi$ and $(j, k) \in \Psi$ imply $(i, k) \in \Psi$.

It is immediate that the morphism $x \mapsto 1+x$ is a scheme automorphism of $\mathfrak{L}$ and that its restriction to $\mathfrak{L}_{ij}$ (for $(i, j) \in \Theta$) is a group scheme isomorphism from $\mathfrak{L}_{ij}$ onto a closed group sub-scheme of $\mathrm{GL}(M)$, which we will denote $\mathfrak{B}_{ij}$.

2.1.3. Proposition. — *Let Ψ be a closed subset of Θ^+ (resp. Θ^-).*

(i) *The image of $\prod_{(i,j)\in\Psi} \mathfrak{L}_{ij}$ by the automorphism $x \mapsto 1 + x$ of $\mathfrak{L}$ is a closed group sub-scheme of $\mathrm{GL}(M)$, which we will denote $\mathfrak{B}_\Psi$.*

(ii) *The product morphism of $\mathrm{GL}(M)$ defines a closed immersion of $\mathfrak{D} \times \mathfrak{B}_\Psi$ onto a closed group sub-scheme (denoted $\mathfrak{D}\mathfrak{B}_\Psi$) of $\mathrm{GL}(M)$, which is the semi-direct product of $\mathfrak{D}$ by $\mathfrak{B}_\Psi$.*

(iii) *Let $(i, j) \in \Theta^+$ (resp. Θ^-) such that $(i, j) \notin \Psi$ and such that $(j, k) \in \Psi$ implies $(i, k) \in \Psi$. Then $\Psi' = \Psi \cup \{(i, j)\}$ is closed and $\mathfrak{B}_{\Psi'}$ is the semi-direct product of $\mathfrak{B}_{ij}$ by $\mathfrak{B}_\Psi$.*

(iv) *The product morphism $\prod_{(i,j)\in\Psi} \mathfrak{B}_{ij} \to \mathfrak{B}_\Psi$ is a scheme isomorphism for any order put on Ψ.*

Proof. Let us prove (i): it suffices to check that for every A-algebra R, $\mathfrak{B}_\Psi(R)$ is a subgroup of $\mathrm{GL}(R\otimes M)$, which results from an immediate matrix calculation. Similarly, let $\mathfrak{X}$ be the closed subscheme of $\mathrm{GL}(M)$ that is the intersection of $\mathrm{GL}(M)$ with the closed subset $\prod_{(i,j)\in\Psi} \mathfrak{L}_{ij} \times \prod_i \mathfrak{L}_{ii}$ of $\mathfrak{L}$; a simple matrix calculation shows that $\mathfrak{X}(R)$ is a subgroup of $\mathrm{GL}(R \otimes M)$ that is the semi-direct product of $\mathfrak{D}(R)$ by $\mathfrak{B}_\Psi(R)$, hence (ii). Under the hypotheses of (iii), one verifies immediately that Ψ' is closed and (iii) again results from a simple matrix calculation in $\mathrm{GL}(R \otimes M)$. □

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Finally, let us prove (iv): it suffices to show that for every A-algebra R, the product map $\prod \mathfrak{B}_{ij}(R) \to \mathfrak{B}_\Psi(R)$ is bijective regardless of the order of the factors. Let us begin by ordering Ψ by numbering its elements $a_1 = (i_1, j_1), \ldots, a_m = (i_m, j_m)$ in such a way that the map $k \mapsto j_k - i_k$ (resp. $i_k - j_k$) is increasing. For $0 \leq r \leq m$, set $\Psi_r = \{a_{r+1}, \ldots, a_m\}$; one immediately verifies that this is a closed subset. Then set $H_r = \mathfrak{B}_{a_r}(R)$ $(1 \leq r \leq m)$ and $K_r = \mathfrak{B}_{\Psi_r}(R)$ $(0 \leq r \leq m)$. Matrix calculations that we leave to the reader show that:

1) The commutator group (H_r, H_s) is contained in K_r for $1 \leq r, s \leq m$;

2) $H_r \cap K_r = \{1\}$ for $1 \leq r \leq m$;

3) $K_r = H_{r+1} \cdot K_{r+1}$ for $0 \leq r \leq m - 1$.

Assertion (iv) then follows from the following lemma:

Lemma 0.3 (2.1.4)**.** *Let K_0 be a group and let H_r and K_r be subgroups of K_0 (for $1 \leq r \leq m$). Suppose that $K_0 \supset K_1 \supset \cdots \supset K_m = \{1\}$ and that conditions 1), 2), and 3) above are satisfied. Then, for every permutation σ of $\{1, \ldots, m\}$, the product map $H_{\sigma(1)} \times \cdots \times H_{\sigma(m)} \to K_0$ is bijective.*

Proof. Proceed by induction on m, the lemma being obvious for $m = 1$. By 3), we have $K_0 = H_1 \dots H_m$ and it then follows from 1) that H_m is central. Let p be the canonical projection from K_0 onto $K_0' = K_0/H_m$. Set $K_r' = p(K_r)$ and $H_r' = p(H_r)$ for $1 \le r \le m-1$. One immediately notices that all the conditions of the lemma remain satisfied with m, K_r, and H_r replaced by $m-1, K_r'$, and H_r'. Now let $k \in K_0$. By the induction hypothesis and given that, by 2) and 3), p is a bijection from H_r onto H_r', there exists, for $1 \le j \le m$ and $\sigma(j) \ne m$, a unique element $v_j \in H_{\sigma(j)}$ such that $p(k) = p(w)$ with $w = \prod_{\substack{1 \le j \le m \\ \sigma(j) \ne m}} v_j$. Then set $v_{\sigma^{-1}(m)} = kw^{-1}$: it is the unique element of H_m such that $k = v_{\sigma^{-1}(m)} w$. But since H_m is central, we also have

$$k = \prod_{1 \le j \le m} v_j \quad \text{with } v_j \in H_{\sigma(j)}.$$

Conversely, this relation implies $p(k) = p(\prod_{\substack{1 \le j \le m \\ \sigma(j) \ne m}} v_j)$, hence the uniqueness of the v_j. □

Proposition 0.4 (2.1.5)**.** *Set $\mathfrak{B}^{\pm} = \mathfrak{B}_{\Psi^{\pm}}$. There exists an element d in the affine algebra of $\mathfrak{GL}(M)$ such that $d(1) = 1$ and such that the product morphism defines an isomorphism of schemes from $\mathfrak{B}^- \times \mathfrak{D} \times \mathfrak{B}^+$ onto the open set $\mathfrak{GL}(M)_d$ consisting of the points of $\mathfrak{GL}(M)$ where d is invertible.*

For $1 \le r \le n$, set $M^{(r)} = \sum_{1 \le i \le r} M_i$ and let p_r be the projection from M onto $M^{(r)}$ parallel to $\sum_{i > r} M_i$. If R is an A-algebra and if $g \in \mathrm{End}(R \otimes M)$, we denote by $g^{(r)}$ the restriction to $M^{(r)}$ of $(1 \otimes p_r) \circ g$: in terms of matrices, if $g = (g_{ij})_{1 \le i,j \le n}$ with $g_{ij} \in \mathfrak{L}_{ij}(R)$, we have $g^{(r)} = (g_{ij})_{1 \le i,j \le r}$. Set $d_r(g) = \det g^{(r)} \in R$. Letting R vary, ...

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we thus obtain an element of the affine algebra of $\mathfrak{GL}(M)$ denoted by d, and we shall prove the proposition by taking

$$d = \prod_{1 \le r \le n} d_r.$$

We indeed have $d(1) = 1$. It is therefore a matter of showing that, for any R, the product map is a bijection from $\mathfrak{B}^-(R) \times \mathfrak{D}(R) \times \mathfrak{B}^+(R)$ onto the set of $g \in \mathrm{End}(R \otimes M)$ such that $d(g)$ is invertible in R. In other words, it is a matter of showing that, if one considers matrices $g = (g_{ij})$, $u = (u_{ij})$, $z = (z_{ij})$ and $v = (v_{ij})$, with $g_{ij}, u_{ij}, z_{ii}, v_{ij} \in \mathrm{Hom}(R \otimes M_j, R \otimes M_i)$, and if one considers the conditions

(1) $u_{ij} = 0$ for $i < j$, $v_{ij} = 0$ for $i > j$, $u_{ii} = v_{ii} = 1$, $z_{ij} = 0$ for $i \ne j$ and z_{ii} is invertible;

(2) $d_r(g)$ is invertible.

Then the following assertions are true:

a) If (1) is satisfied for $1 \le i, j \le n$ and if $g = uzv$, then (2) is satisfied for $1 \le r \le n$.

b) If (2) is satisfied for $1 \le r \le n$, then there exists one and only one triple of matrices (u, z, v) satisfying (1) for $1 \le i, j \le n$ and such that $g = uzv$.

If (1) is satisfied for $1 \le i, j \le n$, one sees immediately that, setting $g = uzv$, one has

$$g^{(r)} = u^{(r)} z^{(r)} v^{(r)} \quad \text{for } 1 \le r \le n, \tag{3}$$

whence $d_r(g) = \det z^{(r)} = \prod_{1 \le i \le r} \det z_{ii}$, which proves a).

Let us prove b) by induction on n, the cases $n = 0$ or 1 being obvious. If (2) is true, there exists by induction hypothesis a unique triple of matrices $u' = (u'_{ij})$, $z' = (z'_{ij})$ and $v' = (v'_{ij})$ satisfying (1) for $1 \le i, j \le n-1$ and such that $g^{(n-1)} = u'z'v'$. Let u'', z'', v'' be the endomorphisms of $R \otimes M$ equal respectively to u', z', v' on $R \otimes M^{(n-1)}$ and to the identity on $R \otimes M_n$. By possibly replacing g by $u''^{-1} g v''^{-1} z''^{-1}$, one sees that one can, taking into account (3), assume that $g^{(n-1)}$ is the identity on $R \otimes M^{(n-1)}$. There then exist $u \in \mathrm{Hom}(R \otimes M^{(n-1)}, R \otimes M_n)$, $v \in \mathrm{Hom}(R \otimes M_n, R \otimes M^{(n-1)})$ and $z \in \mathrm{Hom}(R \otimes M_n, R \otimes M_n)$ such that

$$g = \begin{pmatrix} I & v \\ u & z \end{pmatrix}$$

and the relation

$$\begin{pmatrix} I & v \\ u & z \end{pmatrix} = \begin{pmatrix} I & 0 \\ u & I \end{pmatrix} \begin{pmatrix} I & 0 \\ 0 & z - uv \end{pmatrix} \begin{pmatrix} I & v \\ 0 & I \end{pmatrix}$$

implies the existence and uniqueness of the sought-after matrices, taking into account that it implies $\det(z - uv) = \det g = d_n(g)$.

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2.1.6. — Let V be a finite-dimensional vector space over K and let S be a split torus defined over K of the algebraic group $GL(V)$. Let M be a projective sub-A-module of V (automatically of finite type) such that the canonical map from $K \otimes M$ into V is an isomorphism. Suppose that the canonical A-group scheme $\mathfrak{S}$ extending S (1.2.11) acts on M (cf. 1.4.6). We choose a total order compatible with addition on the character group $X^*(S)$. Let $\alpha_1, \ldots, \alpha_n$ be the weights of S in V, arranged in increasing order, and let V_i be the subspace of V of weight α_i under S. Then M is the direct sum of the $M_i = M \cap V_i$ ([DG], p. 177) and the preceding results can be applied. Note that $V_i = KM_i$.

Since the characters $\alpha_1, \ldots, \alpha_n$ generate $X^*(S)$, they generate with their inverses the affine algebra of $\mathfrak{S}$ and the action of $\mathfrak{S}$ on M is faithful: it allows identifying $\mathfrak{S}$ with a smooth closed subgroup scheme of $\mathfrak{GL}(M)$ (1.4.5).

We know that Lie $GL(V)$ identifies with End $V \simeq \prod_{i,j} \mathrm{Hom}(V_j, V_i)$, the adjoint representation being given by ad $g.x = gxg^{-1}$. The roots of $GL(V)$ relative to S are thus the characters $\alpha_i - \alpha_j$ for $i \neq j$. Let φ be the system of root rays of $GL(V)$ relative to S (1.1.2) (recall that this is the set of open half-lines in $X^*(S) \otimes \mathbf{R}$ that contain a root), and let φ^+ (resp. φ^-) be the set of positive (resp. negative) root rays for the chosen order.

For $a \in \varphi$, the set Θ_a of pairs (i, j) such that $\alpha_j - \alpha_i \in a$ is obviously closed and contained either in Θ^+ or in Θ^-. One can therefore consider the closed A-subgroup scheme $\mathfrak{V}_a = \mathfrak{V}_{\Theta_a}$ of $\mathfrak{GL}(M)$ (2.1.3). More generally, for any closed subset $\Psi \subset \varphi^\pm$, the set Θ_Ψ, the union of the Θ_a for $a \in \Psi$, is closed and contained in $\Theta^\pm$; we set $\mathfrak{V}_\Psi = \mathfrak{V}_{\Theta_\Psi}$. We have seen (2.1.3) that the product map is an isomorphism of schemes from $\prod_{a \in \Psi} \mathfrak{V}_a$ onto $\mathfrak{V}_\Psi$ regardless of the order of the factors.

On the other hand, it is immediate that the root subgroup V_a of $GL(V)$ associated with $a \in \varphi$ (1.1.3) is formed by "matrices" (g_{ij}) such that $g_{ii} = I$ for $1 \leq i \leq n$ and $g_{ij} = 0$ if $\alpha_j - \alpha_i \notin a$. It follows (see 2.1.3) that for every $a \in \varphi$ and more generally for every closed subset $\Psi \subset \varphi^\pm$, we have

$$V_a = (\mathfrak{V}_a)_K, \quad V_\Psi = (\mathfrak{V}_\Psi)_K.$$

Finally, the centralizer of S in $GL(V)$ is the group of "diagonal matrices" $D = \{(g_{ij}) \in GL(V) \mid g_{ij} = 0 \text{ for } i \neq j\}$.

In a general way, if H is a closed subset of $GL(V)$, we denote by $\mathfrak{H}$ its schematic closure in $\mathfrak{GL}(M)$ (1.2.6): it is a closed A-subscheme of $\mathfrak{GL}(M)$ (thus automatically of finite type) and its generic fiber $\mathfrak{H}_K$ identifies with H. If H is a closed subgroup, defined over K, and if $\mathfrak{H}$ is flat, then $\mathfrak{H}$ is a subgroup scheme of $\mathfrak{GL}(M)$ (1.2.7). Conversely, if $\mathfrak{H}'$ is a flat (or simply torsion-free) closed subscheme of $\mathfrak{GL}(M)$ such that $\mathfrak{H}'_K = H$, then $\mathfrak{H}' = \mathfrak{H}$ (1.2.6). The notations $\mathfrak{S}$, $\mathfrak{V}_a$, $\mathfrak{V}_\Psi$, $\mathfrak{D}$ conform to this general convention since they denote flat subschemes of $\mathfrak{GL}(M)$ with generic fiber S, V_a, etc. respectively.

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2.2. Decomposition of a linear group.

2.2.1. — We retain the hypotheses and notation of 2.1.5 and we are further given a connected closed subgroup G of $GL(V)$ defined over K, containing the split torus S. We denote by Φ the system of root rays of G relative to S (1.1.2), and Φ^+ (resp. Φ^-) the set of positive (resp. negative) root rays for the order chosen on $X^*(S)$. In a general way, we take up again the notation of 1.1, notably Z, U_a, U_Ψ, etc.

2.2.2. — Throughout the rest of § 2, we assume that the schematic closures $\mathfrak{Z}$ and $\mathfrak{U}_a$ of Z and the U_a (for $a \in \Phi$) are flat subschemes of $\mathfrak{GL}(M)$ and are thus group subschemes. Note that this is equivalent to saying that there exist flat A-group schemes $\mathfrak{Z}$ and $\mathfrak{U}_a$ (for $a \in \Phi$) such that $\mathfrak{Z}_K = Z$ and $(\mathfrak{U}_a)_K = U_a$ and such that the injection of Z (resp. U_a) extends to a closed immersion of $\mathfrak{Z}$ (resp. $\mathfrak{U}_a$) into $\mathfrak{GL}(M)$. This hypothesis is automatically satisfied when the ring A is Prüferian, the schematic closures being torsion-free (1.2.6). For the different equivalent definitions of a Prüferian ring, see ([NB] AC VII, § 2, exer. 12, p. 93); one of these definitions is that for every prime ideal $\mathfrak{p}$ of A, the local ring $A_\mathfrak{p}$ is a valuation ring; another is that every finitely generated torsion-free A-module is projective, a condition which immediately implies that a torsion-free A-module is flat. Every valuation ring is Prüferian. A ring is a Dedekind ring if and only if it is Prüferian and Noetherian.

Remarquons que $\mathfrak{S}$ is a closed subscheme of $\mathfrak{Z}$ since $Z \supset S$.

Theorem 0.5 (2.2.3)**.** *(i) Let $\Psi \subset \Phi^+$ be a closed subset. The schematic closure $\mathfrak{U}_\Psi$ of U_Ψ in $\mathfrak{GL}(M)$ is a flat closed group subscheme of $\mathfrak{GL}(M)$ and the product morphism $\prod_{a\in\Psi} \mathfrak{U}_a \to \mathfrak{U}_\Psi$ is an isomorphism of A-schemes regardless of the ordering put on Ψ.*

(ii) Let Ψ be as in (i); the schematic closure $\mathfrak{Z}\mathfrak{U}_\Psi$ of ZU_Ψ in $\mathfrak{GL}(M)$ is a flat closed group subscheme of $\mathfrak{GL}(M)$, a semi-direct product of $\mathfrak{Z}$ by $\mathfrak{U}_\Psi$.

(iii) The product morphism defines an isomorphism of A-schemes from $\mathfrak{U}_{\Phi^-} \times \mathfrak{Z} \times \mathfrak{U}_{\Phi^+}$ onto an open set $\mathfrak{C}$ of the schematic closure $\mathfrak{G}$ of G, dense in $\mathfrak{G}$, and there exists a non-zero $d \in A[G]$ such that $\mathfrak{C}$ is the special open set $\mathfrak{G}_d$. Moreover, $\mathfrak{G}$ is the schematic closure of $\mathfrak{C}$.

Proof. Since Lie G is a Lie subalgebra of Lie $GL(V)$, the roots of G relative to S are roots of $GL(V)$ relative to S and $\Phi \subset \varphi$, whence $\Psi \subset \varphi^+$. Consequently $\prod_{a\in\Psi} \mathfrak{U}_a$ is a closed subscheme of $\mathfrak{V}^+ = \prod_{a\in\varphi^+} \mathfrak{V}_a$ (2.1.3 (i)), flat since the $\mathfrak{U}_a$ are. But $(\prod_{a\in\Psi} \mathfrak{U}_a)_K = \prod_{a\in\Psi} U_a = U_\Psi$ is a closed subgroup, defined over K, of G and thus of $GL(V)$, and (i) results from 1.2.6 and 1.2.7.

Similarly, the product morphism identifies $\mathfrak{Z} \times \mathfrak{U}_\Psi$ with a flat closed subscheme $\mathfrak{Z}\mathfrak{U}_\Psi$ of $\mathfrak{D}\mathfrak{B}^+$ and $(\mathfrak{Z}\mathfrak{U}_\Psi)_K = ZU_\Psi$ is a closed subgroup of $GL(V)$, defined over K, whence (ii) follows on one hand from 2.1.3 (ii), and on the other hand from 1.2.6 and 1.2.7. □

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Finally let $d \in A[\mathfrak{GL}(M)]$ satisfying 2.1.5. Let $\mathfrak{U}^\pm = \mathfrak{U}_\pm$. Given (i), $\mathfrak{U}^- \times \mathfrak{Z} \times \mathfrak{U}^+$ is a flat closed sub-scheme of $\mathfrak{B}^- \times \mathfrak{D} \times \mathfrak{B}^+$, the product morphism defines an isomorphism of schemes from $\mathfrak{U}^- \times \mathfrak{Z} \times \mathfrak{U}^+$ onto a flat closed sub-scheme $\mathfrak{C}$ of the open set $\mathfrak{GL}(M)_d$ and $\mathfrak{C}$ is a sub-scheme of $\mathfrak{GL}(M)$. Let $\mathfrak{G}'$ be the schematic closure of $\mathfrak{C}$ in $\mathfrak{GL}(M)$: we will show that $\mathfrak{G} = \mathfrak{G}'$. Indeed, since $\mathfrak{C}$ is flat and $\mathfrak{C}_K \subset G = \mathfrak{G}_K$, we have $\mathfrak{C} \subset \mathfrak{G}$ (1.2.5) ([1]), hence $\mathfrak{G}' \subset \mathfrak{G}$. Conversely, $\mathfrak{C}_K = U^- Z U^+$ is a dense open set of G (1.1.9), hence $\mathfrak{G}'_K \supset G$ and $\mathfrak{G}' \supset \mathfrak{G}$. Consequently, $\mathfrak{C}$ is an open sub-scheme of $\mathfrak{G}$, dense in $\mathfrak{G}$ (1.2.6). We thus have $|\mathfrak{C}| = |\overline{\mathfrak{C}}| \cap \mathfrak{GL}(M)_d = |\mathfrak{G}| \cap \mathfrak{GL}(M)_d$ and $\mathfrak{C}$ is the open sub-scheme $\mathfrak{G}_d$ of $\mathfrak{G}$, noting again by d the canonical image of d in $A[\mathfrak{G}]$, which is not zero since $d(1) = 1$. This completes the proof.

2.2.4. Scholium. — *Suppose that $\mathfrak{G}$ is flat. Then $\mathfrak{G}$ is a closed subgroup sub-scheme of $\mathfrak{GL}(M)$, of finite type, with generic fiber G, containing $\mathfrak{Z}$ and the $\mathfrak{U}_a$ as closed subgroup sub-schemes and possessing an open neighborhood $\mathfrak{C}$ of the unit section isomorphic by the product morphism to*

$$\prod_{a\in\Phi^-} \mathfrak{U}_a \times \mathfrak{Z} \times \prod_{a\in\Phi^+} \mathfrak{U}_a.$$

That $\mathfrak{G}$ is a subgroup sub-scheme results from 1.2.7 and that $\mathfrak{G}$ is of finite type is evident since $\mathfrak{GL}(M)$ is.

Noting that, since $\mathfrak{G}$ is torsion-free, it is automatically flat as soon as the ring A is Prüferian.

2.2.5. Corollary. — *Suppose that $\mathfrak{Z}$ and the $\mathfrak{U}_a$ are smooth and that $\mathfrak{G}$ is flat. Then*

(i) *the schemes $\mathfrak{U}_\Psi$, $\mathfrak{Z}\mathfrak{U}_\Psi$ (for $\Psi \subset \Phi^\pm$ closed) and $\mathfrak{G}$ are smooth;*

(ii) *the neutral component $\mathfrak{G}^0$ of the group scheme $\mathfrak{G}$ is a smooth open subgroup sub-scheme of $\mathfrak{G}$;*

(iii) *if A is a valuation ring of height* 1 *or a Dedekind ring, then* $\mathfrak{G}^0$ *is affine.*

It is clear that $\mathfrak{U}_\Psi$ and $\mathfrak{Z}\mathfrak{U}_\Psi$ are smooth. Consequently $\mathfrak{G}$ is smooth in the neighborhood of the unit section and its fibers are smooth ([DG], p. 238). Moreover $\mathfrak{G}$ is flat of finite type, hence of finite presentation ([26], cor. 3.4.7), thus smooth. From which (i) and (ii) ([SGAD], Exp. $\mathrm{VI_B}$, th. 3.10).

Let us prove (iii). Given that the generic fiber G of $\mathfrak{G}$ is connected, the scheme $\mathfrak{G}^0$ is obtained from $\mathfrak{G}$ by removing from it open and closed parts of closed fibers of $\mathfrak{G}$, in finite number: it is evident if A is a valuation ring of height 1; if A is a Dedekind ring, hence Noetherian, then $\mathfrak{G}$ is of finite presentation, the set of $s \in \operatorname{Spec} A$ such that the fiber $\mathfrak{G}_s$ is connected is a neighborhood of the generic point of Spec A ([EGA] IV, prop. 9.7.8), whose complement consists of a finite number of closed points. Assertion (iii) thus results from the following lemma:

([1]) The relation $\mathfrak{X} \subset \mathfrak{Y}$ between sub-schemes of a scheme $\mathfrak{Z}$ means that the canonical injection $\mathfrak{X} \to \mathfrak{Z}$ factors through $\mathfrak{Y}$. Further on, the notation $|\mathfrak{X}|$ denotes the underlying topological space of $\mathfrak{X}$.

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Lemma 0.6 (2.2.6)**.** *Let us make one of the two following hypotheses:*

a) A is a valuation ring of height 1;

b) A is a Dedekind ring.

Let $\mathfrak{X}$ be a flat affine A-scheme. Let s be a closed point of $\operatorname{Spec} A$ *and let Y be an open and closed subset of the fiber $\mathfrak{X}_s = X$. The open sub-scheme $\mathfrak{X} - Y$ is affine.*

Proof. Let ω denote the valuation of A in case a), that of A_s in case b), and let k be the residue field of A_s. Let $\mathscr{A} = A[\mathfrak{X}]$, and, for all $x \in \mathfrak{X}$, let v_x be the canonical map from $\mathscr{A}$ into the local ring $\mathscr{A}_x$ and also its canonical extension $1 \otimes v_x : K \otimes \mathscr{A} \to K \otimes \mathscr{A}_x$ (recall that $\mathscr{A}$ (resp. $\mathscr{A}_x$) is flat over A and identifies with its canonical image in $K \otimes \mathscr{A}$ (resp. $K \otimes \mathscr{A}_x$)). Since $k[X] = k \otimes \mathscr{A}$ is the direct product of $k[Y]$ and $k[X - Y]$, there exists $f \in \mathscr{A}$ such that $f(x) = 0$ if $x \in X - Y$ and $f(y) = 1$ if $y \in Y$. By considering a covering of $\mathfrak{X}$ by affine open sets contained either in the open set $\mathfrak{X} - Y$, or in the open set $\mathfrak{X} - (X - Y)$, we see that there exist affine open sets $\mathfrak{U}_1, \ldots, \mathfrak{U}_n$ of $\mathfrak{X}$ such that $\mathfrak{X} - Y = (\mathfrak{U}_1 \cup \cdots \cup \mathfrak{U}_n)$. Since $f(x) = 0$ for all $x \in \mathfrak{U}_j \cap X = (\mathfrak{U}_j)_s$, the canonical image of f in $A[\mathfrak{U}_j]$ belongs to $sA[\mathfrak{U}_j]$ and there exists $\alpha_j \in K$ such that $\omega(\alpha_j) < 0$ and $v_x(\alpha_j f) \in \mathscr{A}_x$ for all $x \in (\mathfrak{U}_j)_s$. Let $\alpha \in K$ be such that $\omega(\alpha) = \sup \omega(\alpha_j)$ and that $\alpha \in A_t$ for all $t \in \operatorname{Spec} A, t \neq s$: the existence of such an α is evident in case a) and results from [NB] AC VII, § 2, prop. 2 in case b). We then have $\omega(\alpha) < 0$ and $v_x(\alpha f) \in \mathscr{A}_x$ for $1 \leq j \leq n$ and $x \in \mathfrak{U}_j$.

Let $\mathscr{B} = \mathscr{A}[\alpha f] \subset K \otimes \mathscr{A}$ and let us show that $\mathfrak{X} - Y = \operatorname{Spec} \mathscr{B}$, or more precisely that the morphism $i : \operatorname{Spec} \mathscr{B} \to \mathfrak{X}$ defined by the inclusion $i^* : \mathscr{A} \to \mathscr{B}$ is an open immersion with image $\mathfrak{X} - Y$. Since $A_t \otimes \mathscr{A} = A_t \otimes \mathscr{B}$ for all $t \in \operatorname{Spec} A, t \neq s$, it is clear that the restriction of i to the inverse image open set of $\operatorname{Spec} A - \{s\}$ is an isomorphism onto $\mathfrak{X} - X$. It remains to show that the restriction of i to $(\operatorname{Spec} \mathscr{B})_s$ is an isomorphism onto $X - Y$. Let first $x' \in (\operatorname{Spec} \mathscr{B})_s$. Let $x = i(x') = x' \cap \mathscr{A} \in X$. We have $\mathscr{A}_x \subset \mathscr{B}_{x'} \subset K \otimes \mathscr{A}_x$ and $x\mathscr{A}_x = x'\mathscr{B}_{x'} \cap \mathscr{A}_x$. Consequently x does not belong to Y because $v_y(f) \notin y\mathscr{A}_y$ for all $y \in Y$ (since $f(y) = 1$), while $v_x(f) = \alpha^{-1} v_x(\alpha f) \in x'\mathscr{B}_{x'}$. We have thus shown that $i((\operatorname{Spec} \mathscr{B})_s) \subset X - Y$.

Conversely, let $x \in X - Y$. There exists j such that $x \in \mathfrak{U}_j$, whence $v_x(\mathscr{B}) \subset \mathscr{A}_x$. Let x' be the inverse image of $x\mathscr{A}_x$ by the homomorphism $v_x : \mathscr{B} \to \mathscr{A}_x$. Then $x' \in (\operatorname{Spec} \mathscr{B})_s$, we have $x = x' \cap \mathscr{A} = i(x')$ and v_x factors through an isomorphism from $\mathscr{B}_{x'}$ onto $\mathscr{A}_x$, which proves at once that the morphism $i : (\operatorname{Spec} \mathscr{B})_s \to X - Y$ is surjective and is an isomorphism, and completes the proof of the lemma. □

Remark 1 (2.2.7). It should not be thought that $\mathfrak{G}$ is always flat, nor that it is a group scheme, even if one assumes that $\mathfrak{Z}$ and the $\mathfrak{U}_a$ are smooth and that A is noetherian (for a counter-example, see further down 3.2.14).

It is not true either that the identity component $\mathfrak{G}^0$ is necessarily affine, even if one assumes that $\mathfrak{Z}$, the $\mathfrak{U}_a$ and $\mathfrak{G}$ are smooth and the ring A is noetherian.

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2.2.8. Corollary. — *Keep the hypotheses of 2.2.5 and assume further that the $\mathfrak{U}_a$ are connected. Then $\mathfrak{C}^0 = \mathfrak{U}^-\mathfrak{Z}^0\mathfrak{U}^+$ is an open subscheme of $\mathfrak{G}^0$.*

Indeed, $\mathfrak{C}^0$ is contained in $\mathfrak{G}^0$, and is open in $\mathfrak{C}$, thus in $\mathfrak{G}$.

2.2.9. Corollary. — *Keep the hypotheses of 2.2.8 and assume further that one of the two following conditions is satisfied:*

a) $\mathfrak{Z} = \mathfrak{Z}(A) \cdot \mathfrak{Z}^0$;

b) A is a valuation ring of height 1 or a Dedekind ring.

Then the image $\mathfrak{G}^1 = \mathfrak{Z} \cdot \mathfrak{G}^0$ of $\mathfrak{Z} \times \mathfrak{G}^0$ by the product morphism is a smooth open group subscheme of $\mathfrak{G}$, containing $\mathfrak{Z}$ and the $\mathfrak{U}_a$ as closed group subschemes and $\mathfrak{C}$ as an open subscheme. If b) is satisfied, $\mathfrak{G}^1$ is affine.

The last assertion results from the others and from Lemma 2.2.6. If a) is satisfied, we have $\mathfrak{G}^1 = \mathfrak{Z}(A) \cdot \mathfrak{G}^0$ and the corollary is evident. Suppose b) is satisfied. Since the generic fiber of $\mathfrak{Z}$ is connected, we see as above that there exists a finite number of closed points $s_1, \ldots, s_k$ of $\operatorname{Spec} A$ such that $\mathfrak{Z}_s = \mathfrak{Z}_s^0$ for $s \in \operatorname{Spec} A$ distinct from the s_i, hence

$$\mathfrak{G}^1 = \mathfrak{G}^0 \cup \bigcup_{1 \le i \le k} \mathfrak{Z}_{s_i} \cdot \mathfrak{G}^0_{s_i},$$

and to show that $\mathfrak{G}^1$ is an open group subscheme in $\mathfrak{G}$ it suffices to notice that $\mathfrak{Z}_{s_i} \cdot \mathfrak{G}^0_{s_i}$ is, for $1 \le i \le k$, an open algebraic $k(s_i)$-subgroup in $\mathfrak{G}_{s_i}$. The other assertions of the corollary are evident.

2.2.10. Proposition[8]. — *Suppose that A is Noetherian and that the $\mathfrak{U}_a$ for $a \in \Phi$ are smooth and connected. Let $\mathfrak{H}$ be the image of $\mathfrak{C} \times \mathfrak{C}$ in $\mathfrak{GL}(M)$ by the product morphism π of $\mathfrak{GL}(M)$. Then $\mathfrak{H}$ is open in $\mathfrak{G}$ and is a group subscheme of $\mathfrak{GL}(M)$, with generic fiber G, flat, of finite presentation, not necessarily affine, containing $\mathfrak{Z}$ and the $\mathfrak{U}_a$ as closed group subschemes and containing $\mathfrak{C}$ as open.*

Since $\mathfrak{C} \times \mathfrak{C}$ is flat and $\pi(\mathfrak{C}_K \times \mathfrak{C}_K) \subset G = \mathfrak{G}_K$, we have $\mathfrak{H} \subset \mathfrak{G}$ (1.2.5). The inverse image Γ of $\mathfrak{C}$ under $\pi : \mathfrak{C} \times \mathfrak{C} \to \mathfrak{G}$ is therefore an open subset of $\mathfrak{C} \times \mathfrak{C}$ and it is immediate, since $\mathfrak{U}^-\mathfrak{Z}$ and $\mathfrak{Z}\mathfrak{U}^+$ are groups, that

$$\begin{aligned} \Gamma \supset &((\mathfrak{U}^- \times \mathfrak{Z} \times \{1\}) \times (\mathfrak{U}^- \times \mathfrak{Z} \times \{1\})) \\ &\cup((\{1\} \times \mathfrak{Z} \times \mathfrak{U}^+) \times (\{1\} \times \mathfrak{Z} \times \mathfrak{U}^+)). \end{aligned} \tag{1}$$

Let $\mathfrak{p} \in \operatorname{Spec} A$ and let k be the residue field of the local ring $A_\mathfrak{p}$. As in 1.2.2, we will allow ourselves to play the double game and to consider the fibers at $\mathfrak{p}$ of smooth A-group schemes as algebraic groups defined over k. In particular the fiber $\mathfrak{GL}(M)_\mathfrak{p}$ identifies with the algebraic group $\mathrm{GL}(k \otimes M)$.

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The fiber Γ_p of Γ over p is an open subset of $\mathfrak{C}_p \times \mathfrak{C}_p$, dense in $\mathfrak{C}_p \times \mathfrak{C}_p$: indeed, for any $u, u' \in \mathfrak{U}_p^+$ and $z, z' \in \mathfrak{Z}_p$, the set V of $(v, v') \in \mathfrak{U}_p^+ \times \mathfrak{U}_p^+$ such that $((u, z, v), (u', z', v')) \in \Gamma_p$ is an open subset of $\mathfrak{U}_p^+ \times \mathfrak{U}_p^+$, non-empty since $(1, 1) \in V$ by (1), and thus dense in $\mathfrak{U}_p^+ \times \mathfrak{U}_p^+$ since $\mathfrak{U}_p^+$ is connected.

Let H_p be the closure of $\mathfrak{C}_p$ in $\mathfrak{GL}(k \otimes M)$. Since $\mathfrak{C}_p$ is open in $\mathfrak{G}_p$, a fortiori $\mathfrak{C}_p$ is open in H_p. Moreover, Γ_p is dense in $H_p \times H_p$ and $\pi(H_p \times H_p) \subset H_p$ since by definition $\pi(\Gamma_p) \subset \mathfrak{C}_p$. In other words, H_p is a closed subgroup of $\mathfrak{GL}(k \otimes M)$. Since $\mathfrak{C}_p$ is a dense open subset thereof, we have $H_p = \mathfrak{C}_p \cdot \mathfrak{C}_p$ (cf. [2], prop. 1.3, p. 87), whence

$$H_p = \mathfrak{H} \cap \mathfrak{GL}(M)_p. \tag{2}$$

We also have $H_p = \mathfrak{C}_p^{-1}\mathfrak{C}_p = \mathfrak{U}_p^+\mathfrak{Z}_p\mathfrak{U}_p^-\mathfrak{U}_p^-\mathfrak{Z}_p\mathfrak{U}_p^+ = \mathfrak{U}_p^+\mathfrak{C}_p$ and it follows, since $\mathfrak{U}_p^+$ is defined over k, that if k_s is a separable closure of k, we have

$$H_p \text{ is the union of } a.\mathfrak{C}_p \text{ for } a \in \mathfrak{U}_p^+(k_s). \tag{3}$$

Let us now show that $\mathfrak{H}$ is open in $\mathfrak{G}$ and is flat. To this end, we shall show that, up to localization and base change, one can cover $\mathfrak{H}$ by translates of $\mathfrak{C}$ by sections of $\mathfrak{C}$. Since A is Noetherian, $\mathfrak{C} \times \mathfrak{C}$ and $\mathfrak{G}$ are of finite presentation (that is, A-algebraic schemes in the sense of [DG]) and it follows that $\mathfrak{H}$ is a constructible subset of $\mathfrak{G}$ ([DG], cor. 3.9, p. 78). To show that $\mathfrak{H}$ is open in $\mathfrak{G}$, it suffices therefore, again because A is Noetherian, to show that if $x \in \mathfrak{H}$ and $y \in \mathfrak{G}$ are such that $x \in \overline{\{y\}}$, then $y \in \mathfrak{H}$ ([DG], cor. 3.4, p. 76).

Let p be the projection of x in $\operatorname{Spec} A$. Since the local ring A_p is flat over A, the schematic closures of G and of $\mathfrak{C} \times_A \operatorname{Spec} A_p$ in $\mathfrak{GL}(A_p \otimes M)$ both coincide with the inverse image of $\mathfrak{H}$ in $\mathfrak{GL}(A_p \otimes M)$ ([DG], p. 56). Making the base change $A \to A_p$, we reduce to the case where A is a local ring with maximal ideal p.

Let $\mathcal{O}_x$ then be the local ring of x in $\mathfrak{GL}(M)$ and let $\tilde{A}$ be the strict Henselization of $\mathcal{O}_x$ (cf. [EGA] IV, § 18.8). Let x' be the closed point of $\operatorname{Spec} \tilde{A}$ and let α be the canonical morphism $\operatorname{Spec} \tilde{A} \to \operatorname{Spec} \mathcal{O}_x$: we have $\alpha(x') = x$. The residue field $\tilde{k}$ of $\tilde{A}$ is a separably closed extension of the residue field k of A, and thus contains a separable closure k_s of k. In view of (3), there exists $a \in \mathfrak{U}_p^+(k_s)$ such that $x \in a\mathfrak{C}_p$; since $\mathfrak{U}^+$ is smooth, the "Hensel lemma" implies the existence of a section $\hat{a} \in \mathfrak{U}^+(\tilde{A})$ of $\mathfrak{U}^+$ over $\operatorname{Spec} \tilde{A}$ such that $\hat{a}(x') = a$.

[8] (1) This result will not be used in the remainder of this work, except in 3.9.4.

Let us then make the base change $A \to \tilde{A}$. If $\mathfrak{x}$ (resp. X) is an A-scheme (resp. a K-scheme), let $\tilde{\mathfrak{x}} = \mathfrak{x} \times_{\operatorname{Spec} A} \operatorname{Spec} \tilde{A}$ (resp. $\tilde{X} = X \times_{\operatorname{Spec} K} \operatorname{Spec} \tilde{K}$ with $\tilde{K} = K \otimes \tilde{A}$) and if $f : \mathfrak{x} \to \mathfrak{y}$ is a morphism of A-schemes, let $\tilde{f} = f \times \mathrm{id} : \tilde{\mathfrak{x}} \to \tilde{\mathfrak{y}}$. Finally, let $\beta : \mathfrak{GL}(\tilde{M}) \to \mathfrak{GL}(M)$ be the first projection. Since $\tilde{A}$ is flat over $\mathcal{O}_x$ and $\mathcal{O}_x$ is flat over A (since $\mathfrak{GL}(M)$ is flat), $\tilde{A}$ is flat over A and we see as above that $\tilde{\mathfrak{H}}$ is the schematic closure both of $\tilde{\mathfrak{C}}$ and of $\tilde{G}$ in
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$\mathfrak{GL}(M)$. On the other hand, left translation by the section $\hat{a}$ is an automorphism of the scheme $\tilde{\mathfrak{GL}}(M)$ and the restriction $\hat{a}_k$ of $\hat{a}$ to Spec $\tilde{K}$ belongs to $\mathfrak{U}_k^+(\tilde{\mathfrak{K}}) \subset G(\tilde{K})$; we thus have $\hat{a} \cdot \tilde{G} = \hat{a}_k \cdot \tilde{G} = \tilde{G}$ and $\hat{a} \cdot \tilde{\mathfrak{G}} = \tilde{\mathfrak{G}}$. Since $\tilde{\mathfrak{T}}$ is open in $\tilde{\mathfrak{G}}$, it follows that $\hat{a} \cdot \tilde{\mathfrak{T}}$ is open in $\tilde{\mathfrak{G}}$.

Since $\tilde{A}$ is faithfully flat over $\mathcal{O}_z$, the morphism α is surjective. We have $x \in \overline{\{y\}}$, thus $y \in \operatorname{Spec} \mathcal{O}_z$ and there exists $y' \in \operatorname{Spec} \tilde{A}$ such that $\alpha(y') = y$; furthermore, $x' \in \overline{\{y'\}}$ since the closed point of Spec $\tilde{A}$ belongs to the closure of any point of Spec $\tilde{A}$. Let $x'' = (\alpha \times \mathrm{id})(x') = (x, x')$ and $y'' = (\alpha \times \mathrm{id})(y') = (y, y')$; then $x'' \in \hat{a} \cdot \tilde{\mathfrak{T}}$, $y'' \in \tilde{\mathfrak{G}}$ and $x'' \in \overline{\{y''\}}$. Since $\hat{a} \cdot \tilde{\mathfrak{T}}$ is open in $\tilde{\mathfrak{G}}$, it follows that

$$y'' \in \hat{a} \cdot \tilde{\mathfrak{T}} = \pi \circ (\hat{a} \times \mathrm{id})(\operatorname{Spec} \tilde{A} \times_{\tilde{A}} \tilde{\mathfrak{T}}) \subset \pi(\tilde{\mathfrak{G}} \times_{\tilde{A}} \tilde{\mathfrak{T}}).$$

We deduce that

$$y = \beta(y'') \in \beta \circ \pi(\tilde{\mathfrak{G}} \times_{\tilde{A}} \tilde{\mathfrak{T}}) = \pi \circ (\beta \times \beta)(\tilde{\mathfrak{G}} \times_{\tilde{A}} \tilde{\mathfrak{T}}) \subset \pi(\mathfrak{G} \times \mathfrak{G}) = \mathfrak{H},$$

so that $\mathfrak{H}$ is open in $\mathfrak{G}$.

Furthermore, $\hat{a} \cdot \tilde{\mathfrak{T}}$ is flat over Spec $\tilde{A}$. Since it is a neighborhood of x in $\mathfrak{H}_{\tilde{A}}$, it follows that the local ring of x in $\mathfrak{H}_{\tilde{A}}$ is flat over $\tilde{A}$ and by faithfully flat descent, the local ring of x in $\mathfrak{H}$ is flat over A. Therefore $\mathfrak{H}$ is flat.

Consequently, $\mathfrak{H}$, as an open subset of the subscheme $\mathfrak{G}$, is a subscheme of $\mathfrak{GL}(M)$, which has no reason to be affine. It is a group subscheme of $\mathfrak{GL}(M)$. Indeed, the restriction to $\mathfrak{H}$ of the product map π of $\mathfrak{GL}(M)$ sends $H_K \times H_K$ into G, hence, since $\mathfrak{H}$ is flat, it factors through a morphism $\mathfrak{H} \times \mathfrak{H} \to \mathfrak{G}$ (1.2.5). But we have set-theoretically $\pi(\mathfrak{H} \times \mathfrak{H}) \subset \mathfrak{H}$ since the fibers of $\mathfrak{H}$ are subgroups (cf. (3)). Since $\mathfrak{H}$ is open in $\mathfrak{G}$, it follows that $\pi : \mathfrak{H} \times \mathfrak{H} \to \mathfrak{G}$ factors through $\mathfrak{H}$, and $\mathfrak{H}$ is indeed a group subscheme. It is of finite presentation, since it is an open subset in $\mathfrak{G}$ which is of finite presentation and Noetherian. Finally, the other assertions of the proposition are obvious.

2.2.11. Remark. — If to the hypotheses of 2.2.10 one adds that $\mathfrak{G}$ is smooth, then $\mathfrak{H}$ is smooth; if one adds those of 2.2.9, we have $\mathfrak{H} = \mathfrak{G}^0$.

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§ 3. SCHEMATIC ROOT DATA

We assume henceforth that the connected algebraic group G is reductive.

3.1. Schematic root data.

Definition 0.4 (3.1.1)**.** A schematic root datum on G relative to S is the data of flat A-group schemes of finite type $\mathfrak{Z}$ extending Z and $\mathfrak{U}_a$ extending U_a (for $a \in \Phi$), satisfying the following conditions:

(DRS 0) the injection of S into Z extends to an isomorphism from $\mathfrak{S}$ onto a closed subgroup scheme of $\mathfrak{Z}$;

(DRS 1) let $a, b \in \Phi$, $b \neq -a$; for any nibbling order (see 3.1.2) on $\Phi \cap (a+b)$, the commutator map $\gamma_{a,b} : U_a \times U_b \to \prod_{c \in \Phi \cap (a+b)} U_c$ (1.1.8) extends to a morphism from $\mathfrak{U}_a \times \mathfrak{U}_b$ into $\prod_{e \in \Phi \cap (a+b)} \mathfrak{U}_e$, still denoted $\gamma_{a,b}$;

(DRS 2) let $a \in \Phi$; the map $(z, u) \mapsto zuz^{-1}$ from $Z \times U_a$ into U_a extends to a morphism from $\mathfrak{Z} \times \mathfrak{U}_a$ into $\mathfrak{U}_a$;

(DRS 3) let $a \in \Phi$; there exists an open neighborhood $\mathfrak{W}_a$ of the identity section in $\mathfrak{U}_{-a} \times \mathfrak{U}_a$ and a morphism β_a from $\mathfrak{W}_a$ into $\mathfrak{U}_{-a} \times \mathfrak{Z} \times \mathfrak{U}_a$ such that the composition of $(\beta_a)_K$ with the product map from $U_{-a} \times Z \times U_a$ into G coincides with the restriction to $(\mathfrak{W}_a)_K$ of the product map from $U_a \times U_{-a}$ into G (in other words, $(\mathfrak{W}_a)_K \subset W_a$ and $(\beta_a)_K$ is the restriction of the map denoted $\beta_a : W_a \to U_{-a} \times Z \times U_a$ in 1.1.10).

We say that a schematic root datum $(\mathfrak{Z}, (\mathfrak{U}_a))$ dominates another $(\mathfrak{Z}', (\mathfrak{U}'_a))$ if the identity maps extend to morphisms $\mathfrak{Z} \to \mathfrak{Z}'$ and $\mathfrak{U}_a \to \mathfrak{U}'_a$ $(a \in \Phi)$.

Throughout the remainder of § 3, we denote by $\mathscr{D} = (\mathfrak{Z}, (\mathfrak{U}_a)_{a \in \Phi})$ a schematic root datum on G relative to S.

3.1.2. — Let us recall that if ψ is a positively closed subset of Φ, a *nibbling order* on ψ is a total order on ψ such that each element of ψ is extremal in the set of elements of ψ greater than itself, that is, it is an extremal generator of the convex cone generated by the elements greater than itself (I, 1.3.15). If a is extremal in ψ, then $\psi - \{a\}$ is positively closed. If $(a_1, \ldots, a_r)$ is a nibbling order on ψ, then $(a_i, \ldots, a_r)$ is positively closed for all i.

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3.1.3. — The goal of this paragraph is to show that, given a good number of additional hypotheses which will be progressively introduced, a schematic root datum allows for the definition of an A-group scheme $\mathfrak{G}$ extending G, containing $\mathfrak{Z}$ and the $\mathfrak{U}_a$ as closed subgroup schemes and such that, whatever the system of positive root rays Φ^+ and whatever the orders chosen on $\Phi^\pm$, the product map is an isomorphism of schemes from $(\prod_{a\in\Phi^-} \mathfrak{U}_a) \times \mathfrak{Z} \times (\prod_{a\in\Phi^+} \mathfrak{U}_a)$ onto a dense open subset of $\mathfrak{G}$ ("big cell"). Let us say immediately that this group scheme, when it exists, is generally not unique up to isomorphism (cf. 3.2).

3.1.4. — To this end, we will seek to reduce to the situation of theorem 2.2.3. Let us call a $\mathcal{D}$-module the pair formed by an A-module M projective of finite type and a representation ρ of G in $V = K \otimes M$ such that $\mathfrak{Z}$ and the $\mathfrak{U}_a$ act on M via ρ. Let us say that the $\mathcal{D}$-module (M, ρ) is faithful for $\mathfrak{Z}$ (resp. $\mathfrak{U}_a$) if, moreover, the corresponding morphism from $\mathfrak{Z}$ (resp. $\mathfrak{U}_a$) into $GL(M)$ is a closed immersion, and that (M, ρ) is faithful if it is faithful for $\mathfrak{Z}$ and for each of the $\mathfrak{U}_a$ $(a \in \Phi)$.

If (M, ρ) is a faithful $\mathcal{D}$-module, then ρ is an isomorphism of G onto a subgroup defined over K of $GL(V)$. Indeed, Ker $d\rho$ is stable under S, so it is the sum of its intersections with Lie Z and the Lie U_a. But $d\rho$ is injective on Lie Z and the Lie U_a, hence Ker $d\rho = \{0\}$. Consequently, ρ is separable and Ker ρ is a finite subgroup of G, contained in the center of G and *a fortiori* in Z. Since the restriction of ρ to Z is an immersion, we have Ker $\rho = \{1\}$ and ρ is an immersion.

Consequently, if we manage to construct a faithful $\mathcal{D}$-module (M, ρ), we will find ourselves exactly in the hypotheses of theorem 2.2.3 (after identification of G and $\rho(G)$), and this latter will provide, at least if A is Prüferian, the A-group scheme $\mathfrak{G}$ sought. Different choices of (M, ρ) can moreover give non-isomorphic schemes $\mathfrak{G}$ (3.2).

3.1.5. — Let us note that conversely, if we place ourselves *a priori* in the situation of theorem 2.2.3 (with G reductive), then the schemes $\mathfrak{Z}$ and $\mathfrak{U}_a$ of this theorem form a schematic root datum: (DRS 0) is immediate, (DRS 1) (resp. (DRS 2)) is easily deduced from 1.1.8 and 2.2.3 (i) (resp. 1.1.7 and 2.2.3 (ii)) and (DRS 3) is obtained from 1.1.10 and 2.2.3 (iii) by taking for $\mathfrak{M}_a$ the preimage of $\mathfrak{C}$ by the product map of $\mathfrak{U}_a \times \mathfrak{U}_{-a}$ into $\mathfrak{G}$.

3.1.6. — By combining 3.1.4 and 3.1.5, we see that if (M, ρ) is a $\mathcal{D}$-module faithful for $\mathfrak{Z}$, if ρ is a closed immersion allowing to identify G and $\rho(G)$ and if the schematic closures $\mathfrak{U}'_a$ of the U_a in $GL(M)$ are flat, then $(\mathfrak{Z}, (\mathfrak{U}'_a))$ is a schematic root datum dominated by $\mathcal{D}$.

3.1.7. — Another method for constructing the desired scheme $\mathfrak{G}$ is to construct a "group kernel" structure on the "big cell" $(\prod_{a\in\Phi^-} \mathfrak{U}_a) \times \mathfrak{Z} \times (\prod_{a\in\Phi^+} \mathfrak{U}_a)$.

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and to use the Artin-Weil theorem ([SGAD], Exp. XVIII) on the embedding of such a kernel into a group scheme. This method might perhaps allow one to obtain the result under broader hypotheses (which, moreover, is only of moderate interest to us), but on the one hand, it requires quite complex technical contortions, if only in the very definition of a "group kernel," and on the other hand, it rests on a theorem whose existing accounts have not entirely convinced us that we could apply it.

3.2. An example: the split case.

Until the end of no. 3.2, G is assumed to be split over K and S is a maximal split torus of G, whence $Z = S$. We denote by Φ the root system of G according to S, W the Weyl group of G, and K_0 the prime field of K. Up to replacing G by a K-isomorphic group, we may assume, as we shall do henceforth, that G and S are defined and split over K_0.

In the case which truly interests us in view of the sequel, that of a valuation ring A (and, more generally, for a Prüfer ring), the results that we are going to establish will be found again later, by another method, for any quasi-split group (no. 3.8). We could therefore, in principle, dispense with this paragraph, but it does not seem superfluous to us to treat the split case separately because of the more "elementary" nature of the method, and, secondarily, due to the fact that certain results are easily proven for more general rings. However, we will sometimes be quite brief regarding proofs, and routine calculations will be left to the reader.

3.2.1. — Each U_a $(a \in \Phi)$ is equipped with a structure of a line over K_0 (1.2.10) satisfying the following condition (which characterizes it): for any extension L of K_0 and any maximal torus S' defined and split over L

normalizing U_a, there exists one and only one root a' of G_L according to S' such that U_a is the corresponding root subgroup and, with these notations, we have $sus^{-1} = a'(s)u$ for all $u \in U_a(L)$ and all $s \in S'(L)$.

Let K_1 be a subfield of K. We call a K_1-pinning of U_a a K_1-isomorphism $\mathfrak{D}_{K_1} \to U_a$ (cf. 1.2.10). For every K_1-pinning x_a of U_a, there exists one and only one K_1-pinning x_{-a} of U_{-a}, said to be associated with x_a, characterized by any one of the following equivalent properties:

(i) $m_a = x_a(1)x_{-a}(1)x_a(1)$ belongs to N;

(ii) $m_{-a} = x_{-a}(1)x_a(1)x_{-a}(1)$ belongs to N;

(iii) there exists a K_1-homomorphism ζ_a of SL_2 into G such that $x_a(u) = \zeta_a \begin{pmatrix} 1 & u \\ 0 & 1 \end{pmatrix}$ and $x_{-a}(u) = \zeta_a \begin{pmatrix} 1 & 0 \\ -u & 1 \end{pmatrix}$.

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The homomorphism ζ_a is then unique and one has

$$m_a = m_{-a} = \zeta_a \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

The image of m_a in the Weyl group W of G is the reflection r_a defined by the root a. It is clear that x_a is the K_1-pinning associated with x_{-a}.

We denote by $\tilde{a}$ the K_1-homomorphism $t \mapsto \zeta_a \begin{pmatrix} t & 0 \\ 0 & t^{-1} \end{pmatrix}$ from Mult_{K_1} into G. It is known that $\mathrm{Im}\,\tilde{a} \subset S$ and that, for every character χ of S, the composition $\chi \circ \tilde{a}$ is the character $t \mapsto t^{\langle a^\vee, \chi \rangle}$ of Mult_{K_1}, where $a^\vee$ is the element associated with a in the root system $\Phi^\vee$ inverse to Φ (recall that $\Phi^\vee \subset \mathrm{Hom}(X^*(S), \mathbf{Z})$). In particular, $\tilde{a}$ does not depend on the choice of the pinning x_a. It is clear that $\tilde{a}$ extends in one and only one way into an A-homomorphism, still denoted by $\tilde{a}$, from Mult into $\mathfrak{S}$.

3.2.2. — A K_1-pinning of G (relative to S) is the data of a basis B of Φ and for each $a \in B$, a K_1-pinning x_a of U_a. A K_1-Chevalley system in G (relative to S), is a family $(x_a)_{a \in \Phi}$ of K_1-pinnings of the U_a possessing the two following properties:

(Ch 1) for all $a \in \Phi$, the K_1-pinnings x_a and x_{-a} are associated;

(Ch 2) for $a, b \in \Phi$, such that $r_a(b) \in \Phi$, there exists $\varepsilon = \pm 1$ such that $x_{r_a(b)}(u) = m_a x_b(\varepsilon u) m_a^{-1}$ for all $u \in \mathfrak{D}_{K_1}$.

It is known that for every K_1-pinning of G, there exists a K_1-Chevalley system extending it, which is determined "up to sign", that is, up to possible replacement, for certain roots a, of x_a by $u \mapsto x_a(-u)$.

3.2.3. — In what follows, we shall take $K_1 = K_0$ and suppress the prefix K_0 before "pinning" or "Chevalley system". We now take a Chevalley system $(x_a)_{a\in\Phi}$. It is known that there exist integers $C_{a,b;i,j}$ for $a, b \in \Phi$, $b \neq -a$, $i, j \in \mathbf{N}^*$, $ia + jb \in \Phi$ such that the commutator of $x_a(u)$ and $x_b(v)$ is given by

$$(x_a(u), x_b(v)) = \prod_{i,j\in\mathbf{N}^*, ia+jb\in\Phi} x_{ia+jb}(C_{a,b;i,j}u^i v^j) \tag{1}$$

(the roots $ia+jb$ being arranged in a given arbitrary order). We set $M_{a,b;i} = C_{a,b;i,1}$ and $M_{a,b,0} = 1$. If $a-b \notin \Phi$, one has $M_{a,b,i} = \pm 1$.

3.2.4. — Consider the adjoint group $\operatorname{Ad} G$ of G. It is known that $T = \operatorname{Ad}_G S$ is a split maximal torus of $\operatorname{Ad} G$. The root system of $\operatorname{Ad} G$ with respect to T identifies with Φ. The group $X^*(T)$ of characters of T is the group Q generated by Φ. The adjoint representation defines an isomorphism between the root subgroups associated to $a \in \Phi$ of G and $\operatorname{Ad} G$ and allows them to be identified; the Chevalley system (x_a) "is" then also a Chevalley system for $\operatorname{Ad} G$.

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Let $\mathfrak{g}$ be the Lie algebra of Ad G and $\mathfrak{h}$ the Lie algebra of T. It is known that $\mathfrak{h}$ is identified with $\operatorname{Hom}(Q, K)$ and that Q is therefore identified with an additive subgroup of the dual ${}^*\mathfrak{h}$. Let $\mathfrak{T}$ be the A-split torus scheme, the canonical extension of T. The restriction to S of the adjoint representation of G extends to a surjective morphism from $\mathfrak{S}$ to $\mathfrak{T}$. The Lie algebra $\mathfrak{h}_A$ of $\mathfrak{T}$ is identified with the free sub-A-module $\operatorname{Hom}(Q, A)$ of $\mathfrak{h}$. For $a \in \Phi$, let

$$e_a = (\operatorname{Lie} x_a)(1) \in \operatorname{Lie}_K U_a \subset \operatorname{Lie}_K U_\Phi, \tag{1}$$
$$h_a = -[e_a, e_{-a}], \tag{2}$$

elements that we consider as belonging sometimes to Lie G, sometimes to $\mathfrak{g}$. For $a, b \in \Phi$, we have

$$[e_a, e_b] = \begin{cases} N_{a,b}e_{a+b} & \text{with } N_{a,b} = M_{a,b,1} \in \mathbf{Z} \text{ if } a+b \in \Phi, \\ 0 & \text{if } a+b \notin \Phi \cup \{0\}. \end{cases} \tag{3}$$

Finally, the identification of $\mathfrak{h}$ with $\operatorname{Hom}(Q, K)$ maps h_a to the canonical image of $a^\vee$ and

$$[h_a, e_b] = b(h_a)e_b = \langle a^\vee, b\rangle e_b \quad \text{for } a, b \in \Phi. \tag{4}$$

3.2.5. — The following formulas are consequences of 3.2.3 (1) and of 3.2.4 (2) and (4):

$$\operatorname{Ad} x_a(u).e_b = \sum M_{a,b,j}u^j e_{b+ja} \quad (a, b \in \Phi; b \neq -a), \tag{1}$$

the summation being extended over integers $j \geq 0$ such that $b + ja \in \Phi$;

$$\operatorname{Ad} x_a(u).e_{-a} = e_{-a} - uh_a + u^2 e_a; \tag{2}$$
$$\operatorname{Ad} x_a(u).h = h - a(h)ue_a \quad (h \in \mathfrak{h}). \tag{3}$$

3.2.6. — Let us take for each $a \in \Phi$ an invertible fractional ideal $f(a)$ of A, that is to say, a finitely generated projective sub-A-module of K. We have seen (1.4.1) that $f(a)$ defines a smooth A-group scheme $\mathfrak{Add}_{f(a)}$, with generic fiber Add_K. By transport of structure via x_a, we deduce from it a smooth A-group scheme $\mathfrak{U}_a$ extending U_a.

3.2.7. Proposition. — *With the notations of 3.2.6, the system $\mathfrak{D} = (\mathfrak{S}, (\mathfrak{U}_a)_{a\in\Phi})$ is a schematic root datum of G relative to S if and only if the invertible ideals $f(a)$ satisfy the following two relations:*

$$f(b + ia) \supset M_{a,b,i}f(a)^i f(b) \quad (b, a, b + ia \in \Phi, b \neq -a, i \in \mathbf{N}^*); \tag{1}$$
$$f(a)f(-a) \subset A \quad \text{for all } a \in \Phi. \tag{2}$$

Condition (DRS 0) is tautologically satisfied. In view of 3.2.3 (1), (DRS 1) is fulfilled if and only if, for all $\lambda \in f(ia + jb)$, we have $\lambda C_{a,b;i,j} \in f(a)^i f(b)^j$ [9], that is to say if

$$f(ia + jb) \supset C_{a,b;i,j}f(a)^i f(b)^j \tag{1'}$$

[9] [[UNCERTAIN: $f(a)^i f(b)^j$ — CONFIDENCE: 85% — REASON: There is a small mark before the ideals in the text that could be a notation or a printing artifact, but it is not present in the numbered formula (1').]]

for $a, b, ia+jb \in \Phi, b \neq -a, i, j \in \mathbf{N}^*$. In particular (1) must be satisfied. But, conversely (1) implies (1'). Indeed, the examination of root systems of rank

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two shows that the only case of (1') that is not covered by (1) is the one where a and b form a basis of a direct factor of type G_2 of Φ, the roots of the form $ia + jb$ being then, after a possible exchange of a and b, $a + b, a + 2b, a + 3b, 2a + 3b$; if (1) is satisfied, we have $f(a + 3b) \supset f(a)f(b)^3$ (since $M_{a,b,3} = \pm 1$), $f(2a + 3b) \supset f(a)f(a + 3b)$ (since $M_{a,a+3b,1} = \pm 1$), hence $f(2a + 3b) \supset f(a)^2 f(b)^3$ and, finally, (1'). We have thus established that (DRS 1) is equivalent to (1).

Condition (DRS 2) is always satisfied: the operation of S on U_a extends to the operation composed of the character $a : S \to \mathbf{Mult}$ and the canonical operation of $\mathbf{Mult}$ on $\mathbf{Add}_{f(a)}$.

For the study of (DRS 3), let us identify U_a and U_{-a} with $\mathbf{Add}$ via x_a and x_{-a}, and let u and v be the corresponding coordinates. A simple and classic calculation in SL_2, transported to G via ζ_a, shows that W_a is the open set $(U_a \times U_{-a})_{(1-uv)}$ of $U_a \times U_{-a}$, and that the map β_a from no. 1.1.10 is given by

$$\beta_a(u, v) = \left(\frac{v}{1 - uv}, \tilde{a}(1 - uv), \frac{u}{1 - uv} \right). \tag{3}$$

Now, the local ring R of the neutral section $(0, 0)$ of $U_a \times U_{-a}$ is the set of rational fractions $P(u, v)Q(u, v)^{-1}$ with $Q(0, 0) \in A^\times$. Since the affine algebra of U_b is the symmetric algebra of $f(b)$, the coefficient of $u^n v^m$ in the power series expansion of any element of R belongs to $f(a)^n \cdot f(-a)^m$. For β_a to extend to a neighborhood of $(0, 0)$, it is necessary that for all $\varphi \in A[U_a]$, we have $\varphi\left(\frac{u}{1-uv}\right) \in R$. In particular, we must have $\lambda u(1-uv)^{-1} \in R$ for all $\lambda \in f(a)$. Considering the coefficient of $u^2 v$ in the power series expansion of $\lambda u(1 - uv)^{-1}$, we see that it is necessary that $f(a) \subset f(a)^2 \cdot f(-a)$, hence (2). Conversely, suppose (2) is satisfied; then $1 \in f(a) \cdot f(-a)$, $1 - uv \in A[U_a \times U_{-a}]$, and (DRS 3) is satisfied by taking for W_a the open neighborhood of the neutral section $(0, 0)$ where $1 - uv$ is invertible and by defining the morphism β_a by (3).

3.2.8. — The relations (1) and (2) of 3.2.7 are satisfied if we take $f(a) = A$ for all a. The corresponding schematic root datum is called the *Chevalley schematic root datum* associated with the Chevalley system (x_a).

3.2.9. — Until the end of no. 3.2, we assume conditions (1) and (2) of 3.2.7 are satisfied. Thus, $\mathscr{D}$ is a schematic root datum and our goal is now to construct a faithful $\mathscr{D}$-module. For $a \in \Phi$, Lie U_a identifies with the sub-A-module $f(a)e_a$ of $\mathfrak{g} = \text{Lie Ad } G$. Let us set

$$L_f = \mathfrak{h}_A + \sum_{a \in \Phi} \text{Lie } U_a = \mathfrak{h}_A + \sum_{a \in \Phi} f(a)e_a \subset \mathfrak{g} \tag{1}$$

(cf. 3.2.4). It is a finitely generated projective sub-A-module of $\mathfrak{g}$ and also, given 3.2.4 (3) and the fact that $a(\mathfrak{h}_A) \subset A$, a sub-$A$-Lie algebra.

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Proposition 0.7. *The module L_f is (for the adjoint representation of G in $\mathfrak{g}$) a faithful $\mathscr{D}$-module for the $\mathfrak{U}_a$. It is faithful for $\mathfrak{S}$ if and only if the group G is adjoint.*

It is clear that $\mathfrak{S}$ acts on L_f and that this representation is faithful if and only if $G = \text{Ad } G$.

Showing that $\mathfrak{U}_a$ acts on L_f is equivalent to showing that, for $m \in L_f$ and $m' \in {}^t L_f$, the coefficient $c_{m',m}$ of the restriction to U_a of the adjoint representation of G in $\mathfrak{g}$ belongs to $A[\mathfrak{U}_a]$, that is, is a polynomial in u whose coefficient of u^n belongs to ${}^t f(a)^n$ for all $n \geq 0$. By bilinearity, it suffices to verify this for $m \in \mathfrak{h}_A \cup \bigcup f(b)e_b$ and $m' \in {}^t\mathfrak{h}_A \cup \bigcup f(b)e'_b$, where (e'_b) denotes the dual basis of (e_b) in the orthogonal of $\mathfrak{h}$ in ${}^t\mathfrak{g}$. The calculations are easy and unsurprising; we leave them to the reader.

Since $\mathfrak{h}_A = \text{Hom}(Q, A)$ and since every root a is part of a basis of the $\mathbf{Z}$-module Q, there exists $m \in \mathfrak{h}_A$ such that $a(m) = -1$. It then follows from 3.2.5 (3) that, for $\mu \in {}^t f(a)$ and $m' = \mu e'_a$, we have $c_{m',m} = \mu u$, which implies that the $c_{m',m}$ generate $A[\mathfrak{U}_a]$, and thus that $\mathfrak{U}_a$ acts faithfully on L_f. The proposition is proven.

3.2.10. — We have thus constructed a faithful $\mathscr{D}$-module in the case of an adjoint group G. Let us pass to the general case and first recall ([NB] Lie VIII, § 7, no 3) that a dominant weight ω of G (suppose a system of positive roots Φ^+ has been chosen once and for all) is said to be *minuscule* if $\langle \omega, a^\vee \rangle \in \{0, 1\}$ for every $a \in \Phi^+$. It is known (*loc. cit.*) that the group of characters of S is generated by the roots and a finite number of minuscule dominant weights.

Let ω be a minuscule dominant weight and let ρ be the irreducible K_0-representation of G with dominant weight ω, in a K_0-vector space V_0. Let Π be the set of weights of ρ and V_0^λ the subspace of V_0 corresponding to the weight $\lambda \in \Pi$. It is known (*loc. cit.*) that $\dim V_0^\lambda = 1$, that the elements of Π are the transforms of ω by the Weyl group of G and that $\rho(e_a)^2 = 0$ for all $a \in \Phi$, hence

$$\rho(x_a(u)) = I + u\rho(e_a). \tag{1}$$

Let $L'_f = \text{Lie}\,\mathfrak{S} + \sum_{a\in\Phi} f(a)e_a \subset \text{Lie}\,G$. This is a Lie sub-$A$-algebra of Lie G.

Let $V = K \otimes V_0$ and let $v \in V_0^\omega - \{0\}$ and M_f be the sub-L'_f-module of V generated by v (the Lie algebra L'_f acting on V by $d\rho$). Since $\rho(e_a)v = 0$ if $a \in \Phi^+$, $\rho(h)v = \omega(h)v$ if $h \in \text{Lie}\,\mathfrak{S}$ and $\rho(e_a)^2 = 0$, we have, by standard calculations,

$$M_f = \sum_{I\subset\Phi^-} \left(\prod_{a\in I} f(a)\right)\left(\prod_{a\in I} \rho(e_a)\right) v, \tag{2}$$

the factors of the second product being arranged in an order compatible with addition, chosen once and for all in Φ^-. We thus see that M_f is an A-module of finite type, a direct sum of its intersections with the $V^\lambda = K \otimes V_0^\lambda$.

Applying this result by replacing K by K_0 and A by $A_0 = \mathbf{Z}\cdot 1 \subset K_0$, and setting $f(a) = A_0$ for all $a \in \Phi$, we see (and it is moreover well known) that there exist $v_\lambda \in V_0^\lambda - \{0\}$ for $\lambda \in \Pi$, such that the free sub-A-module generated by the v_λ

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is stable under the $\rho(e_a)$ for $a \in \Phi$. Furthermore, if $a \in \Phi$ and $\lambda, \lambda + a \in \Pi$, we have $\langle \lambda, a^\vee\rangle = -1$ and $\lambda - a \notin \Pi$, hence

$$\rho(e_{-a})\rho(e_a)v_\lambda = \rho(h_a)v_\lambda = \langle \lambda, a^\vee\rangle v_\lambda = -v_\lambda,$$

and consequently,

$$\rho(e_a)v_\lambda = \epsilon v_{\lambda+a}, \tag{3}$$

where ϵ is an invertible element of A_0 (we can in fact choose the v_λ such that $\epsilon = \pm 1$).

For $\lambda \in \Pi$, let $g(\lambda)$ be the preimage of M_f in K under the map $t \mapsto tv_\lambda$. Given (2), this is a fractional ideal of finite type of A, and the stability of M_f under the $\rho(f(a)e_a)$, joined with (3), shows that

$$g(\lambda + a) \supset f(a)g(\lambda) \quad \text{for } a \in \Phi \text{ and } \lambda, \lambda + a \in \Pi. \tag{4}$$

In general, the ideals $g(\lambda)$ have no reason to be invertible, and, consequently, M_f is not necessarily projective. We say that the family $f = (f(a))$ is *good* for ω if there exists a family $(\hat{g}(\lambda))_{\lambda\in\Pi}$ of invertible fractional ideals of A such that

$$\hat{g}(\lambda + a) \supset f(a)\hat{g}(\lambda) \quad \text{for } a \in \Phi \text{ and } \lambda, \lambda + a \in \Pi. \tag{$\hat{4}$}$$

Then, $M_\omega = \sum_{\lambda\in\Pi} \hat{g}(\lambda)v_\lambda$ is a projective A-module of finite type, stable under the $\rho(f(a)e_a)$, and it follows immediately from (1) that M_ω is a $\mathscr{D}$-module.

3.2.11. — Let us give some sufficient conditions for the family f to be good for ω. An obvious condition is that $g(\lambda)$ be invertible for all λ. Or that $f(a) \subset A$ for all a (take $\hat{g}(\lambda) = A$ for all λ). Or again, that every element of Π is of the form $\omega - a$ with $a \in \Phi$ (take $\hat{g}(\omega) = A$ and $\hat{g}(\lambda) = f(\lambda - \omega)$ for $\lambda \neq \omega$), a condition satisfied, for example, for the "standard" representations of SL_n and Sp_{2n}.

More interesting is the case where the ring A possesses the property

(FI) Every fractional ideal of finite type a of A is contained in a smallest invertible ideal $\hat{a}$.

It then suffices, indeed, to take $\hat{g}(\lambda) = \widehat{g(\lambda)}$. Note that condition (FI) is satisfied if A is Prüferian (every ideal of finite type is then invertible), or if every divisorial ideal of A is invertible (since every invertible ideal is divisorial, $\hat{a}$ is then the smallest divisorial ideal containing a, which always exists), notably if A is factorial or is an integrally closed Noetherian ring such that the ring $A_\mathfrak{m}$ is factorial for every maximal ideal $\mathfrak{m}$ of A ([NB] AC VII, §3, prop. 1, p. 34), for example in the case of a regular ring.

3.2.12. — Suppose now that the family f is good for a set of minuscule dominant weights $\omega_1, \ldots, \omega_n$ generating, together with the roots, the character group of S. Then taking the direct sum M of L_f (3.2.9) and the modules M_{ω_i} constructed as in 3.2.10 for $1 \leq i \leq n$, one obtains a faithful $\mathscr{D}$-module: it is faithful for the $\mathfrak{U}_a$ since L_f is (3.2.9), and it is faithful for $\mathfrak{S}$ because the weights of $\mathfrak{S}$ in M generate the character group of $\mathfrak{S}$.

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3.2.13. — If A is a Dedekind ring or, more generally, a Prüfer ring, the preceding results combined with scholium 2.2.4 provide smooth A-group schemes. If we take $f(a) = A$ for all $a \in \Phi$, we recover the "Chevalley

schemes" of C. Chevalley [16] and M. Demazure [17]. Note that our construction is more elementary than those of [16] and [17] in that it does not involve the reduced powers of the e_a.

We conclude this section with two "counter-examples".

3.2.14. — *Non-uniqueness of* $\mathfrak{G}$.

Let A be a discrete valuation ring, with uniformizer π and residue field k, let G be the group SL_2, S the torus $\{\mathrm{diag}(t, t^{-1})\}$ and a the root $(t, t^{-1}) \mapsto t^2$. Let us choose as Chevalley system the one for which ζ_a is the identity map and take $f(a) = f(-a) = \pi A$.

We are going to construct two faithful $\mathcal{D}$-modules leading, by application of Theorem 2.2.3, to two smooth A-group schemes both satisfying the conditions of 3.1.3, but not isomorphic.

1st *solution.* — Let $V = K^2 \times \mathfrak{g}$, $\rho = \mathrm{Id} \times \mathrm{Ad}$ and $M = A^2 \times L_{\mathfrak{f}}$. One immediately verifies, thanks to 3.2.9 and 3.2.12, that M is a faithful $\mathcal{D}$-module. Referring A^2 to its canonical basis and $L_{\mathfrak{f}}$ to the basis $(\pi e_{-a}, h, \pi e_a)$ where h is the element of $\mathfrak{h}_A$ such that $a(h) = 1$, we see that

$$\rho\left(\begin{pmatrix} x & y \\ z & t \end{pmatrix}\right) = \left(\begin{pmatrix} x & y \\ z & t \end{pmatrix}, \begin{pmatrix} t^2 & -\pi^{-1}zt & z^2 \\ -2\pi yt & xt + yz & -2\pi xz \\ y^2 & -\pi^{-1}xy & x^2 \end{pmatrix}\right). \tag{1}$$

It follows that the affine algebra $A[\mathfrak{G}]$ of the schematic closure $\mathfrak{G}$ of $\rho(G)$ in $\mathfrak{GL}(M)$ identifies with the sub-A-algebra of $K[G] = K[x, y, z, t]/(xt - yz - 1)$ generated by $x, y, z, t, \pi^{-1}zt$ and $\pi^{-1}xy$ and is isomorphic to

$$A[X, Y, Z, T, U, V]/(XT - YZ - 1, XY - \pi U, ZT - \pi V).$$

From this, we deduce that $\mathfrak{G}$ is a smooth A-group scheme. Its closed fiber $\mathfrak{G}_k$ has as affine algebra $k[X, Y, Z, T, U, V]/(XT - YZ - 1, XY, ZT)$ and is not connected; its neutral component $\mathfrak{G}_k^\circ$ is defined in $\mathfrak{G}_k$ by the equations $Y = Z = 0$ and "is" the subgroup of $GL(k \otimes M)$ formed by the matrices of the form

$$\left(\begin{pmatrix} x & 0 \\ 0 & x^{-1} \end{pmatrix}, \begin{pmatrix} x^{-2} & -v & 0 \\ 0 & 1 & 0 \\ 0 & -u & x^2 \end{pmatrix}\right). \tag{2}$$

This is a solvable group, an extension of $\mathfrak{Mult}$ by $(\mathfrak{Add})^2$. However $\mathfrak{G}_k$ has another connected component, defined by the equations $X = T = 0$, and the element $w = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$

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of the Weyl group of G "is" a section of $\mathfrak{G}$ (or if one prefers an element of $\mathfrak{G}(A)$), but is not a section of the identity component $\mathfrak{G}^0$. The scheme $\mathfrak{G}$ is the union of $\mathfrak{G}^0$ and $w\mathfrak{G}^0$.

2nd solution. — Let $V = \mathbf{K}^2 \times \mathfrak{g} \times \mathfrak{g}$, $\rho = \mathrm{id} \times \mathrm{Ad} \times \mathrm{Ad}$ and let M be the A-module generated by the canonical basis of $\mathbf{K}^2$, the basis $(\pi e_{-a}, h, e_a)$ of the first copy of $\mathfrak{g}$ and the basis $(e_{-a}, h, \pi e_a)$ of the second copy of $\mathfrak{g}$. We find

$$\rho\left(\begin{pmatrix} x & y \\ z & t \end{pmatrix}\right) = \left(\begin{pmatrix} x & y \\ z & t \end{pmatrix}, \begin{pmatrix} t^2 & -\pi^{-1}zt & \pi^{-1}z^2 \\ -2\pi yt & xt + yz & -2xz \\ \pi y^2 & -xy & x^2 \end{pmatrix}, \begin{pmatrix} t^2 & -zt & \pi z^2 \\ -2yt & xt + yz & -2\pi xz \\ \pi^{-1}y^2 & -\pi^{-1}xy & x^2 \end{pmatrix}\right)$$

and one easily verifies that M is a faithful $\mathscr{D}$-module and that the affine algebra of the schematic closure $\mathfrak{G}'$ of $\rho(G)$ in $\mathbf{GL}(M)$ is generated by $x, y, z, t, \pi^{-1}y$ and $\pi^{-1}z$ (taking into account that e.g. $\pi^{-1}z = \pi^{-1}z^2x - \pi^{-1}zty$) and is isomorphic to $A[X, T, U, V]/(XT - \pi^2 UV - 1)$. Its closed fiber is connected and one easily sees that the identity map of G extends to an isomorphism from $\mathfrak{G}'$ onto the identity component $\mathfrak{G}^0$ of the scheme $\mathfrak{G}$ of the first solution.

3.2.15. — *An example where* $\mathfrak{G}$ *is not flat and is not a group scheme.*

Let A be the (non-Prüferian) polynomial ring $k[\xi, \eta]$ in two indeterminates over an algebraically closed field k and let G, S, a and the Chevalley system be as in 3.2.14. Let $f(a) = \xi A, f(-a) = \eta A, V = \mathbf{K}^2 \times \mathfrak{g}, \rho = \mathrm{id} \times \mathrm{Ad}$ and choose for M the free A-module generated by the canonical basis of $\mathbf{K}^2$ and the elements $\eta e_{-a}, h$ and ξe_a of $\mathfrak{g}$. One sees as above that this is a faithful $\mathscr{D}$-module and that

$$\rho\left(\begin{pmatrix} x & y \\ z & t \end{pmatrix}\right) = \left(\begin{pmatrix} x & y \\ z & t \end{pmatrix}, \begin{pmatrix} t^2 & -\eta^{-1}zt & \xi\eta^{-1}z^2 \\ -2\eta yt & xt + yz & -2\xi xz \\ \eta\xi^{-1}y^2 & -\xi^{-1}xy & x^2 \end{pmatrix}\right) \tag{1}$$

Let $(x_{ij})_{1\le i,j\le 3}$ be the coordinates on End $\mathfrak{g}$ associated with the chosen basis; one sees that $\rho(G)$ is defined in $\mathbf{K}^{13}$ by the equations

$$x_{11} = t^2, \quad x_{21} = -2\eta y t, \quad x_{22} = xt + yz, \quad x_{23} = -2\xi x z, \quad x_{33} = x^2, \tag{2}$$

and

$$\begin{aligned} &xt - yz - 1 = 0, \quad \eta x_{12} + zt = 0, \quad \eta x_{13} - \xi z^2 = 0, \\ &\xi x_{31} - \eta y^2 = 0, \quad \xi x_{32} + xy = 0, \end{aligned} \tag{3}$$

which generate the ideal of $\rho(G)$ in $\mathbf{K}[x, y, z, t, (x_{ij})]$. The equations (2) allow one to eliminate the five coordinates $x_{11}, x_{21}, x_{22}, x_{23}$ and x_{33} and to consider $\rho(G)$ as an irreducible subvariety of dimension 3 of $\mathbf{K}^8$, whose ideal I is generated by the left-hand sides of equations (3). But these do not gener-

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...do not generate the ideal $I_A = I \cap A[x, y, z, t, x_{12}, x_{13}, x_{31}, x_{32}]$: one can show that I_A is generated by the left-hand sides of (3) and those of

$$tx_{13} + \xi x_{12} = 0, \quad x x_{31} + \eta x_{32} = 0, \quad x_{13}x_{31} - y^2 z^2 = 0. \tag{4}$$

Consider then the schematic closure $\mathfrak{G}$ of $\rho(G)$ in $\mathfrak{GL}(M)$ and its fiber $\mathfrak{G}_p = \mathfrak{G}_{(\alpha,\beta)}$ above a closed point $p = (\alpha, \beta)$ of Spec A. If $\alpha, \beta \in k^\times$, we have $\mathfrak{G}_p \cong SL_2$. If $\alpha = 0$ and $\beta \neq 0$, one easily sees that $\mathfrak{G}_p$ "is" the group of matrices of the form

$$\left(\begin{pmatrix} x & 0 \\ z & x^{-1} \end{pmatrix}, \begin{pmatrix} x^{-2} & \beta^{-1} z x^{-1} & 0 \\ 0 & 1 & 0 \\ 0 & u & x^2 \end{pmatrix} \right) \tag{5}$$

and is isomorphic to the semi-direct product of the triangular group $\left\{ \begin{pmatrix} x & 0 \\ z & x^{-1} \end{pmatrix} \right\}$ by $\mathfrak{Add}$.

Finally, if $\alpha = \beta = 0$, then $\mathfrak{G}_{(0,0)}$ has two connected components, defined respectively by the equations

$$y = z = 0, \quad xt = 1, \quad x_{13} = x_{31} = 0 \tag{6}$$

and

$$x = t = 0, \quad yz = -1, \quad x_{13}x_{31} = 1. \tag{7}$$

The first connected component, defined by (6), is the group of dimension 3 already encountered in 3.2.14 (2). But the second is the variety $\mathfrak{G}^1_{(0,0)}$ of dimension 4 of matrices of the form

$$\left(\begin{pmatrix} 0 & y \\ -y^{-1} & 0 \end{pmatrix}, \begin{pmatrix} 0 & u & v \\ 0 & -1 & 0 \\ v^{-1} & w & 0 \end{pmatrix} \right). \tag{8}$$

These two connected components being of different dimensions, the fiber $\mathfrak{G}_{(0,0)}$ cannot be a group and the scheme $\mathfrak{G}$ is not flat (moreover $\dim \mathfrak{G}_{(0,0)} > \dim G$). On the other hand, $\mathfrak{H} = \mathfrak{G} - \mathfrak{G}^1_{(0,0)}$ is open in $\mathfrak{G}$ and is a group scheme, as stated in 2.2.10, but is probably not affine.

Let us remark that to show that the variety (8) of dimension 4 is contained in $\mathfrak{G}_{(0,0)}$, and thus that $\mathfrak{G}$ is not flat, it is not necessary to completely determine the ideal I_A (which we have in fact left to the reader). It suffices, in heuristic terms, to consider the point $\rho\left(\begin{pmatrix} \xi x' & y \\ z & \eta t' \end{pmatrix} \right)$ with $\xi\eta x' t' - yz = 1$, to "let ξ and η tend to 0" while keeping for $\zeta = \xi\eta^{-1}$ a fixed non-zero value and finally to set $u = -zt'$, $v = \zeta z^2$ and $w = -x'y$.

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3.3. Construction of $\mathfrak{U}_\Psi$ and of $\mathfrak{Z}\mathfrak{U}_\Psi$.

Lemma 0.8 (3.3.1). *Let Y be a K-group scheme, X_i (for $i = 1, 2$) a closed K-subgroup scheme of Y, $\mathfrak{Y}$ a flat A-scheme extending the K-scheme Y, $\mathfrak{X}_i$ a flat A-group scheme extending the K-group scheme X_i. Suppose that the product maps $X_1 \times X_2 \to Y$ and $X_2 \times X_1 \to Y$ extend to isomorphisms of A-schemes $\alpha : \mathfrak{X}_1 \times \mathfrak{X}_2 \to \mathfrak{Y}$ and $\beta : \mathfrak{X}_2 \times \mathfrak{X}_1 \to \mathfrak{Y}$. Then, the group law of Y extends to a group law on $\mathfrak{Y}$.*

The product, the identity section, and the inverse of $\mathfrak{Y}$ are obtained respectively as follows:

$$\mathfrak{Y} \times \mathfrak{Y} \xrightarrow{\alpha^{-1}\times\alpha^{-1}} \mathfrak{X}_1 \times \mathfrak{X}_2 \times \mathfrak{X}_1 \times \mathfrak{X}_2 \xrightarrow{\mathrm{id}\times(\alpha^{-1}\circ\beta)\times\mathrm{id}} \mathfrak{X}_1 \times \mathfrak{X}_1 \times \mathfrak{X}_2 \times \mathfrak{X}_2 \xrightarrow{\mathrm{prod}\times\mathrm{prod}} \mathfrak{X}_1 \times \mathfrak{X}_2 \xrightarrow{\alpha} \mathfrak{Y},$$

$$\operatorname{Spec} A \xrightarrow{e\times e} \mathfrak{X}_1 \times \mathfrak{X}_2 \xrightarrow{\alpha} \mathfrak{Y},$$

$$\mathfrak{Y} \xrightarrow{\alpha^{-1}} \mathfrak{X}_1 \times \mathfrak{X}_2 \xrightarrow{\mathrm{inv}\times\mathrm{inv}} \mathfrak{X}_1 \times \mathfrak{X}_2 \xrightarrow{\mathrm{pr}_2\times\mathrm{pr}_1} \mathfrak{X}_2 \times \mathfrak{X}_1 \xrightarrow{\beta} \mathfrak{Y}.$$

It is clear that the morphisms thus defined extend respectively the product, the identity element, and the inverse of Y and thus verify by continuity the axioms of group laws (1.2.8).

Proposition 0.9 (3.3.2). *Let Ψ be a positively closed subset of Φ. There exists an A-scheme $\mathfrak{U}_\Psi$ and only one up to a unique isomorphism, extending U_Ψ and such that, for any nibbling order (3.1.2) or inverse of a nibbling order on Ψ, the K-scheme isomorphism from $\prod_{a\in\Psi} U_a$ onto U_Ψ given by the product map (1.1.7) extends to an isomorphism of A-schemes*

$$\prod_{a\in\Psi} \mathfrak{U}_a \to \mathfrak{U}_\Psi. \tag{1}$$

The scheme $\mathfrak{U}_\Psi$ is flat, the group law of U_Ψ extends to a group law on $\mathfrak{U}_\Psi$, and (1) is the product map. For any positively closed subset Ψ' of Ψ, the injection of $U_{\Psi'}$ into U_Ψ extends to a group A-scheme isomorphism from $\mathfrak{U}_{\Psi'}$ onto a closed A-subgroup scheme of $\mathfrak{U}_\Psi$.

Proof. We will perform the proof by induction on $\operatorname{Card}\Psi$, the case $\operatorname{Card}\Psi = 1$ being included in the hypotheses. The uniqueness and the flatness of $\mathfrak{U}_\Psi$ are evident. To show its existence, it suffices to show that, if a and b are two extremal elements of Ψ, the composed K-isomorphism

$$U_a \times U_{\Psi-\{a\}} \xrightarrow{\mathrm{prod}} U_\Psi \xrightarrow{\mathrm{prod}^{-1}} U_{\Psi-\{b\}} \times U_b \tag{2}$$

extends to an isomorphism $\mathfrak{U}_a \times \mathfrak{U}_{\Psi-\{a\}} \to \mathfrak{U}_{\Psi-\{b\}} \times \mathfrak{U}_b$. If $a \neq b$, this is evident thanks to the induction hypothesis:

$$\mathfrak{U}_a \times \mathfrak{U}_{\Psi-\{a\}} \xrightarrow{\sim} \mathfrak{U}_a \times \mathfrak{U}_{\Psi-\{a,b\}} \times \mathfrak{U}_b \xrightarrow{\sim} \mathfrak{U}_{\Psi-\{b\}} \times \mathfrak{U}_b.$$

□

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Suppose then $a = b$. It suffices to show the existence of a morphism

$$U_a \times U_{\Psi-\{a\}} \to U_{\Psi-\{a\}} \times U_a$$

extending (2): we prove in the same way the existence of a morphism

$$U_{\Psi-\{a\}} \times U_a \to U_a \times U_{\Psi-\{a\}}$$

extending the inverse of (2), and the composition of these two morphisms extends the identity, and thus is the identity. Let then b' be an extremal element of Ψ distinct from a and set $\Psi_1 = \Psi \cap (a+b') \subset \Psi - \{a, b'\}$. According to (DRS 1), the map $(x, y) \to (xyx^{-1}y^{-1}, y, x)$ from $U_a \times U_{b'}$ into $U_{\Psi_1} \times U_{b'} \times U_a$ extends to a morphism

$$\alpha : U_a \times U_{b'} \to U_{\Psi_1} \times U_{b'} \times U_a.$$

The sought-after morphism is then obtained as the composition

$$\begin{aligned} U_a \times U_{\Psi-\{a\}} &\xrightarrow{\mathrm{id}\times\mathrm{prod}^{-1}} U_a \times U_{\Psi-\{a,b'\}} \times U_{b'} \xrightarrow{\mathrm{prod}\times\mathrm{id}} U_{\Psi-\{b'\}} \times U_{b'} \\ &\xrightarrow{\mathrm{prod}^{-1}\times\mathrm{id}} U_{\Psi-\{a,b'\}} \times U_a \times U_{b'} \xrightarrow{\mathrm{id}\times\alpha} U_{\Psi-\{a,b'\}} \times U_{\Psi_1} \times U_{b'} \times U_a \\ &\xrightarrow{\mathrm{prod}\times\mathrm{id}} U_{\Psi-\{a\}} \times U_a. \end{aligned}$$

We have, of course, applied the induction hypothesis several times.

The existence of the group law on U_Ψ then follows from Lemma 3.3.1 and it is clear, by continuity, that (1) is the product map. Finally, the last assertion follows immediately from the following lemma:

Lemma 0.10 (3.3.3)**.** *Let $\Psi' \subset \Psi$ be two positively closed subsets of Φ. Any nibbling order on Ψ' is induced by a nibbling order on Ψ.*

Proof. Let us prove it by induction on $\operatorname{Card}\Psi$. Let a be the first element of Ψ'. If a is extremal in Ψ, we take it as the first element of Ψ and apply the induction hypothesis to the positively closed subsets $\Psi - \{a\}$ and $\Psi' - \{a\}$. If a is not extremal in Ψ, we choose as the first element of Ψ an extremal element $b \notin \Psi'$ and apply the induction hypothesis to $\Psi - \{b\}$ and Ψ'. □

3.3.4. — The composition with the product map of the map $\gamma_{a,b}$ given by (DRS 1) is none other than the commutator map from $U_a \times U_b$ into U_Ψ (for $a, b \in \Psi$). since it extends the commutator map from $U_a \times U_b$ into U_Ψ. It follows that if $\Psi' \subset \Psi$ is such that $a \in \Psi', b \in \Psi'$ imply $\Phi \cap (a + b) \subset \Psi'$, then $U_{\Psi'}$ is a normal group subscheme of U_Ψ. In particular, if a is an extremal element of Ψ, then U_Ψ is the semi-direct product of U_a by $U_{\Psi - \{a\}}$.

On the other hand, in "good cases" (cf. 2.2 and 3.8), the product map $\prod_{a\in\Psi} U_a \to U_\Psi$ is an isomorphism of schemes regardless of the order placed on Ψ.

Proposition 0.11 (3.3.5)**.** *The action of Z on U_Ψ extends to an action of $\mathfrak{Z}$ on U_Ψ, respecting the group structure of U_Ψ.*

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The action of $\mathfrak{Z}$ on $\mathfrak{U}_\Psi$ is obtained by writing $\mathfrak{U}_\Psi$ as a product $\prod_{a\in\Psi} \mathfrak{U}_a$ (3.3.2) and by taking the product of the actions of $\mathfrak{Z}$ on the $\mathfrak{U}_a$ given by (DRS 2). That one indeed obtains an action of $\mathfrak{Z}$ on $\mathfrak{U}_\Psi$ this way and that it respects the group structure of $\mathfrak{U}_\Psi$ results by continuity from the properties of the action of Z on U_Ψ.

3.3.6. — One can therefore consider on $\mathfrak{Z} \times \mathfrak{U}_\Psi$ the corresponding structure of semi-direct product of $\mathfrak{Z}$ by $\mathfrak{U}_\Psi$ (cf. [DG], p. 166): one thus obtains an A-group scheme, denoted $\mathfrak{Z}\mathfrak{U}_\Psi$, extending ZU_Ψ and containing $\mathfrak{Z}$ (resp. $\mathfrak{U}_\Psi$) as a closed subgroup scheme (resp. closed and normal).

3.3.7. Remark. — In addition to the flatness of the schemes $\mathfrak{U}_a$ (resp. $\mathfrak{Z}$ and $\mathfrak{U}_a$), the proof of 3.3.2 (resp. 3.3.5) only uses (DRS 1) (resp. (DRS 1) and (DRS 2)).

3.4. The algebra of distributions associated with a schematic root datum.

Henceforth, we assume that the A-schemes $\mathfrak{U}_a$ for $a \in \Phi$, are smooth.

3.4.1. — We identify the algebra of distributions Dist $\mathfrak{Z}$ (1.3.1) (resp. Dist $\mathfrak{U}_a$, Dist $\mathfrak{U}_\Psi$) with an A-subalgebra of the K-algebra Dist Z (resp. Dist U_a, Dist U_Ψ), which itself is identified with a K-subalgebra of Dist G (1.3.3). We denote by Dist $\mathscr{D}$ the A-subalgebra of Dist G generated by Dist $\mathfrak{Z}$ and the Dist $\mathfrak{U}_a$ for $a \in \Phi$.

Proposition 0.12 (3.4.2)**.** *Let Ψ be a positively closed subset of Φ.*

(i) The product map $\bigotimes_{a\in\Psi}$ Dist $\mathfrak{U}_a \to$ Dist $\mathfrak{U}_\Psi$ is bijective for any nibbling order or inverse of a nibbling order placed on Ψ.

(ii) The product map

$$\textit{Dist } \mathfrak{Z} \otimes \textit{Dist } \mathfrak{U}_\Psi \to \textit{Dist } \mathfrak{Z}\mathfrak{U}_\Psi \quad (\textit{resp. Dist } \mathfrak{U}_\Psi \otimes \textit{Dist } \mathfrak{Z} \to \textit{Dist } \mathfrak{Z}\mathfrak{U}_\Psi)$$

is bijective.

Given 3.3.2 and 3.3.6, this follows from 1.3.5, since the $\mathfrak{U}_a$, and therefore $\mathfrak{U}_\Psi$, are smooth.

Lemma 0.13 (3.4.3)**.** *Let $a \in \Phi$. In the algebra Dist G, we have*

$$\textit{Dist } \mathfrak{U}_a \cdot \textit{Dist } \mathfrak{Z} \cdot \textit{Dist } \mathfrak{U}_{-a} = \textit{Dist } \mathfrak{U}_{-a} \cdot \textit{Dist } \mathfrak{Z} \cdot \textit{Dist } \mathfrak{U}_a.$$

Proof. Let us return to the notation of (DRS 3) and let $\delta \in$ Dist $\mathfrak{U}_a$, $\delta' \in$ Dist $\mathfrak{U}_{-a}$. Since $\mathfrak{W}_a$ is an open neighborhood of the identity section e in $\mathfrak{U}_a \times \mathfrak{U}_{-a}$, the A-module Dist $\mathfrak{W}_a$ is identified with Dist $\mathfrak{U}_a \otimes$ Dist $\mathfrak{U}_{-a}$ (1.3.2 and 1.3.5). Consequently,

$$\delta \otimes \delta' \in \text{Dist } \mathfrak{W}_a, \quad \beta_a(\delta \otimes \delta') \in \text{Dist } \mathfrak{U}_{-a} \otimes \text{Dist } \mathfrak{Z} \otimes \text{Dist } \mathfrak{U}_a \quad (1.3.5)$$

and

$$(\text{prod} \circ \beta_a)(\delta \otimes \delta') \in \text{Dist } \mathfrak{U}_{-a} \cdot \text{Dist } \mathfrak{Z} \cdot \text{Dist } \mathfrak{U}_a.$$

□

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But, according to (DRS 3), $(\mathrm{prod} \circ \beta_a)_K$ is the restriction of the product map $U_a \times U_{-a} \to G$, which implies that $(\mathrm{prod} \circ \beta_a)(\delta \otimes \delta') = \delta\delta'$. In other words, we have

$$\mathrm{Dist}\,\mathfrak{U}_a \cdot \mathrm{Dist}\,\mathfrak{U}_{-a} \subset \mathrm{Dist}\,\mathfrak{U}_{-a} \cdot \mathrm{Dist}\,\mathfrak{Z} \cdot \mathrm{Dist}\,\mathfrak{U}_a,$$

whence the lemma, taking into account 3.4.2 (ii).

Proposition 0.14 (3.4.4)**.** *Let Φ^+ be a system of positive root rays. Let $\Phi^- = -\Phi^+$. The product map from $\mathrm{Dist}\,\mathfrak{U}_{\Phi^-} \otimes \mathrm{Dist}\,\mathfrak{Z} \otimes \mathrm{Dist}\,\mathfrak{U}_{\Phi^+}$ to $\mathrm{Dist}\,\mathscr{D}$ is bijective.*

The injectivity results from that of the product map from $U_{\Phi^-} \times Z \times U_{\Phi^+}$ into G. To establish surjectivity, it suffices to show that the image Δ does not depend on the choice of Φ^+: this will imply $\Delta = \mathrm{Dist}\,\mathscr{D}$ because, on one hand, $\Delta \cdot \mathrm{Dist}\,\mathfrak{Z} = \Delta$ given 3.4.2 (ii), and on the other hand $\Delta \cdot \mathrm{Dist}\,\mathfrak{U}_a = \Delta$ for all $a \in \Phi$, as can be seen by choosing Φ^+ containing a, thus $\Delta \cdot \mathrm{Dist}\,\mathscr{D} = \Delta$ and $\Delta \supset \mathrm{Dist}\,\mathscr{D}$.

Let then Φ_1^+ be another system of positive root rays. We want to show that its substitution for Φ^+ does not change Δ. Let us reason by induction on $\mathrm{Card}(\Phi_1^+ \cap \Phi^-)$. If this cardinality is zero, there is nothing to prove. Otherwise, there exists an extremal element a in Φ_1^+ belonging to Φ^-. Let $\Psi^+ = \Phi_1^+ - \{a\}$ and $\Psi^- = \Phi_1^- - \{-a\}$. Then $\Phi_2^+ = \Psi^+ \cup \{-a\}$ is a system of positive root rays (cf. [NB] Lie VI, p. 157) and, given 3.4.2 (i) and 3.4.3, we have

$$\begin{aligned}
\mathrm{Dist}\,\mathfrak{U}_{\Phi_1^-} \cdot \mathrm{Dist}\,\mathfrak{Z} \cdot \mathrm{Dist}\,\mathfrak{U}_{\Phi_1^+} &= \mathrm{Dist}\,\mathfrak{U}_{\Psi^-} \cdot \mathrm{Dist}\,\mathfrak{U}_{-a} \cdot \mathrm{Dist}\,\mathfrak{Z} \cdot \mathrm{Dist}\,\mathfrak{U}_a \cdot \mathrm{Dist}\,\mathfrak{U}_{\Psi^+} \\
&= \mathrm{Dist}\,\mathfrak{U}_{\Psi^-} \cdot \mathrm{Dist}\,\mathfrak{U}_a \cdot \mathrm{Dist}\,\mathfrak{Z} \cdot \mathrm{Dist}\,\mathfrak{U}_{-a} \cdot \mathrm{Dist}\,\mathfrak{U}_{\Psi^+} \\
&= \mathrm{Dist}\,\mathfrak{U}_{\Phi_2^-} \cdot \mathrm{Dist}\,\mathfrak{Z} \cdot \mathrm{Dist}\,\mathfrak{U}_{\Phi_2^+},
\end{aligned}$$

whence our assertion by induction.

Corollary 0.15 (3.4.5)**.** *The canonical map $K \otimes \mathrm{Dist}\,\mathscr{D} \to \mathrm{Dist}\,G$ is bijective.*

This results from 3.4.4, 1.1.9 and 1.3.4.

3.5. $\mathscr{D}$-modules: preliminary lemmas.

In this section, we denote by $\mathfrak{X}$ a smooth A-group scheme with connected generic fiber $\mathfrak{X}_K$, and by ρ a representation of $\mathfrak{X}_K$ in a vector space V of finite dimension over K.

Lemma 0.16 (3.5.1)**.** *Assume the following hypothesis:*

(C) If $f \in K[\mathfrak{X}]$ is such that $\langle \delta, f \rangle \in A$ for all $\delta \in \mathrm{Dist}\,\mathfrak{X}$, then $f \in A[\mathfrak{X}]$.

Let M be a finitely generated projective sub-A-module of V such that $\rho(\mathrm{Dist}\,\mathfrak{X})M = M$; then $\mathfrak{X}$ acts on M via ρ.

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We have $\rho(\mathrm{Dist}\ \mathfrak{X}_K) \cdot KM = KM$. According to 1.4.8, the vector subspace KM of V is thus invariant under ρ and we can assume $V = KM$. The dual ${}^tM$ of M identifies then with a finitely generated projective sub-A-module of the dual of V. If $m \in M$, $m' \in {}^tM$ and $\delta \in \mathrm{Dist}\ \mathfrak{X}$, we have

$$\langle \delta, c^{\rho}_{m',m} \rangle = \langle m', \rho(\delta)m \rangle \in A$$

(cf. 1.4.7), and condition (C) implies $c^{\rho}_{m',m} \in A[\mathfrak{X}]$, Q.E.D.

Lemma 0.17 (3.5.2)**.** *Let us make the following hypothesis:*

(C') $A[\mathfrak{X}]$ is the union of an increasing sequence of finitely generated projective sub-A-modules M_n such that, for every n, the restriction map r_n from Dist $\mathfrak{X}$ into ${}^tM_n$ is surjective.

Then, condition (C) of 3.5.1 is satisfied and there exists an integer N and an A-linear map q : Dist $\mathfrak{X} \to$ Dist$_N\mathfrak{X}$ such that $\rho(\delta) = \rho(q(\delta))$ for all $\delta \in$ Dist $\mathfrak{X}$.

Proof. For every integer n, one can choose an A-linear map $u_n : {}^tM_n \to \mathrm{Dist}\ \mathfrak{X}$ such that $r_n \circ u_n = \mathrm{id}$ and $\langle u_n \circ r_n(\delta), m\rangle = \langle \delta, m\rangle$ for $\delta \in \mathrm{Dist}\ \mathfrak{X}$ and $m \in M_n$.

Let then $f \in K[\mathfrak{X}]$ be such that $\langle \delta, f\rangle \in A$ for all $\delta \in \mathrm{Dist}\ \mathfrak{X}$. There exists an integer n such that $f \in K \otimes M_n$ and for every $\lambda \in {}^tM_n$, we have

$$\langle \lambda, f\rangle = \langle r_n \circ u_n(\lambda), f\rangle = \langle u_n(\lambda), f\rangle \in A,$$

whence $f \in M_n$, which proves (C).

Furthermore, let M be the A-module generated by a basis of V. Its image under the coaction $c^\rho : V \to K[\mathfrak{X}] \otimes_K V = A[\mathfrak{X}] \otimes_A V$ is contained in $M_n \otimes_A V$ for a sufficiently large integer n. For $u \in {}^tV$, $m \in M$ and $\delta \in \mathrm{Dist}\ \mathfrak{X}$, we have

$$\langle u, \rho(\delta)m\rangle = \langle \delta \otimes u, c^\rho(m)\rangle = \langle u_n \circ r_n(\delta) \otimes u, c^\rho(m)\rangle,$$

whence the second assertion by taking $q = u_n \circ r_n$ and N large enough so that $\mathrm{Dist}_N\mathfrak{X}$ contains the finitely generated sub-A-module that is the image of q. □

Remark 2 (3.5.3). 1) One can show that if A is a Dedekind ring, then condition (C) is satisfied if and only if $\mathfrak{X}$ is connected (1.2.12).

2) One can express conditions (C) and (C') for any smooth A-scheme $\mathfrak{X}$ with connected generic fiber, equipped with a section s, by replacing Dist $\mathfrak{X}$ with ${}^s\mathrm{Dist}\ \mathfrak{X}$. Let $d \in A[\mathfrak{X}]$ be invertible on s. If $\mathfrak{X}$ equipped with s satisfies (C), then the special open $\mathfrak{X}_d$ equipped with s also satisfies (C). Indeed, we have ${}^s\mathrm{Dist}\ \mathfrak{X}_d = {}^s\mathrm{Dist}\ \mathfrak{X}$ and if $g \in K[\mathfrak{X}]$ is such that $\langle \delta, d^{-n}g\rangle \in A$ for all $\delta \in {}^s\mathrm{Dist}\ \mathfrak{X}$, then $\langle \delta, g\rangle = \sum \langle \delta_i, d^n\rangle \cdot \langle \delta'_i, d^{-n}g\rangle \in A$ for all $\delta \in {}^s\mathrm{Dist}\ \mathfrak{X}$ (1.3.5 (1)), whence $g \in A[\mathfrak{X}]$.

Similarly, if $\mathfrak{X}$ equipped with s satisfies (C'), then so does $\mathfrak{X}_d$. Indeed, up to replacing (M_n) by a subsequence, we can assume $dM_n \subset M_{n+1}$. Let $M'_n = d^{-n}M_n$; then $A[\mathfrak{X}_d]$ is the union of the increasing sequence of the M'_n, which are finitely generated projective, and a linear form on M'_n is of the form $f \mapsto \langle \delta, d^n f\rangle$ with $\delta \in {}^s\mathrm{Dist}\ \mathfrak{X}$. It then suffices to apply 1.3.5 (1).

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Moreover, it is immediate that the A-scheme $\mathfrak{X}$ associated with a finitely generated projective A-module M, equipped with an arbitrary section s, satisfies (C'): one can assume that s is the zero section and it suffices to take for M_n the A-submodule of $A[\mathfrak{X}] = \mathrm{Sym}\ {}^tM$ generated by the homogeneous elements of degree $\leq n$.

In particular, we see that every A-group scheme $\mathfrak{X}$ whose underlying scheme is a special open set of a scheme associated with a finitely generated projective A-module satisfies (C') and therefore (C). This applies in particular to $\mathfrak{Mult}$.

3) One can show that condition (C') is equivalent to:

(C'') There exists a sequence of finitely generated projective A-submodules $(P_n)_{n\geq 0}$ of $A[\mathfrak{X}]$ such that P_n is a complement of $I(s)^{n+1}$ in $I(s)^n$ and that $A[\mathfrak{X}]$ is the direct sum of the P_n.

Lemma 0.18 (3.5.4)**.** *Suppose an operation of $\mathfrak{Mult}$ on the scheme $\mathfrak{X}$ is given, fixing the neutral element of $\mathfrak{X}$ and such that the characters of $\mathfrak{Mult}_K$ occurring in the associated representation of $\mathfrak{Mult}_K$ in the tangent space to $\mathfrak{X}_K$ at the neutral element are strictly positive. Then, condition (C') is satisfied.*

Proof. The operation of $\mathfrak{Mult}$ on $\mathfrak{X}$ is expressed by the existence of a grading $(R_j)_{j\in\mathbb{Z}}$ of the A-algebra $A[\mathfrak{X}]$ such that the coaction

$$A[\mathfrak{X}] \to A[\mathfrak{Mult}] \otimes A[\mathfrak{X}] = A[T, T^{-1}] \otimes A[\mathfrak{X}]$$

is given by $\sum r_j \mapsto \sum T^j \otimes r_j$ (for $r_j \in R_j$). In other words, $\mathfrak{Mult}$ operates on R_j via the character $-j : t \mapsto t^{-j}$. Since $\mathfrak{Mult}$ fixes the neutral element, the augmentation ideal $I = I(e)$ is graded. By hypothesis, the characters of $\mathfrak{Mult}_K$ occurring in KI/KI^2 are strictly negative, which implies that the characters of $\mathfrak{Mult}_K$ occurring in KI^n/KI^{n+1} are $\leq -n$. Since $\cap I^n = \{0\}$ (1.3.7), it follows that $R_j = \{0\}$ for $j < 0$ and that

$$I^n \subset \sum_{j\geq n} R_j. \tag{1}$$

Let then $M_n = \sum_{0\leq j\leq n-1} R_j$. Then $A[\mathfrak{X}]$ is the union of the M_n. Moreover, (1) shows that the finitely generated projective A-module $A[\mathfrak{X}]/I^n$ is identified with the direct sum of M_n and $\sum_{j\geq n} R_j/I^n$. Consequently, M_n is finitely generated projective and if $u \in {}^tM_n$, the linear form on $A[\mathfrak{X}]$ equal to u on M_n and to 0 on $\sum_{j\geq n} R_j$ is zero on I^n according to (1), thus belongs to Dist $\mathfrak{X}$, which completes the proof. □

Proposition 0.19 (3.5.5)**.** *Let Ψ be a positively closed subset of Φ. The group scheme $\mathfrak{U}_\Psi$ satisfies condition (C') of 3.5.2, and therefore condition (C) of 3.5.1.*

Indeed, one can choose an isomorphism j of $\mathfrak{Mult}$ onto a closed group subscheme of $\mathfrak{S}$ such that, for every $a \in \Psi$ and every root α of G according to S belonging to a, the character $\alpha \circ j$ of $\mathfrak{Mult}$ is strictly positive. The composition of j with the action of $\mathfrak{Z}$ on $\mathfrak{U}_\Psi$ (3.3.5) provides an action of $\mathfrak{Mult}$ on $\mathfrak{U}_\Psi$ satisfying the condition of 3.5.4.

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Lemma 0.20. ***3.5.6.*** — *Let $v \in V$ and set $M = \rho(Dist\ \mathfrak{X}) \cdot v$. Suppose that M is projective of finite type, that $\mathfrak{X}$ acts on M and that $A[\mathfrak{X}]$ possesses a generating set formed of coefficients of ρ of the form $c_{x,v}$, with $x \in {}^tV$. Then, $\mathfrak{X}$ acts faithfully on M.*

Proof. Let $x \in {}^tV$ be such that $c_{x,v} \in A[\mathfrak{X}]$ and let y be the restriction of x to M. For $\delta \in \mathrm{Dist}\ \mathfrak{X}$, we have

$$\langle y, \rho(\delta) \cdot v\rangle = \langle x, \rho(\delta) \cdot v\rangle = \langle \delta, c_{x,v}\rangle \in A,$$

hence $y \in {}^tM$. One then easily verifies that $c_{x,v}$ is the coefficient $c^{\bar\rho}_{y,v}$ of the representation $\bar\rho$ of $\mathfrak{X}$ in M defined by ρ. Consequently $A[\mathfrak{X}]$ is generated by the coefficients of $\bar\rho$, Q.E.D. □

Lemma 0.21. ***3.5.7.*** — *Suppose $\mathfrak{X}_K$ is unipotent and let $v \in V$ be such that the K-group scheme stabilizer of v in $\mathfrak{X}_K$ is trivial. For an integer $N \geq 1$, let $\rho(N)$ denote the representation $\bigotimes^N \rho$ of $\mathfrak{X}$ in $V_N = \bigotimes^N V$ and set $v_N = \bigotimes^N v$. If N is sufficiently large, the algebra $A[\mathfrak{X}]$ is generated by coefficients of $\rho(N)$ of the form $c^{(N)}_{y,v_N}$ for $y \in {}^tV_N$.*

We immediately reduce to the case where V is the smallest vector subspace containing the orbit $\rho(\mathfrak{X}_K) \cdot v$. Since $\mathfrak{X}_K$ is unipotent, this orbit is moreover closed ([DG], p. 488) and the map $x \mapsto \rho(x) \cdot v$ is a closed immersion of $\mathfrak{X}_K$ into V ([2], prop. 6.7). Consequently, the K-algebra $K[\mathfrak{X}]$ is generated by the coefficients $c_{x,v}$ for $x \in {}^tV$.

The set Γ_N of coefficients $c^{(N)}_{y,v_N}$ of $\rho(N)$ is exactly the set of homogeneous polynomials of degree N with coefficients in K in the coefficients $c_{x,v}$ for $x \in {}^tV$. On the other hand, since $\mathfrak{X}_K$ is unipotent, it possesses a fixed point $z \neq 0$ in ${}^tV$ and the coefficient $c_{z,v}$ is a non-zero constant (otherwise we would have $\mathfrak{X}_K \cdot v \subset \mathrm{Ker}\ z$), which we can assume to be equal to 1. It follows that Γ_N is the set of all polynomials of degree N, homogeneous or not, in the $c_{x,v}$.

Now, $A[\mathfrak{X}]$ is generated by a finite number of elements $f_1, \ldots, f_r$ of $K[\mathfrak{X}]$ and each f_i is a polynomial with coefficients in K in the $c_{x,v}$. It then suffices to take the integer N larger than the largest of the degrees of these polynomials.

3.6. Construction of a faithful $\mathscr{D}$-module for $\mathfrak{U}_\Psi$.

Henceforth, we assume that the ring A is Prüfer (cf. 2.2.2).

Proposition 0.22. ***3.6.1.*** — *Let ρ be a representation of G in a finite-dimensional vector space V over K and let M_0 be a sub-A-module of finite type of V. Let Ψ be a positively closed subset of Φ and set $M = \rho(Dist\ \mathfrak{U}_\Psi) \cdot M_0$. Then,*

(i) M is of finite type (hence projective) and $\mathfrak{U}_\Psi$ acts on M;

(ii) if $\mathfrak{Z}$ acts on M_0, then $\mathfrak{Z}$ acts on M.

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In view of 3.5.5 and 3.5.2, there exists an integer N such that $M = \rho(\mathrm{Dist}_N\, \mathfrak{U}_\Psi) \cdot M_0$. Consequently, M is of finite type, hence projective since it is torsion-free, and $\mathfrak{U}_\Psi$ acts on M according to 3.5.1.

Suppose now that $\mathfrak{Z}$ acts on M_0. The action of $\mathfrak{Z}$ on $\mathfrak{U}_\Psi$ (3.3.5) defines an action of $\mathfrak{Z}$ on the module $\mathrm{Dist}_N\, \mathfrak{U}_\Psi$ (1.4.9), whence an action of $\mathfrak{Z}$ on $M' = \mathrm{Dist}_N\, \mathfrak{U}_\Psi \otimes M_0$. The map $\varphi : M' \to M$ defined by $\varphi(\delta \otimes m) = \rho(\delta) \cdot m$ is surjective and $K \otimes \varphi : K \otimes M' \to K \otimes M \subset V$ is compatible with the action of Z (for all $z \in Z(K)$, we have $(K \otimes \varphi)(z \cdot (\delta \otimes m)) = \rho(z\delta z^{-1})\rho(z) \cdot m = \rho(z)\rho(\delta) \cdot m$, that is, it is a morphism of $K[Z]$-comodules, or even a morphism of $A[\mathfrak{Z}]$-comodules (note that for example $K[Z] \otimes_K (K \otimes_A M') = A[\mathfrak{Z}] \otimes_A (K \otimes_A M')$). It follows that $\mathrm{Ker}\, K \otimes \varphi$ and M' are two $A[\mathfrak{Z}]$-subcomodules of $K \otimes M'$, and the same holds for their intersection $\mathrm{Ker}\, \varphi$ [30]. By passing to the quotient by $\mathrm{Ker}\, \varphi$, we thus obtain an $A[\mathfrak{Z}]$-comodule structure on M, that is, an action of $\mathfrak{Z}$ on M, which recovers the representation ρ of Z on KM by tensoring with K.

3.6.2. Corollary. — *Let ρ be a representation of G on a finite-dimensional vector space V over K and let M_0 be a sub-A-module of finite type of V such that $\mathfrak{Z}$ acts on M_0. Let $M = \rho(\mathrm{Dist}\, \mathfrak{U}_\Psi)M_0$. Then, M is an A-module of finite type, KM is invariant under $\rho(G)$ and the pair (M, ρ_1), where ρ_1 is the restriction of ρ to KM, is a $\mathfrak{G}$-module.*

Since KM is invariant under $\rho(\operatorname{Dist} G)$ (3.4.5), it is invariant under ρ (1.4.8) and we may assume $V = KM$. Let us choose a system of positive root rays Φ^+, let $\mathfrak{U}^\pm = \mathfrak{U}_{\Phi^\pm}$ and $M_1 = \rho(\operatorname{Dist}\mathfrak{U}^+)M_0$; in view of 3.6.1, M_1 is of finite type and $\mathfrak{Z}$ acts on M_1. It follows, taking 3.4.4 into account, that $M = \rho(\operatorname{Dist}\mathfrak{U}^-)M_1$ and 3.6.1 implies that $\mathfrak{Z}$ and $\mathfrak{U}^-$ act on M. Finally, $M = \rho(\operatorname{Dist}\mathfrak{U}^+)M$ and, again by 3.6.1, $\mathfrak{U}^+$ acts on M.

3.6.3. Let $\tilde{G}$ be a simply connected covering of the adjoint group $\operatorname{Ad} G$ of G. It is known that there exists a unique K-split torus $\tilde{S}$ of $\tilde{G}$ whose image in $\operatorname{Ad} G$ coincides with that of S and the system of root rays of $\tilde{G}$ relative to $\tilde{S}$ is canonically identified with Φ. For $\Psi \subset \Phi$ positively closed, the canonical projection $\tilde{G} \to G$ defines an isomorphism from the root subgroup $\tilde{\mathfrak{U}}_\Psi$ of $\tilde{G}$ associated to Ψ onto $\mathfrak{U}_\Psi$, which allows us to identify these two groups.

3.6.4. — Let Φ^+ be a system of positive root rays and let $\mathfrak{U}^\pm = \mathfrak{U}_{\Phi^\pm}$, etc. Let L be the Lie algebra of G and L^- that of U^-. Let $d = \dim U^-$, $V = \bigwedge^d L$ and $W = \bigwedge^d L^-$, so that $W \subset V$ and $\dim W = 1$; let v be a non-zero element of W and let ρ (resp. $\tilde{\rho}$) be the adjoint representation of G (resp. $\tilde{G}$) on L. For an integer $N \geq 1$, we denote by $\rho(N)$ the representation $\bigotimes^N \rho$ on $V_N = \bigotimes^N V$, we set $W_N = \bigotimes^N W \subset V_N$ and $v_N = \bigotimes^N v$. Finally, let

$$M_N = \rho(N)(\operatorname{Dist}\mathfrak{G}) \cdot v_N.$$

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3.6.4. Proposition. — *(i) The A-module M_N is of finite type and $\mathfrak{Z}$, $\mathfrak{U}^+$, and $\mathfrak{U}^-$ act on M_N via $\rho(N)$.*

(ii) If N is sufficiently large, $\mathfrak{U}^+$ acts faithfully on M_N. In other words, the pair $(M_N, \rho(N))$ is a $\mathscr{D}$-module, which is faithful for $\mathfrak{U}^+$ (hence for the $\mathfrak{U}_a$ for $a \in \Phi^+$) as soon as N is large enough.

Indeed, it is clear that Z leaves W invariant and acts on it via a character χ. Given (DRS 2) (or 3.3.5), $\mathfrak{Z}$ acts on the submodule $\sum_{a\in\Phi^-} \operatorname{Lie}\mathfrak{U}_a$ of L^-, which implies that χ extends to a homomorphism (still denoted χ) from $\mathfrak{Z}$ to Mult and $\mathfrak{Z}$ acts via χ^N on any submodule of finite type of W_N, in particular on Av_N.

Assertion (i) then follows from 3.6.2 and assertion (ii) follows from Lemmas 3.5.6, 3.5.7 and...

3.6.5. Lemma. — *The stabilizer of v in U^+ is trivial.*

It amounts to the same to show that the stabilizer of v in $\tilde{U}^+$ is trivial; in other words, we may assume that G is semi-simple and simply connected. Let $\overline{K}$ be an algebraic closure of K and let π be the map $x \mapsto \rho(x).v$ from $U^+(\overline{K})$ to $\overline{K} \otimes V$. We need to show (cf. [2], prop. 6.7) that

$$P = \pi^{-1}(v) = \{e\} \quad \text{and} \quad Q = d\pi_e^{-1}(0) = \{0\}.$$

Let T be a maximal torus defined over K of G, containing S, let $\tilde{\Phi}$ be the root system of $G_{\overline{K}}$ relative to T, and let Ψ^+ (resp. Ψ^-) be the set of roots $\alpha \in \tilde{\Phi}$ whose restriction to S belongs to an element of Φ^+ (resp. Φ^-). The subsets Ψ^+ and Ψ^- are positively closed, we have $\Psi^- = -\Psi^+$, and the unipotent subgroup of $G_{\overline{K}}$ associated with Ψ^+ (resp. Ψ^-) is nothing other than $(U^+)_{\overline{K}}$ (resp. $(U^-)_{\overline{K}}$).

Since $T \subset Z$, the closed subgroup P of U^+ is normalized by T and is therefore generated by the root subgroups U_α that it contains ([4], 3.11). But $U_\alpha \subset P$ implies $\operatorname{Lie} U_\alpha \subset Q$: it thus suffices for us to show that $Q = \{0\}$.

Being normalized by T, the algebra Q is the sum of the Lie U_α it contains (with $\alpha \in \Psi^+$). Ultimately, we only need to show that if we choose a Chevalley pinning $(x_\alpha)_{\alpha\in\tilde{\Phi}}$ of $G_{\overline{K}}$ relative to T and if we set $e_\alpha = dx_\alpha(1)$, then $\alpha \in \Psi^+$ implies $\rho_{\overline{K}}(e_\alpha).v \neq 0$.

Now, we have $\overline{K} \otimes L^- = \sum_{\beta\in\Psi^+} \overline{K} e_{-\beta}$, from which $v = k \bigwedge_{\beta\in\Psi^+} e_{-\beta}$ with $k \in \overline{K}^\times$, and

$$\begin{aligned}\rho_{\overline{K}}(e_\alpha)v &= k\Big(\pm [e_\alpha, e_{-\alpha}] \wedge \bigwedge_{\beta\in\Psi^+-\{\alpha\}} e_{-\beta} + \sum_{\gamma\in\Psi^+-\{\alpha\}} \pm[e_\alpha, e_{-\gamma}] \wedge \bigwedge_{\beta\in\Psi^+-\{\gamma\}} e_{-\beta}\Big)\\ &= k\Big(\pm h_\alpha \wedge \bigwedge_{\beta\in\Psi^+-\{\alpha\}} e_{-\beta} + \sum_{\gamma\in\Psi^+-\{\alpha\}} \pm N_{\alpha,-\gamma} e_{\alpha-\gamma} \wedge \bigwedge_{\beta\in\Psi^+-\{\gamma\}} e_{-\beta}\Big).\end{aligned}$$

This demonstrates our assertion, since $h_\alpha = [e_\alpha, e_{-\alpha}]$ is a non-zero element of Lie T.

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3.7. Construction of a faithful $\mathscr{D}$-module for $\mathfrak{Z}$.

From now on, we assume that G is quasi-split over K and that S is a maximal split torus over K. It follows that Z is a maximal torus of G. We further assume that the group A-scheme $\mathfrak{Z}$ has a faithful linear representation (a condition always satisfied if A is a Dedekind ring (1.4.5)).

3.7.1. — Let K_s be a separable closure of K. We denote by $\tilde{\Phi}$ the root system of G_{K_s} relative to the maximal torus Z_{K_s}, and by $\tilde{\Phi}^+$ the set of $\alpha \in \tilde{\Phi}$ whose restriction to S belongs to an element of Φ^+: this is a system of positive roots in $\tilde{\Phi}$ and $(U^+)_{K_s}$ is the maximal unipotent subgroup of G_{K_s} associated with $\tilde{\Phi}^+$. It follows that the character χ of $\mathfrak{Z}$ introduced in 3.6.4, i.e., the representation of $\mathfrak{Z}$ in the module $D = \det \operatorname{Lie} \mathfrak{U}^+$, is the product of the positive roots in the group of characters $X^*(Z)$ noted multiplicatively:

$$\chi = \prod_{\alpha \in \tilde{\Phi}^+} \alpha.$$

It is therefore a dominant weight of Z, equal to the square of the product of the fundamental dominant weights ([NB] Lie VI, § 1, prop. 29) and for any character λ of Z, the character $\lambda\chi^n$ is a dominant weight for a sufficiently large integer n.

3.7.2. — Let M_0 be a finitely generated projective A-module and let σ be a faithful linear representation of $\mathfrak{Z}$ in M_0. Let $V = K \otimes M_0$. Since Z is a torus, the representation σ_K of Z in V is completely reducible: let $V = \sum_{1 \le j \le r} V_j$ be a decomposition of σ_K into irreducible K-components $\sigma_j : Z \to GL(V_j)$. For $1 \le j \le r$, the representation $(\sigma_j)_{K_s}$ of Z_{K_s} in $K_s \otimes V_j$ is a direct sum of characters $\lambda_{j,1}, \ldots, \lambda_{j,n_j}$ which are distinct and permuted transitively by the Galois group Γ of K_s over K acting naturally on $X^*(Z)$ (note that, since G is quasi-split over K, the two classical actions of Γ on $X^*(Z)$ (cf. [36], 3.1) are identical).

Let n be an integer sufficiently large so that the characters $\lambda_{j,k}\chi^n$ are all dominant weights for $1 \le j \le r$ and $1 \le k \le n_j$. The representation $\sigma' = (\sigma \otimes \chi^n) \oplus \chi$ of $\mathfrak{Z}$ in $M_0' = (M_0 \otimes (\bigotimes^n D)) \oplus D$ is still faithful. The image under $(\sigma')^*$ of $A[GL(M_0')]$ indeed contains, on the one hand, χ and χ^{-1}, and on the other hand, the products by χ^n of the coefficients of σ and the product by $\chi^{-n \dim V}$ of the inverse of $\det \sigma$, thus it contains the image under σ^* of $A[GL(M_0)]$, which is the whole of $A[\mathfrak{Z}]$.

Consequently, by possibly replacing σ with σ', we can assume that all weights $\lambda_{j,k}$ are dominant weights.

3.7.3. — For $1 \le j \le r$, the weights $\lambda_{j,1}, \ldots, \lambda_{j,n_j}$ then form an orbit of Γ in the cone of dominant weights. It is known ([36], 3.3, 7.2 and 7.4) that there exists an irreducible K-representation ρ_j (unique up to isomorphism) of G in a K-vector space W_j which decomposes over K_s into the direct sum of the representations

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irreducible $\rho_{j,k}$ of G_{K_s}, of dominant weights $\lambda_{j,k}$ (for $1 \le k \le n_j$). Let $W_{j,k}$ be the subspace of $K_s \otimes W_j$ on which $(\rho_j)_{K_s}$ induces $\rho_{j,k}$ and let $W_{j,k}^0$ be the line of dominant vectors of $W_{j,k}$. For a given j, the $W_{j,k}$ (resp. $W_{j,k}^0$) are transitively permuted by Γ; there exists one and only one K-subspace W_j^0 of W_j, of dimension n_j, such that $\sum_{1 \le k \le n_j} W_{j,k}^0 = K_s \otimes W_j^0$ and the K-representation of Z in W_j^0 is isomorphic to σ_j.

3.7.4. — Taking the direct sum of the ρ_j for $1 \le j \le r$, one obtains a K-representation ρ of G in $W = \sum W_j$ and a K-subspace $W^0 = \sum W_j^0$ of W, invariant by Z, formed of dominant vectors, hence fixed by U^+, and such that the representation of Z in W^0 induced by ρ is isomorphic to σ_K. One can therefore identify M_0 with a sub-A-module of W^0 and $\mathfrak{Z}$ acts faithfully by ρ in M_0.

3.7.5. — Let $M = \rho(\operatorname{Dist} \mathfrak{D})M_0$. Corollary 3.6.2 implies that M is of finite type and that $\mathfrak{Z}$, $\mathfrak{U}^+$ and $\mathfrak{U}^-$ act on M, that is, M is a $\mathfrak{D}$-module.

Moreover, elementary considerations on weights show that

$$\rho(\operatorname{Dist}^+ U^+)W^0 = \{0\}; \tag{1}$$

$$W^0 \cap \rho(\operatorname{Dist}^+ U^-)W^0 = \{0\}. \tag{2}$$

From (1) and 3.4.4, it follows that $M = \rho(\operatorname{Dist} \mathfrak{U}^-)M_0$ and from (2) it then results that M is the direct sum of the sub-modules M_0 and $\rho(\operatorname{Dist}^+\mathfrak{U}^-)M_0$, which are invariant by $\mathfrak{Z}$. Since $\mathfrak{Z}$ acts faithfully on M_0, we see that $\mathfrak{Z}$ acts faithfully on M.

In conclusion, we have demonstrated the following proposition:

3.7.6. Proposition. — *There exists a faithful $\mathfrak{D}$-module for $\mathfrak{Z}$.*

3.7.7. — Taking then the direct sum of a faithful $\mathfrak{D}$-module for $\mathfrak{Z}$, a faithful $\mathfrak{D}$-module for $\mathfrak{U}^+$ and a faithful $\mathfrak{D}$-module for $\mathfrak{U}^-$ (3.6.4), we finally obtain:

3.7.8. Proposition. — *There exists a faithful $\mathfrak{D}$-module.*

3.8. Harvest.

Gathering the existence of a faithful $\mathfrak{D}$-module and the results of 2.2, we obtain:

3.8.1. Theorem. — *Let A be a Prüferian integral domain, K its field of fractions, G a connected reductive algebraic group defined over K, quasi-split over K, S a split maximal torus of G over K and $(\mathfrak{Z}, (\mathfrak{U}_a)_{a \in \Phi})$ a*

schematic root datum on G relative to S. Suppose that the A-group schemes $\mathfrak{U}_a$ are smooth and that $\mathfrak{Z}$ possesses a faithful linear representation (this last condition being a consequence of the others when A is a Dedekind ring).

Then, the K-group scheme G extends to a flat A-group scheme $\mathfrak{G}$, of finite type, possessing a faithful linear representation, and satisfying the following conditions:

(S 1) the injection of Z (resp. U_a for $a \in \Phi$) into G extends to an isomorphism of $\mathfrak{Z}$ (resp. $\mathfrak{U}_a$) onto a closed A-subgroup scheme of $\mathfrak{G}$;

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(S 2) for every system of positive root rays Φ^+ of Φ and for every order on Φ^+ (resp. $\Phi^- = -\Phi^+$), the product map is an isomorphism of schemes from $\prod_{a\in\Phi^+} \mathfrak{U}_a$ (resp. $\prod_{a\in\Phi^-} \mathfrak{U}_a$) onto a closed subgroup scheme $\mathfrak{U}^+$ (resp. $\mathfrak{U}^-$) of $\mathfrak{G}$;

(S 3) the product map is an isomorphism of schemes from $\mathfrak{U}^- \times \mathfrak{Z} \times \mathfrak{U}^+$ onto an open sub-scheme $\mathfrak{C}$ of $\mathfrak{G}$. If in addition $\mathfrak{Z}$ is smooth, then the scheme $\mathfrak{G}$ is smooth.

3.8.2. — We have not repeated in the preceding theorem all the properties of $\mathfrak{G}$ already seen. Let us briefly recall some of them.

a) The open set $\mathfrak{C}$ of (S 3) is a special open set, defined by an element $d \in A[\mathfrak{G}]$ such that $d(1) = 1$.

b) For every positively closed subset Ψ of Φ and for every order on Ψ, the product map is an isomorphism of schemes from $\prod_{a\in\Psi} \mathfrak{U}_a$ onto a closed subgroup scheme $\mathfrak{U}_\Psi$ of $\mathfrak{G}$ (2.2.3 (i): compare with 3.3.2).

c) $\mathfrak{Z}$ normalizes $\mathfrak{U}^+$ and the product map is an isomorphism of the semi-direct product $\mathfrak{Z} \ltimes \mathfrak{U}^+$ onto a closed subgroup scheme $\mathfrak{Z}\mathfrak{U}^+$ of $\mathfrak{G}$ (2.2.3 (ii)).

3.8.3. — We have already seen that the scheme $\mathfrak{G}$ provided by the preceding theorem depends not only on the schematic root datum but also on the faithful $\mathfrak{D}$-module M chosen, even if A is a complete discrete valuation ring (3.2.14). However, when $\mathfrak{Z}$ is smooth, the neutral component $\mathfrak{G}^0$ of $\mathfrak{G}$ is independent of the choice of M. Let us content ourselves with stating the result which will be useful to us in what follows:

Theorem. — *Let us keep the hypotheses of Theorem 3.8.1 and suppose further that A is a valuation ring of height 1 or a Dedekind ring, that $\mathfrak{Z}$ is smooth and that the $\mathfrak{U}_a$ are connected.*

(i) *The group scheme $\mathfrak{G}^0$ is (up to unique isomorphism) the only connected group scheme with generic fiber G possessing the three properties (S^0 1), (S^0 2) and (S^0 3) obtained by replacing $\mathfrak{Z}$ by $\mathfrak{Z}^0$ in the conditions (S 1), (S 2) and (S 3) of 3.8.1.*

(ii) *The group scheme $\mathfrak{G}^1 = \mathfrak{Z}\cdot\mathfrak{G}^0$ (cf. 2.2.9) is (up to unique isomorphism) the only group scheme with generic fiber G possessing the three properties (S 1), (S 2) and (S 3) of 3.8.1 and such that $\mathfrak{G}^1 = \mathfrak{Z}\cdot(\mathfrak{G}^1)^0$.*

One can even in (i) (resp. (ii)) replace the condition (S^0 2) (resp. (S 2)) by that obtained by replacing twice the words "for every" by "there exists a".

This follows immediately from 1.2.13 and 1.2.14, taking into account that the imposed conditions imply the smoothness of the schemes considered.

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3.9. Complements.

3.9.1. — Reductive groups in characteristic zero.

In the statement of Theorem 3.8.1, one can remove the two hypotheses G quasi-split over K and S maximal among the K-split tori provided that one assumes that $\mathfrak{Z}$ possesses a faithful linear representation σ such that the K-representation σ_K of Z is completely reducible (a condition automatically satisfied if $\operatorname{char} K = 0$, since Z is reductive).

As this result will not be used in the following, we will be content with sketching its proof. It is a matter of proving 3.7.6 and, by using the notation and the arguments of 3.7, we see that it is sufficient to show the following assertion:

a) There exists a K-representation of G in a vector space W and a sub-A-module of finite type M_0 of the subspace W^0 of dominant vectors of W, generating W^0, such that $\mathfrak{Z}$ acts faithfully in M_0.

Let T be a maximal torus of G defined over K, with $S \subset T$ hence $T \subset Z$. Let $\overline{K}$ be a separable closure of K and Ψ the root system of G according to T. The root system Ψ_Z of Z according to T is the set of $\alpha \in \Psi$ trivial on S and one can choose a basis B of Ψ such that $B_Z = B \cap \Psi_Z$ is a basis of Ψ_Z and that a root $\alpha \in \Psi$ whose restriction to S belongs to an element of Φ^+ is positive. We denote by Λ_G^+ (resp. Λ_Z^+) the set of dominant weights of G (resp. Z) relative to B (resp. B_Z); one has $\Lambda_G^+ \subset \Lambda_Z^+$.

The character χ of $\mathfrak{Z}$ in $D = \det \mathrm{Lie}\,\mathfrak{U}^+$ (3.6.4) restricted to T is the product of the roots $\alpha \in \Psi^+ - \Psi_Z^+$, from which it follows that $\langle \chi, \alpha^\vee \rangle = 0$ if $\alpha \in B_Z$ and $\langle \chi, \alpha^\vee \rangle \geq 1$ if $\alpha \in B - B_Z$. Consequently, if λ is a dominant weight of Z, then $\lambda \chi^N$ is a dominant weight of G for N a sufficiently large integer.

Let us then start from a faithful representation σ of $\mathfrak{Z}$ in a projective A-module M_0 of finite type such that the K-representation σ_K of Z is completely reducible. We see as in 3.7 that, up to replacing σ with the representation $(\sigma \otimes (\bigotimes^N \chi)) \oplus \chi$ and considering each K-irreducible component separately, it is sufficient, to prove *a*), to show:

b) let σ be a K-irreducible K-representation of Z in V such that the dominant weights of the $\overline{K}$-irreducible components of $\sigma_{\overline{K}}$ are also dominant weights of G; there exists a K-representation ρ of G in a vector space $W \supset V$ such that the restriction of ρ to Z leaves V stable and induces σ on it, and that the space of dominant vectors for G of W coincides with that of the dominant vectors for Z of V.

Recall that the Galois group $\Gamma = \mathrm{Gal}(\overline{K}/K)$ acts on Λ_G^+ (resp. Λ_Z^+): if $\lambda \mapsto {}^\gamma\lambda$ is the "natural" operation of $\gamma \in \Gamma$ on the group of characters of T and if $\lambda \in \Lambda_G^+$ (resp. $\lambda \in \Lambda_Z^+$), then $\gamma_G(\lambda)$ (resp. $\gamma_Z(\lambda)$) is the unique dominant weight of G (resp. Z) transformed from ${}^\gamma\lambda$ by an element of the Weyl group W_G of G (resp. W_Z of Z). Since $W_Z \subset W_G$, we see that if λ and $\gamma_Z(\lambda)$ are dominant weights of G (then

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of Z), we have $\gamma_G(\lambda) = \gamma_Z(\lambda)$. In particular, on the one hand *every dominant weight of G which is fixed by the action of Γ on Λ_G^+ is also fixed for the action of Γ on Λ_Z^+* and on the other hand the hypotheses of *b*) imply that the set P of dominant weights of the $\bar{K}$-irreducible components of $\sigma_{\bar{K}}$ is an orbit of Γ for its action on Λ_G^+ as well as on Λ_Z^+.

Using the results of $n°7$ of [36], one easily reduces to the case where P is reduced to a single weight λ. It then suffices to adapt the proof of Theorem 3.3 of [36], whose notation we adopt (up to some obvious changes, notably from k to K...). We can suppose that the $\bar{K}$-isomorphism φ of G onto a K-split group G^d maps T to a maximal K-split torus T^d of G^d and consequently maps Z onto a K-split subgroup Z^d of G^d. The 1-cocycle $\xi : \gamma \mapsto \varphi \circ \gamma \circ \varphi^{-1} \circ \gamma^{-1}$ then maps Γ into the group of automorphisms of $G^d(\bar{K})$ preserving $Z^d(\bar{K})$. The $\bar{K}$-irreducible representation ρ^d of G^d in W^d of dominant weight $\lambda \circ \varphi^{-1}$ restricted to Z^d contains once and only once the $\bar{K}$-irreducible representation σ^d of Z^d of the same dominant weight ([36], p. 204). Let V^d be the corresponding subspace: one immediately sees that the 1-cocycle ζ defined in [36], p. 204, with values in $PGL(W)$, preserves V and provides a 1-cocycle ζ' with values in $PGL(V)$. It then suffices to twist simultaneously on the one hand G^d and Z^d by ξ, and on the other hand $GL(W)$ by ζ and $GL(V)$ by ζ' to deduce from ρ^d, as in [36], the representation ρ of G sought (one will notice that the division algebras canonically associated to G and λ on one side, and to Z and λ on the other, are isomorphic: this also follows from Prop. 5.5.2 of [36]).

3.9.2. — To avoid technical complications, we assumed from the beginning of this chapter that K is the field of fractions of A. However, the method can be adapted to apply to the descent from a field to a subfield. One can thus deduce the existence of Chevalley schemes over $\mathbf{Z}$ from the existence of the corresponding groups over $\mathbf{C}$: one begins by descending from $\mathbf{C}$ to $\mathbf{Q}$ by making the latter play the role of A, then from $\mathbf{Q}$ to $\mathbf{Z}$ as in 3.2.

3.9.3. — *Case of Chevalley schematic root data.*

Suppose that A is a Dedekind ring, G a split group and $\mathscr{D}$ a Chevalley schematic root datum. The following proposition then shows that a faithful $\mathscr{D}$-module for $\mathfrak{S} = 3$ is "almost always" faithful.

a) **Proposition.** — *Suppose that at least one of the following conditions is satisfied:*

(i) *the ideal (2) is totally unramified in A, that is to say $2 \neq 0$ and every ideal whose power is divisible by 2 is itself divisible by 2.*

(ii) *the group G has no normal subgroup that is adjoint of type C_n ($n \geqq 1$).*

Then, every schematic root datum dominated by $\mathscr{D}$ coincides with $\mathscr{D}$. In particular, every $\mathscr{D}$-module faithful for $\mathfrak{S}$ is faithful.

That the last assertion is a consequence of the preceding one follows from 3.1.6.

b) It is easy to see that if a smooth group scheme $\mathfrak{G}$ with generic fiber G has as "identity component" $\mathfrak{G}^0$ a Chevalley scheme, then $\mathfrak{G} = \mathfrak{G}^0$ (for the case of a Henselian valuation ring, from which the general case follows immediately, cf. 4.6). From this we deduce:

Corollary 0.23. *Suppose one of the conditions (i) and (ii) is satisfied, let $\rho : G \to GL(V)$ be a faithful linear representation of G and let M be a finitely generated sub-A-module of V stable under Dist $\mathscr{D}$ and such that $KM = V$. Then the schematic closure $\mathfrak{G}$ of $\rho(G)$ in $\mathfrak{GL}(M)$ is the Chevalley scheme associated with $\mathscr{D}$.*

Indeed, 3.5.3 (2) implies that $\mathfrak{G}$ and the $\mathfrak{U}_a$ operate on M and 2.1.6 shows that $\mathfrak{G}$ operates faithfully on it. Given the proposition, M is thus a faithful $\mathscr{D}$-module. Finally, 3.8.3 then, again, the proposition imply that $\mathfrak{G}^0$, thus $\mathfrak{G}$, is the Chevalley scheme associated with $\mathscr{D}$.

For $A = \mathbf{Z}$, this is essentially a result of C. Chevalley [16]; moreover, our proof is similar to his.

c) The preceding corollary and, a fortiori, the proposition are not true without hypotheses on A and G as shown by the following example: $A = \mathbf{Z}[\sqrt{2}]$, $G = \mathrm{PGL}_2$, $\rho = \mathrm{Ad}$, $\Phi = \{\pm a\}$, $M = \sqrt{2}Ae_{-a} + Ah_a + \sqrt{2}Ae_a$, with the notations $e_{\pm a}$, h_a of 3.2.2, which we preserve in the proof below.

d) *Proof of the proposition.*

The condition (ii) is equivalent to

(ii') for $a \in \Phi$, the subgroup of G generated by U_a and U_{-a} is isomorphic to SL_2.

Let $\mathscr{D}' = (\mathfrak{T}, (\mathfrak{U}'_a))$ be a schematic root datum dominated by $\mathscr{D}$. We must show that, for $a \in \Phi$, the morphism $\mathfrak{U}_a \to \mathfrak{U}'_a$ extending the identity of U_a is an isomorphism. Replacing G by the subgroup G_a generated by U_a and U_{-a}, which is permissible given the reformulation (ii') of (ii), we can suppose that $G = G_a$. Let then $e = e_a$, $f = e_{-a}$, $h = h_a$. Identify $K[U_a]$ with a polynomial ring $K[X]$ by means of the pinning used to define the datum $\mathscr{D}$ (cf. 3.2.2). Thus, $A[\mathfrak{U}_a] = A[X]$. Since $\mathfrak{U}'_a$ is normalized by $\mathfrak{G}$, one can write

$$A[\mathfrak{U}'_a] = A + \sum \mathfrak{a}_i X^i,$$

where $\mathfrak{a}_i$ is an ideal of A, and our assertion reduces to $\mathfrak{a}_1 = A$. Assume the contrary and seek to deduce a contradiction from it. By localizing at a prime ideal containing $\mathfrak{a}_1$, we can suppose that A is a discrete valuation ring and, denoting by π a uniformizer, that π divides $\mathfrak{a}_1$, which implies that $\pi^{-1}e \in \Delta' = \mathrm{Dist}\ \mathscr{D}'$. Since the identity of U_{-a} extends into a morphism $\mathfrak{U}_{-a} \to \mathfrak{U}'_{-a}$, Δ' also contains the divided powers $f^{(n)}$ of f. But a simple calculation (using, if one wishes, [NB] Lie VIII, no 12.5, Lemma 4) shows that

$$h = [[f^{(2)}, e] - f.[f, e], e].$$

Consequently, $\pi^{-2}h \in \Delta'$, and this implies that G is isomorphic to PGL_2 (otherwise $\mathfrak{G}$ would have Ah as Lie algebra) and that π^2 divides 2, in contradiction with conditions (i) and (ii).

3.9.4. — Rings of homological dimension $\leqslant 2$. (Complement added in proof.) Theorem 3.8.1 generalizes to the case where the ring A is no longer assumed to be Prüferian but is Noetherian, integrally closed, and of homological dimension $\leqslant 2$, for example for $A = k[x, y]$, where k is a field, or for $A = A_1[[z]]$, where A_1 is a Dedekind ring (this case could be useful in the study of groups associated with Kac-Moody algebras). We must however assume then that the schemes $\mathfrak{U}_\alpha$ are connected, and the obtained A-scheme $\mathfrak{G}$ is *perhaps not affine* (cf. 3.2.15): it perhaps no longer possesses a faithful linear representation in the sense of 1.4.5, but it still identifies with a group subscheme (not necessarily closed) of a $GL(M)$, and it possesses the other properties stated in 3.8.1. To establish this result, it suffices to show the existence of a faithful $\mathcal{D}$-module and to apply proposition 2.2.10. In other words, it suffices to generalize 3.6 and 3.7.

To this end, we note that if M is a finitely generated A-submodule of a K-vector space V, generating V (in other words, a lattice of V), then M is contained in a smallest finitely generated projective A-submodule of V, which we call the projective envelope of M and denote by $\hat{M}$. Indeed, as A is a Krull ring, it is known that M is contained in a smallest reflexive lattice, which is the intersection of the localizations $M_\mathfrak{p}$ for $\mathfrak{p}$ describing the set P of prime ideals of height 1 of A ([NB] AC VII, p. 50). But as A is Noetherian of homological dimension $\leqslant 2$, an A-module is a reflexive lattice if and only if it is finitely generated projective ([NB] AC X, § 8, exer. 13, p. 203), whence our assertion. Furthermore, there exists $x \in K^\times$ such that $M \subset \hat{M} \subset xM$ ([NB] AC VII, § 4, prop. 2) and x belongs to only a finite number of prime ideals $\mathfrak{p}_1, \ldots, \mathfrak{p}_n \in P$, whence

$$\hat{M} = xM \cap M_{\mathfrak{p}_1} \cap \cdots \cap M_{\mathfrak{p}_n}. \tag{1}$$

From (1) it follows immediately that if $u \in \operatorname{End} V$ is such that $u(M) \subset M$, then $u(\hat{M}) \subset \hat{M}$ and that if V is equipped with a comodule structure

$$d : V \to C \otimes_A V$$

over a flat A-coalgebra C such that M is a sub-comodule of V, then $\hat{M}$ is also a sub-comodule of V, as a finite intersection of sub-comodules ([30]).

One then easily sees that 3.6.1 remains valid, provided M_0 is assumed to be finitely generated projective and replacing in (i) and (ii) the submodule $M = \rho(\operatorname{Dist} \mathfrak{U}_\gamma) \cdot M_0$ by its projective envelope $\hat{M}$. One similarly modifies 3.6.2 (resp. 3.6.4, resp. 3.7.5) by assuming M_0 to be finitely generated projective and replacing $M = \rho(\operatorname{Dist} \mathcal{D}) \cdot M_0$ by $\hat{M}$ (resp. $M_N = \rho(N)(\operatorname{Dist} \mathcal{D}) \cdot v_N$ by $\hat{M}_N$, resp. $M = \rho(\operatorname{Dist} \mathcal{D}) \cdot M_0$ by $\hat{M}$) and the existence of a faithful $\mathcal{D}$-module (3.7.8) results from 3.6.4 and 3.7.5 thus modified.

§ 4. QUASI-SPLIT GROUPS OVER A VALUED FIELD

In this section, unless explicitly stated otherwise, the letter G denotes a connected reductive group quasi-split over K, and S is a maximal torus of it defined and split over K. Starting from No. 4.2, K will be assumed to be valued.

4.1. Generalities[10]. Structure of the U_a.

4.1.1. — Let us recall that a reductive group defined over K is said to be *quasi-split* (over K) if it possesses the following equivalent properties:

— the centralizer Z of a maximal split K-torus is a torus;

— G possesses a Borel subgroup B defined over K.

One can always assume that $B \supset Z$. To help the reader remember that Z is a (maximal) torus, we will also denote it by T.

4.1.2. — Let K_s be a separable closure of K. It is known that T and G are split over K_s. The Galois group $\operatorname{Gal}(K_s/K)$ operates on the character group $X^*(T)$ of T, preserves the root system $\widetilde{\Phi}$ of G relative to T as well as the base $\widetilde{\Delta}$ of $\widetilde{\Phi}$ associated with the K-Borel subgroup B. There exists a smallest sub-extension $\widetilde{K}$ of K_s splitting G: it is also the smallest sub-extension splitting T; it is Galois and is the fixed field of the kernel of the representation of $\operatorname{Gal}(K_s/K)$ in $X^*(T)$.

[10] The results of this No. 4.1 are valid for any field K, possibly finite.

We denote by Φ the root system of G relative to S, by ${}_a\Phi$ the system of root rays of G relative to S (in the sense of 1.1.2), and by V the dual of the vector subspace generated by Φ in $\mathbf{R} \otimes X^*(S)$. It is known that an element $a \in {}_a\Phi$ contains one or two elements of Φ. In the first case, we also denote by a the unique element of Φ belonging to a. In the second case, we say that a is *plural* and we denote by a (resp. $2a$) the non-divisible (resp. divisible) element of Φ belonging to a. It is known that the elements of Φ are the restrictions to S of the elements of $\widetilde{\Phi}$, that the restrictions to S of the elements of $\widetilde{\Delta}$ form a base Δ of Φ, and that the elements of $\widetilde{\Delta}$ whose restriction to S is a given element of Δ form an orbit of $\mathrm{Gal}(K_s/K)$ (or of $\mathrm{Gal}(\widetilde{K}/K)$) in $\widetilde{\Delta}$ (for all these assertions, see [4] *passim*, especially § 6).

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4.1.3. — For $\alpha \in \tilde{\Phi}$, let $\tilde{U}_\alpha$ be the root subgroup corresponding to $G_{\tilde{K}}$. The Galois group $\mathrm{Gal}(\tilde{K}/K)$ permutes the $\tilde{U}_\alpha$ and $\gamma(\tilde{U}_\alpha) = \tilde{U}_{\gamma(\alpha)}$. Let Σ_α be the stabilizer of the root α in $\mathrm{Gal}(\tilde{K}/K)$: it is also the stabilizer of $\tilde{U}_\alpha$; it depends therefore only on the subgroup $\tilde{U}_\alpha$ and not on the choice of the torus S normalizing $\tilde{U}_\alpha$. The field of invariants of Σ_α will be denoted L_α and called the *field of definition* of the root α (or of the subgroup $\tilde{U}_\alpha$). The operation of Σ_α on $\tilde{U}_\alpha$ provides it, by Galois descent, with a line structure (1.2.10) over L_α. If $\tilde{x}_\alpha : \mathbb{D}_\alpha \to \tilde{U}_\alpha$ is an L_α-pinning of $\tilde{U}_\alpha$ (3.2.1), and if $\gamma \in \mathrm{Gal}(\tilde{K}/K)$, we have $\gamma(L_\alpha) = L_{\gamma(\alpha)}$ and $\gamma \circ \tilde{x}_\alpha \circ \gamma^{-1} : \mathbb{D}_{\gamma(\alpha)} \to \tilde{U}_{\gamma(\alpha)}$ is an $L_{\gamma(\alpha)}$-pinning of $\tilde{U}_{\gamma(\alpha)}$, depending only on $\gamma(\alpha)$.

It follows that one can choose a $\tilde{K}$-pinning $(\tilde{x}_\alpha)_{\alpha \in \tilde{\Phi}}$ of $G_{\tilde{K}}$ (3.2.2) such that:

1) $\tilde{x}_\alpha : \mathbb{D} \to \tilde{U}_\alpha$ is an L_α-isomorphism (for $\alpha \in \tilde{\Phi}$).
2) $\tilde{x}_{\gamma(\alpha)} = \gamma \circ \tilde{x}_\alpha \circ \gamma^{-1}$ (for $\alpha \in \tilde{\Phi}$ and $\gamma \in \mathrm{Gal}(\tilde{K}/K)$).

We shall say that this is a *Steinberg pinning* of G (relative to S). One then shows (cf. [31]) that there exists a $\tilde{K}$-Chevalley system $(\tilde{x}_\alpha)_{\alpha \in \tilde{\Phi}}$ (3.2.2) extending this pinning and possessing the following properties:

3) If the restriction a of $\alpha \in \tilde{\Phi}$ to S is a non-divisible element of Φ, then $\tilde{x}_\alpha$ is an L_α-isomorphism of $\mathbb{D}$ onto $\tilde{U}_\alpha$ and we have $\tilde{x}_{\gamma(\alpha)} = \gamma \circ \tilde{x}_\alpha \circ \gamma^{-1}$ for all $\gamma \in \mathrm{Gal}(\tilde{K}/K)$.
4) If the restriction a of $\alpha \in \tilde{\Phi}$ to S is a divisible element of Φ, then there exist two distinct roots $\beta, \beta' \in \tilde{\Phi}$ of restriction $a/2$ to S such that $\alpha = \beta + \beta'$; we have $L_\beta = L_{\beta'}$, L_β is a quadratic (separable) extension of L_α and for all $\gamma \in \mathrm{Gal}(\tilde{K}/K)$, there exists $\epsilon = \pm 1$ such that $\gamma \circ \tilde{x}_\alpha(u) \circ \gamma^{-1} = \tilde{x}_{\gamma(\alpha)}(\epsilon u)$; if $\gamma \in \mathrm{Gal}(\tilde{K}/L_\alpha)$, we have $\epsilon = -1$ if and only if γ induces on L_β the unique non-trivial automorphism.

Such a system will be called a *Chevalley-Steinberg system* of G.

4.1.4. — We are now going to describe the possible structures for the root subgroups U_a for $a \in \Phi$. For this, one can obviously suppose that a contains an element of Δ. We denote by $\tilde{A}_a$ the corresponding orbit of $\mathrm{Gal}(\tilde{K}/K)$ in $\tilde{\Phi}$. Let further $\pi : G^a \to \langle U_a, U_{-a} \rangle$ be a universal covering of the (semi-simple) group generated by U_a and U_{-a}: it is a semi-simple group, simply connected, defined and quasi-split over K, split over $\tilde{K}$, of relative rank one, admitting $\pi^{-1}(S)$ (resp. $\pi^{-1}(T)$) as maximal split torus (resp. maximal torus). The orbit $\tilde{A}_a$ is identified with a base of the system of roots of G^a relative to $\pi^{-1}(T)$ and π induces isomorphisms of the two maximal unipotent subgroups U_+ and U_- of G^a normalized by $\pi^{-1}(S)$ onto U_a and U_{-a} respectively. The group $\mathrm{Gal}(\tilde{K}/K)$ transitively permutes the elements of the orbit $\tilde{A}_a$. A glance at the classification of Dynkin diagrams then shows that only two cases are possible:

— Case I: $G^a_{\tilde{K}}$ is isomorphic to a product of groups $\mathfrak{SL}_2$ indexed by $\tilde{A}_a$ and transitively permuted by $\mathrm{Gal}(\tilde{K}/K)$, the field of definition of the factor with index α is L_α and $G^a \simeq \Pi_{L_\alpha/K}\mathfrak{SL}_2$ (cf. 1.5.16).

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— Case II: $G^a_{\tilde{K}}$ is isomorphic to a product of groups $\mathfrak{SL}_3$ indexed by the set I of subsets of two elements $\{\alpha, \alpha'\}$ of $\tilde{\Delta}_a$ such that $\alpha + \alpha'$ is a root. We then have $L_\alpha = L_{\alpha'}$, L_α is a quadratic (separable) extension of $L_{\alpha+\alpha'}$, the simple factor G' of index $\{\alpha, \alpha'\}$ is defined over $L_2 = L_{\alpha+\alpha'}$, split over $L = L_\alpha$ and is isomorphic over L_2 to the special unitary group of the Hermitian form $h : (x_{-1}, x_0, x_1) \mapsto x_{-1}^\sigma x_1 + x_0^\sigma x_0 + x_1^\sigma x_{-1}$ on L^3 (where σ is the non-trivial element of $\mathrm{Gal}(L/L_2)$), considered as an algebraic group over L_2 and which we denote by $L^L_{SU_3}$, or simply $\mathfrak{SU}_3$, when no confusion is to be feared. We then have $G^a \simeq \Pi_{L_2/K}\mathfrak{SU}_3$.

Note that we find ourselves in Case II or in Case I depending on whether the root ray a is plural or not.

4.1.5. — Let us place ourselves in Case I. The group $(U_a)_{\tilde{K}}$ is then the direct product of the $\tilde{U}_\alpha$ for $\alpha \in \tilde{\Delta}_a$, and these are permuted transitively by $\mathrm{Gal}(\tilde{K}/K)$. It follows that U_a is K-isomorphic to $\Pi_{L_\alpha/K}\tilde{U}_\alpha$ (for $\alpha \in \tilde{\Delta}_a$). If $\tilde{x}_\alpha$ is an L_α-pinning of $\tilde{U}_\alpha$, then $x_a = \Pi_{L_\alpha/K}\tilde{x}_\alpha$ is a K-isomorphism of $\Pi_{L_\alpha/K}\mathbb{D}$ onto U_a; an isomorphism obtained in this way (or rather the pair (L_α, x_a)) will be called a *pinning* of U_a.

Note that $\Pi_{L_\alpha/K}\mathbb{D}$ is none other than the K-scheme $\mathfrak{L}_\alpha$ associated to the L_α-module L_α (1.4.1), defined by $\mathfrak{L}_\alpha(R) = L_\alpha \otimes_K R$ for any K-algebra R. The isomorphism x_a then provides an isomorphism of groups from the additive group $L_\alpha = \mathfrak{L}_\alpha(K)$ onto $U_a(K)$, which we still denote by x_a, by abuse of language. If $(\tilde{x}_\beta)_{\beta\in\tilde{\Delta}}$ is a Steinberg pinning of G, we have, for $u \in L_\alpha$,

$$x_a(u) = \prod_{\beta\in\tilde{\Delta}_a} \tilde{x}_\beta(u_\beta) \quad \text{with } u_{\gamma(\beta)} = \gamma(u) \quad \text{for } \gamma \in \mathrm{Gal}(\tilde{K}/K). \tag{1}$$

The pinning of U_{-a} associated to x_a is the one obtained starting from the L_α-pinning of $\tilde{U}_{-\alpha}$ associated to $\tilde{x}_{-\alpha}$ (3.2.1) (note that $L_{-\alpha} = L_\alpha$!). One easily sees that it can be characterized by the following property (cf. 3.2.1 (i)): for $u \in L_\alpha^\times$, we have

$$\begin{aligned} m_a(u) &= x_a(u)x_{-a}(u^{-1})x_a(u) \in N \\ (\text{resp. } m_{-a}(u) &= x_{-a}(u)x_a(u^{-1})x_{-a}(u) \in N). \end{aligned} \tag{2}$$

The image of $m_a(u)$ in the Weyl group $W = N(K)/T(K)$ is the reflection with respect to the root a. We set $m_a = m_a(1)$. Then,

$$x_{-a}(u) = m_a x_a(u) m_a^{-1}. \tag{3}$$

Finally, let G^α be the simple factor of $G^a_{\tilde{K}}$ of index α. In view of 3.2.1, there exists a unique L_α-isomorphism ζ_α of $\mathfrak{SL}_2$ onto G^α such that $\tilde{x}_{\pm\alpha} = \pi \circ \zeta_\alpha \circ y_\pm$ where y_+ (resp. y_-) is the homomorphism $u \mapsto \begin{pmatrix} 1 & u \\ 0 & 1 \end{pmatrix}$ (resp. $u \mapsto \begin{pmatrix} 1 & 0 \\ -u & 1 \end{pmatrix}$) of $\mathbb{D}$ into $\mathfrak{SL}_2$. We denote by ζ_a the K-homomorphism $\pi \circ \Pi_{L/K}\zeta_\alpha$ from $\Pi_{L/K}\mathfrak{SL}_2$ into G and by $\tilde{\zeta}$ the homomorphism

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$t \mapsto \zeta_a\left(\begin{pmatrix} t & 0 \\ 0 & t^{-1} \end{pmatrix}\right)$ from $\Pi_{L/K}\mathrm{Mult}$ into T (cf. 3.2.1). We have $m_a = \zeta_a\left(\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}\right)$ and $m_a(u) = \tilde{a}(u)m_a = m_a\tilde{a}(u^{-1})$ for all $u \in L_a^\times$.

One should be careful that, contrary to what the notation might suggest, the objects $x_{\pm a}$, m_a, ζ_a depend not only on a, but on the choice of $\alpha \in \tilde{\Delta}_a$ and the pinning $\tilde{x}_\alpha$.

4.1.6. — Let us now determine the open set W_a of $U_a \times U_{-a}$ and the morphism $\beta_a : W_a \to U_{-a} \times T \times U_a$ introduced in 1.1.10. From the formulas

$$\begin{pmatrix} 1 & u \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ -u' & 1 \end{pmatrix} = \begin{pmatrix} 1-uu' & u \\ -u' & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 \\ -v' & 1 \end{pmatrix}\begin{pmatrix} t & 0 \\ 0 & t^{-1} \end{pmatrix}\begin{pmatrix} 1 & v \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} t & tv \\ -v't & t^{-1} - tvv' \end{pmatrix}$$

it follows immediately that W_a is the special open set of $U_a \times U_{-a}$ defined by the function

$$d_a = \mathrm{Norm}_{L/K}(1 - uu') \tag{1}$$

and that

$$\beta_a(x_a(u), x_{-a}(u')) = (x_{-a}((1-uu')^{-1}u'), \tilde{a}((1-uu')), x_a((1-uu')^{-1}u)). \tag{2}$$

4.1.7. — When one is given the subgroup U_a, but it alone, several choices must be made to define a pinning of U_a:

(i) Choice of a maximal split torus S normalizing U_a;

(ii) Choice of a root subgroup $\tilde{U}_\alpha \subset U_a$;

(iii) Choice of an L_α-isomorphism $\mathfrak{D} \to \tilde{U}_\alpha$.

The first choice is of no consequence: indeed, the identity component of the normalizer of U_a contains T, so is generated by T and the root subgroups $\tilde{U}_\beta$ that it contains ([4], 3.4). However, a root subgroup $\tilde{U}_\beta$ cannot normalize U_a unless it centralizes it, whence $\mathrm{Norm}^0 U_a = T \cdot \mathrm{Cent}^0 U_a$. It follows that the root subgroups $\tilde{U}_\alpha$ of U_a are exactly the subgroups invariant under the action of $\mathrm{Norm}^0 U_a$ on U_a, non-trivial and minimal. They thus do not depend on the choice of $S \subset \mathrm{Norm}^0 U_a$. On the other hand, it is known that the structure of a line and the notion of pinning of a $\tilde{U}_\alpha$ are intrinsic (3.2.1).

Replacing $\tilde{U}_\alpha$ by $\tilde{U}_{\gamma(\alpha)}$ amounts to replacing L_α by the conjugate $\gamma(L_\alpha)$ and the L_α-pinning $\tilde{x}_\alpha$ of $\tilde{U}_\alpha$ by $\gamma \circ \tilde{x}_\alpha \circ \gamma^{-1}$. The pinning x_a is then replaced by $x_a \circ i$, where i is the isomorphism from $\Pi_{L_{\gamma(\alpha)}/K}\mathfrak{D}$ onto $\Pi_{L_\alpha/K}\mathfrak{D}$ induced by the K-isomorphism $\gamma^{-1} : L_{\gamma(\alpha)} \to L_\alpha$.

Finally, the L_α-automorphisms of $\mathfrak{D}$ are given by the multiplications $u \mapsto cu$ for $c \in L_\alpha^\times$. Consequently, changing the L_α-pinning of $\tilde{U}_\alpha$ amounts to replacing the pinning x_a with a composition of x_a with a map $u \mapsto cu$ for some $c \in L_a^\times$.

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In summary, we see that a pinning of U_a is the data of a suitable sub-extension L of $\tilde{K}$ and a suitable isomorphism x of $\Pi_{L/K}\mathfrak{D}$ onto U_a. If (L', x') is another pinning, what precedes shows that there exists a K-isomorphism $j : L' \to L$ and a $c \in L^\times$ such that x' is obtained by successively performing the K-isomorphism $j' : \mathfrak{D}_{L'} \to \mathfrak{D}_L$ defined by j, the automorphism $u \mapsto cu$ of $\mathfrak{D}$ and the isomorphism $x : \mathfrak{D}_L \to U_a$. Moreover, c is well-determined since $x'(1) = cx(1)$ and j is well-determined, because j' is and j is none other than the corresponding isomorphism of $L' = \mathfrak{D}_{L'}(K)$ onto $L = \mathfrak{D}_L(K)$.

4.1.8. — From this discussion, the following facts easily follow:

(1) It is known that U_a is canonically equipped with a K-vector space structure ([4], 3.17); every pinning (L, x) is an isomorphism of K-vector spaces from $\Pi_{L/K}\mathfrak{D}$ onto U_a.

(2) The set of endomorphisms of the K-vector space U_a of the form $\mu_x(t) : x(u) \mapsto x(tu)$ (for $t \in L$) does not depend on the choice of the pinning (L, x). It is a field isomorphic to L via the mapping $t \mapsto \mu_x(t)$ and containing the homotheties of the K-vector space U_a. It is denoted by L_a and is called the *field attached to the root ray a*.

(3) The non-zero elements of L_a are exactly the automorphisms of U_a induced by the elements rational over K of the identity component of the normalizer of U_a in the adjoint group of G.

(4) Let $\alpha \in \tilde{\Delta}_a$ and let $\tilde{x}_\alpha, x_\alpha$ be as above. The isomorphism μ_{x_α} of (2) depends only on α and not on the choice of the pinning $\tilde{x}_\alpha$. Its inverse is an embedding of the K-extension L_a into $\tilde{K}$ and one thus obtains, as α ranges over $\tilde{\Delta}_a$, all the embeddings of L_a into $\tilde{K}$ exactly once.

(5) The isomorphism $x_a(u) \mapsto x_{-a}(u) = m_a x_a(u) m_a^{-1}$ from U_a onto U_{-a} (with the notation of 4.1.6) defines an isomorphism from L_a onto L_{-a}, independent of the choice of pinning. It is called the *canonical isomorphism* from L_a onto L_{-a}.

4.1.9. — Let us now place ourselves in case II, and let us take up the notation of 4.1.4, case II. Let us choose an L_a-pinning $\tilde{x}_\alpha$ of $\tilde{U}_\alpha$, let $\tilde{x}_{\alpha'} = \sigma \circ \tilde{x}_\alpha \circ \sigma^{-1}$, $\tilde{x}_{\alpha+\alpha'} = \operatorname{int} m_{\alpha'}^{-1} \circ \tilde{x}_\alpha$ (where $m_{\alpha'}$ is the element of N associated to $\tilde{x}_{\alpha'}$ (3.2.1)) and let $\tilde{x}_{-\alpha}, \tilde{x}_{-\alpha'}$ and $\tilde{x}_{-(\alpha+\alpha')}$ be the associated pinnings. It is then easily shown that there exists one and only one L-isomorphism ζ of $\mathfrak{S}\mathfrak{L}_3$ onto G' such that

$$\zeta \begin{pmatrix} 1 & w & -v \\ 0 & 1 & u \\ 0 & 0 & 1 \end{pmatrix} = \tilde{x}_\alpha(u)\tilde{x}_{\alpha+\alpha'}(-v)\tilde{x}_{\alpha'}(-w),$$

$$\zeta \begin{pmatrix} 1 & 0 & 0 \\ w & 1 & 0 \\ -v & u & 1 \end{pmatrix} = \tilde{x}_{-\alpha'}(w)\tilde{x}_{-(\alpha+\alpha')}(v)\tilde{x}_{-\alpha}(-u).$$

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If we transport the action of σ to $\mathbf{SL}_3$ thanks to ζ, we observe that

$$\sigma \begin{pmatrix} 1 & w & -v \\ 0 & 1 & u \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & -u^\sigma & u^\sigma w^\sigma + v^\sigma \\ 0 & 1 & -w^\sigma \\ 0 & 0 & 1 \end{pmatrix}$$

and that by twisting $\mathbf{SL}_3$ by σ, we obtain an algebraic group defined over L_2 which is none other, as already mentioned, than the group $\mathbf{SU}_3$ of the hermitian form $h : (x_{-1}, x_0, x_1) \mapsto x_{-1}^\sigma x_1 + x_0^\sigma x_0 + x_1^\sigma x_{-1}$ on L^3, and $\zeta_a = \pi \circ \Pi_{L_2/K}\zeta$ is a homomorphism from $\Pi_{L_2/K}\mathbf{SU}_3$ into G.

The diagonal and antidiagonal matrices belonging to the group of rational points over L_2 of this group $\mathbf{SU}_3$ are the matrices

$$j(t) = \begin{pmatrix} t & 0 & 0 \\ 0 & t^{-1}t^\sigma & 0 \\ 0 & 0 & (t^\sigma)^{-1} \end{pmatrix} \quad (\text{for } t \in L^\times), \tag{1}$$

$$m(t) = \begin{pmatrix} 0 & 0 & -(t^\sigma)^{-1} \\ 0 & -t^{-1}t^\sigma & 0 \\ -t & 0 & 0 \end{pmatrix} \quad (\text{for } t \in L^\times). \tag{2}$$

The unipotent upper triangular matrices of this same group are the matrices

$$\mu(u,v) = \begin{pmatrix} 1 & -u^\sigma & -v \\ 0 & 1 & u \\ 0 & 0 & 1 \end{pmatrix} \tag{3}$$

for $u, v \in L$ satisfying

$$v + v^\sigma = u^\sigma u \tag{4}$$

(the notations were chosen in such a way that $\mu(u,v)$ corresponds to the element denoted $u_1(z,k)$ in I, 10.1.2, with $z = u$ and $k = v$).

Consider then the subvariety $H_0(L, L_2)$ of $L \times L$, considered as a 4-dimensional vector space over L_2, defined by equation (4) and let us equip it with the law of composition

$$(u,v)(u',v') = (u + u', v + v' + u^\sigma u') \tag{5}$$

obtained by transporting via μ the group law induced by that of $\mathbf{SU}_3$. It follows then from what precedes that $H_0(L, L_2)$ is an L_2-group and that

$$\zeta \circ \mu : (u,v) \mapsto \tilde{x}_\alpha(u)\tilde{x}_{\alpha+\alpha'}(-v)\tilde{x}_{\alpha'}(u^\sigma) \tag{6}$$

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is an L_2-isomorphism of groups from $H_0(L, L_2)$ to the maximal unipotent subgroup $U' = \tilde{U}_\alpha \tilde{U}_{\alpha+\alpha'} \tilde{U}_{\alpha'}$ of G'. Since $U_a = \prod_{L_2/K} U'$, we see that

$$x_a = \pi \circ \prod_{L_2/K} (\zeta \circ \mu)$$

is a K-isomorphism of groups from $H(L, L_2) = \prod_{L_2/K} H_0(L, L_2)$ to U_a. An isomorphism thus obtained (or rather the triple (L, L_2, x_a)) is called a *pinning* of U_a. As above, we also denote by x_a the corresponding isomorphism of the group of rational points over K of $H(L, L_2)$ (a group which is identified with the set of $(u,v) \in L \times L$ satisfying (4) equipped with the group law (5)) onto $U_a(K)$, an isomorphism which is given by

$$x_a(u,v) = \prod \tilde{x}_\beta(u_\beta)\tilde{x}_{\beta+\beta'}(-v_\beta)\tilde{x}_{\beta'}(u_\beta), \tag{7}$$

where β runs through a subset of $\tilde{\Delta}_a$ containing one and only one point of each element of I, β' being the other point of this element of I, and where the various terms occurring in the right-hand side are defined as follows: for each β, one chooses $\gamma \in \mathrm{Gal}(K_s/K)$ such that $\beta = \gamma(\alpha)$; one then has $\beta' = \gamma(\alpha')$; we set $\tilde{x}_\beta = \gamma \circ \tilde{x}_\alpha \circ \gamma^{-1}$, $\tilde{x}_{\beta'} = \gamma \circ \tilde{x}_{\alpha'} \circ \gamma^{-1}$, $\tilde{x}_{\beta+\beta'} = \gamma \circ \tilde{x}_{\alpha+\alpha'} \circ \gamma^{-1}$, $u_\beta = \gamma(u)$, $v_\beta = \gamma(v)$ (note that $u_\beta, v_\beta \in L_\beta$ and that $v_\beta + v_\beta^\tau = u_\beta^\tau u_\beta$, where τ is the unique non-trivial $L_{\beta+\beta'}$-automorphism of L_β).

4.1.10. — It follows from 4.1.9 (7) that the root subgroup U_{2a} associated with the root $2a$ consists of the $x_a(0,v)$, where v runs through the set L^0 of elements of trace zero of the quadratic extension L of L_2, and the map $v \mapsto x_a(0,v)$ is an isomorphism of K-vector spaces from L^0 onto U_{2a}. Moreover, the image of $x_a(u,v)$ in U_a/U_{2a} depends only on u; if we denote it by $\bar{x}_a(u)$, the map $u \mapsto \bar{x}_a(u)$ is an isomorphism of K-vector spaces from L onto the quotient group U_a/U_{2a}.

4.1.11. — The pinning x_{-a} of U_{-a} associated with x_a is the one obtained by the same method starting from the pinning $\tilde{x}_{-\alpha'}$ of $\tilde{U}_{-\alpha'}$. It is an isomorphism from $H(L, L_2)$ onto U_{-a} given by $x_{-a} = \pi \circ \prod_{L_2/K}(\zeta \circ \mu_-)$, with

$$\mu_-(u,v) = \begin{pmatrix} I & 0 & 0 \\ u & I & 0 \\ -v & -u^\sigma & I \end{pmatrix} \quad (\text{for } (u,v) \in H(L, L_2)),$$

characterized by the following property: for $(u,v) \in L \times L$ satisfying (4) with $v \neq 0$, the elements $y', y'' \in U_a(K)$ such that the product

$$m_a(u,v) = y' x_{-a}(u,v) y''$$

belongs to N (and thus has as its image in W the reflection with respect to the root a) are given by

$$y' = x_a(uv^{-1}, (v^\sigma)^{-1}) \quad \text{and} \quad y'' = x_a(u(v^\sigma)^{-1}, (v^\sigma)^{-1}) \tag{1}$$

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(I, 10.1.11 (3)). Let us resume the notation of 4.1.9 and set $m_a = \zeta_a(m(1))$. Then we have

$$m_a(u, v) = \zeta_a(m(v)) = m_a\tilde{a}(v), \tag{2}$$

$$x_{-a}(u, v) = m_a x_a(u, v) m_a^{-1}, \tag{3}$$

where we denote by $\tilde{a}$ the homomorphism $t \mapsto \pi \circ \Pi_{L_2/K}(\zeta \circ j)$ from $\Pi_{L_2/K}\mathrm{Mult}$ into T.

4.1.12. — Simple matrix calculations in SU_3, which would be somewhat unaesthetic to print and which we leave to the reader, imply that the open set W_a of $U_a \times U_{-a}$ from 1.1.10 is the special open set defined by the function

$$d_a(x_a(u, v), x_{-a}(u', v')) = \mathrm{Norm}_{L/K}(1 - u^\sigma u' + vv') \tag{1}$$

and that the morphism $\beta_a : W_a \to U_{-a} \times T \times U_a$ from 1.1.10 is given by

$$\beta_a(x_a(u, v), x_{-a}(u', v')) = (x_{-a}(t^{-1}(u' - uv'), t^{-1}v'), \tilde{a}(t), x_a((t^\sigma)^{-1}(u - v^\sigma u'), t^{-1}v)), \tag{2}$$

with $t = 1 - u^\sigma u' + vv'$.

4.1.13. — Let us now study how the various choices made (choices of S, of $\tilde{U}_a$, of $\tilde{x}_a$) intervene. If S' is another maximal split torus normalizing U_a, there exists $g \in (\mathrm{Norm}^0 U_a)(K)$ such that $S' = gSg^{-1}$. One easily sees that

$$\mathrm{Norm}^0 U_a = T \cdot \mathrm{Cent}^0 U_a \cdot U_a.$$

But the intersection $T \cdot \mathrm{Cent}^0 U_a \cap U_a$ is the center of U_a, which, in view of 4.1.9 (4) and (5), is K-isomorphic to the vector subspace $\{v \in L \mid v + v^\sigma = 0\}$ of L and thus has trivial Galois cohomology. It follows that $g = g_1 g_2$ with $g_1 \in (T \cdot \mathrm{Cent}^0 U_a)(K)$ and $g_2 \in U_a(K)$. Then int g_1 preserves the subgroup $\tilde{U}_a$ and one sees that changing the split torus normalizing U_a amounts to composing the pinning with an inner automorphism of $H(L, L_2)$, defined by a rational element over K. Such an automorphism is of the form

$$(u, v) \mapsto (u, v + w^\sigma u - u^\sigma w) \quad \text{for some } w \in L. \tag{1}$$

If S is fixed, replacing the root a by $\gamma(a)$ and $\tilde{x}_a$ by $\gamma \circ \tilde{x}_a \circ \gamma^{-1}$ amounts to replacing x_a by $x_a \circ H(\gamma^{-1})$, where $H(\gamma^{-1}) : H(\gamma(L), \gamma(L_2)) \to H(L, L_2)$ is the isomorphism deduced from γ^{-1}. Finally, another pinning $\tilde{x}'_a$ of $\tilde{U}_a$ is of the form $u \mapsto \tilde{x}_a(cu)$ with $c \in L^\times$ and replacing $\tilde{x}_a$ by $\tilde{x}'_a$ amounts to composing x_a with the automorphism $(u, v) \mapsto (cu, c^\sigma cv)$ of $H(L, L_2)$.

In summary, if (L', L'_2, x'_a) is another pinning of U_a, there exists a K-isomorphism $j : L' \to L$, such that $j(L'_2) = L_2$, a $c \in L^\times$ and a $w \in L$ such that x'_a is obtained by performing successively

— the isomorphism $H(j) : H(L', L'_2) \to H(L, L_2)$ defined by j,

— the automorphism $\phi_{c,w} : (u, v) \mapsto (cu, c^\sigma c(v + w^\sigma u - u^\sigma w))$ of $H(L, L_2)$,

— the pinning x_a.

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Moreover, j, c and w are well determined by these conditions. Indeed, it is immediate that $H(j)$ determines j and that $\varphi_{c,w}$ determines c and w. On the other hand, the automorphisms of $H(L, L_2)$ of the form $\varphi_{c,w}$ form a group which contains no non-trivial automorphism of the form $H(\gamma)$ for a K-automorphism γ of L preserving L_2. The composite $\varphi_{c,w} \circ H(j) = x_a^{-1} \circ x'_a$ thus well determines j, c and w.

4.1.14. — The assertions (1) to (5) of 4.1.8 remain exact, provided that U_a is replaced by the quotient group U_a/U_{2a} and that, instead of the pinnings x_a, the corresponding isomorphisms $\bar{x}_a$ of L onto U_a/U_{2a} (4.1.10) are considered.

Moreover, the image in L_a of the subfield L_2 of L under the isomorphism of L into End U_a/U_{2a} defined by $\bar{x}_a$ does not depend on the choice of the pinning; it is denoted by L_{2a} and one thus obtains the quadratic extension $L_a \supset L_{2a}$ attached to the plural root ray a. It can also be characterized as follows: a K-rational element of the connected normalizer of U_a in the adjoint group of G defines an element of norm one of L_a if and only if it induces the identity on U_{2a}.

4.1.15. — It is often convenient to use another description of the group $H(L, L_2)$ defined in 4.1.9. Let L^0 be the L_2-vector subspace of L formed of elements of trace zero and let $\lambda \in L$ be such that $\lambda + \lambda^\sigma = 1$. The map

$$j_\lambda : (u, v) \mapsto (u, v - \lambda u^\sigma u) \tag{1}$$

is an isomorphism of the K-algebraic group $H(L, L_2)$ onto the K-algebraic group H^λ constituted by the K-scheme obtained by restriction of scalars from L_2 to K of the L_2-additive group $L \times L^0$, equipped with the group law

$$(x, y) \cdot (x', y') = (x + x', y + y' - \lambda x x'^\sigma + \lambda^\sigma x^\sigma x'). \tag{2}$$

4.1.16. — Let us fix a Chevalley-Steinberg system $(\tilde{x}_\alpha)_{\alpha \in \tilde{\Phi}}$ of G (4.1.2). For any non-divisible root $a \in \Phi$, choose a root $\alpha(a) \in \tilde{\Phi}$ whose restriction to S is equal to a. Then 4.1.5 and 4.1.9 provide us, starting from the $\tilde{x}_{\alpha(a)}$, with pinnings x_a of the root groups U_a (still denoting by a the root ray containing a). We can choose the $\alpha(a)$ in such a way that x_{-a} is the pinning associated with x_a and we shall then say that $(x_a)_{a \in \Phi}$ is a *coherent system of pinnings* of the U_a, deduced from the given Chevalley-Steinberg system.

Suppose that G is K-simple and simply connected (we leave it to the reader to generalize). Let $\tilde{\Delta}_0$ be a component of the Dynkin diagram of G over $\tilde{K}$, let $\tilde{\Phi}_0$ be the subsystem of roots generated by $\tilde{\Delta}_0$ and let Σ be the quotient of the stabilizer by the fixator of $\tilde{\Delta}_0$ in $\mathrm{Gal}(\tilde{K}/K)$. Every orbit of $\mathrm{Gal}(\tilde{K}/K)$ in $\tilde{\Phi}$ meets $\tilde{\Phi}_0$ and one can choose the roots $\alpha(a)$ in $\tilde{\Phi}_0$. Examination of the Dynkin diagrams and their automorphisms then provides the following results:

a) If $\Sigma = \{1\}$, in other words if G is K-isomorphic to a group $\Pi_{L/K} H$, where L is the field of invariants of the stabilizer of $\tilde{\Delta}_0$ and where H is simple, split over L,

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of Dynkin diagram $\widetilde{\Delta}_0$, then all roots are non-divisible and all fields L_α for $\alpha \in \widetilde{\Phi}_0$ are equal to L;

b) If Card $\Sigma = 2$ and there exists no plural root ray, the fields L_α for $\alpha \in \widetilde{\Phi}_0$ such that its restriction to S is a long (resp. short) root are all equal to the same field L (resp. L_2) and L is a (separable) quadratic extension of L_2;

c) If Card $\Sigma = 2$ and there exist plural root rays, the L_α for $\alpha \in \widetilde{\Phi}_0$ such that its restriction to S is non-divisible (resp. divisible) are all equal to the same field L (resp. L_2) and L is a (separable) quadratic extension of L_2; the group G is K-isomorphic to $\Pi_{L_2/K} \mathfrak{SU}_{2n+1}$ $(n \geqq 1)$;

d) If Card $\Sigma = 3$ (hence $\widetilde{\Delta}_0$ is of type D_4), the assertions of *b*) are valid, with the difference that L is then a Galois cubic extension of L_2;

e) If Card $\Sigma = 6$ (hence $\widetilde{\Delta}_0$ is of type D_4), the L_α for $\alpha \in \widetilde{\Phi}_0$ whose restriction to S is a long root are all equal to the same field L_2; there exists a non-Galois cubic extension L of L_2 contained in $\widetilde{K}$ such that, for each short root a, the three fields associated with the three roots $\alpha \in \widetilde{\Phi}_0$ whose restriction to S is a are the three conjugates of L in $\widetilde{K}$; one can choose the $\alpha(a)$ such that $L_{\alpha(a)} = L$ for every short root a.

Remark. — In cases *b*) to *e*), the field L_2 is the field of invariants of the stabilizer of $\widetilde{\Delta}_0$, the group G is K-isomorphic to a group $\Pi_{L_2/K} H$, where H is absolutely simple, quasi-split over L_2, with Dynkin diagram $\widetilde{\Delta}_0$, and $\widetilde{K}$ is the smallest Galois extension of K containing L.

4.1.17. — Let us keep the hypotheses and notation of 4.1.16. Let $a \in \Phi$. The stabilizer W_a of a in the Weyl group W of G is generated by the reflections with respect to the roots b orthogonal to a. By considering the groups of relative rank 2 corresponding to the pair a, b and by using for example the formulas of I, 10.1.2, one easily sees that every element $n \in N(K)$ stabilizing a induces the identity on the field L_a attached to a, except in the following two cases:

(E 1) we are in case *b*) of 4.1.16 and a is a short root;

(E 2) we are in case *c*) of 4.1.16 and a is non-plural.

In each of these two exceptional cases, the field L_a attached to a is a separable quadratic extension of a subfield isomorphic to L_2 and n induces the identity (resp. conjugation) on L_a if and only if its image in W_a is the product of an even (resp. odd) number of reflections.

One then sees that, except in the two exceptional cases above, the fields (resp. quadratic extensions) attached to the root rays of an orbit of the Weyl group W in Φ are canonically isomorphic, the canonical isomorphism from L_a to $L_{w(a)}$ being given by the isomorphism $u \mapsto nun^{-1}$ from U_a to $U_{w(a)}$ for $n \in N(K)$ with image w in W. In the two exceptional cases, the orbit of a under W is also its orbit under the sub-

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group W_+ of even elements of W and the above assertion is true by replacing W by W_+. On the other hand, if $w \in W - W_+$, the corresponding isomorphism of L_a onto $L_{w(a)}$ is the composition of the canonical isomorphism and the conjugation. In particular, the canonical isomorphism of L_a onto L_{-a} that we have just introduced is the composition with the conjugation of the one introduced in 4.1.8 (5)!

4.1.18. — Let K' be an extension of K, linearly disjoint from $\tilde{K}$, that is to say such that $\tilde{K}' = K' \otimes_K \tilde{K}$ is a field (cf. 1.6.8 (1)).

Then all the preceding considerations are preserved by the base change $K \to K'$. Indeed, it is known that $\tilde{K}'$ is a Galois extension of K', whose Galois group is canonically identified with $\mathrm{Gal}(\tilde{K}/K)$ ([NB] A V, p. 68). It follows immediately that $\tilde{K}'$ "is" the smallest splitting extension of $G' = G_{K'}$, that $S' = S_{K'}$ is a maximal split K'-torus of G', that the root system of G' according to S' is identified with Φ, etc. In particular each pinning $x_a : \Pi_{L/K}\mathbf{D} \to U_a$ (resp. $x_\alpha : H(L, L_\alpha) \to U_\alpha$) canonically provides a pinning $x'_a : \Pi_{K'\otimes L/K'}\mathbf{D} \to U'_a$ (resp. $x'_\alpha : H(K' \otimes L, K' \otimes L_\alpha) \to U'_\alpha$). We leave to the reader the task of developing the sorites if they deem it necessary.

4.1.19. — To the algebraic group G are naturally associated two *root data* (in the sense of I, 6.1.1, taking into account 6.1.2 (8)):

(i) In the group $\tilde{G} = G(\tilde{K})$ of rational points of G over $\tilde{K}$, the one formed by the subgroup $\tilde{T} = T(\tilde{K})$ and the subgroups $\tilde{U}_\alpha = \tilde{U}_\alpha(\tilde{K})$ for $\alpha \in \tilde{\Phi}$: we have seen (I, 6.1.3 b)) that $(\tilde{T}, (\tilde{U}_\alpha, \tilde{M}_\alpha)_{\alpha\in\tilde{\Phi}})$ is a generative root datum of type $\tilde{\Phi}$ in $\tilde{G}$, the class $\tilde{M}_\alpha$ being equal to $\tilde{T}\tilde{m}_\alpha$ (where $\tilde{m}_\alpha$ is defined as in 3.2.1 from a pinning $\tilde{x}_\alpha$ of $\tilde{U}_\alpha$);

(ii) In the group $G = G(K)$ of rational points of G over K, the one formed by the subgroup $T = T(K)$ and the subgroups $U_a = U_a(K)$ for $a \in \Phi$: we have already asserted (I, 6.1.3 c)) that $(T, (U_a, M_a)_{a\in\Phi})$ is a generative root datum of type Φ in G, with $M_a = Tm_a$ (where m_a is defined as in 4.1.5 and 4.1.11). Let us briefly indicate that axiom (DR 1) of I, 6.1.1 is obvious, that (DR 2) results from ([4], 3.8), that (DR 3) is obvious, that (DR 4) results from 4.1.5 (2) and 4.1.11, (DR 5) from the fact that the image of m_a in the Weyl group W is the reflection r_a and (DR 6) from ([4], 3.22 c)). The group $N = N(K)$ of rational points over K of the normalizer of S in G coincides with the group also denoted N in I, 6.1.2 (10): this follows from the fact that the quotient N/T in one or the other sense "is" the Weyl group of Φ.

Furthermore, these two root data are compatible (I, 9.1.9 (DDR 1)) and satisfy conditions (DDR 2) of I, 9.1.10 and (DDR 3) of I, 9.2.9, by replacing the notations G, U, etc. by $\tilde{G}, \tilde{U}$, etc. and the notations $G^\natural, U^\natural$, etc. by G, U, etc., and by taking for V the vector space $\tilde{V}$ dual to the vector subspace generated by $\tilde{\Phi}$ in $X^*(T) \otimes \mathbf{R}$ and for $V^\natural$ the subspace V of invariants of $\mathrm{Gal}(\tilde{K}/K)$ in $\tilde{V}$, a subspace which identifies...

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identifies with the dual of the vector space generated by Φ in $X^*(S) \otimes \mathbf{R}$. This is clear for (DDR 1) and even for the stronger condition (DDR 1 bis) of I, 9.1.13 (a condition which requires that $U_a(K)$ be contained in the product $\prod \widetilde{U}_\alpha(\widetilde{K})$ for $\alpha \in \widetilde{\Phi}$ with restriction to S equal to a or to $2a$, which is true by the very definition of the U_a), evident for (DDR 2) (which requires that $\mathrm{Card}\, a \cap \Phi \leqq 2$ for all $a \in \Phi$) and for (DDR 3) (which means that $T \subset \widetilde{T}$, $M_a \subset N$ and that the inner automorphism defined by m_a permutes the algebraic groups U_a and U_{-a} for any $a \in \Phi$).

4.2. Valuations.

From now on, we assume that the field K is equipped with a non-trivial valuation ω (with real values) and that the minimal Galois extension $\widetilde{K}$ of K splitting G is univalent (1.6.1), that is to say that ω extends in a unique way to a valuation of $\widetilde{K}$, still denoted ω.

4.2.1. — We defined in chapter I the notion of a valuation of a root datum $(Z, (V_b)_{b\in\Psi})$: let us recall that it is a family $\psi = (\psi_b)_{b\in\Psi}$ of mappings $\psi_b : V_b - \{1\} \to \mathbf{R}$, satisfying the axioms (V 0) to (V 5) which we do not copy here (I, 6.2.1).

We also defined (I, 6.2.3 b)) a valuation $\widetilde{\varphi} = (\widetilde{\varphi}_\alpha)_{\alpha\in\widetilde{\Phi}}$ of the root datum $(\widetilde{T}, (\widetilde{U}_\alpha)_{\alpha\in\widetilde{\Phi}})$ (4.1.19 (i)) of the group $\widetilde{G} = G(\widetilde{K})$ of rational points over $\widetilde{K}$ of the $\widetilde{K}$-split group $G_{\widetilde{K}}$ in the following way: one chooses a $\widetilde{K}$-Chevalley system $(\widetilde{x}_\alpha)_{\alpha\in\widetilde{\Phi}}$ of $G_{\widetilde{K}}$ (relative to the $\widetilde{K}$-split maximal torus $\widetilde{T}_{\widetilde{K}}$) and one sets

$$\widetilde{\varphi}_\alpha(\widetilde{x}_\alpha(u)) = \omega(u) \quad \text{for all } u \in \widetilde{K}^\times. \tag{1}$$

One says that $\widetilde{\varphi}$ is the Chevalley valuation associated with the chosen Chevalley system.

In what follows, we shall assume that $(\widetilde{x}_\alpha)_{\alpha\in\tilde{\Phi}}$ is a Chevalley-Steinberg system of G (4.1.3); we shall denote by $(x_a)_{a\in\Phi}$ a coherent system of pinnings deduced from $(\widetilde{x}_\alpha)$ and L_a (resp. (L_a, L_{2a})) the field (resp. the quadratic extension) attached to a non-multipliable (resp. multipliable) root ray a (4.1.8 and 14).

4.2.2. — We are going to show that the valuation $\widetilde{\varphi}$ "descends" to the root datum $(T, (U_a)_{a\in\Phi})$ of the group $G = G(K)$ (4.1.19 (ii)) and thus defines a valuation of the latter.

To this end, we shall apply the results of chapter I, § 9, and notably theorem 9.2.10, but the reader who is put off by the technique of this § 9 may construct a more down-to-earth proof, following for example the indications of 4.2.11.

Let us first explain the definition of the mappings $\varphi_a : U_a - \{1\} \to \mathbf{R}$ starting from $\widetilde{\varphi}$ (I, 9.1.6): for $a \in \Phi$, let A (resp. B) be the set of $\alpha \in \tilde{\Phi}$ with restriction a (resp. $2a$) to S; for $u \in U_a$, it is known that there exist unique $\widetilde{u}_\alpha \in \widetilde{U}_\alpha$ such that (for an arbitrarily chosen order on $A \cup B$)

$$u = \prod_{A\cup B} \widetilde{u}_\alpha,$$

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and we set

$$\varphi_a(u) = \inf\left(\inf_{\alpha\in A} \tilde{\varphi}_\alpha(\tilde{u}_\alpha), \inf_{\alpha\in B} \frac{1}{2}\tilde{\varphi}_\alpha(\tilde{u}_\alpha)\right), \tag{1}$$

the result not depending on the order chosen on $A \cup B$ (loc. cit.). Given the invariance of ω under $\mathrm{Gal}(\tilde{K}/K)$, we then see that

— if the root $a \in \Phi$ is neither divisible nor multipliable, we have

$$\varphi_a(x_a(u)) = \omega(u) \quad \text{for all } u \in L_a^\times; \tag{2}$$

— if the root $a \in \Phi$ is multipliable, we have

$$\varphi_a(x_a(u,v)) = \frac{1}{2}\omega(v) \leqq \omega(u) \quad \text{for } (u,v) \in H(L_a, L_{2a}), (u,v) \neq (0,0); \tag{3}$$

$$\varphi_{2a}(x_a(0,v)) = \omega(v) \quad \text{for } v \in L_a^\times, \mathrm{Tr}_{L_a/L_{2a}} v = 0. \tag{4}$$

Indeed, (2) follows from 4.1.5 (1). If a is multipliable, (1) and 4.1.9 (7) show that $\varphi_a(x_a(u,v)) = \inf\big(\omega(u), \frac{1}{2}\omega(v)\big)$; but 4.1.9 (4) implies $\omega(v) \leqq 2\omega(u)$ for $(u,v) \in H(L_a, L_{2a})$, whence (3). Finally, (4) follows from (3), since φ_{2a} is the restriction of $2\varphi_a$ to U_{2a} (I, 9.1.10).

4.2.3. — Let us denote by $\tilde{\mathscr{I}}$ the building of $\tilde{G}$ defined by $\tilde{f}$ (I, 7.4.2) and $\tilde{A}$ the corresponding apartment, or alternatively the set of valuations equipollent to $\tilde{f}$ (I, 6.2.5). Recall that $\tilde{A}$ is a Euclidean affine space under the vector space $\tilde{V}$, dual to the space generated by $\tilde{\Phi}$ in $X^*(\tilde{T}) \otimes \mathbf{R}$ (4.1.19), equipped with a scalar product invariant under the Weyl group ${}^v\tilde{W}$ of $\tilde{\Phi}$. The subgroup $\tilde{N} = N(\tilde{T})$ of $\tilde{G}$ operates on $\tilde{A}$ via a homomorphism $\tilde{\nu}$ from $\tilde{N}$ into the group of affine automorphisms of $\tilde{A}$ (I, 6.2.10). If $t \in \tilde{T}$, then $\tilde{\nu}(t)$ is the translation by the vector v given by

$$\langle v, \alpha\rangle = -\omega(\alpha(t)) \quad \text{for all } \alpha \in \tilde{\Phi}, \tag{1}$$

and for all $\tilde{n} \in \tilde{N}$, the image ${}^v\tilde{\nu}(\tilde{n})$ of $\tilde{\nu}(\tilde{n})$ in $GL(\tilde{V})$ is the element of the Weyl group ${}^v\tilde{W}$ of $\tilde{G}$ which is the canonical image of $\tilde{n}$.

Given 4.1.3 (3) and (4), the Galois group $\mathrm{Gal}(\tilde{K}/K)$ acting on $\tilde{G}$ permutes up to sign the pinnings $\tilde{x}_\alpha$ of the given Chevalley-Steinberg system and consequently preserves the valuation $\tilde{f}$: we have $\tilde{\varphi}_{\gamma(\alpha)}(\gamma u) = \tilde{\varphi}_\alpha(u)$ for all $\alpha \in \tilde{\Phi}$, $u \in \tilde{U}_\alpha(\tilde{K})$ and $\gamma \in \mathrm{Gal}(\tilde{K}/K)$.

By transport of structure, it follows that $\mathrm{Gal}(\tilde{K}/K)$ operates by automorphisms on $\tilde{\mathscr{I}}$, preserves the point $\tilde{f} \in \tilde{\mathscr{I}}$ and the apartment $\tilde{A}$, and operates on $\tilde{A}$ by affine automorphisms.

Let us then denote by $\mathscr{I}$ (resp. A) the set of fixed points of $\mathrm{Gal}(\tilde{K}/K)$ in $\tilde{\mathscr{I}}$ (resp. $\tilde{A}$): it is a convex subset (resp. an affine subspace with direction V) of $\tilde{\mathscr{I}}$ (resp. $\tilde{A}$). Furthermore, since $\mathrm{Gal}(\tilde{K}/K)$ preserves the base $\tilde{\Delta}$ of $\tilde{\Phi}$ (4.1.2), it preserves the chamber...

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vector space $\tilde{D} \subset \tilde{V}$ associated (I, 1.3.12), thus the sector $\tilde{\phi} + \tilde{D}$ of $\tilde{A}$ and consequently the chamber $C = C_{\tilde{\phi},\tilde{D}}$ (I, 7.2.4). It immediately follows (as we have already said in I, 9.2.3) that the conditions (DI 1), (DI 2) and (DI 3) of I, 9.2.1 are satisfied by taking $F = C$.

We have already seen in 4.1.19 that the conditions (DDR 1), (DDR 2) and (DDR 3) were also satisfied. Finally, the conditions (DV 1) (i.e., $\tilde{\phi} \in A$) and (DV 2) (i.e., Card $\phi_a(U_a) \geq 3$ for all $a \in \Phi$, which is evident given 4.2.2 (2), (3) and (4) since the valuation ω of K is not improper) are likewise satisfied. All the conditions of theorem 9.2.10 of chapter I are thus fulfilled, whence:

Theorem 0.24. *The valuation ϕ descends to G; in other words the family $\phi = (\phi_a)_{a\in\Phi}$ defined by the relations (2), (3) and (4) of 4.2.2 is a valuation of the root datum $(T, (U_a)_{a\in\Phi})$ of the group $G = G(K)$.*

4.2.4. — One can therefore apply to $G = G(K)$ equipped with the root datum $(T, (U_a)_{a\in\Phi})$ and the valuation ϕ thereof the results of chapter I, §§ 6 to 9, whose notations we will use freely, with the exception of the use of the symbol $\natural$. In particular, one can consider the building $\mathcal{I}$ of G (I, 7.4.2), which identifies canonically as a metric space, with a part of $\mathcal{I}_\natural \subset \tilde{\mathcal{I}}$ (which may be distinct from $\mathcal{I}_\natural$ when $\tilde{K}$ is ramified), in a manner compatible with the laws of operation of $G(K) \subset G(\tilde{K})$ on $\mathcal{I}$ and $\tilde{\mathcal{I}}$ (I, 9.2.14). The apartment of $\mathcal{I}$ formed by the valuations equipollent to ϕ (I, 6.2.5) then identifies with A (by the map $\phi + v \mapsto \tilde{\phi} + v$ for $v \in V$), and is an affine space under V. An affine root of A (I, 6.2.6) is always the intersection with A of an affine root of $\tilde{A}$, but the converse may be inaccurate when $\tilde{K}$ is ramified (cf. 4.2.2).

4.2.5. — Let H be an algebraic group defined over K and let $X_K^*(H)$ be the group of rational characters over K of H. Let $c \in X_K^*(H) \otimes \mathbb{Q}$; there exists an integer $N > 0$ tel que $Nc \in X_K^*(H)$ and one sets

$$(\omega \circ c)(t) = \frac{1}{N}\omega((Nc)(t)) \quad \text{for } t \in H(K), \tag{1}$$

a valid definition because it is immediate that the right-hand side of (1) does not depend on the chosen integer N. One thus obtains a homomorphism $\omega \circ c$ from $H(K)$ into $\mathbf{R}$.

4.2.6. — The injection $S \to T$ provides a homomorphism from $X_K^*(T)$ into $X_K^*(S) = X^*(S)$ which is injective and has finite cokernel since S is the maximal K-split sub-torus of T. It thus allows for the canonical identification of $X^*(S) \otimes \mathbb{Q}$ and $X_K^*(T) \otimes \mathbb{Q}$ and to define as in 4.2.5 the homomorphism $\omega \circ c : T(K) \to \mathbf{R}$ for all $c \in X^*(S) \otimes \mathbb{Q}$.

Let $\alpha \in \tilde{\Phi}$, with restriction a to S. Then a identifies with the element $(1/[\tilde{K} : K]) \sum_{\sigma\in\mathrm{Gal}(\tilde{K}/K)} {}^\sigma\alpha$ of $X^*(S) \otimes \mathbb{Q}$ and, for all $t \in T(K)$, one has

$$(\omega \circ a)(t) = \omega(\alpha(t)). \tag{1}$$

4.2.7. — It follows immediately, from either formulas (2), (3) and (4) of 4.2.2, or formula (1) of 4.2.3, that for all $\alpha \in \widetilde{\Phi}$ with restriction a to S, one has

$$\phi_a(tut^{-1}) = \phi_a(u) + \omega(\alpha(t)) \quad (u \in U_a(K), t \in T(K)), \tag{1}$$

or also, in view of 4.2.6,

$$\phi_a(tut^{-1}) = \phi_a(u) + (\omega \circ a)(t) \quad (u \in U_a(K), t \in T(K)). \tag{2}$$

As in I, 6.2.10, let ν denote the homomorphism from $N(K)$ to the group of automorphisms of the apartment A defined by the natural action of $N(K)$ on the valuations of $(T, (U_a))$. Formula (2) then means that, for $t \in T(K)$, the automorphism $\nu(t)$ is the translation by the vector denoted again by $\nu(t) \in V$ defined by

$$\langle \nu(t), a\rangle = -(\omega \circ a)(t) \quad \text{for } a \in \Phi. \tag{3}$$

4.2.8. Definition. — *A valuation of the root datum $(T, (U_a))$ of G is said to be compatible with ω if it satisfies the equivalent conditions (1), (2), (3) of 4.2.7.*

4.2.9. Proposition. — *The valuations of the root datum $(T, (U_a)_{a\in\Phi})$ of G compatible with ω are the valuations equipollent to ϕ.*

It is clear that every valuation equipollent to ϕ (I, 6.2.5) (that is to say, a point of the apartment A, or also a valuation of the form $\phi + v$ with $v \in V$) is compatible with ω. Conversely, let ϕ' be a valuation of $(T, (U_a))$ compatible with ω and let $A' = \phi' + V$ be its equipollence class. In view of I, 6.2.12 b), it suffices to show that there exists an affine isomorphism $j : A \to A'$ transforming the homomorphism $\nu : N(K) \to \mathrm{Aut}\ A$ defined by ϕ into $\nu' : N(K) \to \mathrm{Aut}\ A'$ defined by ϕ'.

Now, it is known that there exists a finite subgroup N_1 of $N(K)$ such that $N(K) = N_1 \cdot T(K)$ (e.g. the subgroup of $N(K)$ generated by the m_a for $a \in \Delta$). Then, N_1 possesses a fixed point $p \in A$ (resp. $p' \in A'$) and the isomorphism $j : A \to A'$ defined by $j(p+v) = p' + v$ for all $v \in V$ answers the question: on the one hand, we have $\nu'(n) \circ j = j \circ \nu(n)$ for all $n \in N_1$ since n then preserves p and p' and since $\nu(n)$ and $\nu'(n)$ have the same image in Aut V, namely the natural operation of the image of n in the Weyl group W of G; on the other hand, $\nu'(t) \circ j = j \circ \nu(t)$ since $\nu(t)$ and $\nu'(t)$ are, in view of 4.2.7 (3), translations by the same vector.

4.2.10. — To the family of Chevalley-Steinberg pinnings there corresponds therefore a family of equipollent valuations of the root datum $(T, (U_a))$, which we will sometimes call the Chevalley-Steinberg valuations of the latter. It is easy to determine them among the valuations equipollent to ϕ. Indeed, one obtains the Chevalley-Steinberg pinnings from the given pinning in the following way: one chooses a homomorphism θ from $\widetilde{\Phi}$ into $\widetilde{K}^\times$ such that $\gamma \circ \theta = \theta \circ \gamma$ for all $\gamma \in \mathrm{Gal}(\widetilde{K}/K)$ and one sets

$$\widetilde{x}'_\alpha(u) = \widetilde{x}_\alpha(\theta(\alpha)u) \quad \text{for } \alpha \in \widetilde{\Phi} \text{ and } u \in \widetilde{K}.$$

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For $a \in \Phi$ non-divisible, let $\theta(a) = \theta(\alpha(a))$. Then $\theta(a) \in L_a^\times$ and the valuation φ' associated with the pinning $(\tilde{x}'_a)$ is given by

$$a(\varphi' - \varphi) = -\omega(\theta(a)) \text{ for all } a \in \Phi, \text{ non-divisible.}$$

Conversely, if one chooses for each $a \in \Delta$ an element $\omega_a \in \omega(L_a^\times)$, there exists a unique Chevalley-Steinberg valuation such that $a(\varphi' - \varphi) = -\omega_a$ for all $a \in \Delta$.

4.2.11. — It will be noted that the proof of the equipollence of two valuations compatible with ω of the root datum $(T, (U_a))$ is independent of that of the existence of such a valuation.

It is moreover possible to prove the existence of a valuation of $(T, (U_a))$ compatible with ω without using the descent theorem of I, § 9. One can, for example, show directly that the family $\varphi = (\varphi_a)$ defined by formulas (2), (3), and (4) of 4.2.2 satisfies axioms (V 0) to (V 5) of valuations: properties (V 0) and (V 4) are obvious, (V 1) is also obvious when the root a is non-multipliable and results easily from the determination of the group law on U_a when $2a \in \Phi$ (4.1.9); condition (V 2) results from the change-of-basis formulas from one pinning of U_a to another (4.1.7 and 4.1.13), because the composite of a pinning and conjugation by an element of $N(K)$ is still a pinning; condition (V 5) results from 4.1.5 (2) and 4.1.11 (1).

It remains then to prove the "commutation" axiom (V 3). This can be done by simple verification from the explicit calculation of the commutator maps $\gamma_{a,b} : U_a \times U_b \to \prod U_c$ defined in 1.1.8 (1) (for $a, b \in \Phi, b \neq -a$). We shall give the results of this calculation in the Appendix.

One can also note that the verification of (V 3) only involves a maximal unipotent subgroup of a semi-simple subgroup of relative rank two of G. One can then reduce for this verification to the case where G is semi-simple, simply connected, of relative rank two. If one further supposes that Φ has no factor of type G_2, one is then reduced, up to a central isogeny and a restriction of scalars, to 'classical groups' (moreover of a particular type), for which the existence of a valuation compatible with ω and in particular (V 3) are proven in § 10 of Chapter I.

4.2.12. — Recall that the building $\mathscr{I}$ of G is perfectly defined by the valued root datum, with all its structures (apartments, with their affine space structure, chambers, facets, etc.) except for its metric, which depends on the choice of a scalar product on V invariant under the Weyl group (such a scalar product is unique only up to multiplication by a scalar on each of the vector subspaces generated by an irreducible component of Φ). In particular, let K' be another valued field, $\sigma : K \to K'$ an isomorphism of valued fields, G' an algebraic group defined over K', and j a σ-isomorphism from G onto G', hence an isomorphism also denoted j from $G(K)$ onto $G'(K')$. Let $S' = j(S)$ and

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let us denote by a $'$ the objects attached to the triple (K', G', S'). Let us equip V' with the scalar product deduced from that chosen on V by transport of structure. Still by transport of structure, we deduce an isometry, also denoted by j, from $\mathscr{I}$ onto $\mathscr{I}'$, compatible with all the structures carried by the buildings and such that

$$j(g \cdot x) = j(g) \cdot j(x) \quad \text{for } g \in G(K) \text{ and } x \in \mathscr{I}. \tag{1}$$

This bijection is the unique map $j : \mathscr{I} \to \mathscr{I}'$ satisfying (1) and at least one of the following two conditions:

(2) j is isometric;

(3) for all $x, y \in \mathscr{I}$, the restriction of j to the segment $[xy]$ is an affine bijection from the latter onto the segment $[j(x)j(y)]$.

It suffices to show this uniqueness when σ and $j : G \to G'$ are the identity. Under (2), it then follows from I, 9.2.15, taking $G^\natural = G$ and $\mathscr{I}_\natural = \mathscr{I}$ there. Suppose (3); then j permutes the segments, hence permutes the convex subsets of $\mathscr{I}$. Furthermore, (1) implies that j permutes the subsets stable under a given subgroup of $G(K)$. It then follows from I, 7.4.32 that $j(A) = A$, then from I, 7.4.3 that j is the identity.

Note that if $\dim A \geq 2$, one can replace (3) by

$$\text{the image under } j \text{ of a segment is a segment.} \tag{4}$$

If ω is dense, condition (1) alone suffices to ensure the uniqueness of j (I, 8.2.1).

Remark. — More generally, the above proof shows that, for any building $\mathscr{I}$ of a valued root datum, the identity is the unique map j from $\mathscr{I}$ into itself which satisfies (1) and either (2) or (3), this second condition being superfluous in the case of a dense valuation.

4.2.13. — The building $\mathscr{I}$ does not depend, up to unique isomorphism, on the choice of S: this follows from what precedes, taking into account the conjugacy by an element of $G(K)$ of the maximal K-split tori.

4.2.14. — The center C of $G(K)$ acts trivially on $\mathscr{I}$: if $g \in C$ then $g \in T$ and $\nu(g) = 0$ by the very definition of ν (I, 6.2.5), whence $ghx = hgx = hx$ for all $x \in A$ and $h \in G(K)$.

4.2.15. — The building of G is invariant under central epimorphism: let G' be another algebraic group defined over K and let $j : G \to G'$ be a surjective homomorphism with central kernel. Then G' is reductive, quasi-split over K, split over $\tilde{K}$; the image $S' = j(S)$ is a maximal K-split torus thereof, with centralizer $T' = j(T)$; the map $a' \mapsto a' \circ j$ is an isomorphism from $\Phi' = \Phi(S', G')$ onto Φ and if we identify Φ and Φ' by this map, then for all $a \in \Phi_{\text{red}}$ the restriction of j to U_a is an isomorphism from U_a onto U'_a (for all these assertions, see ([5], th. 2.20). One can then define φ'_a by $\varphi_a = \varphi'_a \circ j$, whence a valuation φ' of the root datum $(T', (U'_a))$

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of $G'(K)$ and it is easily verified that there exists a unique isomorphism from $\mathcal{I}$ onto the building $\mathcal{I}'$ of G' compatible with the actions of $G(K)$ and $G'(K)$ and mapping φ to φ'.

4.2.16. — It is sometimes preferable to introduce the enlarged building $\mathcal{I}^1$ of G defined as follows (cf. [27], [40]). Let us consider the group $X_K^*(G)$ of rational characters of G over K, the vector space $X_K^*(G) \otimes \mathbf{R}$ and its dual V^1, which we equip with a Euclidean metric. We set $\mathcal{I}^1 = \mathcal{I} \times V^1$ and we make $G(K)$ act on $\mathcal{I}^1$ by setting

$$g.(x, v) = (g.x, v + \theta(g)) \quad (g \in G(K), x \in \mathcal{I}, v \in V^1), \tag{1}$$

where $\theta(g)$ is the element of V^1 defined by

$$\langle \theta(g), \chi \rangle = -\omega(\chi(g)) \quad \text{for all } \chi \in X_K^*(G). \tag{2}$$

One defines the apartments, chambers, facets, ... of $\mathcal{I}^1$ as the products by V^1 of the apartments, chambers, facets, ... of $\mathcal{I}$ and the metric of $\mathcal{I}^1$ by taking, for every apartment A' of $\mathcal{I}$, the Euclidean metric of the affine space $A' \times V^1$ which is the Euclidean sum of the metrics given on A' and V^1. It is easy to see that almost all of the geometric properties of the building $\mathcal{I}$ remain valid for $\mathcal{I}^1$. However, the uniqueness assertion of 4.2.12 is no longer true: the maps $j : \mathcal{I}^1 \to \mathcal{I}^1$ commuting with the action of G and satisfying one of the conditions (2) or (3) of 4.2.12 are the maps of the form $(x, v) \mapsto (x, v + v_0)$ where v_0 is a constant belonging to V^1. Furthermore, let DG be the derived group of G and let CG be its connected center. The isogeny $DG \times CG \to G$ (resp. $(T \cap DG) \times CG \to T$) allows one to identify $X_K^*(G) \otimes \mathbf{Q}$ and $X_K^*(CG) \otimes \mathbf{Q}$ (resp. $X_K^*(T) \otimes \mathbf{Q}$ and the direct sum of $X_K^*(T \cap DG) \otimes \mathbf{Q}$ and $X_K^*(CG) \otimes \mathbf{Q}$), thus $X_K^*(T) \otimes \mathbf{R}$ with the dual of $V \times V^1$, the direct factor $X_K^*(T \cap DG) \otimes \mathbf{R}$ (resp. $X_K^*(CG) \otimes \mathbf{R} = X_K^*(G) \otimes \mathbf{R}$) being the orthogonal of V^1 (resp. V).

We then have, for all $t \in T(K)$ and all $c \in X_K^*(T) \otimes \mathbf{Q}$,

$$\langle \nu(t) + \theta(t), c \rangle = -(\omega \circ c)(t). \tag{3}$$

Indeed, (3) is true for c if it is true for a multiple nc of c, with $n \neq 0$. It therefore suffices, taking into account the additivity in c of both sides, to verify it

— for $c \in X_K^*(T \cap DG)$, and we then have $\langle \theta(t), c \rangle = 0$ by definition of the duality between $V \times V^1$ and $X_K^*(T) \otimes \mathbf{R}$ and $\langle \nu(t), c \rangle = -(\omega \circ c)(t)$ according to 4.2.7 (3);

— for $c \in X_K^*(G)$, and we then have $\langle \nu(t), c \rangle = 0$ by definition of the duality and $\langle \theta(t), c \rangle = -(\omega \circ c)(t)$ by definition of θ.

We identify $\mathcal{I}$ and the subset $\mathcal{I} \times \{0\}$ of $\mathcal{I}^1$. Then, the stabilizer of $\mathcal{I}$ for the action of G on $\mathcal{I}^1$ is the normal subgroup

$$G^1 = \theta^{-1}(0) = \{g \in G(K) \mid \omega(\chi(g)) = 0 \text{ for all } \chi \in X_K^*(G)\}.$$

We have $U_a(K) \subset G^1$, from which it follows immediately that $(T \cap G^1, (U_a))$ is a generating root datum in G^1 (I, 6.1.2 (12)), of which φ is a valuation, and the building of G^1 identifies canonically with $\mathscr{I}$. If X is a subset of $G(K)$, we shall set

$$X^1 = X \cap G^1.$$

If X is the stabilizer (resp. pointwise stabilizer) of a subset M (or of a germ of subsets) of $\mathscr{I}$ for the action of G on $\mathscr{I}$, then X^1 is the stabilizer of $M \times \{0\}$ in G for the action of G on $\mathscr{I}^1$. In particular N^1 is the stabilizer of $A \times \{0\}$. We set

$$W^1 = \nu(N^1).$$

4.2.17. — We are going to give an example of an enlarged building which will be useful to us in § 5. For this, we shall use the results of I, 7.6, which unfortunately we must complete. Let us resume *for this section exclusively* the sole hypotheses and notation of I, 7.6. The following result should have been stated as corollary 7.6.6 in Chapter I:

Proposition 0.25. *Let $x_0 \in A$ and set $\mathscr{I}_1' = G_1^0.(x_0 + Ker\ p)$.*

(i) $\mathscr{I}_1' \cap A = x_0 + Ker\ p$.

(ii) The restriction π' of $\tilde{\pi}$ to $\mathscr{I}_1'$ is an isometry from $\mathscr{I}_1'$ onto $\mathscr{I}_1$ commuting with the actions of G_1^0.

(iii) Let j be the inverse of π'; the map $(y, v) \mapsto j(y) + v$ is a bijection from $\mathscr{I}_1 \times L_1$ onto $G_1.A$.

(iv) There exists a homomorphism θ_1 from G_1 to L_1 such that

$$g.(j(y) + v) = j(g.y) + v + \theta_1(g)$$

for all $g \in G_1, y \in \mathscr{I}_1$ and $v \in L_1$. We have $\theta_1(g) = 0$ for $g \in G_1^0$ and $\theta_1(t) = (p \circ \nu)(t)$ for $t \in T_1$.

Proof. To prove (i), it suffices to show that if $x \in A, g \in G_1^0$ and $x' = g.x \in A$, then $x' \in x + \mathrm{Ker}\ p$ (recall that p is the orthogonal projection of V onto L_1). Now it follows from I, 7.6.4 (3) and 7.6.5 (ii) that there exists $n \in N_1^0$ such that $x' = n.x$, whence $x' \in x + \mathrm{Ker}\ p$ since, by definition, $\nu(N_1^0)$ is generated by orthogonal reflections with respect to hyperplanes stable under L_1.
Let us prove (ii). The map $\pi' : x_0 + \mathrm{Ker}\ p \to A_1$ is bijective. Since $\mathscr{I}_1 = G_1^0.A_1$, it follows that π' is surjective. Furthermore, if $x, x' \in \mathscr{I}_1'$, there exists $g \in G_1^0$ such that $\pi'(x), \pi'(x') \in g.A_1$. In view of (i), $(g^{-1} \circ \pi')(x)$ and $(g^{-1} \circ \pi')(x')$ belong to $x_0 + \mathrm{Ker}\ p$; it follows immediately that π' is isometric, whence (ii).
Consequently, j is a section of the map $\tilde{\pi} : G_1.A \to \mathscr{I}_1$, whence (iii), since the fibers of this map are the $x + L_1$ for $x \in G_1.A$ (I, 7.6.4 (iii)).
Finally, let us prove (iv). In view of I, 7.6.4 (iii), there exists a function $\theta : G_1 \times \mathscr{I}_1 \to L_1$ such that $g.(j(x)+v) = j(g.x) + \theta(g, x) + v$ for $x \in \mathscr{I}_1, g \in G_1$ and $v \in L_1$. A simple and classical calculation gives

$$\theta(gh, x) = \theta(g, h.x) + \theta(h, x) \quad \text{for } g, h \in G_1, x \in \mathscr{I}_1. \tag{1}$$

□

But $\theta(g, x) = 0$ if $g \in G_1^0$, from which

$$\theta(gt, x) = \theta(t, x) = \theta(t, t^{-1}gt.x) \quad \text{for } g \in G_1^0 \text{ and } t \in T_1. \tag{2}$$

Furthermore, if $x \in A_1$ and $t \in T_1$, we have $t.j(x) = j(x) + \nu(t)$ and $j(t.x) \in j(x) + \mathrm{Ker}\ p$, from which

$$\theta(t, x) = (p \circ \nu)(t) \quad \text{for } x \in A_1 \text{ and } t \in T_1. \tag{3}$$

But (2) shows that $\theta(t, g.x) = \theta(t, x)$ for $g \in G_1^0$, $t \in T_1$ and $x \in \mathcal{I}_1$. Since $\mathcal{I}_1 = G_1^0 \cdot A_1$, we therefore have (3) for all $t \in T_1$ and all $x \in \mathcal{I}_1$ and (iv) follows immediately from (2).

4.2.18. — Let us now return to our general conventions. Let Φ_1 be a quasi-closed root subsystem of Φ (I, 7.6.1) and let G_1 be the algebraic subgroup of G generated by T and the U_a for $a \in \Phi_1$. Then G_1 is reductive connected, defined and quasi-split over K, split over $\widetilde{K}$ and the root system of G_1 with respect to the maximal K-split torus S is Φ_1. One can therefore apply to it, on the one hand, the results proven for G in the preceding pages, and on the other hand, the results of I, 7.6 whose notation we resume (with $G_1 = G_1(K)$). The dual V_1^1 of $X_K^*(G_1) \otimes \mathbf{R}$ identifies with $L_1 \times V^1$ thanks to the identifications already seen:

$$X_K^*(G_1) \otimes \mathbf{R} = X_K^*(CG_1) \otimes \mathbf{R} = (X_K^*(CG_1 \cap DG) \otimes \mathbf{R}) \times (X_K^*(G) \otimes \mathbf{R}).$$

Consider then, with the notations of 4.2.16 and 4.2.17, the map

$$j_1 : (x, v, v') \mapsto (j(x) + v, v')$$

from $\mathcal{I}_1 \times L_1 \times V^1 = \mathcal{I}_1 \times V_1^1$ into $G_1 \cdot A \times V^1 = G_1 \cdot (A \times V^1) \subset \mathcal{I}^1$. It is clear, given 4.2.17, that this is a surjective isometry, and j_1 is compatible with the actions of G_1 on the enlarged building $\mathcal{I}_1 \times V_1^1$ of G_1 on the one hand, and on the enlarged building $\mathcal{I}^1$ of G on the other hand: this follows from 4.2.15, taking into account the relations $G_1 = G_1^0 \cdot T$ and $G_1^0 \subset DG$. In other words, $G_1 \cdot (A \times V^1)$, that is to say the smallest subset of the enlarged building of G stable under G_1 and containing the apartment $A \times V^1$, identifies with the enlarged building of G_1, and this in one and only one way up to a translation by an element of V_1^1.

4.2.19. — **Bornologies.** The group $G = G(K)$ is naturally equipped with a bornology $\mathcal{B}(\omega)$: a subset X of G is bounded if and only if, for all $f \in K[G]$, the set of $\omega(f(x))$ remains bounded below as x ranges over X. This bornology is compatible with the group law of G.

On the other hand, the valuation ϕ defines on G two bornologies:

— the bornology $\mathcal{B}(\phi)$ called defined by ϕ in I, 8.1.7, which is the preimage of the natural bornology of the isometry group Isom $\mathcal{I}$ of the building $\mathcal{I}$ (I, 3.1.2 b));

— the bornology $\mathcal{B}^1(\phi)$ preimage of the natural bornology of the isometry group Isom $\mathcal{I}^1$ of the enlarged building $\mathcal{I}^1$.

Proposition. — (i) $\mathcal{B}(\omega) = \mathcal{B}^1(\phi)$.

(ii) The bornologies induced on the subgroup G^1 by $\mathcal{B}(\omega)$ and by $\mathcal{B}(\phi)$ are equal.

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Assertion (ii) follows from (i) since G^1 acts trivially on V^1. Let us prove (i). It is clear that, for $a \in \Phi$ and $k \in \mathbf{R}$, the subgroup $U_{a,k}$ is bounded for $\mathscr{B}(\omega)$ and that U_a is not. In other words, $\mathscr{B}(\omega)$ is weakly compatible with φ in the sense of I, 8.1.1. According to I, 8.1.5, it suffices, to prove (i), to show that $\mathscr{B}(\omega)$ and $\mathscr{B}^1(\varphi)$ induce the same bornology on T. Now, saying that $X \subset T$ is bounded for $\mathscr{B}^1(\varphi)$ means that $x \mapsto (\omega \circ a)(x)$ remains bounded below on X on the one hand for $a \in \Phi$, and on the other hand for $a \in X_K^*(G)$. But $\Phi \cup X_K^*(G)$ generates $X_K^*(T) \otimes \mathbf{Q}$ (cf. 4.2.6 and 4.2.16) and our assertion follows from the following lemma:

Lemma 0.26. *Let T be a torus defined over K, split over $\tilde{K}$. A subset X of $T(K)$ is bounded for the natural bornology $\mathscr{B}(\omega)$ if and only if the map $x \mapsto \omega(\chi(x))$ is bounded below on X for all $\chi \in X_K^*(T)$.*

The condition is obviously necessary. Suppose it is satisfied. For all $f \in K[T]$, there exist $\chi_1, \ldots, \chi_n \in X^*(T)$ and $c_1, \ldots, c_n \in \tilde{K}$ such that $f = \sum c_i \chi_i$. It thus suffices to show that $\omega(\chi(x))$ remains bounded below on X for all $\chi \in X^*(T)$. Let

$$\chi^\natural = \sum_{\gamma \in \mathrm{Gal}(\tilde{K}/K)} \gamma \cdot \chi \in X_K^*(T).$$

We have $\omega((\gamma \cdot \chi)(x)) = \omega((\chi(x))^\gamma) = \omega(\chi(x))$ for all $\gamma \in \mathrm{Gal}(\tilde{K}/K)$ and all $x \in T(K)$, hence

$$\omega(\chi(x)) = [\tilde{K} : K]^{-1} \omega(\chi^\natural(x)),$$

which proves the lemma and the proposition.

4.2.20. — *Value sets.* Let φ be the valuation of the root datum $(T, (U_a)_{a \in \Phi})$ given by 4.2.2. Recall that, for $a \in \Phi$, we set (I, 6.2.2):

$$\Gamma_a = \varphi_a(U_a - \{1\}), \tag{1}$$

$$\Gamma'_a = \{\varphi_a(u) \mid u \in U_a - \{1\}, \varphi_a(u) = \sup \varphi_a(uU_{2a})\} \tag{2}$$

(if $2a \notin \Phi$, we set $U_{2a} = \{1\}$). We will determine these value sets. To do this, let L_a (resp. (L_a, L_{2a})) denote, as above, the field (resp. the quadratic extension) attached to a non-multipliable (resp. multipliable) root ray a. In a general way, if $L \supset L_a$ is a quadratic extension, with $K \subset L_a \subset L \subset \tilde{K}$, we denote by L^0 (resp. L^1) the set of elements of L with trace zero (resp. equal to 1) and set

$$L^1_{\max} = \{\lambda \in L^1 \mid \omega(\lambda) = \sup\{\omega(x) \mid x \in L^1\}\}. \tag{3}$$

If the residue field of L is of characteristic $\neq 2$, we have $\frac{1}{2} \in L^1_{\max}$. With these notations:

Lemma 0.27. *(i) If $L^1_{\max} = \emptyset$, the valuation ω does not reach its maximum on any coset of L modulo L^0 not containing 0.*
(ii) If $\lambda \in L^1_{\max}$, we have, for $x \in L_a$ and $y \in L^0$,

$$\omega(x\lambda + y) = \inf(\omega(x\lambda), \omega(y)). \tag{4}$$

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Let $z \in L^\times$ be such that $\omega(z) = \sup \omega(z + L^0)$. Then $z \notin L^0$, $\operatorname{Tr} z \neq 0$ and $\lambda = (\operatorname{Tr} z)^{-1} \cdot z$ belongs to $L^1_{\max}$, whence (i).

Let us now adopt the notation of (ii). It suffices to prove (4) when $x \neq 0$; we then have $\lambda + x^{-1}y \in L^1$, whence $\omega(\lambda) \geq \omega(\lambda + x^{-1}y)$ and $\omega(x\lambda) \geq \omega(x\lambda + y)$. We deduce $\omega(y) \geq \inf(\omega(x\lambda + y), \omega(-x\lambda)) = \omega(x\lambda + y)$, whence (4).

Let us note that $\lambda \notin L^0$ and that $L = L_2\lambda + L^0$; formula (4) thus allows one to calculate the valuation of any element of L.

4.2.21. — Let $a \in \Phi$. One has

(1) $\Gamma_a = \Gamma'_a = \omega(L_a^\times)$ if a is not multipliable;

(2) $\Gamma_{2a} = \Gamma'_{2a} = \omega(L_a^0 - \{0\})$ if a is multipliable;

(3) $\Gamma'_a = \emptyset$ and $\Gamma_a = \frac{1}{2}\omega(L_a^0 - \{0\})$ if a is multipliable and $(L_a^1)_{\max} = \emptyset$;

(4) $\Gamma'_a = \frac{1}{2}\omega(\lambda) + \omega(L_a^\times)$ and $\Gamma_a = \frac{1}{2}\omega(L_a^\times)$ if a is multipliable and $\lambda \in (L_a^1)_{\max}$.

Indeed, formulas (1) and (2) result immediately from 4.2.2 (2) and (4) respectively. On the other hand, 4.2.2 (3) and 4.1.9 (5) show that, if a is multipliable, Γ_a (resp. Γ'_a) is the set of $\phi_a(x_a(u,v)) = \frac{1}{2}\omega(v)$ for $(u,v) \in H(L_a, L_{2a})$ (resp. and $\omega(v) = \sup \omega(v + L_a^0)$).

When $(L_a^1)_{\max} = \emptyset$, (3) follows immediately, in view of 4.2.20 (i) and I, 6.2.2. Suppose then that $\lambda \in (L_a^1)_{\max}$, whence $v = \lambda u^\sigma u + v_0$, with $v_0 \in L_a^0$. The first relation (4) then results from 4.2.20 (ii). Since the inclusion $\Gamma_a \subset \frac{1}{2}\omega(L_a^\times)$ is obvious, it remains to establish the reverse inclusion. If $e = 1$, it results from the fact that $\omega(L_a^\times) = \omega(L_{2a}^\times) = \omega(L_{2a}^0 - \{0\}) \subset 2\Gamma_a$, and if $e = 2$, 4.2.20 (ii) implies that

$$\begin{aligned}\omega(L_a^\times) &= (\omega(\lambda) + \omega(L_{2a}^\times)) \cup \omega(L_a^0 - \{0\}) \\ &= (\omega(\lambda) + 2\omega(L_a^\times)) \cup \omega(L_a^0 - 0) \subset 2\Gamma_a,\end{aligned}$$

whence (4). Let us further note that, still under the hypothesis where $\lambda \in (L_a^1)_{\max}$, 4.2.20 (ii) also implies

$$\phi_a(x_a(u,v)) \leq \omega(u) + \frac{1}{2}\omega(\lambda), \tag{5}$$

with equality if and only if $\phi_a(x_a(u,v)) = \sup \phi_a(x_a(u,v)U_{2a})$ (e.g., if $v = \lambda u^\sigma u$).

If ω is discrete, we obviously have $(L_a^1)_{\max} \neq \emptyset$ (since $x \in L_a^1$ implies $\omega(x) \leq 0$), whence $\Gamma'_a \neq \emptyset$ for every multipliable root a. Furthermore, it is clear that the valuation ϕ is discrete (I, 6.2.21).

If ω is dense and if $(L_a^1)_{\max} \neq \emptyset$ for every multipliable root a (which is the case if $\tilde{K}$ is admissible, see further on 4.3.3), then Γ'_a is dense in $\mathbf{R}$ for every $a \in \Phi$.

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4.2.22. — Proposition. — *Suppose G is simply connected. Then, the group $W = \nu(N)$ is generated by the reflections $r_{a,k}$ with respect to the hyperplanes*

$$\partial\alpha_{a,k} = \{x \in A \mid a(x - \varphi) + k = 0\} \quad \textit{for } a \in \Phi \textit{ and } k \in \Gamma_a.$$

If, in addition, ω is discrete, W is the affine Weyl group of the affine root system $\alpha_{a,k} = \{x \in A \mid a(x-\varphi)+k \geq 0\}$ for $a \in \Phi$ and $k \in \Gamma'_a$.

It is known that $G(K)$ is generated by the union of the $U_a(K)$ for $a \in \Phi$ (cf. for example [33], lemma 64, p. 183). The first assertion then follows from I, 6.2.11 and the second follows immediately.

4.2.23. — Gradings. — Suppose ω is discrete. From the formulas in 4.2.21, the determination of the grading $\mathscr{E} \subset \Phi \times \Sigma$ attached to the affine root system of A (I, 1.4) follows immediately. It is sufficient to do this when G is K-simple. There then exists a field extension L of K contained in $\tilde{K}$ such that $G = \Pi_{L/K}H$, where H is an absolutely simple L-group; the building of G over K identifies with that of H over L and the gradings are the same. We can therefore limit ourselves to the case where G is absolutely simple. We then have the following results, where we resume the notation of I, 1.4.6:

a) If G is split of type X_n, the grading $\mathscr{E}$ is of type X_n;

b) If G is not split and if $\tilde{K}$ is an *unramified* extension of K, then

— for G of type ${}^2A_{2n}$ $(n \geq 1)$, $\mathscr{E}$ is of type $C - BC_n^{\mathrm{IV}}$,

— for G of type ${}^2A_{2n+1}$ $(n \geq 1)$, $\mathscr{E}$ is of type C_{n+1},

— for G of type ${}^2D_n$ $(n \geq 4)$, $\mathscr{E}$ is of type B_{n-1},

— for G of type ${}^2E_6$, $\mathscr{E}$ is of type F_4,

— for G of type ${}^3D_4$ or ${}^6D_4$, $\mathscr{E}$ is of type G_2;

c) If G is not split and if $\tilde{K}$ is a *ramified* extension of K, then

— for G of type ${}^2A_{2n}$ $(n \geq 1)$, $\mathscr{E}$ is of type $C - BC_n^{\mathrm{III}}$,

— for G of type ${}^2A_{2n+1}$ $(n \geq 1)$, $\mathscr{E}$ is of type $B - C_{n+1}$,

— for G of type ${}^2D_n$ $(n \geq 4)$, $\mathscr{E}$ is of type $C - B_{n-1}$,

— for G of type ${}^2E_6$, $\mathscr{E}$ is of type F_4^I,

— for G of type ${}^3D_4$ or ${}^6D_4$, $\mathscr{E}$ is of type G_2^I.

4.2.24. — Let E be an algebraically closed extension of $\tilde{K}$, let K' be a subfield of E containing K equipped with a valuation ω' extending ω and let $\tilde{K}' = K'\tilde{K}$: this is the smallest Galois extension of K' splitting G. Suppose that $\tilde{K}'$ is a univalent extension of K' and let $G' = G_{K'}$, $T' = T_{K'}$, etc. Note that by identifying $X^*(T)$ and $X^*(T')$, we have $\tilde{\Phi} = \tilde{\Phi}'$. Let S'' be the maximal K'-split subtorus of T' and Φ'' the root system of G' with respect to S''. It is clear that the Chevalley-Steinberg system $(\tilde{x}_\alpha)$ "is" still a Chevalley-Steinberg system of G', thus it defines a valuation $\tilde{\phi}'$ of the root datum $(T'(\tilde{K}'), (U'_\alpha(\tilde{K}'))_{\alpha \in \tilde{\Phi}})$ of $G(\tilde{K}')$, which descends to a valuation ϕ' of the root datum $(T'(K'), (U'_b(K'))_{b \in \Phi''})$ of $G(K')$. It is immediate that $\tilde{\phi}'$ descends to $\tilde{\phi}$ (cf. I, 9.1.19 a)), from which it follows by transitivity that ϕ' descends to ϕ.

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Applying I, 9.1.17 (with $S^h = T(K)$), we see that there exists an isometry j, and only one, from the building $\mathcal{I}$ of $G(K)$ into the building $\mathcal{I}'$ of $G(K')$ such that $j(\varphi) = \varphi'$ and $j(g \cdot x) = g \cdot j(x)$ for $x \in \mathcal{I}$ and $g \in G(K)$. In view of 4.2.10, j is independent of the choice of the Chevalley-Steinberg system and allows us to identify $\mathcal{I}$ with a subset of $\mathcal{I}'$. The image $j(A)$ is contained in the apartment A' of $\mathcal{I}'$ associated with the maximal K'-split torus S''. Finally, in view of the conjugacy of maximal K-split tori and 4.2.10, we see that j also does not depend on the choice of S; we can therefore call it canonical. In particular, if u is a K-automorphism of K' preserving ω', then u operates on $\mathcal{I}'$ and we have $u \circ j = j$.

Definition 0.5 (4.2.25)**.** Suppose that $\tilde{K}$ is an admissible extension of K (1.6.1)[11], and let K' be an extension of K, equipped with a valuation ω' extending ω. We say that the base change $K \to K'$ is *conservative with respect to* $\tilde{K}$ if ω' is an unramified extension of ω (i.e., $\omega'(K') = \omega(K)$) and if the residue fields of K' and $\tilde{K}$ are linearly disjoint over the residue field of K (cf. 1.6.8).

Such a base change does not, in fact, change much. In view of 1.6.8, K' and $\tilde{K}$ are linearly disjoint over K, which allows us to apply 4.1.18, and $\tilde{K}'$ is an admissible extension of K'. With the notation of 4.1.18 and 4.2.24, we have $S'' = S'$, $\Phi'' = \Phi$, the root subgroups of G and G' are "the same", etc. The restriction of j to A is then an isomorphism from A onto the apartment A' of $\mathcal{I}'$ corresponding to S', for all structures of A (metric, affine root system, etc.).

This will often allow us to reduce either to the case where the residue field of K is infinite thanks to 1.6.9 a), or to the one where K is henselian thanks to 1.6.9 b).

Of course, the isometry $j : \mathcal{I} \to \mathcal{I}'$ is not in general surjective, except however if K' is the henselization or the completion of K (cf. [27], prop. 2.3.5): by considering the retraction of $\mathcal{I}'$ onto A' with respect to a sector germ

[11] Which we shall assume from the following number onwards.

having for direction the vector chamber D associated with Φ^+ (I, 7.2.3), we see that every $x \in \mathcal{I}'$ is the transform of a point $y \in A'$ by an element $u \in U^+(K')$ (I, 7.4.25 (1)); but the density of K in K' immediately implies that there exist $u' \in U^+(K)$ and $v \in U_{y+D}(K')$ with $u = u'v$, from which $x = u' \cdot y \in j(\mathcal{I})$. Even in this case, there exist in general apartments of $\mathcal{I}'$ which are not images of apartments of $\mathcal{I}$.

4.3. The schemes $\mathcal{U}_a$.

From now on we assume that the extension $\tilde{K}$ of K is admissible (1.6.1).

4.3.1. — We keep the notation of 4.2. In particular, $\varphi = (\varphi_a)$ is the valuation of the root datum $(T, (U_a)_{a\in\Phi})$ of $G = G(K)$ associated with the coherent system of pin-

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pinnings (x_a) (4.2.4). Each of the groups $U_a = U_a(K)$ is filtered by φ_a: recall (I, 6.2.1) that we set, for $a \in \Phi$ and $k \in \mathbf{R}$,

$$U_{a,k} = \{u \in U_a(K) \mid \varphi_a(u) \geq k\}. \tag{1}$$

The $U_{a,k}$ are subgroups of U_a by the very definition of valuations (I, 6.2.1 (V1)). We will see that each $U_{a,k}$ for $k \in \Gamma'_a$ is the group of integer points of a smooth $\mathcal{O}$-group scheme with generic fiber U_a. These are the $\mathcal{O}$-schemes that we will use in n° 4.5 to construct "schematic root data" to which the results of § 3 will be applied.

4.3.2. — If $a \in \Phi$ is non-multipliable, these schemes are obvious: the group U_a is identified, via the pinning x_a, with the additive group of the field L_a attached to a (4.1.8) and for every $k \in \Gamma_a = \Gamma'_a = \omega(L_a^\times)$ (4.2.20 (1)), the subgroup $U_{a,k}$ is the image under x_a of the free $\mathcal{O}$-module of finite type $L_{a,k} = \{u \in L_a \mid \omega(u) \geq k\}$. Let $\mathfrak{L}_{a,k}$ denote the smooth $\mathcal{O}$-group scheme canonically associated with this module (1.4.1). By transport of structure via x_a, one obtains a smooth $\mathcal{O}$-group scheme $\mathfrak{U}_{a,k}$ with generic fiber U_a such that

$$\mathfrak{U}_{a,k}(\mathcal{O}) = U_{a,k}. \tag{1}$$

Now let $k \in \mathbf{R}$ be arbitrary; if the valuation ω is discrete, there exists a smallest $k' \in \Gamma_a$ such that $k' \geq k$ and we have $U_{a,k} = U_{a,k'}$. We then set

$$\mathfrak{U}_{a,k} = \mathfrak{U}_{a,k'} \quad (\omega \text{ discrete}, k' = \inf\{h \in \Gamma_a \mid h \geq k\}). \tag{2}$$

Relation (1) is obviously still exact.

On the other hand, if ω is dense and if $k \notin \Gamma_a$, the group $U_{a,k}$ is identified with a sub-$\mathcal{O}$-module of L_a which is not of finite type, and there is no $\mathcal{O}$-scheme having $U_{a,k}$ as its group of integer points and possessing the properties we would desire (for example, that the image of the group of integer points in the closed fiber is not reduced to $\{0\}$). We will therefore not define a scheme $\mathfrak{U}_{a,k}$ when ω is dense and $k \notin \Gamma_a$.

To treat the case of a multipliable root ray, we will need the following lemma:

Lemma 0.28 (4.3.3)**.** *Let $L_2 \subset L$ be two sub-extensions of $\tilde{K}$ such that $[L : L_2] = 2$.*

(i) The extension L of L_2 is admissible, hence separable, and one of the following two conditions is satisfied:

(nr) the residue extension $\bar{L}$ of $\bar{L}_2$ is quadratic and $\omega(L) = \omega(L_2)$;

(r) ω is discrete and the extension L of L_2 is ramified.

(ii) There exists $t \in L$ such that $L = L_2[t]$ and that the coefficients $\alpha, \beta \in L_2$ of the equation $t^2 - \alpha t + \beta = 0$ satisfied by t possess the following properties:

a) $\omega(\beta) = 0$ (case (nr)) or else ω is discrete and β is a uniformizer of L_2 (case (r));

b) either $\alpha = 0$, or $\omega(\beta) \leq \omega(\alpha) < \omega(2)$, or even $0 < \omega(\beta) \leq \omega(\alpha) = \omega(2)$.

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Let us assume henceforth these conditions are satisfied and let us define $\lambda \in \mathbf{L}$ as follows:

— if $\alpha = 0$ (which implies $2 \neq 0$), we set $\lambda = 1/2$;

— if $\alpha \neq 0$, we set $\lambda = t\alpha^{-1}$.

Then $\lambda \in \mathbf{L}^1_{\max}$, which is therefore not empty.

(iii) We have $\omega(\lambda) \leq 0$ and $\omega(\lambda) = 0$ if and only if either char $\bar{K} \neq 2$, or the extension $\mathbf{L}$ of $\mathbf{L}_2$ is étale.

(iv) Let σ be the unique non-trivial $\mathbf{L}_2$-automorphism of $\mathbf{L}$ and let $k \in \mathbf{R}$. We have

$$\inf\{\omega(u - u^\sigma) \mid \omega(u) \geq k\} \geq k - \omega(\lambda)$$

with equality if $k \in \omega(\mathbf{L}^\times)$, except in the case (r) when $k \in 2\mathbf{Z}\omega(t)$, in which case the lower bound is $k - \omega(\lambda) + \omega(t)$.

Assertion (i) follows from 1.6.3 and 1.6.1 c). Let us prove (ii). Let us take for t in the case (nr) an element of $\mathcal{O}_{\mathbf{L}}$ whose image in $\bar{\mathbf{L}}$ generates the extension $\bar{\mathbf{L}}$ of $\bar{\mathbf{L}}_2$ and in the case (r) a uniformizer of $\mathbf{L}$: it is immediate that a) is true and that $\omega(\alpha) \geq \omega(\beta) \geq 0$. If $\alpha \neq 0$ and if we have, either $\omega(\alpha) > \omega(2)$, or $\omega(\alpha) = \omega(2)$ and $\omega(\beta) = 0$, then $2 \neq 0$ and one can replace t by $t - \frac{\alpha}{2}$, which cancels the linear term of the equation for t and allows us to satisfy a) and b). Before continuing the proof of (ii), let us note two immediate consequences of a) and b). We have

$$\omega(x + ty) = \inf(\omega(x), \omega(ty)) \quad \text{for } x, y \in \mathbf{L}_2; \tag{1}$$

indeed, in the contrary case, we would have $\omega(x) = \omega(ty)$, which excludes the case (r), and the image of t in $\bar{\mathbf{L}}$ would coincide with that of $-xy^{-1}$, which is impossible in the case (nr). On the other hand, since $\omega(t) = \omega(\beta)/2$, b) implies immediately that

$$\text{if } \alpha \neq 0, \quad \omega(\alpha t^{-1}) < \omega(2). \tag{2}$$

Let us return to the proof of (ii). If $\alpha = 0$, we have $\mathbf{L}^1 = \frac{1}{2} + t\mathbf{L}_2$, and (1) implies that $\frac{1}{2} \in \mathbf{L}^1_{\max}$. Suppose that $\alpha \neq 0$, whence $t\alpha^{-1} \in \mathbf{L}^1$, and that $t\alpha^{-1} \notin \mathbf{L}^1_{\max}$. Since $\mathbf{L}^0 = (1 - 2t\alpha^{-1})\mathbf{L}_2$, this means that there exists $u \in \mathbf{L}_2$ such that

$$\omega(t\alpha^{-1}) < \omega(t\alpha^{-1} + u(1 - 2t\alpha^{-1})),$$

that is, in view of (1),

$$\omega(t\alpha^{-1}) < \inf(\omega(u), \omega(t\alpha^{-1}) + \omega(1 - 2u)). \tag{3}$$

This inequality implies that $\omega(1 - 2u) > 0$. We successively deduce that $\omega(u) = \omega\left(\frac{1}{2}\right)$ and then, taking (3) into account, that $\omega(t\alpha^{-1}) < \omega\left(\frac{1}{2}\right)$, in contradiction with (2), whence (ii).

Let us prove (iii). By virtue of (2), we have

$$\omega(\lambda) = -\omega(2 - \alpha t^{-1}). \tag{4}$$

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Consequently, $\omega(\lambda) \leqq 0$. When $\alpha = 0$, $\omega(\lambda)$ is zero if and only if char $\overline{K} \neq 2$. When $\alpha \neq 0$, whence char $\overline{K} = 2$ according to b), we have $\omega(\lambda) = 0$ if and only if $\omega(\alpha) = \omega(t)$, which is equivalent to L being unramified and $\overline{L}$ separable over $\overline{L}_2$, that is to say L is étale.

Let us finally prove (iv). Let $u = x + ty$ with $x, y \in L_2$; we have $u - u^\sigma = (2t - \alpha)y$. In view of (4), it follows that

$$\omega(u - u^\sigma) = \omega(y) + \omega(t) - \omega(\lambda) \geqq \omega(u) - \omega(\lambda)$$

with equality if $u = ty$. Assertion (iv) results immediately from this, taking (1) into account.

4.3.4. — Let us keep the hypotheses and notations of 4.3.3. We easily see that:

- in the case (nr), we have $\omega(L^0) = \omega(L_2) = \omega(L)$ (cf. 4.3.3 (1));
- in the case (r), ω is discrete, which allows us to normalize ω such that $\omega(L^\times) = \mathbf{Z}$, and it is then necessary to distinguish two sub-cases:
  - if $\alpha \neq 0$, we have $\omega(2t\alpha^{-1}) > 1$ (cf. 4.3.3 (2)), whence $\omega(L^0) = \omega(L_2) = 2\mathbf{Z} \cup \{\infty\}$, $\omega(t\alpha^{-1}) = 1 - \omega(\alpha) \in 1 + 2\mathbf{Z}$ and $\frac{1}{2}\omega(t\alpha^{-1}) + \omega(L^\times) = \frac{1}{2} + \mathbf{Z}$;
  - if $\alpha = 0$, then $\omega(L^{0\times}) = 1 + 2\mathbf{Z}$, $\omega(\frac{1}{2}) \in 2\mathbf{Z}$ and $\frac{1}{2}\omega(\frac{1}{2}) + \omega(L^\times) = \mathbf{Z}$.

Applying this to the case of a plural root radius a, we see, by using 4.2.21, that if the quadratic extension attached to a is unramified, $2\Gamma'_a$ is contained in Γ_{2a} and is a class modulo $2\Gamma_{2a}$, whereas in the ramified case, Γ_{2a} and $2\Gamma'_a$ are the two classes of the (discrete) group $\omega(L_a^\times)$ modulo $2\omega(L_a^\times)$, and are disjoint. We also see, by applying formulas (1) to (4) of 4.2.21, that the valuation φ is always special, that is to say that $0 \in \Gamma_a$ for every non-divisible a (I, 6.2.13).

4.3.5. — Let a be a plural root radius, $L \supset L_2$ the quadratic extension attached to a, σ the unique non-trivial L_2-automorphism of L and λ an element of $L^1_{\max}$; let $\gamma = -\frac{1}{2}\omega(\lambda)$. Let $k, \ell \in \mathbf{R}$ be such that

$$k \in \Gamma'_a = -\gamma + \omega(L^\times), \quad \ell \in \Gamma'_{2a} = \Gamma_{2a} = \omega(L^0 - \{0\}), \tag{1}$$

$$\ell \leq \inf \Gamma_{2a} \cap [2k, +\infty[. \tag{2}$$

Let us note that (2) is equivalent to $\ell \leq 2k$, except if ω is discrete and the extension $L \supset L_2$ is ramified. In this latter case, we easily see from 4.3.4 that if $c = \omega(t)$, whence $\omega(L^\times) = c\mathbf{Z}$, relation (2) is equivalent to

$$\ell \leq 2k + c \tag{2 bis}$$

(this remains true in the unramified case, for that matter!).

Let us denote by $\mathfrak{L}_{k+\gamma}$ (resp. $\mathfrak{L}^0_\ell$) the smooth $\mathcal{O}$-group scheme canonically associated with the $\mathcal{O}$-module (free of finite type in view of (1))

$$L_{k+\gamma} = \{u \in L \mid \omega(u) \geq k + \gamma\} \quad (\text{resp. } L^0_\ell = \{v \in L^0 \mid \omega(v) \geq \ell\}).$$

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In view of (2) and 4.3.3 (iv), the bilinear map $(u, u') \mapsto \lambda^\sigma u u'^\sigma - \lambda u^\sigma u'$ from $L \times L$ to L^0 maps $\mathfrak{l}_{k+\gamma} \times \mathfrak{l}_{k+\gamma}$ into $\mathfrak{l}^0_\ell$ and therefore defines a morphism of schemes from $\mathfrak{L}_{k+\gamma} \times \mathfrak{L}_{k+\gamma}$ to $\mathfrak{L}^0_\ell$. It follows that the product law of 4.1.15 (2) equips the scheme $\mathfrak{L}_{k+\gamma} \times \mathfrak{L}^0_\ell$ with a structure of *group scheme* that we denote by $\mathfrak{H}^\lambda_{(k,\ell)}$. By transport of structure via the isomorphism j_λ^{-1} (defined in 4.1.15 (1)), one obtains a smooth $\mathcal{O}$-group scheme $\mathfrak{H}(L, L_2)_{(k,\ell)}$ with generic fiber $H(L, L_2)$ and which does *not* depend on the choice of λ: indeed, if λ' is another element of $L^1_{\max}$, the isomorphism $j_{\lambda'} \circ j_\lambda^{-1}$ is given by

$$(u, v) \mapsto (u, v + (\lambda - \lambda')u^\sigma u)$$

and extends to an isomorphism of group schemes from $\mathfrak{H}^\lambda_{(k,\ell)}$ to $\mathfrak{H}^{\lambda'}_{(k,\ell)}$.

The pinning $x_a : H(L, L_2) \to U_a$ then provides, by transport of structure, a smooth $\mathcal{O}$-group scheme $\mathfrak{U}_{a,(k,\ell)}$ with generic fiber U_a. When L is unramified (resp. ramified), we further set $\mathfrak{U}_{a,k} = \mathfrak{U}_{a,(k,2k)}$ (resp. $\mathfrak{U}_{a,k} = \mathfrak{U}_{a,(k,2k+c)}$).

Let us show that

$$\begin{aligned} \mathfrak{U}_{a,(k,\ell)}(\mathcal{O}) &= \{x_a(u, v) \mid (u, v) \in H(L, L_2), \omega(u) \geqq k + \gamma, \omega(v - \lambda u^\sigma u) \geqq \ell\} \\ &= U_{a,k} \cdot U_{2a,\ell}. \end{aligned} \tag{3}$$

Indeed, let $(u, v) \in H(L, L_2)$ and $x = x_a(u, v)$. We have $x = x'x''$ with

$$\begin{aligned} x' &= x_a(u, \lambda u^\sigma u) = x_a(j_\lambda^{-1}(u, 0)) \in U_a, \\ x'' &= x_a(0, v - \lambda u^\sigma u) = x_a(j_\lambda^{-1}(0, v - \lambda u^\sigma u)) \in U_{2a}. \end{aligned}$$

The equality of the first two sides of (3) follows immediately from these formulas and the very definition of the scheme $\mathfrak{U}_{a,(k,\ell)}$. On the other hand, we have

$$\begin{aligned} \phi_a(x') &= \frac{1}{2}\omega(\lambda u^\sigma u) = \omega(u) - \gamma, \\ \phi_{2a}(x'') &= \omega(v - \lambda u^\sigma u), \end{aligned}$$

which shows that the second side of (3) is contained in the third. But Lemma 4.2.20 implies that

$$\sup \phi_a(xU_{2a}) = \sup \frac{1}{2}\omega(\lambda u^\sigma u + L^0) = \omega(u) - \gamma.$$

Consequently, if $x \in U_{a,k} \cdot U_{2a,\ell}$, we have $\omega(u) \geqq k + \gamma$, hence

$$x' \in U_{a,k} \quad \text{and} \quad x'' \in U_{a,k} \cdot U_{2a,\ell} \cap U_{2a} = U_{2a,\ell}$$

(taking into account (2) and I, 6.2.1 (V4)), hence $\omega(v - \lambda u^\sigma u) \geqq \ell$. Finally, we see that x belongs to the second side of (3), which completes the proof.

4.3.6. — Let us make explicit the group law of the scheme $\mathfrak{U}_{a,(k,\ell)}$, following the notations of 4.3.3 with $L = L_a$ and $L_2 = L_{2a}$. We set $c = \omega(t)$ and choose $u \in L$ of valuation $k + \gamma$.

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1st case: $\alpha = 0$. We have $\lambda = \frac{1}{2}$, $2\gamma = \omega(2)$ and $L^0 = tL_2$. Let $d = 2k + 2\gamma - \ell + c$. We have $d \geq \omega(2)$ and $d \in \omega(L_2^\times)$. We can therefore choose $\delta \in \mathcal{O}_L$, of valuation d. A simple calculation then shows that the map

$$(x, y, z) \mapsto x_a\left((x + ty)u, \left(\frac{1}{2}(x^2 + \beta y^2) + \delta^{-1}tz\right)uu^\sigma\right) \tag{1}$$

is an isomorphism of schemes from $\mathcal{O}_L^3$ to $\mathfrak{U}_{a,(k,\ell)}$ and that the group law of $\mathfrak{U}_{a,(k,\ell)}$ transported by this isomorphism to $\mathcal{O}_L^3$ is given by

$$(x, y, z)(x', y', z') = (x + x', y + y', z + z' - \delta(yx' - xy')). \tag{2}$$

Note that $d = \omega(\delta) > 0$ except if char $\bar{K} \neq 2$ and $\ell = 2k + c$.

2nd case: $\alpha \neq 0$. We have $\lambda = t\alpha^{-1}$, $2\gamma = \omega(\alpha) - c$ and $L^0 = (1 - 2\alpha^{-1}t)L_2$. Let $d = 2k + 2\gamma - \ell$. We have $d \in \omega(L_2^\times)$ and $d \geq \omega(\alpha) - 2c \geq 0$. We can therefore choose $\delta \in \mathcal{O}_L$ of valuation d and everything proceeds as in the first case, provided that we replace (1) and (2) by

$$(x, y, z) \mapsto x_a\left((x + ty)u, \left(t\alpha^{-1}(x^2 + \alpha xy + \beta y^2) + \delta^{-1}(1 - 2t\alpha^{-1})z\right)uu^\sigma\right) \tag{1 bis}$$

$$(x, y, z)(x', y', z') = (x + x', y + y', z + z' + \delta(xx' + \alpha yx' + \beta yy')) \tag{2 bis}$$

Note that here we have $\omega(\delta) > 0$ except if $\omega(\alpha) = 2c$ and $\ell = 2k + c$.

We observe that the group law on $(\bar{L}_2)^3$ deduced from (2) or (2 bis) by reduction modulo the maximal ideal of $\mathcal{O}$ (in other words, the group law of the closed fiber of the group scheme $\mathfrak{U}_{a,(k,\ell)}$)

- is non-commutative if $\ell = 2k + c$ and if either char $\bar{K} \neq 2$, or char $\bar{K} = 2$ and the extension $L \supset L_2$ is étale;
- is commutative of exponent 4 if ω is discrete, L is ramified, α is a uniformizer of L_2 (whence char $\bar{K} = 2$) and $\ell = 2k + c$;
- is the addition of $(\bar{L}_2)^3$ in all other cases.

4.3.7. — By transport of structure by the isomorphism $v \mapsto x_a(0, v)$ from L^0 onto U_{2a}, we deduce from the additive group scheme $\mathfrak{L}_\ell^0$ (for $\ell \in \omega(L^0) - \{0\}$) an $\mathcal{O}$-group scheme, smooth with generic fiber U_{2a}, denoted $\mathfrak{U}_{2a,\ell}$. It is clear that if k, ℓ satisfy 4.3.5 (1) and (2), the injection of U_{2a} into U_a extends to an isomorphism of $\mathfrak{U}_{2a,\ell}$ onto a closed normal subgroup scheme of $\mathfrak{U}_{a,(k,\ell)}$. The quotient group scheme $\mathfrak{U}_{a,(k,\ell)}/\mathfrak{U}_{2a,\ell}$ exists and is isomorphic to $\mathfrak{L}_{k+\gamma}$. The canonical map from $\mathfrak{U}_{a,(k,\ell)}(\mathcal{O})$ to $(\mathfrak{U}_{a,(k,\ell)}/\mathfrak{U}_{2a,\ell})(\mathcal{O})$ is surjective.

4.3.8. — Let $k, \ell \in \mathbf{R}$ now be arbitrary. If ω is dense, we see as in 4.3.2 that it is not reasonable to attempt to define a scheme $\mathfrak{U}_{a,(k,\ell)}$ when the conditions (1) and (2) of 4.3.5 are not satisfied. On the other hand, if ω is discrete, let

$$k' = \inf\{h \in \Gamma'_a \mid h \geq k\}, \tag{1}$$

$$\ell' = \inf\{h \in \Gamma'_{2a} \mid h \geq \inf(2k, \ell)\}. \tag{2}$$

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Then, $k' \in \Gamma'_a$, $l' \in \Gamma'_{2a}$ and 4.3.5 (2) is satisfied by k' and l': if L_a is unramified, we have $2\Gamma'_a \subset \Gamma'_{2a}$ (4.3.4), whence $l' \leq 2k'$; if L_a is ramified, we have $\Gamma'_{2a} = c + 2\Gamma'_a$ (4.3.4), and one sees immediately that $l' \leq 2k' + c$. One then sets

$$\mathfrak{U}_{a,(k,l)} = \mathfrak{U}_{a,(k',l')} \tag{3}$$

and one has

$$\mathfrak{U}_{a,(k,l)}(\mathcal{O}) = U_{a,k'} \cdot U_{2a,l'} = U_{a,k} \cdot U_{2a,l}. \tag{4}$$

The first equality results from the definition, and one obviously has

$$U_{a,k'} \subset U_{a,k} \quad \text{and} \quad U_{2a,l'} \subset U_{2a,\inf(2k,l)} \subset U_{a,k} \cdot U_{2a,l}.$$

Conversely, if $x \in U_{a,k}$, there exists by the very definition of Γ'_a and k' an $x' \in U_{a,k'}$ and an $x'' \in U_{2a}$ with $x = x'x''$. One then has $x'' \in U_{2a,2k}$ and it remains only to see that $U_{2a,2k} \cdot U_{2a,l} \subset U_{2a,l'}$, which is obvious.

4.3.9. — The underlying schemes of the $\mathfrak{U}_{a,k}$, the $\mathfrak{U}_{a,(k,l)}$ and the $\mathfrak{U}_{a,(k,l)}/\mathfrak{U}_{2a,l}$ are schemes associated to free $\mathcal{O}$-modules of finite type. In particular, the canonical maps $\mathfrak{U}_{a,k}(\mathcal{O}) \to \mathfrak{U}_{a,k}(\overline{K})$ and $\mathfrak{U}_{a,(k,l)}(\mathcal{O}) \to \mathfrak{U}_{a,(k,l)}(\overline{K})$ are surjective. If $\overline{K}$ is infinite, these are fattened schemes (1.7), characterized among the fattened schemes with generic fiber U_a by the relations 4.3.2 (1) or 4.3.5 (3).

4.3.10. — It is immediate that if $k' \leq k$ and $l' \leq l$, the identity of U_a extends to a morphism of group schemes from $\mathfrak{U}_{a,(k,l)}$ into $\mathfrak{U}_{a,(k',l')}$ (or from $\mathfrak{U}_{a,k}$ into $\mathfrak{U}_{a,k'}$ when a is non-multipliable).

4.3.11. — A conservative base change $K \to K'$ (4.2.25) commutes with all of the preceding considerations: from the coherent system of pinnings (x'_a) of $G_{K'}$ deduced from the system (x_a) as in 4.2.25, one constructs schemes $\mathfrak{U}'_{a,k}$ or $\mathfrak{U}'_{a,(k,l)}$ which are none other than the schemes deduced from the $\mathfrak{U}_{a,k}$ or the $\mathfrak{U}_{a,(k,l)}$ by the base change $\mathcal{O} \to \mathcal{O}' = \mathcal{O}_{K'}$. This follows immediately from 4.2.25 and from the relation $\mathcal{O}_{K'} \otimes_{\mathcal{O}} \mathcal{O}_L = \mathcal{O}' \otimes_{\mathcal{O}} \mathcal{O}_L$ (1.6.8), valid for any field L with $K \subset L \subset \tilde{K}$, which, joined to the fact that K' is unramified over K, implies that, for all $k \in \mathbb{R}$, one has

$$\{u \in K' \otimes_K L \mid \omega(u) \geq k\} = \{\sum \xi_i u_i \mid \xi_i \in \mathcal{O}_{K'}, u_i \in L, \omega(u_i) \geq k\}.$$

Finally if $K \subset L_2 \subset L \subset \tilde{K}$ with $[L : L_2] = 2$ and $L = L_2[t]$, then $K' \otimes L$ is the admissible quadratic extension of $K' \otimes L_2$ generated by $1 \otimes t$ and $(K' \otimes L)_{\max} \supset L_{\max}$ according to lemma 4.3.3: one can therefore, in the preceding constructions, keep the same element λ after base change.

4.3.12. — The family of schemes $\mathfrak{U}_{a,k}$ or $\mathfrak{U}_{a,(k,l)}$ does not depend (up to indexing) on the choice of the maximal split torus S normalizing U_a nor on the choice of the coherent system of pinnings: this follows easily from the explicit formulas for change of pinning of U_a seen in 4.1.7 and 4.1.13.

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4.4. The scheme $\mathfrak{T}$.

In this section, up to and including 4.4.14, the group G plays no role and the letter T denotes any torus defined over K. We denote by $\tilde{K}$ an admissible Galois extension splitting T (not necessarily minimal) and we set $\tilde{\mathcal{O}} = \mathcal{O}_{\tilde{K}}$. From 4.4.15 onwards, we return to the previous conventions.

4.4.1. — Let L be an extension of K, equipped with a valuation ω_L extending ω. We set

$$T_b(L) = \{t \in T(L) \mid \omega_L(\chi(t)) = 0 \quad \text{for all } \chi \in X^*_L(T)\},$$

where $X^*_L(T) \subset L[T]$ is the group of rational characters of T over L.

4.4.2. Proposition. — *(i) $T_b(K)$ is the set of $t \in T(K)$ such that $\omega(\chi(t)) = 0$ for all $\chi \in X^*_{\tilde{K}}(T) = X^*(T)$; in other words*

$$T_b(K) = T_b(\tilde{K}) \cap T(K).$$

(ii) $T_b(K)$ is the largest bounded subgroup of $T(K)$ (for the natural bornology $\mathcal{B}(\omega)$ defined by ω on $T(K)$ (cf. 4.2.19)). (iii) Let u be a K-morphism of groups from T into a torus T' defined over K and split over $\tilde{K}$. Then

$$u(T_b(K)) \subset T'_b(K).$$

(iv) Let $\mathfrak{T}$ be a flat $\mathcal{O}$-group scheme with generic fiber T. Then

$$\mathfrak{T}(\mathcal{O}) \subset T_b(K).$$

Proof. Let us prove (ii). According to Lemma 4.2.19, a subgroup X of $T(K)$ is bounded if and only if the function $t \mapsto \omega(\chi(t))$ remains bounded from below on X for all $\chi \in X^*_{\tilde{K}}(T)$. But, since $\omega(\chi(t^n)) = n\omega(\chi(t))$ for all $n \in \mathbf{Z}$, this is equivalent to $X \subset T_b(K)$, whence (ii). We deduce $T_b(K) = T_b(\tilde{K}) \cap T(K)$, that is (i). Assertion (iii) is immediate, since $\chi \circ u \in X^*_{\tilde{K}}(T)$ for all $\chi \in X^*_{\tilde{K}}(T')$, and (iv) also, because $\mathfrak{T}(\mathcal{O})$ is a bounded subgroup of $T(K)$: for any $f \in K[T]$, there exists $c \in K^\times$ such that $cf \in \mathcal{O}[\mathfrak{T}]$, whence $f(t) \in c^{-1}\mathcal{O}$ for all $t \in \mathfrak{T}(\mathcal{O})$. □

4.4.3. — Suppose that T is an induced torus (1.5.17), that is, a direct product of tori of the form $\Pi_{L_i/K}\mathrm{Mult}_{L_i}$ $(1 \leq i \leq r)$, where the L_i are sub-extensions of $\tilde{K}$, and thus admissible. The valuation rings $\mathcal{O}_{L_i}$ are free $\mathcal{O}$-modules of finite type and one can consider the $\mathcal{O}$-group scheme

$$\mathfrak{T} = \prod_{1 \leq i \leq r} (\Pi_{\mathcal{O}_{L_i}/\mathcal{O}}\mathrm{Mult}_{\mathcal{O}_{L_i}}).$$

Since $K \otimes_{\mathcal{O}} \mathcal{O}_{L_i} = L_i$, the generic fiber of $\mathfrak{T}$ is identified with T (1.5.3). It is known that $\mathfrak{T}$ is smooth (1.5.8). On the other hand, we have seen (1.5.13) (1) that the underlying scheme

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to $\mathfrak{T}$ is a special open subset of a free $\mathcal{O}$-module of finite type; it is therefore a full scheme as soon as the residue field $\overline{K}$ of K is infinite (1.7.3 e)). Furthermore:

Proposition. — $\mathfrak{T}(\mathcal{O}) = T_b(K) = \{t \in T(K) \mid \omega(\chi(t)) = 0 \text{ for all } \chi \in X_K^*(T)\}$.

The second equality has been seen (4.4.2 (i)). One can obviously assume $r = 1$ and set $L_1 = L$. It is known that $T(K)$ then identifies with $L^\times$ and that $X_K^*(T)$ is generated by the norm $\mathrm{Norm}_{L/K}$ (1.5.17). Consequently, $T_b(K)$ is the set of $t \in L^\times$ such that $\mathrm{Norm}_{L/K}t \in \mathcal{O}_K^\times$, a relation which implies $t \in \mathcal{O}_L^\times$. The proposition follows, since $\mathfrak{T}$ is the special open subset of the $\mathcal{O}$-module $\mathcal{O}_L$ defined by the norm (1.5.13 (1)).

4.4.4. — Let T and T' be two induced tori, $\mathfrak{T}$ and $\mathfrak{T}'$ the $\mathcal{O}$-schemes associated as in 4.4.3.

Proposition. — Any K-morphism f of algebraic groups from T to T' extends (in one and only one way) to an $\mathcal{O}$-morphism of group schemes from $\mathfrak{T}$ to $\mathfrak{T}'$.

We have $f(\mathfrak{T}(\mathcal{O})) \subset f(T_b(K)) \subset T'_b(K) = \mathfrak{T}'(\mathcal{O})$ by 4.4.2 and 4.4.3. The proposition follows if the residue field $\overline{K}$ is infinite, since $\mathfrak{T}$ is then full (1.7.1). If $\overline{K}$ is finite, one can find an étale extension K' of K such that the residue fields $\overline{K}$ and $\overline{K}'$ are linearly disjoint over $\overline{K}$ and $\overline{K}'$ is infinite (1.6.9 a)). The base change $K \to K'$ then entirely respects the situation: let us take up the notations of 4.4.3 and set $L'_i = K' \otimes_K L_i$; then L'_i is an admissible extension field of K', $\mathcal{O}_{L'_i} = \mathcal{O}_{K'} \otimes_{\mathcal{O}} \mathcal{O}_{L_i}$ (1.6.8 (ii)) and one has (1.5.3)

$$T_{K'} = \prod_{1 \leq i \leq r} \Pi_{L'_i/K'}\mathfrak{Mult}_{L'_i},$$

$$\mathfrak{T}_{\mathcal{O}_{K'}} = \prod_{1 \leq i \leq r} \Pi_{\mathcal{O}_{L'_i}/\mathcal{O}_{K'}}\mathfrak{Mult}_{\mathcal{O}_{L'_i}}.$$

Consequently, $f_{K'}$ extends to an $\mathcal{O}_{K'}$-morphism of $\mathfrak{T}_{\mathcal{O}_{K'}}$ into $\mathfrak{T}'_{\mathcal{O}_{K'}}$, whence the conclusion by descent (1.2.4).

4.4.5. — It follows in particular from this proposition that the scheme $\mathfrak{T}$ depends only on T and not on its notation as a product of tori of the form $\Pi_{L/K}\mathfrak{Mult}_L$ (which was not evident a priori). We shall call it the canonical scheme of generic fiber T. It is clear that if T is split over K, one recovers the canonical scheme introduced in 1.2.11. More generally:

Proposition. — Let S be the maximal K-split sub-torus of T. The schematic closure of S in $\mathfrak{T}$ is the canonical scheme of generic fiber S.

This follows from 1.5.13 (1) and 1.5.17.

Note further that $\mathfrak{T}$ does not depend on the choice of the admissible Galois extension splitting T (as soon as one exists!): one can replace $\widetilde{K}$ by the smallest extension $\widetilde{K}_0$

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splitting T, which is Galois admissible and automatically contains the fields L_i of 4.4.2, and this does not change the definition of $\mathfrak{T}$ at all.

4.4.6. — Now let T be any torus, not necessarily induced (but always defined over K and split over $\tilde{K}$). Consider the canonical closed immersion (1.5.11)

$$T \to T^{\#} = \Pi_{\tilde{K}/K}T_{\tilde{K}}. \tag{1}$$

Since $T_{\tilde{K}}$ is split, $T^{\#}$ is an induced torus and one can consider the canonical scheme $\mathfrak{T}^{\#}$ with generic fiber $T^{\#}$. Let $\mathfrak{T}^a$ be the schematic closure of T in $\mathfrak{T}^{\#}$: it is known that it is an $\mathcal{O}$-scheme with generic fiber T that is

torsion-free (1.2.6), hence flat since $\mathcal{O}$ is Prüferian, thus an $\mathcal{O}$-group scheme (1.2.7), of finite type since $\mathfrak{T}^{\#}$ is of finite type, and we have, by 1.2.5,

$$\mathfrak{T}^a(\mathcal{O}) = \mathfrak{T}^{\#}(\mathcal{O}) \cap T(K) = T_b(\tilde{K}) \cap T(K) = T_b(K), \tag{2}$$

the last equality resulting from the surjectivity of the canonical map $X^*(T^{\#}) \to X^*(T)$.

4.4.7. — If T' is another torus and $\mathfrak{T}'^a$ the scheme obtained from T' as in 4.4.6, and if $f : T \to T'$ is a K-morphism of algebraic groups from T to T', then f extends (in one and only one way) to an $\mathcal{O}$-morphism of group schemes f^a from $\mathfrak{T}^a$ to $\mathfrak{T}'^a$: indeed, $\Pi_{\tilde{K}/K} f_{\tilde{K}}$ extends, by 4.4.4, to a morphism from $\mathfrak{T}^{\#}$ to $\mathfrak{T}'^{\#}$ inducing f on T.

If f is a closed immersion, then f^a is a closed immersion. Indeed $T'_{\tilde{K}}$ decomposes into a direct product of the torus $T_{\tilde{K}} = f_{\tilde{K}}(T_{\tilde{K}})$ and a torus T'' defined and split over $\tilde{K}$ ([2], p. 206). Consequently $T'^{\#} = T^{\#} \times (\Pi_{\tilde{K}/K} T'')$, hence $\mathfrak{T}'^{\#} = \mathfrak{T}^{\#} \times \mathfrak{T}''^{\#}$, where $\mathfrak{T}''^{\#}$ is the canonical scheme with generic fiber $T''^{\#}$, which proves our assertion.

4.4.8. — If T is an induced torus, $\mathfrak{T}^a$ is nothing other than the canonical scheme with generic fiber T. It is obviously sufficient to show this when $T = \Pi_{L/K}\mathrm{Mult}_L$. Since $\tilde{K} \otimes_K L = \tilde{K}^n$ (with $n = [L : K]$), we have

$$\begin{aligned} T^{\#} &= \Pi_{\tilde{K}/K}(\Pi_{L/K}\mathrm{Mult}_L)_{\tilde{K}} = \Pi_{\tilde{K}/K}(\Pi_{\tilde{K}^n/\tilde{K}}\mathrm{Mult}_{\tilde{K}^n}) \\ &= \Pi_{\tilde{K}^n/K}\mathrm{Mult}_{\tilde{K}^n} = (\Pi_{\tilde{K}/K}\mathrm{Mult}_{\tilde{K}})^n, \\ \text{and} \quad \mathfrak{T}^{\#} &= (\Pi_{\tilde{\mathcal{O}}/\mathcal{O}}\mathrm{Mult}_{\tilde{\mathcal{O}}})^n = \Pi_{\tilde{\mathcal{O}}^n/\mathcal{O}}\mathrm{Mult}_{\tilde{\mathcal{O}}^n}. \end{aligned}$$

Then consider the canonical closed immersions

$$\begin{aligned} \mathrm{Mult}_{\mathcal{O}_L} &\to \Pi_{\tilde{\mathcal{O}}^n/\mathcal{O}_L}\mathrm{Mult}_{\tilde{\mathcal{O}}^n}, \\ \mathrm{Mult}_L &\to \Pi_{\tilde{K}^n/L}\mathrm{Mult}_{\tilde{K}^n}. \end{aligned}$$

By 1.5.9, we deduce the closed immersions

$$\begin{aligned} \mathfrak{T} = \Pi_{\mathcal{O}_L/\mathcal{O}}\mathrm{Mult}_{\mathcal{O}_L} &\to \Pi_{\mathcal{O}_L/\mathcal{O}}(\Pi_{\tilde{\mathcal{O}}^n/\mathcal{O}_L}\mathrm{Mult}_{\tilde{\mathcal{O}}^n}) = \Pi_{\tilde{\mathcal{O}}^n/\mathcal{O}}\mathrm{Mult}_{\tilde{\mathcal{O}}^n} = \mathfrak{T}^{\#}, \\ T = \Pi_{L/K}\mathrm{Mult}_L &\to \Pi_{L/K}(\Pi_{\tilde{K}^n/L}\mathrm{Mult}_{\tilde{K}^n}) = \Pi_{\tilde{K}^n/K}\mathrm{Mult}_{\tilde{K}^n} = T^{\#}, \end{aligned}$$

and the second line is deduced from the first by the base change $\mathcal{O} \to K$, i.e., by passing to the generic fibers. Since $\mathfrak{T}$ is flat, it indeed follows that $\mathfrak{T} = \mathfrak{T}^a$.

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We can therefore extend the terminology and say that the $\mathcal{O}$-scheme $\mathfrak{T}^a$ which we shall henceforth denote by $\mathfrak{T}$ is the *canonical scheme* of generic fiber T.

Here again, it does not depend on the choice of $\tilde{K}$: if $\tilde{K}_1 \subset \tilde{K}_2$, the identity map of $T_{\tilde{K}_1} = (T_{\tilde{K}_1})_{\tilde{K}_1}$ provides a closed immersion of $\Pi_{\tilde{K}_1/K} T_{\tilde{K}_1}$ into $\Pi_{\tilde{K}_2/K} T_{\tilde{K}_2}$, whence, etc.

4.4.9. — Note that the group scheme $\mathfrak{T}$ always admits a faithful linear representation. This is evident if T is an induced torus, since $\mathrm{Mult}_{\mathcal{O}_L}$ possesses a faithful linear representation ρ in $\mathcal{O}_L$ and $\Pi_{\mathcal{O}_L/\mathcal{O}}\rho$ is then a faithful linear representation of $\Pi_{\mathcal{O}_L/\mathcal{O}}\mathrm{Mult}_{\mathcal{O}_L}$ in the $\mathcal{O}$-module underlying $\mathcal{O}_L$ (1.5.13 (2)). The general case follows from this, since $\mathfrak{T}$ is by definition a closed sub-scheme of the canonical scheme of an induced torus.

4.4.10. — Let $K \to K'$ be a base change conservative with respect to $\tilde{K}$ (4.2.25). Let $\mathcal{O}' = \mathcal{O}_{K'}$. The $\mathcal{O}'$-canonical scheme of generic fiber $T_{K'}$ (which is a torus defined over K' and split over the admissible Galois extension $\tilde{K}' = K' \otimes_K \tilde{K}$ of K') is none other than $\mathfrak{T}_{\mathcal{O}'}$: this follows immediately from 1.5.3 and the commutation of the operation of schematic closure with flat base change $\mathcal{O} \to \mathcal{O}'$ (1.2.6).

4.4.11. — Similarly, let ${}^hK$ be a henselization of K and let $K' = {}^{hs}K$ be a strict henselization of K containing the maximal étale sub-extension of ${}^hK \otimes_K \tilde{K}$ (considered as an extension of ${}^hK$) (1.6.7 *b*)). Let $\tilde{K}'$ be a composite extension of ${}^{hs}K$ and $\tilde{K}$: it is unique up to isomorphism and is an admissible extension of K' (1.6.8 and 9 *c*)). Furthermore, T is split over $\tilde{K}'$ and, using 1.6.8 and 1.5.3, one easily shows that, if T is an induced torus, then $T_{K'}$ is also an induced torus and the $\mathcal{O}'$-canonical scheme of generic fiber $T_{K'}$ is $\mathfrak{T}_{\mathcal{O}'}$ (where $\mathcal{O}' = {}^{hs}\mathcal{O}$ is the valuation ring of $K' = {}^{hs}K$, that is, a strict henselization of $\mathcal{O}$). This last assertion remains true for any torus T thanks to 1.2.6, and it follows that

$$\mathfrak{T}({}^{hs}\mathcal{O}) = T_b({}^{hs}K). \tag{1}$$

Let us set

$$\mathcal{H}_K(T) = \{f \in K[T] \mid f(T_b({}^{hs}K)) \subset {}^{hs}\mathcal{O}\}. \tag{2}$$

One obviously has

$$\mathcal{O}[\mathfrak{T}] \subset \mathcal{H}_K(T) \tag{3}$$

(4.4.2 (iv)). If $\mathfrak{T}$ satisfies condition (ET 1) of 1.7.2 (for example if ω is discrete or if T is an induced torus) and is smooth, then (1) implies $\mathcal{O}[\mathfrak{T}] = \mathcal{H}_K(T)$ (1.7.6) and characterizes $\mathfrak{T}$ among the group schemes with generic fiber T that are smooth and satisfy (ET 1).

4.4.12. — Definition. — *We call a smoothening of $\mathfrak{T}$ an $\mathcal{O}$-group scheme $\mathfrak{T}^R$ of generic fiber T, smooth, satisfying (ET 1), admitting a faithful linear representation, such that the identity on T extends to a morphism $i : \mathfrak{T}^R \to \mathfrak{T}$ and such that, for every smooth $\mathcal{O}$-scheme $\mathfrak{X}$ satisfying (ET 1) and every morphism $v : \mathfrak{X} \to \mathfrak{T}$, there exists a unique morphism $v^R : \mathfrak{X} \to \mathfrak{T}^R$ such that $v = i \circ v^R$.*

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Such a smoothening, if it exists, is obviously unique up to unique isomorphism. If $K \to K'$ is a conservative and étale base change and if $\mathfrak{T}^R$ is a smoothening of $\mathfrak{T}$, then $(\mathfrak{T}^R)_{o'}$ is a smoothening of $\mathfrak{T}_{o'}$ (it satisfies (ET 1) according to 1.7.5).

Proposition 0.29. *(i) If $\mathfrak{T}^R$ is a smoothening of $\mathfrak{T}$, one has*

$$\mathfrak{T}^R(\mathcal{O}) = \mathfrak{T}(\mathcal{O}) = T_b(K),$$
$$\mathfrak{T}^R(h^s\mathcal{O}) = \mathfrak{T}(h^s\mathcal{O}) = T_b(h^sK),$$
$$\mathcal{O}[\mathfrak{T}^R] = \mathscr{H}_K(T).$$

(ii) Conversely, if the affine algebra $\mathcal{O}$-scheme $\mathscr{H}_K(T)$ is a smooth group $\mathcal{O}$-scheme, satisfying (ET 1) and possessing a faithful linear representation, then it is a smoothening of $\mathfrak{T}$.

Assertion (i) is obvious: the very definition of the smoothening implies that $\mathfrak{T}^R(\mathcal{O}) = \mathfrak{T}(\mathcal{O})$ and $\mathfrak{T}^R(h^s\mathcal{O}) = \mathfrak{T}(h^s\mathcal{O})$, and the last assertion of (i) results from 1.7.6.

Assertion (ii) also results from 1.7.6: if $v : \mathfrak{X} \to \mathfrak{T}$ is as in the preceding definition, we have $v(\mathfrak{X}(h^s\mathcal{O})) \subset \mathfrak{T}(h^s\mathcal{O})$, hence $v^*(\mathscr{H}_K(T)) \subset \mathcal{O}[\mathfrak{X}]$ (1.7.6), whence the existence and uniqueness of v^R (1.2.4).

4.4.13. — If ω is discrete (which implies that every flat $\mathcal{O}$-scheme satisfies (ET 1) and that every smooth group $\mathcal{O}$-scheme possesses a faithful representation), a theorem of M. Raynaud ([24], p. 345) ensures that the smoothening $\mathfrak{T}^R$ of $\mathfrak{T}$ always exists (let us note that Raynaud's smoothening theorem is valid for any flat group $\mathcal{O}$-scheme, of finite type, with smooth generic fiber, whether the latter is a torus or not).

When ω is dense, we do not know if a smoothening of $\mathfrak{T}$ always exists. Let us show however this existence when $[\tilde{K} : K] = 2$ (or, which amounts practically to the same, if T splits over a quadratic extension of K). We will obtain by the same occasion an explicit construction of Raynaud's smoothening when, in addition, ω is discrete.

It is easily shown that it suffices to study the case of the "torus of elements of norm 1" of $\tilde{K}$, as a torus defined over K and split over a quadratic extension $\tilde{K}$ is a direct product of a split torus, of induced tori $\Pi_{\tilde{K}/K}\mathbf{Mult}$ and of tori of this form. Using the notation of 4.3.3, with $L_2 = K$, we have

$$K[T] = K[X, Y]/(X^2 + \alpha XY + \beta Y^2 - 1)$$

and one easily verifies that the canonical scheme $\mathfrak{T}$ is given by

$$\mathcal{O}[\mathfrak{T}] = \mathcal{O}[X, Y]/(X^2 + \alpha XY + \beta Y^2 - 1).$$

Note that T is anisotropic over K and that

$$\mathfrak{T}(\mathcal{O}) = T(K) = T_b(K).$$

If $\operatorname{char} \bar{K} \neq 2$ or if $\omega(\alpha) = 0$, the scheme $\mathfrak{T}$ is smooth and there is nothing to prove. Otherwise, $\mathfrak{T}$ is not smooth and we shall distinguish two cases:

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a) $\operatorname{char}\bar{K} = 2$ and $\alpha = 0$ (hence $\operatorname{char} K \neq 2$). Let $X = 1 - 2U$ and $Y = 2V$, from which $K[T] = K[U,V]/(U - U^2 - \beta V^2)$, and let us consider the scheme $\mathfrak{T}^R$ with generic fiber T defined by

$$\mathcal{O}[\mathfrak{T}^R] = \mathcal{O}[U,V]/(U - U^2 - \beta V^2).$$

One easily verifies that this is a smooth group scheme, whose closed fiber is isomorphic to $\mathbf{Z}/2\mathbf{Z} \times \mathfrak{Add}$ (hence not connected) when $\tilde{K}$ is ramified, and to a non-split connected unipotent group of dimension 1 when $\tilde{K}$ is unramified. Let ${}^{hs}K$ be a strict henselization of K. If $u, v \in {}^{hs}K$, we have

$$\omega(u^2 + \beta v^2) = \inf(\omega(u^2), \omega(\beta v^2)),$$

because either $\tilde{K}$ is ramified (which implies that ω is discrete and that β is a uniformizer of K) and $\omega(u^2)$ and $\omega(\beta v^2)$ have different "parity", or $\tilde{K}$ is unramified, which implies that the residual extension $\bar{\tilde{K}}/\bar{K}$ is purely inseparable quadratic and that the polynomial $U^2 + \bar{\beta}V^2$ remains irreducible over the separable closure of $\bar{K}$. It follows that the equation $u = u^2 + \beta v^2$ implies $u, v \in {}^{hs}\mathcal{O}$. We easily deduce that

$$\mathfrak{T}^R({}^{hs}\mathcal{O}) = T({}^{hs}K) = \mathfrak{T}({}^{hs}\mathcal{O}). \tag{1}$$

b) $\operatorname{char}\bar{K} = 2, 0 < \omega(\alpha) \leq \omega(2)$ (cf. 4.3.3 b)). Let $X = 1 + \alpha U$ and $Y = \alpha V$, hence $K[T] = K[U,V]/(V + \delta U + U^2 + \alpha UV + \beta V^2)$ with $\delta = 2\alpha^{-1}$, and we take

$$\mathcal{O}[\mathfrak{T}^R] = \mathcal{O}[U,V]/(V + \delta U + U^2 + \alpha UV + \beta V^2).$$

One then sees as above that this is a smooth group scheme with generic fiber T, with closed fiber isomorphic to $\mathfrak{Add}$ when $\tilde{K}$ is ramified and to a non-split connected unipotent group when $\tilde{K}$ is unramified, and one proves (1) as above.

In both cases, $\mathcal{O}[\mathfrak{T}^R]$ is a free $\mathcal{O}$-module, admitting a basis consisting of the monomials V^n and UV^n for $n \in \mathbf{N}$, so (ET 1) is satisfied. Consequently, 1.7.6 implies

$$\mathcal{O}[\mathfrak{T}^R] = \mathcal{H}_K(T). \tag{2}$$

Finally, $\mathfrak{T}^R$ admits a faithful linear representation: one easily verifies that the (free) $\mathcal{O}$-submodule M generated by U and V is a sub-comodule of $\mathcal{O}[\mathfrak{T}^R]$ for the regular representation and the restriction to M of this representation is faithful (cf. [DG], p. 183).

In conclusion, (2) and Proposition 4.4.12 (ii) imply that $\mathfrak{T}^R$ is indeed the smoothening of $\mathfrak{T}$.

Proposition 0.30 (4.4.14)**.** *Let $K \to K'$ be a base change that is conservative with respect to $\tilde{K}$ (4.2.25). Suppose at least one of the two following conditions is satisfied:*

a) T is an induced torus (1.5.17);

b) the extension $\tilde{K}$ of K is unramified.

Then, the canonical injection of $T(K)/T_b(K)$ into $T(K')/T_b(K')$ is surjective.

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Suppose *a*). — It is sufficient to examine the case $T = \Pi_{L/K}\mathrm{Mult}$, where L is a field with $K \subset L \subset \tilde{K}$. According to 1.6.8 (ii), $L' = L \otimes_K K'$ is an *unramified* extension of L. Now the canonical identification of $T(K)$ (resp. $T(K')$) with $L^\times$ (resp. $L'^\times$) identifies $T_b(K)$ with $\mathcal{O}_L^\times$ (resp. $T_b(K')$ with $\mathcal{O}_{L'}^\times$), hence the assertion.

Suppose *b*). — Let $\tilde{K}' = \tilde{K} \otimes_K K'$. In view of *b*) and 1.6.8 (ii), we have $\omega(K) = \omega(\tilde{K}) = \omega(\tilde{K}')$. Let $\gamma = \mathrm{Gal}(\tilde{K}'/K') = \mathrm{Gal}(\tilde{K}/K)$ and identify $T(K)$ with $\mathrm{Hom}_\gamma(X^*(T), \tilde{K}^\times)$ and $T(K')$ with $\mathrm{Hom}_\gamma(X^*(T), \tilde{K}'^\times)$.

Let $(e_i)_{1 \leq i \leq n}$ be a basis of $X^*(T)$ and let $x \in T(K')$. Let $(\omega_j)_{1 \leq j \leq k}$ be a basis of the $\mathbb{Z}$-submodule Λ of $\mathbb{R}$ generated by the values $\omega(x(e_i))$ for $1 \leq i \leq n$ and let π_j be an element of K with valuation ω_j $(1 \leq j \leq k)$. The map $\omega_j \mapsto \pi_j$ extends to an isomorphism r from Λ onto a subgroup of $K^\times$, with inverse ω. Let us consider then the homomorphism $y = r \circ \omega \circ x$ from $X^*(T)$ into $K^\times$; it is immediate that $y \in \mathrm{Hom}_\gamma(X^*(T), K^\times) \subset T(K)$ and that $y^{-1}x \in T_b$. Consequently,

$$T(K') = T(K)T_b(K'),$$

hence the proposition.

4.4.15. — Let us now return to our general conventions: T is the maximal torus of the quasi-split group G, centralizer of the maximal K-split torus S, and $T_b(K)$ is the stabilizer H^1 of the apartment $A \times V^1$ of the enlarged building of G (4.2.16).

Proposition 0.31 (4.4.16)**.** *If G is adjoint (resp. simply connected), then T is an induced torus.*

In fact, the elements of the basis $\tilde{\Delta}$ of the root system $\tilde{\Phi}$ of G according to T (resp. the fundamental weights relative to $\tilde{\Delta}$) defined by a K-Borel subgroup containing T (cf. 4.1.2) form a basis of $X^*(T)$ invariant under the Galois group $\mathrm{Gal}(\tilde{K}/K)$.

Proposition 0.32 (4.4.17)**.** *Suppose that at least one of the following two conditions is satisfied:*

a′) G is simply connected or adjoint or split;

b′) The extension $\tilde{K}$ of K is unramified.

Then, the image W of $N(K)$ in Aut A by the homomorphism ν (4.2.7) is invariant under any conservative base change.

In fact, the image of W in $GL(V)$ is the Weyl group of the root system Φ and the subgroup of translations of W identifies canonically with a quotient of $T(K)/T_b(K)$ (4.2.7). The proposition thus follows from 4.4.14, taking 4.4.16 into account.

4.4.18. — The canonical scheme $\mathfrak{T}$ with generic fiber T possesses the following properties:

(I) $\mathfrak{T}$ is a flat $\mathcal{O}$-group scheme of finite type, possessing a faithful linear representation;

(II) the inclusion of S in T extends to a closed immersion of $\mathfrak{S}$ into $\mathfrak{T}$;

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(III) the Weyl group ${}^vW$ of G over K, considered as a group of automorphisms of T, extends to a group of automorphisms of $\mathfrak{T}$;

(IV) the action of the Galois group $\mathrm{Gal}(\tilde{K}/K)$ on T extends to a Galois action of $\mathrm{Gal}(\tilde{K}/K)$ on $\mathfrak{T}$;

(V) let T_{Ad} be the image of T in the adjoint group $\mathrm{Ad}\, G$; the restriction to T of the adjoint representation extends to a morphism from $\mathfrak{T}$ into the canonical scheme $\mathfrak{T}_{\mathrm{Ad}}$ with generic fiber T_{Ad};

(VI) let $\pi : \tilde{G} \to G$ be a simply connected covering of the derived group of G and let $\tilde{T}$ be the inverse image of T by π; the restriction of π to $\tilde{T}$ extends to a morphism of the canonical scheme $\tilde{\mathfrak{T}}$ with generic fiber $\tilde{T}$ into $\mathfrak{T}$;

(VII) $\mathfrak{T}(\mathcal{O}) = T_b(K) = \{t \in T(K) \mid \omega(\chi(t)) = 0 \text{ for all } \chi \in X_K(T)\}$ and $\mathfrak{T}({}^{hs}\mathcal{O}) = T_b({}^{hs}K) = \{t \in T({}^{hs}K) \mid \omega(\chi(t)) = 0 \text{ for all } \chi \in X_{{}^{hs}K}(T)\}$;

(VIII) $\pi^{-1}(\mathfrak{T}(\mathcal{O})) \cap \tilde{T}(K) = \tilde{\mathfrak{T}}(\mathcal{O})$ and, when G is semi-simple, $\mathrm{Ad}^{-1}(\mathfrak{T}_{\mathrm{Ad}}(\mathcal{O})) \cap T(K) = \mathfrak{T}(\mathcal{O})$.

In fact, (I), (II) and (VII) have been explicitly seen; (III), (IV), (V) and (VI) result from the functorial properties of $\mathfrak{T}$ (4.4.7). Finally, (VIII) results from (VII) and the following fact: the map $\chi \mapsto \chi \circ \mathrm{Ad}$ and, when G is semi-simple, the map $\chi \mapsto \chi \circ \pi$ send the group of rational characters over K of T_{Ad} (which is the group generated by Φ) onto a subgroup of finite index of the group of rational characters over K of $\tilde{T}$ (which is the weight group of Φ) resp. of T (which is an intermediate group between the root group and the weight group of Φ).

Furthermore, 4.4.16 implies (IX) If G is simply connected or adjoint, then $\mathfrak{T}$ is smooth and connected and one has (IX 1) $\mathcal{O}[\mathfrak{T}] = \mathscr{A}_K(T) = \{f \in K[T] \mid f(T_b({}^{hs}K)) \subset {}^{hs}\mathcal{O}\}$. If in addition the residue field $\bar{K}$ is infinite, $\mathfrak{T}$ is stuffed and one has (IX 2) $\mathcal{O}[\mathfrak{T}] = \{f \in K[T] \mid f(T_b(K)) \subset \mathcal{O}\}$.

Finally, if the smoothening $\mathfrak{T}^R$ of $\mathfrak{T}$ exists, for example if ω is discrete, one checks immediately that $\mathfrak{T}^R$ possesses properties (I) to (VIII) (where one replaces $\mathfrak{T}$ by $\mathfrak{T}^R$), with in addition that of being smooth (but not necessarily connected), even if G is neither simply connected nor adjoint.

4.4.19. — Let $a \in \Phi$ and let $\mathfrak{U}_a$ be one of the schemes $\mathfrak{U}_{a,k}$ or $\mathfrak{U}_{a,(k,l)}$ defined in 4.3.

Proposition 0.33. *Let $\mathfrak{T}$ be a flat $\mathcal{O}$-group scheme with generic fiber T, possessing property (V) of 4.4.18. The map $(t, x) \mapsto txt^{-1}$ from $T \times U$ into U_a extends to a morphism from $\mathfrak{T} \times \mathfrak{U}_a$ into $\mathfrak{U}_a$.*

In view of (V), it suffices to show it when G is adjoint semi-simple and $\mathfrak{T}$ is the canonical scheme with generic fiber T. Furthermore, we can assume, up to a conservative base change and then applying 1.2.4, that the field[12]

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[12] [[ILLEGIBLE — REASON: The text is cut off at the bottom of the page.]]

residual field $\overline{K}$ is infinite. The schemes $\mathfrak{T}$ and $\mathfrak{U}_a$ are then padded (4.4.18 (IX) and 4.3.9) and it suffices to show that $txt^{-1} \in \mathfrak{U}_a(\mathcal{O})$ for all $t \in \mathfrak{T}(\mathcal{O})$ and $x \in \mathfrak{U}_a(\mathcal{O})$, which is immediate. Indeed, taking up the notation of 4.1.5 and 9, we have $\alpha(t) \in \mathcal{O}_{L_a}^{\times}$ for all $t \in \mathfrak{T}(\mathcal{O}) = T_b(K)$ and

$$tx_a(u)t^{-1} = x_a(\alpha(t)u) \quad \text{if } a \text{ is non-plural and } u \in L_a, \tag{1}$$

$$tx_a(u,v)t^{-1} = x_a(\alpha(t)u, \alpha(t)\alpha(t)^{\sigma}v) \tag{2}$$

$$\text{if } a \text{ is plural and } (u,v) \in H(L_a, L_{2a}),$$

whence our assertion.

4.5. Quasi-concave functions and schematic root data. **4.5.1.** — We shall, in this section, construct families $(\mathfrak{U}_a)_{a\in\Phi}$ of $\mathcal{O}$-schemes forming, with the canonical scheme $\mathfrak{T}$ with generic fiber T (resp. the smoothed version $\mathfrak{T}^R$ of $\mathfrak{T}$ when it exists), a schematic root datum in the sense of §3 (recall that $Z = T$). We will obviously take for $\mathfrak{U}_a$ one of the schemes defined in section 4.3: in other words, *one is given a function* $f : \Phi \to \mathbf{R}$, *satisfying, when* ω *is dense, the condition*

(Sch) $f(a) \in \Gamma'_a$ *and* $f(2a) \leq 2f(a)$ *for all* $a \in \Phi$,

(where we take $f(2a) = 2f(a)$ when $2a$ is not a root) and *we set, for* $a \in \Phi$,

$\mathfrak{U}_{f,a} = \mathfrak{U}_{a,f(a)}$ if a is non-plural (4.3.2);

$\mathfrak{U}_{f,a} = \mathfrak{U}_{a,(f(a),f(2a))}$ if a is plural (4.3.5, 6 and 8).

We have seen that

$$\mathfrak{U}_{f,a}(\mathcal{O}) = U_{a,f(a)}U_{2a,f(2a)}$$

(with the usual convention $U_{2a} = \{1\}$ when a is non-plural).

Suppose ω is discrete and let us set $g(a) = f(a)$ for $a \in \Phi, a/2 \notin \Phi$ and $g(a) = \inf(f(a), 2f(a/2))$ for $a, a/2 \in \Phi$. It follows from the very definitions (4.3.8) that $\mathfrak{U}_{f,a} = \mathfrak{U}_{g,a}$ for all $a \in \Phi$. In other words, one can always reduce to the case where f satisfies the following condition (cf. 4.3.5 (2)), which is moreover a consequence of (Sch) when ω is dense:

(QC 0) $f(2a) \leq \inf \Gamma_{2a} \cap [2f(a), +\infty[$ for all $a \in \Phi$.

4.5.2. — Let f be as above. For $a \in \Phi$, set

$$f'(a) = \inf\left\{k \in \Gamma'_a \mid k \geq f(a) \text{ or, when } \frac{a}{2} \in \Phi, k \geq 2f\left(\frac{a}{2}\right)\right\}. \tag{1}$$

We say that f' is the *optimization* of f and that f is *optimal* if $f = f'$. When f is optimal, $k = f(a)$ and $\ell = f(2a)$ satisfy the relations (1) and (2) of 4.3.5, and f satisfies (QC 0).

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When ω is dense, it follows immediately from (Sch) that $f' = f$. If ω is discrete, $f'(a)$ can be different from $f(a)$, but one has, by the very definition,

$$\mathfrak{U}_{f,a} = \mathfrak{U}_{f',a} \quad \text{for all } a \in \Phi, \tag{2}$$

$$U_{a,f(a)}U_{2a,f(2a)} = U_{a,f'(a)}U_{2a,f'(2a)}, \tag{3}$$

(cf. I, 6.4.12 a)). If f satisfies (QC 0), one also has

$$U_{2a,f(2a)} = U_{2a,f'(2a)}. \tag{4}$$

Let us note that definition (1) appears different from the one given in I, 6.4.10, where the infimum was taken in the ordered set $\tilde{\mathbf{R}} \supset \mathbf{R}$; but the hypotheses made on f actually entail the coincidence of the two definitions: this results from (Sch) when ω is dense (and in this case, (Sch) means that the function $f' : \Phi \to \tilde{\mathbf{R}}$ from (I, 6.4.10) satisfies $f'(a) \in \Gamma'_a$ for all $a \in \Phi$), and from the fact that the Γ'_a are discrete when ω is discrete.

4.5.3. — Recall that f is said to be *concave* (I, 6.4.3 and 5) if it satisfies the two conditions

(C 1) $f(a) + f(b) \geqq f(a+b)$ for a, b and $a + b \in \Phi$,

(C 2) $f(a) + f(-a) \geqq 0$ for all $a \in \Phi$,

and *quasi-concave* (cf. I, 6.4.7 and addendum E2 below, p. 173) if it satisfies (QC 0) and the following conditions, where $N_f^{(a)}$ denotes the intersection with $N = N(K)$ of the subgroup $U_f^{(a)}$ generated by the $U_{ia,f(ia)}$ for $i = \pm 1, \pm 2$ (I, 6.4.2):

(QC 1) $U_{-2a,f(-2a)}U_{-a,f(-a)}U_{a,f(a)}U_{2a,f(2a)}N_f^{(a)}$ is a subgroup of $G = G(K)$ for every root $a \in \Phi$.

(QC 2) If $a, b \in \Phi$ are not proportional, the commutator group $(U_{a,f(a)}, U_{b,f(b)})$ is contained in the group generated by the $U_{pa+qb,f(pa+qb)}$ for $p, q \in \mathbf{N}^*$ and $pa + qb \in \Phi$.

It is known (I, 6.4.7) that every concave function is quasi-concave. In view of 4.5.2 (3) and (4), the optimization of a function f satisfying (QC 0), and (Sch) when ω is dense, is quasi-concave if and only if f is. On the other hand, the optimization of a concave function is not necessarily concave: this is why it is useful to consider functions which are not optimal.

Let us note that the condition (QC 1) decomposes into conditions (QC 1)$_a$ for $a \in \Phi$, involving only integer multiples of a, and that (QC 2) decomposes into conditions (QC 2)$_{a,b}$ for $a, b \in \Phi$ non-proportional, involving only the roots of the form $pa + qb$ with $p, q \in \mathbf{N}$.

4.5.4. — The main goal of this section is to establish the following theorem, the proof of which will be completed in 4.5.11:

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Theorem 0.34. — *Let $f : \Phi \to \mathbf{R}$, satisfying (QC 0) and (Sch) when ω is dense. Let $\mathfrak{Z}$ be an $\mathcal{O}$-group scheme of generic fiber T, possessing properties (I), (II), (V) and (VI) of 4.4.18. For $\mathfrak{Z}$ and the $\mathfrak{U}_{f,a}$ for $a \in \Phi$ to form a schematic root datum on G relative to S, it is necessary and sufficient that f be quasi-concave.*

It will be noted that one can take for $\mathfrak{Z}$ the canonical scheme of generic fiber T, or its smoothed version when it exists (for example if ω is discrete or if $[\tilde{K} : K] = 2$), or else the identity component of the latter when it possesses a faithful linear representation (same examples).

4.5.5. — Let us begin by studying condition (QC 1)$_a$.

Proposition 0.35. — *Let $f : \Phi \to \mathbf{R}$, satisfying (Sch) if ω is dense. Suppose f is optimal; then the following conditions are equivalent:*

(i) f satisfies (QC 1)$_a$;

(ii) f satisfies the inequalities

$$f(a) + f(-a) \geq 0, \quad f(2a) + f(-2a) \geq 0, \tag{1}$$

$$f(a) + f(-2a) \geq f(-a), \quad f(-a) + f(2a) \geq f(a), \tag{2}$$

$$2f(a) + f(-2a) \geq 0, \quad 2f(-a) + f(2a) \geq 0; \tag{3}$$

(iii) f is the optimized version of a function $f_1 : \Phi \to \mathbf{R}$, satisfying (Sch) when ω is dense, and whose restriction to the root system $\{\pm a, \pm 2a\}$ is concave.

Proof. It is clear that inequalities (1) and (2) imply (3) and that (ii) implies (iii): it suffices to take for f_1 the function defined by $f_1(2a) = \inf(f(2a), 2f(a))$, $f_1(-2a) = \inf(f(-2a), 2f(-a))$ and $f_1(b) = f(b)$ for $b \neq \pm 2a$. If ω is dense, we have moreover $f_1 = f$. On the other hand, we know from (I, 6.4.7) that (iii) implies (i). It remains to show that (i) implies (1) and (2).

Suppose then f is optimal satisfying (QC 1)$_a$. Inequalities (1) and (3) are assertion (i) of lemma 6.4.11 of I, which completes the proof if $2a \notin \Phi$, or if $f(a) + f(-a) = 0$, because (3) then implies (2).

It remains to examine the case where a is plural and $f(a) + f(-a) > 0$. If $N_f^{(a)} \notin T(K)$, assertion (ii) of the same lemma 6.4.11 of I shows that $f(2a) + f(-2a) = 0$ and that the reflection $r_{a,k}$ with $2k = f(2a)$ belongs to the canonical image of $N_f^{(a)}$ in the group of affine automorphisms of the apartment A, set of equipollent valuations to ϕ. Formula (1) of I, 6.4.10 then gives

$$f(-a) = f(a) + a(r_{a,k}(\phi) - \phi) = f(a) - f(2a),$$

and similarly $f(-a) = f(a) + f(-2a)$, whence (2).

On the other hand, if $f(-2a) + 2f(a) = 0$, then $2f(a) \in 2\Gamma'_a \cap (-\Gamma_{-2a}) = 2\Gamma'_a \cap \Gamma_{2a}$ and $f(2a) = 2f(a)$, which brings us back to the previous case. □

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Finally, it remains to prove (2) when a is plural, that $N_f^{(a)} \subset T(K)$ and that consequently

$$f(a) + f(-a) > 0, \quad f(2a) + f(-2a) > 0, \quad f(\pm 2a) + 2f(\mp a) > 0.$$

According to I, 6.4.11 (ii), we then have $N_f^{(a)} \subset H$ (recall that H is the kernel of the homomorphism $\nu : N \to \mathrm{Aut}\, A$ and that H normalizes all the subgroups $U_{b,k}$ for $b \in \Phi$ and $k \in \mathbf{R}$), whence

$$U_f^{(a)} \subset U_{-2a,f(-2a)} U_{-a,f(-a)} H U_{a,f(a)} U_{2a,f(2a)},$$

and consequently, by using the notations of 4.1.12, $U_{f,a} \times U_{f,-a} \subset W_a(K)$ and

$$\beta_a(U_{f,a} \times U_{f,-a}) \subset U_{f,-a} \times T(K) \times U_{f,a}.$$

Let us then choose $u \in L_a$ such that $\omega(u) = f(a) + \gamma$ (with $\gamma = -\frac{1}{2}\omega(\lambda)$: cf. 4.3.5) and $v' \in L_a^0$ such that $\omega(v') = f(-2a)$, which is possible since $f(a) \in \Gamma'_a = \gamma + \omega(L_a^\times)$ and that $f(2a) \in \Gamma'_{-2a} = \omega(L_a^0 - \{0\})$. Then $x_a(u, \lambda u^\sigma u) \in U_{f,a}$, $x_{-a}(0, v') \in U_{f,-a}$, and, as $\omega(\lambda u^\sigma u v') = 2f(a) + f(-2a) > 0$, we have $t = 1 + \lambda u^\sigma u v' \in \mathcal{O}_L^\times$ and

$$\omega(-t^{-1} u v') = f(a) + \gamma + f(-2a). \tag{4}$$

But according to 4.1.12 (2), we must have $x_{-a}(-t^{-1}uv', t^{-1}v') \in U_{f,-a}$; 4.2.21 (4) and (5) then imply that $f(a) + f(-2a) \geq f(-a)$, which is the first of the inequalities (2). The second is proved similarly, by exchanging a and $-a$.

4.5.6. Remarks. — Let f be as in 4.5.5, satisfying (QC I)$_a$ for an $a \in \Phi$.
1) If ω is dense or if a is not plural or if a is plural and the extension L_a/L_{2a} is unramified, then $f = f_{\bar{1}}$ and the restriction of f to $\{\pm a, \pm 2a\}$ is concave.
2) Suppose that at least one of the inequalities (1) is an equality. Then the restriction of f to $\{\pm a, \pm 2a\}$ is concave, unless ω is discrete, a is plural, the extension L_a/L_{2a} is ramified and if $f(a) + f(-a) = 0$, $f(2a) = 2f(a) + c$ and $f(-2a) = -2f(a) + c$ (where c is as above the valuation of a uniformizer of L_a).
3) It has been seen in the course of the proof that if a is plural and if $f(-2a) + 2f(a) = 0$, then $f(a) + f(-a) = f(2a) + f(-2a) = 0$ and $f(\pm 2a) = 2f(\pm a)$. This is moreover only possible if the extension L_a/L_{2a} is ramified.

4.5.7. Corollary. — Let $K \to K_1$ be a conservative change of base, and let $f : \Phi \to \mathbf{R}$ (still satisfying (QC o), and (Sch) when ω is dense). For f to satisfy (QC I)$_a$ for G, it is necessary and sufficient that it satisfy it for G_{K_1}.

Indeed, the sets Γ'_a remain unchanged by the change of base $K \to K_1$, thus the condition (Sch) and the optimized f are preserved, as are moreover the inequalities of 4.5.5 (ii).

4.5.8. — The scheme $\mathfrak{W}_{f,a}$. Let $a \in \Phi$.

Proposition 0.36. *Let f be as in 4.5.5, satisfying* (QC I)$_a$*, and let $\mathfrak{T}$ be an $\mathcal{O}$-group scheme with generic fiber T possessing property* (VI) *of 4.4.18.*

(i) The function d_a on $U_a \times U_{-a}$ defined in 4.1.6 (1) and 4.1.12 (1) belongs to $\mathcal{O}[\mathfrak{U}_{f,a} \times \mathfrak{U}_{f,-a}]$.

(ii) Let $\mathfrak{W}_{f,a}$ be the special open subset of the scheme $\mathfrak{U}_{f,a} \times \mathfrak{U}_{f,-a}$ defined by d_a. The generic fiber of $\mathfrak{W}_{f,a}$ is equal to W_a and the morphism $\beta_a : W_a \to U_{-a} \times T \times U_a$ extends to a morphism (still denoted by β_a) from $\mathfrak{W}_{f,a}$ into $\mathfrak{U}_{f,-a} \times \mathfrak{T} \times \mathfrak{U}_{f,a}$. We have $(\mathfrak{U}_{f,a} \times \{1\}) \cup (\{1\} \times \mathfrak{U}_{f,-a}) \subset \mathfrak{W}_{f,a}$. If $f(a) + f(-a) > 0$ and $f(2a) + f(-2a) > 0$, then $\mathfrak{W}_{f,a}$ contains the closed fiber of $\mathfrak{U}_{f,a} \times \mathfrak{U}_{f,-a}$.

Proof. Suppose first of all that G is simply connected and that $\mathfrak{T}$ is the canonical scheme with generic fiber T. In view of 1.2.4, 4.4.10, and 4.5.7, we may suppose the residue field $\bar{k}$ is infinite. The schemes $\mathfrak{U}_{f,a}$ and $\mathfrak{U}_{f,-a}$ are then thickened and to demonstrate (i), it suffices to prove that

$$d_a(\mathfrak{U}_{f,a}(\mathcal{O}) \times \mathfrak{U}_{f,-a}(\mathcal{O})) \subset \mathcal{O}. \tag{1}$$

This done, we can consider $\mathfrak{W}_{f,a}$, whose generic fiber is W_a, since the latter is the special open subset of $U_a \times U_{-a}$ defined by d_a, and the extendability of β_a is equivalent to

$$\beta_a(\mathfrak{W}_{f,a}(\mathcal{O})) \subset \mathfrak{U}_{f,-a}(\mathcal{O}) \times \mathfrak{T}(\mathcal{O}) \times \mathfrak{U}_{f,a}(\mathcal{O}), \tag{2}$$

since $\mathfrak{W}_{f,a}$ is thickened (1.7.3 e)). Let us therefore demonstrate (1) and then (2).
If a is not plural and if $u, u' \in L_a$ with $\omega(u) \geq f(a)$ and $\omega(u') \geq f(-a)$, we have $\omega(uu') \geq 0$ and $t = 1 - uu' \in \mathcal{O}_{L_a}$ whence

$$d_a(x_a(u), x_{-a}(u')) = \mathrm{Norm}_{L_a/K}(1 - uu') \in \mathcal{O},$$

whence (1). If moreover $(x_a(u), x_{-a}(u')) \in \mathfrak{W}_{f,a}(\mathcal{O})$, then $d_a(x_a(u), x_{-a}(u')) \in \mathcal{O}^\times$, whence $\omega(t) = 0$, $\widetilde{a}(t) \in \mathfrak{T}(\mathcal{O})$ by 4.4.7, and $\omega(t^{-1}u') \geq f(-a)$, $\omega(t^{-1}u) \geq f(a)$, whence (2) by 4.1.6 (2).
If a is plural, consider the concave function f_1 of which f is the optimized version, defined in 4.5.5. For $u, v \in L_a$ with $v + v^\sigma = u^\sigma u$, we have $x_a(u, v) \in \mathfrak{U}_{f,a}(\mathcal{O})$ if and only if $\omega(u) \geq f_1(a) + \gamma$ and $\omega(v) \geq f_1(2a)$ and similarly for $-a$. It follows immediately from the concavity of f_1 that, if $x_a(u, v) \in \mathfrak{U}_{f,a}(\mathcal{O})$ and $x_{-a}(u', v') \in \mathfrak{U}_{f,-a}(\mathcal{O})$, then $t = 1 - u^\sigma u' + vv' \in \mathcal{O}_{L_a}$, whence (1) since

$$d_a(x_a(u, v), x_{-a}(u', v')) = \mathrm{Norm}_{L_a/K}\, t \in \mathcal{O}.$$

If moreover $(x_a(u, v), x_{-a}(u', v')) \in \mathfrak{W}_{f,a}(\mathcal{O})$, then $\omega(t) = 0$, $\widetilde{a}(t) \in \mathfrak{T}(\mathcal{O})$ by 4.4.7. On the other hand, $\omega(u' - uv') \geq \inf(f_1(-a) + \gamma, f_1(a) + \gamma + f_1(-2a)) = f_1(-a) + \gamma$, given the concavity of f_1, and similarly $\omega(u - v^\sigma u') \geq f_1(a) + \gamma$. Formula (2) then follows from 4.1.12 (2), taking into account 4.2.21 (5).
On the other hand, it is obvious that $d_a = 1$ on $\mathfrak{U}_{f,a} \times \{1\}$ and on $\{1\} \times \mathfrak{U}_{f,-a}$. □

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Finally, if $f(a) + f(-a) > 0$ and $f(2a) + f(-2a) > 0$, we have seen in the proof of 4.5.5 that one necessarily had $f(\pm 2a) + 2f(\mp a) > 0$, whence

$$\alpha = \inf(f_1(a) + f_1(-a), f_1(2a) + f_1(-2a)) > 0$$

and it follows from the preceding calculations that, with the notation above, $\omega(1 - t) \geq \alpha$. We deduce from this that there exists $\xi \in \mathcal{O}$ with $\omega(\xi) > 0$ such that $\xi^{-1}(1 - d_a) \in \mathcal{O}[\mathfrak{U}_{f,a} \times \mathfrak{U}_{f,-a}]$, which proves the last assertion of the proposition.

In the general case, let us consider a simply connected covering $\pi : \tilde{G} \to G$ of the derived group of G and the maximal torus $\tilde{T} = \pi^{-1}(T)$ of $\tilde{G}$; let us identify the root subgroups $U_{\pm a}$ of G and of $\tilde{G}$ via π. The morphism $\beta_a : W_a \to U_{-a} \times T \times U_a$ factors as $\beta_a : W_a \to U_{-a} \times \tilde{T} \times U_a$ followed by $\pi \times \pi \times \pi$ and it suffices to apply 4.4.18 (VI).

4.5.9. Remarks. — a) $\mathfrak{W}_{f,a}$ is the largest open set of $\mathfrak{U}_{f,a} \times \mathfrak{U}_{f,-a}$ with generic fiber W_a since its complement is the closure of the complement of W_a in $U_a \times U_{-a}$.
b) We have

$$\begin{aligned} \mathfrak{W}_{f,a}(\mathcal{O}) &= \{(x, y) \in \mathfrak{U}_{f,a}(\mathcal{O}) \times \mathfrak{U}_{f,-a}(\mathcal{O}) \mid xy \in U_{-a}(K)\mathfrak{T}(\mathcal{O})U_a(K)\} \\ &= \{(x, y) \in U_a(K) \times U_{-a}(K) \mid xy \in \mathfrak{U}_{f,-a}(\mathcal{O})\mathfrak{T}(\mathcal{O})\mathfrak{U}_{f,a}(\mathcal{O})\}. \end{aligned} \tag{1}$$

Indeed, $\mathfrak{W}_{f,a}(\mathcal{O})$ is contained in the 2nd and 3rd members of (1). Conversely, if (x, y) belongs to the 2nd member of (1), then $(x, y) \in W_a(K)$ and formulas 4.1.6 (2) and 4.1.12 (2) show that $d_a(x, y) \in \mathcal{O}^\times$, because the relation $\tilde{a}(t) \in \mathfrak{T}(\mathcal{O})$ for $t \in L_a$ implies $t \in \mathcal{O}_{L_a}^\times$. One sees similarly that the 3rd member of (1) is contained in $\mathfrak{W}_{f,a}(\mathcal{O})$.

On the other hand, the relations $x \in \mathfrak{U}_{f,a}(\mathcal{O})$, $y \in \mathfrak{U}_{f,-a}(\mathcal{O})$ and $xy \in U_{-a}(K)T(K)U_a(K)$ (that is to say $(x, y) \in W_a(K)$) do *not* imply $(x, y) \in \mathfrak{W}_{f,a}(\mathcal{O})$.

4.5.10. — Let us now move on to the study of condition (QC 2).

Proposition 0.37. *Let $f : \Phi \to \mathbf{R}$ be satisfying (QC 0), and (Sch) if ω is dense, and let $K \to K'$ be a conservative base change. In order for f to satisfy (QC 2) relative to G, it is necessary and sufficient that it satisfy it relative to $G_{K'}$.*

Proof. We can suppose f is optimal (cf. proof of 4.5.7). Let $a, b \in \Phi$ be non-proportional; let Ψ denote the set of $c \in \Phi$ of the form $pa + qb$ with $p, q \in \mathbf{N}^*$ coprime, and let X (resp. X') denote the group generated by the $U_{c,f(c)}$ (resp. the $U'_{c,f(c)} = \{x \in U_c(K') \mid \varphi'_c(x) \geq f(c)\}$) for $c \in (\Psi \cup 2\Psi) \cap \Phi$. Recall that $U_{f,c} = U_{c,f(c)}U_{2c,f(2c)}$ and $U'_{f,c} = U'_{c,f(c)}U'_{2c,f(2c)}$.
If f satisfies (QC 2) for $G_{K'}$, then, according to I, 6.4.9 (ii), we have

$$X' = \prod_{c \in \Psi} U'_{f,c}. \tag{1}$$

□

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Now, $U'_{f,c} \cap G(K) = U_{f,c}$ by 4.3.2 and 4.3.5 (3), and, since the product map is an isomorphism of K-algebraic varieties from $\prod U_c$ onto U_Ψ (cf. 1.1.7), we deduce from (1) that

$$X \subset X' \cap G(K) = \prod_{c \in \Psi} U_{f,c} \subset X, \tag{2}$$

whence $X = X' \cap G(K)$. But the commutator group $(U_{a,f(a)}, U_{b,f(b)})$ is contained in $X' \cap G(K)$, thus in X, and (2) shows that f satisfies (QC 2) relative to G.

Conversely, suppose that f possesses property (QC 2) for G and let us show that f satisfies $(\text{QC } 2)_{a,b}$ for $G_{K'}$.

We will reason by induction on $\mathrm{Card}(\Psi \cup 2\Psi) \cap \Phi$, the assertion being trivial if this cardinality is zero. The induction hypothesis implies that (1) and (2) are exact and also that the commutator groups $(U'_{2a,f(2a)}, U'_{b,f(b)})$ and $(U'_{a,f(a)}, X')$ are contained in X'. Let us show that $(U'_{a,f(a)}, U'_{b,f(b)}) \subset X'$; taking into account the classical relation

$$(x'x, y) = (x, y)((y, x), x')(x', y)$$

(where $(x, y) = xyx^{-1}y^{-1}$, according to the convention adopted in I), it suffices, in view of the above, to show that there exists a subset V of $U'_{a,f(a)}$ telle que $U'_{a,f(a)}$ is the monoid generated by V and $U'_{2a,f(2a)}$ and that the commutator (x, y) belongs to X' for all $v \in V$ and $y \in U'_{b,f(b)}$.

Let $\Phi_1 = \Phi \cap (\mathbf{Z}a + \mathbf{Z}b)$: this is a closed symmetric subset of Φ and the subgroup G_{Φ_1} generated by T and the U_c for $c \in \Phi_1$ is reductive, quasi-split with maximal K-split torus S and relative root system Φ_1 (cf. [4], 3.8 and 3.13). Now, the condition $(\text{QC } 2)_{a,b}$ involves only a maximal unipotent subgroup of G_{Φ_1} containing U_a, U_b and the U_c for $c \in \Psi$. We can therefore replace G by G_{Φ_1} and even by its adjoint group.

Suppose then G adjoint and $\Phi \subset \mathbf{Z}a + \mathbf{Z}b$. Then, $\{a, b\}$ is a basis of the character group of S and for all $\xi \in \mathcal{O}'^{\times}$, there exists $s \in S(K')$ such that $a(s) = \xi$ and $b(s) = 1$.

Suppose for the sake of definiteness that $2a \in \Phi$ (the case where $2a \notin \Phi$ being treated in an analogous but simpler way: we will leave it to the reader to adapt the following proof). Let $u, v \in L_a$, with $v + v^\sigma = uu^\sigma$. We have

$$\begin{aligned} sx_a(u, v)s^{-1} &= x_a(a(s)u, a(s)^2 v) = x_a(\xi u, \xi^2 v), \\ sys^{-1} &= y \quad \text{for all } y \in U_b(K'), \\ sX's^{-1} &= X', \end{aligned}$$

from which we deduce, taking into account the validity of (QC 2) for G, that the commutator $(x_a(\xi u, \xi^2 v), y)$ belongs to X' for $u, v \in L_a$ with $v + v^\sigma = uu^\sigma$, $x_a(u, v) \in U_{a,f(a)}$ and $y \in U_{b,f(b)}$. Since

$$\begin{aligned} &\{u' \in K' \otimes L_a \mid \omega(u') \geq f(a) + \gamma\} \\ &\qquad = \{\sum \xi_i u_i \mid \xi_i \in \mathcal{O}'^{\times}, u_i \in L_a, \omega(u_i) \geq f(a) + \gamma\} \end{aligned}$$

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(cf. 4.3.11), one sees that the set V of $x_a(\xi u, \xi^2 v)$ for $x_a(u, v) \in U_{a,f(a)}$ and $\xi \in \mathcal{O}'^{\times}$ satisfies the desired conditions and that consequently

$$(U'_{a,f(a)}, U_{b,f(b)}) \subset X'.$$

Interchanging then the roles of a and b, one similarly sees that $(U'_{a,f(a)}, U'_{b,f(b)}) \subset X'$, which completes the proof.

4.5.11. — Let us finally prove Theorem 4.5.4. Suppose f is quasi-concave and let us verify the (DRS) conditions of 3.1.1, taking as the open set $\mathfrak{W}_a$ in (DRS 3) the open set $\mathfrak{W}_{f,a}$ defined in 4.5.8. Condition (DRS 0) is 4.4.18 (II). Condition (DRS 3) is assertion (ii) of 4.5.8. Condition (DRS 2) is nothing other than 4.4.19.

Finally, to prove (DRS 1), one can, in view of 1.2.4, 4.5.7, and 4.5.10, suppose the residue field $\overline{K}$ of K is infinite, hence the schemes $\mathfrak{U}_{f,a}$ are stout, and it then suffices to show that for $a, b \in \Phi$, $b \neq -a$, the commutator map sends $\mathfrak{U}_{f,a}(\mathcal{O}) \times \mathfrak{U}_{f,b}(\mathcal{O})$ into the product of the $\mathfrak{U}_{f,c}(\mathcal{O})$ for $c \in \Phi \cap (a + b)$: this is obvious for $b = a$ and follows from (QC 2) for $b \neq a$, taking into account I, 6.4.9.

Conversely, suppose the (DRS) conditions are satisfied. Then, it is immediate that (QC 2) is satisfied. Let us prove $(\text{QC1})_a$ for $a \in \Phi$. In view of 4.5.7, one can suppose f is optimal and $\overline{K}$ is infinite; the canonical image of $\mathfrak{W}_a(\mathcal{O})$ in the closed fiber of $\mathfrak{U}_{f,a} \times \mathfrak{U}_{f,-a}$ is then dense. Suppose a is plural and let $u_+ \in L_a$ with $\omega(u_+) = f(a) + \gamma$ and $v_+ \in L_a^0$ with $\omega(v_+) = f(2a)$. The map $(\xi, \eta) \mapsto j_+(\xi, \eta) = x_a(\xi u_+, \lambda \xi^\sigma \xi u_+^q u_+ + \eta v_+)$ is

then an isomorphism of $\mathcal{O}$-schemes from $\mathfrak{X} = \prod_{\mathcal{O}_{L_a}/\mathcal{O}} \mathrm{Add} \times \prod_{\mathcal{O}_{L_{2a}}/\mathcal{O}} \mathrm{Add}$ onto $\mathfrak{U}_{f,a}$. We define u_-, v_- and j_- in a similar way. If $(j_+(\xi,\eta), j_-(\xi',\eta')) \in \mathfrak{W}_a(\mathcal{O})$, we have $t = d_a(j_+(\xi,\eta), j_-(\xi',\eta')) \in \mathcal{O}^\times$, from which

$$\omega(1 - \xi^\sigma \xi' u_+^\sigma u_- + (\lambda \xi^\sigma \xi u_+^q u_+ + \eta v_+)(\lambda \xi'^\sigma \xi' u_-^q u_- + \eta' v_-)) = 0 \tag{1}$$

(cf. 4.1.12 (1)), and $pr_1 \circ \beta_a(j_+(\xi,\eta), j_-(\xi',\eta')) \in \mathfrak{U}_{f,-a}(\mathcal{O})$, from which

$$\omega(\xi' u_- - \xi u_+(\lambda \xi'^\sigma \xi' u_-^q u_- + \eta' v_-)) \geq f(-a) + \gamma \tag{2}$$

(cf. 4.1.12 (2)). However, one has

$$\begin{aligned}
\omega(u_+ u_-) &= f(a) + f(-a) + 2\gamma, \\
\omega(\lambda^2 u_+^q u_+ u_-^q u_-) &= 2(f(a) + f(-a)), \\
\omega(\lambda u_+^q u_+ v_-) &= 2f(a) + f(-2a), \\
\omega(\lambda u_-^q u_- v_+) &= 2f(-a) + f(2a), \\
\omega(v_+ v_-) &= f(2a) + f(-2a), \\
\omega(u_-) &= f(-a) + \gamma, \\
\omega(\lambda u_+ u_-^q u_-) &= f(a) + 2f(-a) + \gamma, \\
\omega(u_+ v_-) &= f(a) + f(-2a) + \gamma,
\end{aligned}$$

and one easily verifies that (1) cannot be "generically true", that is to say for (ξ,η,ξ',η') describing the set of integral points of a non-empty open set of $\mathfrak{X} \times \mathfrak{X}$.

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only if the inequalities (1) and (3) of 4.5.5 (ii) are satisfied, and that (2) can be generically true only if, in addition, the first inequality (2) of 4.5.5 (ii) is satisfied. By interchanging the roles of a and $-a$, one similarly finds the second inequality (2) of 4.5.5 (ii). Consequently (DRS 3) implies that f satisfies (QC 1)$_a$. Analogous calculations, but clearly simpler and which we leave to the reader, provide the same conclusion in the non-plural case.

This completes the proof of Theorem 4.5.4.

4.5.12. — One should not believe that the schematic root data associated by Theorem 4.5.3 to quasi-concave functions on Φ are always the only schematic root data on G relative to S, even if one requires that $\mathfrak{T}$ is the canonical scheme with generic fiber T and if one assumes it to be smooth. This is however very often the case, but it is possible to construct for example in $G = \Pi_{L/K}\mathrm{SL}_2$ with $K = \mathbf{Q}_2$ and $L = K(\sqrt{2})$ schematic root data for which the schemes $\mathfrak{U}_a$ are not of the type studied in 4.3. We hope to return later to this classification problem.

4.6. The schemes $\mathfrak{G}_f$.

One preserves the previous hypotheses and notations. If $\mathfrak{X}$ is an $\mathcal{O}$-scheme and $u : \mathfrak{X} \to \mathfrak{Y}$ a morphism of $\mathcal{O}$-schemes, we denote by $\overline{\mathfrak{X}}$ the closed fiber of $\mathfrak{X}$ and we set $\overline{u} = u_{\overline{k}} : \overline{\mathfrak{X}} \to \overline{\mathfrak{Y}}$.

4.6.1. — *We assume henceforth that the canonical $\mathcal{O}$-scheme with generic fiber T (4.4.8) possesses a smoothing (4.4.12), which we denote by $\mathfrak{T}$ (and not $\mathfrak{T}^R$ as in 4.4). Let us recall that this hypothesis is fulfilled in each of the following cases:*

(A) G is split over K, or more generally, there exists a sub-extension $L \subset \tilde{K}$ and a group G' split over L such that $G = \Pi_{L/K} G'$;

(B) G is simply connected or adjoint;
(in these two cases, T is an induced torus and $\mathfrak{T}$ is the canonical scheme with generic fiber T);

(C) ω is discrete;

(D) $[\tilde{K} : K] = 2$.

Recall also that $\mathfrak{T}$ then possesses properties (I) to (VIII) and (IX 1) of 4.4.18. In cases (A) and (B), $\mathfrak{T}$ is connected; it is not always the same in cases (C) and (D) even if $\mathcal{O}$ is a complete discrete valuation ring with algebraically closed or finite residue field (cf. 4.4.13).

Recall further that the additional hypothesis we have just made is stable by conservative étale base change (4.4.12), and even by a change

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of any arbitrary conservative base change in each of the four cases above. This will allow us to sometimes reduce to the case where the residue field $\overline{K}$ is infinite (cf. 1.6.9 a)).

Finally, we are given a function $f : \Phi \to \mathbf{R}$, satisfying condition (Sch) of 4.5.1 when ω is dense, which is optimal (4.5.2) and quasi-concave (4.5.3).

4.6.2. — The combination of theorems 4.5.4 on the one hand, and 3.8.1 and 3.8.3 (where, of course, we take $\mathfrak{Z} = \mathfrak{T}$ and $\mathfrak{U}_a = \mathfrak{U}_{f,a}$) on the other, provides a smooth connected (affine) $\mathcal{O}$-group scheme with generic fiber G, which we denote by $\mathfrak{G}_f^0$, characterized (up to unique isomorphism) by properties (S^0 1), (S^0 2) and (S^0 3) of 3.8.3, and likewise a smooth (affine) $\mathcal{O}$-group scheme with generic fiber G, denoted $\mathfrak{G}_f$, admitting $\mathfrak{G}_f^0$ as identity component, and characterized (up to unique isomorphism) by properties (S 1), (S 2), (S 3) of 3.8.3 and the relation $\mathfrak{G}_f = \mathfrak{T} \cdot \mathfrak{G}_f^0$. We denote by $\mathfrak{C}_f$ (resp. $\mathfrak{C}_f^0$) the "big cell" of $\mathfrak{G}_f$ (resp. $\mathfrak{G}_f^0$), i.e., the open subscheme of $\mathfrak{G}_f$ (resp. $\mathfrak{G}_f^0$) defined in (S 3) (resp. (S^0 3)).

Let Φ^+ be a system of positive roots; condition (S 2) or (S^0 2) states that the product map is an isomorphism of schemes from $\prod_{a\in\Phi^+} \mathfrak{U}_{f,a}$ onto a smooth closed subgroup scheme of $\mathfrak{G}_f^0$, which we will denote by $\mathfrak{U}_f^+$ (the notation $\mathfrak{U}_f^-$ is then self-explanatory). More generally, let Ψ be a quasi-closed subset of Φ^+ and let Ψ_{nd} be the set of $a \in \Psi$ such that $(1/2)a \notin \Psi$.

We know ([4], 3.11) that the product map is an isomorphism of K-varieties from $\prod_{a\in\Psi_{nd}} U_a$ onto the algebraic group generated by the U_a. If $a \in \Psi_{nd} \cap 2\Phi$, $\mathfrak{U}_{f,a}$ is a closed subgroup scheme of $\mathfrak{U}_{f,(1/2)a}$. We then deduce from 1.2.7 that the product map is an isomorphism of $\mathcal{O}$-schemes from $\prod_{a\in\Psi_{nd}} \mathfrak{U}_{f,a}$ onto a smooth closed subgroup scheme of $\mathfrak{U}_f^+$, which we denote by $\mathfrak{U}_{f,\Psi}$. The scheme underlying $\mathfrak{U}_{f,\Psi}$ is that associated with a finitely generated free $\mathcal{O}$-module (4.3.9) and the canonical map $\mathfrak{U}_{f,\Psi}(\mathcal{O}) \to \mathfrak{U}_{f,\Psi}(\overline{K})$ is surjective.

Let Ψ be a closed subset of Φ^+ and let $g : \Psi \to \mathbf{R}$ be a concave function (i.e. such that $g(a+b) \le g(a) + g(b)$ for $a, b, a+b \in \Psi$), satisfying (Sch) for $a \in \Psi$ when ω is dense. Each half-space $\{x \in A \mid a(x-\phi) + g(a) > 0\}$ contains a sector whose direction is the vectorial chamber associated with Φ^+, and the same is true for their intersection. We can thus choose a Chevalley-Steinberg valuation x belonging to this intersection and a real number $N \in \omega(K^\times)$ such that

$$-a(x-\phi) < g(a) < -a(x-\phi) + N \quad \text{for all } a \in \Psi.$$

Let us set $f_0(a) = g(a)$ for $a \in \Psi$ and $f_0(a) = -a(x-\phi) + N$ for $a \in \Phi - \Psi$. One easily verifies that $f_0 : \Phi \to \mathbf{R}$ is concave and satisfies (Sch) when ω is dense. Let f be its optimized version. For all $a \in \Psi$, we have

$$\mathfrak{U}_{f,a} = \mathfrak{U}_{a,(g(a),g(2a))}.$$

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Consequently, the group scheme $\mathfrak{U}_{g,\Psi}$ does not depend on the choice of x and N. We will denote it by $\mathfrak{U}_{g,\Psi}$. The product map is an isomorphism of $\prod_{a\in\Psi_{\mathrm{nd}}} \mathfrak{U}_{g,a}$ onto $\mathfrak{U}_{g,\Psi}$.

Let $\Psi'' = \Phi \cap (\Psi' + \Psi')$, denote again by g the restriction of g to Ψ'' and consider the scheme $\mathfrak{U}_{g,\Psi''}$. It follows from the above that $\mathfrak{U}_{g,\Psi''}$ is a closed normal subgroup scheme of $\mathfrak{U}_{g,\Psi}$, normal because the commutator map sends $\mathfrak{U}_{g,a} \times \mathfrak{U}_{g,b}$ into $\mathfrak{U}_{g,\Psi''}$ for any $a, b \in \Psi$. The quotient group scheme $\mathfrak{U}_{g,\Psi}/\mathfrak{U}_{g,\Psi''}$ exists and is isomorphic to the group scheme associated with a free $\mathcal{O}$-module of finite type according to 4.3.9. Let furthermore $h : \Psi \to \mathbf{R}$ satisfying the same conditions as g. Suppose $h \ge g$; the identity of U_Ψ extends to a morphism of group schemes from $\mathfrak{U}_{h,\Psi}$ to $\mathfrak{U}_{g,\Psi}$ (cf. 4.3.10 and 4.6.24), sending $\mathfrak{U}_{h,\Psi''}$ into $\mathfrak{U}_{g,\Psi''}$, thus defining by passing to the quotients a homomorphism from $\mathfrak{U}_{h,\Psi}/\mathfrak{U}_{h,\Psi''}$ to $\mathfrak{U}_{g,\Psi}/\mathfrak{U}_{g,\Psi''}$. If, in addition, $h(a) = g(a)$ for all $a \in \Psi - \Psi''$, then this latter morphism is an isomorphism.

4.6.3. — The groups of integral points $\mathfrak{G}_f^0(\mathcal{O})$ and $\mathfrak{G}_f(\mathcal{O})$ are closely linked to the subgroups of G that we associated with the quasi-concave function f in Chapter I, 6.4.2, whose notation we adopt. Recall that U_f (resp. $U_f^\pm$) is the subgroup generated by the $U_{f,a}$ for $a \in \Phi$ (resp. $a \in \Phi^\pm$), that $N_f = U_f \cap N(K)$ and that $P_f = HU_f$, where $H \subset T(K)$ is the kernel of the homomorphism $\nu : N(K) \to W \subset \mathrm{Aut}\, A$ (I, 6.2.11). Recall also that $G^1 = \theta^{-1}(\{0\})$ is the subgroup of $t \in G(K)$ such that $\omega(\chi(t)) = 0$ for all $\chi \in X_k^*(G)$ (4.2.16). We have seen (4.3.2 (1) and 4.3.5 (3)) that

$$\mathfrak{U}_{f,a}(\mathcal{O}) = U_{f,a} = U_{a,f(a)} \cdot U_{2a,f(2a)} \quad (a \in \Phi), \tag{1}$$

whence by 3.8.3 (S 2) and I, 6.4.9:

$$\mathfrak{U}_f^{\pm}(\mathcal{O}) = U_f^{\pm}. \tag{2}$$

On the other hand, we deduce from 4.4.18 (VII), taking into account the isomorphism of $X_k^*(T) \otimes \mathbf{Q}$ with the direct sum $X_k^*(G) \otimes \mathbf{Q} + X^*(S) \otimes \mathbf{Q}$ (cf. 4.2.16), that

$$\mathfrak{T}(\mathcal{O}) = \operatorname{Ker} \nu \cap \operatorname{Ker} \theta = H \cap G^1 = H^1. \tag{3}$$

We set

$$H^0 = \mathfrak{T}^0(\mathcal{O}), \quad P_f^0 = H^0 U_f \subset \mathfrak{G}_f^0(\mathcal{O}), \quad P_f^1 = H^1 U_f \subset \mathfrak{G}_f(\mathcal{O}), \tag{4}$$

(we will see in 4.6.7 that these two inclusions are equalities). Since $U_f \subset G^1$, we have $P_f^1 = P_f \cap G^1$, in accordance with the conventions of 4.2.16. We finally set

$$N_f^0 = P_f^0 \cap N(K) = H^0 N_f, \quad N_f^1 = P_f^1 \cap N(K) = H^1 N_f^0 = H^1 N_f, \tag{5}$$

$$W_f = \nu(N_f^0) = \nu(N_f^1) = \nu(N_f). \tag{6}$$

Note that (unfortunately) we do not have in general $N_f^1 = N_f \cap G^1$!

Proposition 0.38 (4.6.4). — *(i) $\overline{\mathfrak{S}}$ is a maximal $\overline{k}$-split torus of $\overline{\mathfrak{G}}_f^0$.*
(ii) $\mathfrak{T}^0$ is the centralizer of $\mathfrak{S}$ in $\mathfrak{G}_f^0$; in other words, $\overline{\mathfrak{T}}^0$ is the centralizer of $\overline{\mathfrak{S}}$ in $\overline{\mathfrak{G}}_f^0$.

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Let $\hat{K}$ and $\hat{\mathcal{O}}$ be the respective completions of K and $\mathcal{O}$. Since the base change $K \to \hat{K}$ is conservative, S is still a maximal $\hat{K}$-split torus of G. If $\overline{\mathfrak{S}}$ (which is a $\overline{K}$-split torus of $\overline{\mathfrak{G}}_f^0$) were not maximal, there would exist an integer $n > \dim S$ and a closed immersion $\bar{u} : (\mathfrak{Mult}_{\overline{K}})^n \to \overline{\mathfrak{G}}_f^0$. According to [SGAD], Exp. XI, cor. 4.2, $\bar{u}$ lifts to an $\mathcal{O}$-morphism $u : (\mathfrak{Mult}_{\mathcal{O}})^n \to (\mathfrak{G}_f^0)_{\hat{\mathcal{O}}}$ and, according to [SGAD], Exp. IX, cor. 2.5 and cor. 6.6, u is a closed immersion. But then $\operatorname{Im} u_{\hat{K}}$ is a $\hat{K}$-split torus of G of dimension $> \dim S$, which is impossible. Whence (i).

Quant to assertion (ii), it follows immediately from the decomposition $\mathfrak{C}_f^0 = \mathfrak{U}_f^- \times \mathfrak{T}^0 \times \mathfrak{U}_f^+$ of the open neighborhood $\mathfrak{C}_f^0$ of the identity section, taking into account that the centralizer of a sub-torus in a smooth connected group scheme is smooth and connected ([SGAD], Exp. XI, cor. 5.3 and [2], 11.12).

Corollary. — (i) *For all $a \in \Phi$, the root subgroup of $\overline{\mathfrak{G}}_f^0$ associated with a is the canonical image of $\overline{\mathfrak{U}}_{f,a}$.*
(ii) *For every quasi-closed subset $\Psi \subset \Phi^+$, the unipotent subgroup of $\overline{\mathfrak{G}}_f^0$ associated with Ψ is the canonical image of $\overline{\mathfrak{U}}_{f,\Psi}$.*

This is evident.

4.6.5. Proposition. — (i) $\overline{\mathfrak{G}}_f^0 = \overline{\mathfrak{U}}_f^+(\overline{K}) \cdot \overline{\mathfrak{C}}_f^0 = \mathfrak{U}_f^+(\mathcal{O}) \cdot \overline{\mathfrak{C}}_f^0$.
(ii) $\overline{\mathfrak{G}}_f = \mathfrak{U}_f^+(\mathcal{O}) \cdot \overline{\mathfrak{C}}_f$.

Since $\overline{\mathfrak{C}}_f^0$ is an open neighborhood of the identity element, and since $\mathfrak{U}_f^- \mathfrak{T}^0$ is a closed subgroup of $\mathfrak{G}_f^0$, we have

$$\overline{\mathfrak{G}}_f^0 = (\overline{\mathfrak{C}}_f^0)^{-1} \cdot \overline{\mathfrak{C}}_f^0 = (\overline{\mathfrak{U}}_f^+ \overline{\mathfrak{T}}^0 \overline{\mathfrak{U}}_f^-) \cdot (\overline{\mathfrak{U}}_f^- \overline{\mathfrak{T}}^0 \overline{\mathfrak{U}}_f^+) = \overline{\mathfrak{U}}_f^+ \cdot \overline{\mathfrak{C}}_f^0.$$

If $\overline{K}$ is infinite, $\overline{\mathfrak{U}}_f^+(\overline{K})$ is dense in $\overline{\mathfrak{U}}_f^+$ (which is a $\overline{K}$-vector space), whence the first equality of (i). If $\overline{K}$ is finite, it is perfect, the unipotent radical R of $\overline{\mathfrak{G}}_f^0$ is defined and split over $\overline{K}$ and $\overline{\mathfrak{C}}_f^0 \cdot R = \overline{\mathfrak{C}}_f^0$ (cf. 1.1.11). According to [4], 6.25, we have

$$\overline{\mathfrak{G}}_f^0 / R = (\overline{\mathfrak{U}}_f^+ / (R \cap \overline{\mathfrak{U}}_f^+))(\overline{K}) \cdot (\overline{\mathfrak{C}}_f^0 / R).$$

But R being connected split unipotent, the canonical map from $\overline{\mathfrak{U}}_f^+(\overline{K})$ into $(\overline{\mathfrak{U}}_f^+ / (R \cap \overline{\mathfrak{U}}_f^+))(\overline{K})$ is surjective, whence again the first equality of (i). The second follows from it, since the canonical map from $\mathfrak{U}_f^+(\mathcal{O})$ into $\overline{\mathfrak{U}}_f^+(\overline{K})$ is surjective (4.6.2). Finally (ii) is an immediate consequence of (i) since by definition itself we have $\mathfrak{G}_f = \mathfrak{T} \cdot \mathfrak{G}_f^0$.

4.6.6. Corollary. — *We have $\mathfrak{G}_f^0 = \mathfrak{U}_f^+(\mathcal{O}) \cdot \mathfrak{C}_f^0$ and $\mathfrak{G}_f = \mathfrak{U}_f^+(\mathcal{O}) \cdot \mathfrak{C}_f$. If $\overline{K}$ is infinite and if $\mathfrak{T}^0$ (resp. $\mathfrak{T}$) is plump (for example in cases (A) or (B), or in case (C) if in addition K is strictly henselian), then $\mathfrak{G}_f^0$ (resp. $\mathfrak{G}_f$) are plump.*

Indeed, as above, we see that $G = (\mathfrak{U}_f^+)_K \cdot (\mathfrak{C}_f^0)_K$ and $\mathfrak{U}_f^+(\mathcal{O})$ is dense in $(\mathfrak{U}_f^+)_K$ since the scheme $\mathfrak{U}_f^+$ is associated with a free $\mathcal{O}$-module of finite type. Whence the first two assertions. The last then follows from 1.7.

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Corollary 0.39 (4.6.7). *(i)* $\mathfrak{G}_I^0(\mathcal{O}) = \mathfrak{U}_I^+(\mathcal{O})\mathfrak{C}_I^0(\mathcal{O}) = \mathrm{U}_I^+\mathrm{U}_I^-\mathrm{U}_I^+\mathrm{H}^0 = \mathrm{P}_I^0 = \mathrm{U}_I^+\mathrm{U}_I^-\mathrm{N}_I^0$ *and* $\mathfrak{G}_I(\mathcal{O}) = \mathfrak{U}_I^+(\mathcal{O})\mathfrak{C}_I(\mathcal{O}) = \mathfrak{G}_I^0(\mathcal{O})\mathrm{H}^1 = \mathrm{P}_I^1 = \mathrm{U}_I^+\mathrm{U}_I^-\mathrm{N}_I^1$.

(ii) $\mathfrak{G}_I^0(\overline{K}) = \mathfrak{G}_I^0(\mathcal{O}) \cdot \overline{\mathfrak{T}}^0(\overline{K})$ *and* $\mathfrak{G}_I(\overline{K}) = \mathfrak{G}_I(\mathcal{O}) \cdot \overline{\mathfrak{T}}(\overline{K})$.

(iii) If the canonical map $\mathfrak{T}^0(\mathcal{O}) \to \overline{\mathfrak{T}}^0(\overline{K})$ (resp. $\mathfrak{T}(\mathcal{O}) \to \overline{\mathfrak{T}}(\overline{K})$) is surjective (which is true in cases (A) and (B) or if $\mathcal{O}$ is henselian), then the canonical map $\mathfrak{G}_I^0(\mathcal{O}) \to \overline{\mathfrak{G}}_I^0(\overline{K})$ (resp. $\mathfrak{G}_I(\mathcal{O}) \to \overline{\mathfrak{G}}_I(\overline{K})$) is surjective.

Proof. Let $x \in \mathfrak{G}_I^0(\mathcal{O})$ (resp. $x \in \mathfrak{G}_I(\mathcal{O})$); in view of 4.6.5 there exists $y \in \mathfrak{U}_I^+(\mathcal{O})$ such that the closed point of the section $y^{-1}x$ belongs to $\overline{\mathfrak{C}}_I^0$ (resp. $\overline{\mathfrak{C}}_I$). Since $\mathfrak{C}_I^0$ (resp. $\mathfrak{C}_I$) is open, it follows that $y^{-1}x$ is a section of $\mathfrak{C}_I^0$ (resp. $\mathfrak{C}_I$) and one has $\mathfrak{G}_I^0(\mathcal{O}) = \mathfrak{U}_I^+(\mathcal{O})\mathfrak{C}_I^0(\mathcal{O})$ (resp. $\mathfrak{G}_I(\mathcal{O}) = \mathfrak{U}_I^+(\mathcal{O})\mathfrak{C}_I(\mathcal{O})$). The other equalities of (i) follow, taking into account for the last ones part I, 6.4.9 (iii).
Assertion (ii) follows from 4.6.4 and from the surjectivity of the canonical map $\mathfrak{U}_{I,a}(\mathcal{O}) \to \overline{\mathfrak{U}}_{I,a}(\overline{K})$ and (iii) follows trivially from (ii). □

4.6.8. — If $\mathfrak{X}$ is an $\mathcal{O}$-scheme in groups, let us denote by ${}^1\mathfrak{X}(\mathcal{O})$ the kernel of the canonical map from $\mathfrak{X}(\mathcal{O})$ to $\overline{\mathfrak{X}}(\overline{K})$. With this notation:

Corollary 0.40. *The product map is a bijection from ${}^1\mathfrak{U}_I^-(\mathcal{O}) \times {}^1\mathfrak{T}^0(\mathcal{O}) \times {}^1\mathfrak{U}_I^+(\mathcal{O})$ (resp. ${}^1\mathfrak{U}_I^-(\mathcal{O}) \times {}^1\mathfrak{T}(\mathcal{O}) \times {}^1\mathfrak{U}_I^+(\mathcal{O})$) to ${}^1\mathfrak{G}_I^0(\mathcal{O})$ (resp. ${}^1\mathfrak{G}_I(\mathcal{O})$).*

Indeed, we see as above that ${}^1\mathfrak{G}_I^0(\mathcal{O}) \subset \mathfrak{C}_I^0(\mathcal{O})$ (resp. ${}^1\mathfrak{G}_I(\mathcal{O}) \subset \mathfrak{C}_I(\mathcal{O})$).

4.6.9. — Let us recall that in I, 6.4.1, we embedded the ordered additive monoid $\mathbf{R}$ into an ordered monoid $\widetilde{\mathbf{R}}$, which is none other than the additive monoid of intervals of $\mathbf{R}$ of the form $[k, +\infty[$ or $]k, +\infty[$ [with $-\infty < k \leq +\infty$], equipped with the inverse order of inclusion: we identify $k \in \mathbf{R}$ with the closed interval $[k, +\infty[$. We denote by $k+$ the open interval $]k, +\infty[$ and by ∞ the empty interval. If g is a function on Φ with values in $\mathbf{R}$ and if $a \in \Phi$, we set, as in I, 6.4.23:

$$g^*(a) = g(a) \quad \text{if } g(a) + g(-a) > 0,$$
$$g^*(a) = g(a) + \quad \text{if } g(a) + g(-a) \leq 0.$$

Proposition 0.41 (4.6.10). *(i) The function $f^* : \Phi \to \widetilde{\mathbf{R}}$ is quasi-concave (in the sense of I, 6.4.8).*

(ii) For every $a \in \Phi$, the intersection of the split unipotent radical R of $\overline{\mathfrak{G}}_I^0$ with $\overline{\mathfrak{U}}_{I,a}(\overline{K})$ is the canonical image I_a of the subgroup $U_{f^*,a}$ of $U_{f,a} = \mathfrak{U}_{I,a}(\mathcal{O})$.

(iii) $R(\overline{K})$ is the subgroup of $\mathfrak{G}_I^0(\overline{K})$ generated by the canonical image of the subgroup $U_{f^*} \subset \mathfrak{U}_I$ of $\mathfrak{G}_I^0(\mathcal{O})$ and the group of points rational over $\overline{K}$ of the split unipotent radical of $\overline{\mathfrak{T}}^0$.

Proof. Let us first show that for every $a \in \Phi$, the function f^* satisfies (QC 1)$_a$. For this we may place ourselves in the group G_a generated by U_a, U_{-a} and T and suppose □

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that $\Phi = \{\pm a\}$. If ω is dense or if a is non-multipliable, then f is concave (4.5.6), thus f^* is concave (I, 6.4.23) and satisfies (QC 1). Suppose ω is discrete and a is multipliable and let us return to the concave function f_1 of which f is the optimized (4.5.5). Let us show that the optimized $(f^*)'$ of f^* is equal to the optimized $(f_1^*)'$ of f_1^* (defined as in I, 6.4.10). To this end, let us distinguish several cases:

a) If $f(a) + f(-a) > 0$ and $f(2a) + f(-2a) > 0$, then $f = f^* = f' = f_1'$ and $f_1 \leq f_1^* \leq f^*$, whence $(f^*)' = f_1' \leq (f_1^*)' \leq (f^*)'$;

b) If $f(2a)+f(-2a)=0$, then $f=f_1$ (4.5.5);

c) If $f(2a)+f(-2a)>0$ and $f(a)+f(-a)=0$, then the extension L_a of L_{2a} is ramified, $f(\pm 2a)=2f(\pm a)+c$ (4.3.5), $f_1(\pm a)=f(\pm a)$ and $f_1(\pm 2a)=2f(\pm a)$; furthermore, $f(a)\in\Gamma'_a$. We then verify without difficulty that

$$(f_1^*)'(\pm a)=f_1(\pm a)+c=f(\pm a)+c=(f^*)'(\pm a)$$
$$(f_1^*)'(\pm 2a)=2f(\pm a)+c=f(\pm 2a)=(f^*)'(\pm 2a)$$

whence our assertion.

Since f_1 is concave, the same holds for f_1^* (I, 6.4.23), its optimized is quasi-concave, thus f^* is quasi-concave. Returning to the group G itself, this shows that f^* satisfies (QC 1)$_a$ for all $a\in\Phi$. Moreover, according to I, 6.4.23 and 27, there exists a subgroup X of H^1 such that $U_{f^*,-a}XU_{f^*,a}$ is a subgroup of G^1 normalized by $U_{f,a}$ and $U_{f,-a}$. It follows that $J_a=\overline{\mathfrak{U}}_{f,-a}(\overline{K})\overline{\mathfrak{T}}^0(\overline{K})I_a$ is a subgroup of $\mathfrak{G}_f^0(K)$.

Let us now prove (ii). Here again we may suppose $G=G_a$ (1.1.11). We shall carry out the proof in the case where a is multipliable, the non-multipliable case being analogous but significantly simpler. Let q be the canonical map of $\overline{\mathfrak{G}}_f^0$ onto its quotient by R and let $\xi\in I_a$. If $q(\xi)\neq 1$, it follows from the existence of the root datum of $\overline{\mathfrak{G}}_f^0$ recalled in 1.1.13, from the properties of root data (axioms (DR 4) and (DR 5) of I, 6.1.1) and from the surjectivity of $q:\overline{\mathfrak{U}}_{f,a}(\overline{K})\to q(\overline{\mathfrak{U}}_{f,a})(\overline{K})$ (cf. 1.1.12) that there exist $\xi',\xi''\in\overline{\mathfrak{U}}_{f,-a}(\overline{K})$ such that $q(\xi'\xi\xi'')$ normalizes $q(\overline{\mathfrak{S}})$ and induces therein the reflection with respect to a. But ξ' and ξ'' belong to the subgroup J_a introduced above,

$$q(\xi'\xi\xi'')\in q(\overline{\mathfrak{U}}_{f,-a})(\overline{K})\cdot q(\overline{\mathfrak{T}}^0)(\overline{K})\cdot q(\overline{\mathfrak{U}}_{f,a})(\overline{K})$$

and the properties of the Bruhat decomposition of $(\overline{\mathfrak{G}}_f^0/R)(\overline{K})$ associated with the root datum of $\overline{\mathfrak{G}}_f^0/R$ (I, 6.1.15) then imply that $q(\xi'\xi\xi'')\subset q(\overline{\mathfrak{T}}^0)$, so $q(\xi'\xi\xi'')$ centralizes $q(\overline{\mathfrak{S}})$, whence a contradiction. Consequently

$$I_a\subset\overline{\mathfrak{U}}_{f,a}(\overline{K})\cap R. \tag{1}$$

It remains to show that if $\xi\in\overline{\mathfrak{U}}_{f,a}(\overline{K})$ and $\xi\notin I_a$, then $\xi\notin R$. Given the surjectivity of the canonical map $\mathfrak{U}_{f,a}(\mathcal{O})\to\overline{\mathfrak{U}}_{f,a}(\overline{K})$, there exists $x\in\mathfrak{U}_{f,a}(\mathcal{O})$ with image ξ, and $x\notin U_{f^*,a}$. If $x\notin U_{f^*,a}\cdot U_{2a,f(2a)}$ (which requires $f^*(a)>f(a)$, thus $f(a)+f(-a)=0$), we have, with the notation of 4.1.9 and 4.3.5, $x=x_a(u,v)$ with $\omega(u)=f(a)+\gamma$ and $\omega(v)=2f(a)$.

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The elements $x'=x_{-a}(uv^{-1},v^{-\sigma})$ and $x''=x_{-a}(uv^{-\sigma},v^{-\sigma})$ then belong to $U_{f,-a}$ and $x'xx''\in\mathfrak{G}_I^0(\mathcal{O})$. But 4.1.11 implies that $x'xx''$ normalizes S and induces the reflection with respect to a on it. Consequently, the image of $x'xx''$ in $\overline{\mathfrak{G}}_I^0/R$ normalizes $q(\overline{S})$ and does not centralize it, which implies $q(\overline{\xi})\neq 1$ and $\overline{\xi}\notin R$. If now $x\in U_{f^*,a}\setminus U_{2a,f(2a)}$, one is reduced, by multiplying it by an element of $U_{f^*,a}$ (which according to (1) multiplies ξ by an element of R), to the case where $x\in U_{2a,f(2a)}$ and $x\notin U_{2a,f^*(2a)}$, which requires $f(2a)+f(-2a)=0$. We then have $x=x_a(0,v)$ with $\omega(v)=f(2a)$ and we finish as in the first case by considering the elements $x'=x''=x_{-a}(0,v^{-\sigma})$. This completes the proof of (ii).

Assertion (iii) follows from (ii) since R is the product of its intersections with the $\overline{\mathfrak{U}}_{f,a}$ and the split unipotent radical of $\overline{\mathfrak{T}}^0$ (1.1.11).

Finally, let us finish the proof of (i). We have already seen that f^* satisfies $(QC\,1)_a$ for all $a\in\Phi$. Moreover, one easily checks that $U_{f^*,a}$ contains the kernel of the canonical map $\mathfrak{U}_{f,a}(\mathcal{O})\to\mathfrak{U}_{f,a}(\overline{K})$, hence is the inverse image of I_a in $U_{f,a}$. For any positively closed subset Ψ of Φ, the product of the I_a for $a\in\Psi$ is a group (1.1.7); it is then the same for the product of the $U_{f^*,a}$ for $a\in\Psi$, from which $(QC\,2)$ follows easily.

4.6.11. Remark. — If $\overline{K}$ is infinite (and one can reduce to this case by a conservative étale base change), $R(\overline{K})$ is dense in R and 4.6.10 completely determines R.

Corollary 0.42 (4.6.12)**.** *(i) The root system of $\overline{\mathfrak{G}}_I^0/R$ with respect to the maximal $\overline{K}$-split torus $q(\overline{S})$ (identified with $\overline{S}$) is the set Φ_f of $a\in\Phi$ such that $f(a)+f(-a)=0$.*
(ii) The canonical map from Aut A into Aut V^* induces an isomorphism of W_f onto the Weyl group of Φ_f (acting naturally on the subspace of V^* generated by Φ_f and trivially on its orthogonal complement).
(iii) Suppose $\overline{K}$ is separably closed and let R' be the unipotent radical of $\overline{\mathfrak{G}}_I^0$. The root system of the reductive group $\overline{\mathfrak{G}}_I^0/R'$ (defined over the algebraic closure of $\overline{K}$) is the set Φ_f^{nm} of non-multipliable elements of Φ_f.

Assertion (i) results from 4.6.10 and 1.1.11; (ii) results from I, 6.4.10 and (iii) from [7], th. 3, taking into account that, when $\overline{K}$ is separably closed, $q(\overline{S})$ is a maximal $\overline{K}$-split torus.

Corollary 0.43 (4.6.13)**.** *Let $\mathcal{Q}^0$ (resp. $\mathcal{Q}$) be the normalizer of $\overline{S}$ in $\overline{\mathfrak{G}}_I^0$ (resp. $\overline{\mathfrak{G}}_I$). Every coset of $\mathcal{Q}^0$ (resp. $\mathcal{Q}$) according to $\overline{\mathfrak{T}}^0$ (resp. $\overline{\mathfrak{T}}$) contains the canonical image of an element of N_I^0.*

Indeed, we know that $\mathcal{Q}^0 = \mathcal{Q}^0(\overline{K}).\overline{\mathfrak{T}}^0$ (1.1.15), from which $\mathcal{Q} = \mathcal{Q}^0(\overline{K}).\overline{\mathfrak{T}}$, and that $\mathcal{Q}^0(\overline{K})/\overline{\mathfrak{T}}^0(\overline{K}) = W_f$ (1.1.13), from which the corollary follows since $N_I^0/H^0 = W_f$.

It will be noted that the scheme of connected components of the normalizer of $\overline{S}$ in $\mathfrak{G}_I^0$ is in general not constant; its generic fiber is the Weyl group of Φ and its closed fiber that of Φ_f (it can be reduced to the identity element).

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Corollary 0.44 (4.6.14)**.** *Let $a \in \Phi$. The root subgroup V_a of the quasi-reductive quotient ${}^q\overline{\mathfrak{G}}_f^0$ of $\overline{\mathfrak{G}}_f^0$ (1.1.12) associated to a is $\overline{K}$-isomorphic to:*

- *$\{1\}$ if $a \cap \Phi_f = \emptyset$;*
- *$\Pi_{\overline{L}_a/\overline{K}}\mathbb{A}$ if $a \in \Phi_f$ and $2a \notin \Phi_f$ (which implies, when a is plural, that the extension L_a of L_{2a} is ramified);*
- *$\Pi_{\overline{L}_{2a}/\overline{K}}\mathbb{A}$ if $a \notin \Phi_f$ and $2a \in \Phi_f$;*
- *$H(\overline{L}_a, \overline{L}_{2a})$ (for the notation, see 4.1.9) if a and $2a \in \Phi_f$ (which implies that the extension L_a of L_{2a} is unramified) and if the residue extension $\overline{L}_a$ of $\overline{L}_{2a}$ is separable;*
- *$\Pi_{\overline{L}_a/\overline{L}_{2a}}\mathbb{A} \times \Pi_{\overline{L}_{2a}/\overline{K}}\mathbb{A}$ if a and $2a \in \Phi_f$ (which implies that the extension L_a of L_{2a} is unramified) and if the residue extension $\overline{L}_a$ of $\overline{L}_{2a}$ is inseparable.*

The proof is easy and we leave it to the reader.

Corollary 0.45 (4.6.15)**.** *Assume $\overline{K}$ is separably closed. Then $\overline{\mathfrak{G}}_f^0$ is reductive if and only if G is split over K and if there exists a special point $x \in A$ such that $f(a) = -a(x - \phi)$ for all $a \in \Phi$. The scheme $\overline{\mathfrak{G}}_f^0$ is then a Chevalley scheme (3.2.13).*

Recall that $x \in A$ is said to be *special* if $a(x - \phi) \in \Gamma_a$ for all $a \in \Phi$ (I, 6.2.13). When G is split over K, this means that x is a Chevalley valuation of G (4.2.1); the corresponding Chevalley scheme is the scheme $\overline{\mathfrak{G}}_f^0$ with $f(a) = f_x(a) = -a(x - \phi)$ ($a \in \Phi$): the reader will verify that the two definitions are identical.

Assume $\overline{K}$ is separably closed; 4.6.12 (iii) then implies that $\overline{\mathfrak{G}}_f^0$ is reductive if and only if $\dim \overline{\mathfrak{G}}_f^0 = \operatorname{Card} \Phi_f^{nm} + \dim \overline{\mathfrak{S}}$, that is to say

$$\dim G = \operatorname{Card} \Phi_f^{nm} + \dim S. \tag{1}$$

But, $\dim G = \sum_{a \in \Phi} \dim U_a + \dim T$ and (1) is therefore equivalent to the set of three relations

$$\dim S = \dim T, \tag{2}$$
$$\Phi_f^{nm} = \Phi, \tag{3}$$
$$\dim U_a = 1 \quad \text{for all } a \in \Phi. \tag{4}$$

The relation (2) says that G is split over K and implies (4), and (3) then says that

$$f(a) + f(-a) = 0 \quad \text{for all } a \in \Phi. \tag{5}$$

Let B be a base of Φ and let x be the point of A defined by the relations $a(x - \phi) + f(a) = 0$ for $a \in B$. Since f is optimal, one has $f(a) \in \Gamma_a$ for all $a \in B$ and x is a special point of A (I, 6.2.14). It then follows from I, 6.4.10 (1) that (5) is equivalent to $f = f_x$, which completes the proof.

4.6.16. — The scheme $\mathfrak{G}_f$ and the other notions that we have associated with f depend, of course, not only on the function $f : \Phi \to \mathbf{R}$ but also on the valuation of

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Chevalley-Steinberg valuation φ which we fixed in 4.2. Let $\varphi' = \varphi + v$ be another Chevalley-Steinberg valuation; we denote by a left superscript the notions attached to φ or to φ'. Since $a(v) \in \omega(L_a^\times)$ for all $a \in \Phi$

(4.2.10), we see immediately that if $k \in \mathbb{R}$ (resp. $(k,\ell) \in \mathbb{R}^2$) satisfies the conditions of 4.3.2 (resp. 4.3.5), the same is true of $k + a(v)$ (resp. $(k + a(v), \ell + 2a(v))$) and that

$$^{\varphi}\mathfrak{U}_{a,k} = {}^{\varphi'}\mathfrak{U}_{a,k+a(v)} \quad \text{if } a \text{ is non-plural,}$$
$$^{\varphi}\mathfrak{U}_{a,(k,\ell)} = {}^{\varphi'}\mathfrak{U}_{a,(k+a(v),\ell+2a(v))} \quad \text{if } a \text{ is plural,}$$

whence

$$^{\varphi}\mathfrak{G}_f = {}^{\varphi'}\mathfrak{G}_{f'} \quad \text{with } f'(a) = f(a) + a(\varphi' - \varphi) \quad \text{for } a \in \Phi. \tag{1}$$

This allows us to extend our notations to the case of any valuation φ' equipollent to φ, by taking (1) as a definition. We shall say that the pairs (φ, f) and (φ', f') are equivalent.

4.6.17. — The group $N = N(K)$ acts by transport of structure on the set of pairs (ψ, g) formed by a valuation equipollent to φ and a function $g : \Phi \to \mathbb{R}$ and this operation is compatible with the equivalence relation. Since each equivalence class contains one and only one element of the form (φ, g), we deduce from this an action of N on the set F of functions $g : \Phi \to \mathbb{R}$ and a simple calculation shows that for $n \in N$ and $a \in \Phi$, we have

$$(n.g)(a) = g({}^t({}^v\nu(n)).a) - a(\nu(n).\varphi - \varphi). \tag{1}$$

In a general way, if u is an automorphism of the affine space A such that ${}^t({}^vu)$ preserves Φ, we define $u.g$ by

$$(u.g)(a) = g({}^t({}^vu).a) - a(u.\varphi - \varphi) \quad (a \in \Phi). \tag{1 bis}$$

The subgroup H of N then acts trivially on F, whence an action of $W = N/H$ on F. It is clear that the inner automorphism of G defined by n extends to an isomorphism of $\mathfrak{G}_f$ onto $\mathfrak{G}_{n.f}$, and in a general way transforms all the objects that we have attached to f into the analogs attached to $n.f$. For example, we have $nU_fn^{-1} = U_{n.f}$, $nN_f^0n^{-1} = N_{n.f}^0$, etc. We set

$$N_f^\dagger = \{n \in N \cap G^1 \mid n.f = f\}$$
$$W_f^\dagger = \nu(N_f^\dagger) = N_f^\dagger/H^1.$$

It is immediate, given (1), that $W_f^\dagger$ contains no non-trivial translation. The canonical map $w \to {}^vw$ from W onto the Weyl group of Φ thus allows us to identify $W_f^\dagger$ with a subgroup of ${}^vW$.

The group N_f^0 is contained in $N_f^\dagger$, and is a normal subgroup of it: this is almost obvious, particularly from the fact that the inner automorphism defined by an element of N_f^0 permutes the subgroups $U_{f,a}$ among themselves, but this follows even more easily from I, 6.4.10 (1). Consequently, W_f is a normal subgroup of $W_f^\dagger$.

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4.6.18. Proposition. — *There exists a smooth $\mathcal{O}$-group scheme $\mathfrak{G}_J^\dagger$ with generic fiber G, containing $\mathfrak{G}_J$ as a normal open group sub-scheme, such that $N^\dagger \subset \mathfrak{G}_J^\dagger(\mathcal{O})$, that $\mathfrak{G}_J^\dagger = N^\dagger \cdot \mathfrak{G}_J$, and that $\mathfrak{G}_J^\dagger/\mathfrak{G}_J \simeq N^\dagger/N_J^\dagger$.*

Let us first show a lemma, whose statement and proof result from a discussion with P. Deligne: his observations allowed us to simplify our original text.

Lemma. — *Let $\mathfrak{X}$ be an $\mathcal{O}$-group scheme, not necessarily affine, flat and separated, and let Y be a subgroup of $\mathfrak{X}(K)$ such that, for every $y \in Y$, the inner automorphism of $\mathfrak{X}_K$ defined by y extends (in a necessarily unique way) to an automorphism of $\mathfrak{X}$, denoted by i_y. There exists an $\mathcal{O}$-group scheme $\tilde{\mathfrak{X}}$, not necessarily affine, flat and separated, with generic fiber $\mathfrak{X}_K$, containing $\mathfrak{X}$ as a normal open group sub-scheme, such that $Y \subset \tilde{\mathfrak{X}}(\mathcal{O})$ and that $\tilde{\mathfrak{X}} = \mathfrak{X} \cdot Y$.*

Let Y' be a system of representatives of $Y/Y \cap \mathfrak{X}(\mathcal{O})$ in Y, containing 1. For $y \in Y'$, let $\mathfrak{X}_y$ be a copy of $\mathfrak{X}$ (which will be the translate of $\mathfrak{X}$ by y). Let $x \mapsto x_y$ denote the chosen isomorphism from $\mathfrak{X}$ to $\mathfrak{X}_y$ and let $\tilde{\mathfrak{X}}$ be the $\mathcal{O}$-scheme, not necessarily affine but flat, obtained by gluing the $\mathfrak{X}_y$ for $y \in Y'$ along the generic fibers in the natural way, that is, by identifying $(\mathfrak{X}_y)_K$ with $(\mathfrak{X}_z)_K$ via the isomorphism $x \mapsto xy^{-1}z$. We identify $\mathfrak{X}$ and $\mathfrak{X}_1$ and we identify $\mathfrak{X}_y$ with its (open) image in $\tilde{\mathfrak{X}}$.

The maps

$$(x_y, x'_z) \mapsto (xi_y(x')yz(z')^{-1})_{z'}$$

from $\mathfrak{X}_y \times \mathfrak{X}_z$ to $\mathfrak{X}_{z'}$ (where $y, z \in Y'$ and where z' is the representative of yz in Y') glue together and define a morphism of schemes $\tilde{\mathfrak{X}} \times \tilde{\mathfrak{X}} \to \tilde{\mathfrak{X}}$, and it is easily verified that this morphism is a group law on $\tilde{\mathfrak{X}}$, whose neutral element is the identity section e of $\mathfrak{X}_1$, whose inverse map is obtained by gluing the morphisms

$x_y \mapsto (i_{y^{-1}}(x^{-1}) \cdot (y'y)^{-1})_{y'}$, where y' is the representative of y^{-1} in Y', which induces on $\mathfrak{X} = \mathfrak{X}_1$ the given group law and which makes $\mathfrak{X}$ a normal open group sub-scheme of $\tilde{\mathfrak{X}}$.

The only thing that remains for us to demonstrate is that $\tilde{\mathfrak{X}}$ is separated, or equivalently that the identity section is closed ([DG], p. 165). Suppose the contrary; there then exists $y \in Y'$, $y \neq 1$ such that the generic point e_K of e is not closed in $\mathfrak{X}_y$ or equivalently, since $e_K = (y^{-1})_y$, such that y^{-1} (considered as a closed point of the generic fiber of $\mathfrak{X}$) is not closed in $\mathfrak{X}$. Let then $\mathfrak{F}$ be the schematic closure of y^{-1} in $\mathfrak{X}$ and let $\mathfrak{F}_0$ be an affine open subset of $\mathfrak{F}$ such that $\mathfrak{F}_0 \cap \mathfrak{X}_s \neq \emptyset$. Since $\mathfrak{F}$, and therefore $\mathfrak{F}_0$, is flat (1.2.6), the affine algebra R of $\mathfrak{F}_0$ injects into $R \otimes K$. On the other hand, we have $R \otimes K = K$ since the generic fiber of $\mathfrak{F}_0$ (or of $\mathfrak{F}$) is reduced to the point $y^{-1} \in \mathfrak{X}(K)$. But, a sub-$\mathcal{O}$-algebra of K is equal either to $\mathcal{O}$ or to K, and we cannot have $R = K$ since $\mathfrak{F}_0 \cap \mathfrak{X}_s \neq \emptyset$. Therefore, $R = \mathcal{O}$ and $\mathfrak{F}_0$ is a section of $\mathfrak{X}$. In other words $y^{-1} \in \mathfrak{X}(\mathcal{O})$ and $y = 1$, whence a contradiction. The lemma is proven.

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4.6.19. — From the preceding lemma results the existence of an $\mathcal{O}$-scheme $\mathfrak{G}_I^\dagger$, a priori not necessarily affine, but quasi-compact since $N_I^\dagger/N_I^0$ is finite and possessing the other required properties, and notably smoothness since $\mathfrak{G}_I$ is smooth. It remains therefore only to demonstrate that $\mathfrak{G}_I^\dagger$ is affine.

Let us first show that $\mathfrak{G}_I^\dagger$ is quasi-affine (i.e. isomorphic to a quasi-compact open subset of an affine scheme). Since $\mathfrak{G}_I^\dagger$ is smooth, quasi-compact, separated and with affine generic fiber, it suffices, according to a theorem of M. Raynaud ([25], Th. X.10, p. 153), to show that the centralizer of $\mathfrak{S}$ in $\mathfrak{G}_I^\dagger$ is quasi-affine. We will even show that this centralizer is contained in $\mathfrak{G}_I$, and thus is affine. It suffices to show that, for every extension k of $\bar{K}$ and every $x \in \mathfrak{G}_I^\dagger(k)$ centralizing $\mathfrak{S}$, one has $x \in \mathfrak{G}_I(k)$. Now one can write $x = yx'$, with $y \in N_I^\dagger$ and $x' \in \mathfrak{G}_I(k)$; then x' normalizes $\mathfrak{S}$ and induces on $X^*(\mathfrak{S}) = X^*(S)$ an element of W_I (4.6.12). But x centralizes $\mathfrak{S}$; consequently, y induces on $\mathfrak{S}$ an element of W_I, thus $y \in N_I^0$ and $x \in \mathfrak{G}_I(k)$, which was to be demonstrated.

Another theorem of M. Raynaud ([25], prop. VII. 3.1, p. 117) then implies that the canonical morphism from $\mathfrak{G}_I^\dagger$ into $\mathfrak{G}' = \operatorname{Spec} \mathcal{O}[\mathfrak{G}_I^\dagger]$ endowed with its natural structure of (affine) group $\mathcal{O}$-scheme ([SGAD], Exp. VI_B, 11.2, p. 396) is an isomorphism onto an open subgroup of $\mathfrak{G}'$. Moreover, $K[\mathfrak{G}'] = K \otimes \mathcal{O}[\mathfrak{G}_I^\dagger] = K[\mathfrak{G}_I^\dagger]$, because one sees immediately by considering a covering of $\mathfrak{G}_I^\dagger$ by a finite number of affine open sets, that for all $f \in K[\mathfrak{G}_I^\dagger]$, there exists $a \in \mathcal{O}$ such that $af \in \mathcal{O}[\mathfrak{G}_I^\dagger]$. Lemma 2.2.6 then implies that $\mathfrak{G}_I^\dagger$ is affine (and consequently equal to $\mathfrak{G}'$).

The proof of 4.6.18 is complete.

4.6.20. Remarks. — a) If ω is discrete, a theorem of Raynaud (unpublished) ensures that every separated, flat group $\mathcal{O}$-scheme of finite type and with affine generic fiber is itself affine: this makes 4.6.19 unnecessary in this case.

b) The schemes $\mathfrak{G}_I^\dagger$ and $\mathfrak{G}_I$ are stable by conservative étale base change, and even by any conservative base change in each of the four cases (A) to (D) of 4.6.1. We do not know if the same is always true for $\mathfrak{G}_I^\dagger$. It is however exact in each of the cases (A) and (B), or if the extension $\tilde{K}$ of K is unramified: it results indeed from 4.4.17 that $W_I^\dagger$ then remains stable.

4.6.21. — Let $\mathscr{Y}$ (resp. $\mathscr{Y}^1$) be the set of subgroups of $N_I^\dagger/N_I^0$ (resp. N_I^1/N_I^0). For $Y \in \mathscr{Y}$ (resp. $Y \in \mathscr{Y}^1$), let YN_I^0 (resp. YN_I^1) be its inverse image in $N_I^\dagger$ and let us set $Y\mathfrak{G}_I^0 = YN_I^0 \cdot \mathfrak{G}_I^0$ (resp. $Y\mathfrak{G}_I = YN_I^1 \cdot \mathfrak{G}_I$). Since $\mathfrak{G}_I$ is open in $\mathfrak{G}_I^\dagger = N_I^\dagger \cdot \mathfrak{G}_I$, we have $\mathfrak{G}_I^\dagger(\mathcal{O}) = N_I^\dagger \mathfrak{G}_I(\mathcal{O})$ and, given 4.6.7 (i), $\mathfrak{G}_I^\dagger(\mathcal{O}) = N_I^\dagger \mathfrak{G}_I^0(\mathcal{O})$. The following proposition results immediately:

Proposition 0.46. *(i) The map $Y \mapsto Y\mathfrak{G}_I$ is an isomorphism of ordered sets (by inclusion) from $\mathscr{Y}^1$ onto the set of open group sub-schemes of $\mathfrak{G}_I^\dagger$ containing $\mathfrak{G}_I$.*
(ii) The map $Y \mapsto Y\mathfrak{G}_I^0$ is an isomorphism of ordered sets (by inclusion) from $\mathscr{Y}$ onto the set of open group sub-schemes $\mathfrak{G}'$ of $\mathfrak{G}_I^\dagger$ such that $\mathfrak{G}' = \mathfrak{G}'(\mathcal{O}) \cdot \mathfrak{G}_I^0$.

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Let us remark that if $\mathfrak{T} = \mathfrak{T}(\mathcal{O}) \cdot \mathfrak{T}^0$ (for example if $\mathfrak{T}$ is connected, notably in cases (A) and (B) of 4.6.1, or if $\mathcal{O}$ is strictly Henselian), then $\mathfrak{G}_f = \mathfrak{T}(\mathcal{O}) \cdot \mathfrak{G}_f^0$ and $\mathfrak{G}_f^\dagger = \mathbf{N}_f^\dagger \cdot \mathfrak{G}_f^0$. Consequently, every open subgroup scheme $\mathfrak{G}'$ of $\mathfrak{G}_f^\dagger$ satisfies the condition of (ii), and we have $\mathfrak{G}_f^\dagger = (\mathbf{N}_f^\dagger/\mathbf{N}_f^0)\mathfrak{G}_f^0$ and $\mathfrak{G}_f = (\mathbf{N}_f/\mathbf{N}_f^0)\mathfrak{G}_f^0$. Recall that $\mathbf{N}_f^\dagger/\mathbf{N}_f^0 \simeq \mathfrak{T}(\mathcal{O})/\mathfrak{T}^0(\mathcal{O}) = H^1/H^0$.

Corollary 0.47. *Let $\mathfrak{H}$ be a smooth $\mathcal{O}$-group scheme with generic fiber G, such that $\mathfrak{H} = \mathfrak{H}(\mathcal{O}) \cdot \mathfrak{H}^0$ (a condition automatically satisfied if $\mathcal{O}$ is strictly Henselian). Suppose that the canonical injection of $(\mathfrak{C}_f^\circ)_K$ (resp. $(\mathfrak{C}_f)_K$) into G extends to an isomorphism of schemes from $\mathfrak{C}_f^\circ$ (resp. $\mathfrak{C}_f$) onto an open subscheme of $\mathfrak{H}$. Then, there exists a unique subgroup Y of $\mathbf{N}_f^\dagger/\mathbf{N}_f^0$ (resp. of $\mathbf{N}_f/\mathbf{N}_f^0$) such that the identity of G extends to an isomorphism of group schemes from $\mathfrak{H}$ onto $Y\mathfrak{G}_f^0$ (resp. onto $Y\mathfrak{G}_f$).*

This follows from 1.2.13 and 14.

4.6.22. — Suppose G is split over K and let x be a special point of A. Let us take $f = f_x$, so that $\mathfrak{G}_f^0$ is the Chevalley scheme associated with x (4.6.15). Since $\Phi_f = \Phi$, the image of W_f in ${}^vW$ is the whole of ${}^vW$ and one has $W_f^\dagger = W_f$. Since, moreover, $\mathfrak{T}$ is connected, we see that $\mathfrak{G}_f^0 = \mathfrak{G}_f = \mathfrak{G}_f^\dagger$.

It follows that if $\mathfrak{H}$ is a smooth $\mathcal{O}$-group scheme with generic fiber G, whose identity component is a Chevalley scheme, then $\mathfrak{H} = \mathfrak{H}^0$: it suffices to apply the preceding corollary, after possibly performing the base change $K \to {}^hK$, which is conservative since $\tilde{K} = K$.

4.6.23. — When ω is discrete, one can in all of the preceding remove the hypothesis that f is optimal: if $f : \Phi \to \mathbf{R}$ is any quasi-concave function, one considers its optimization f', which is optimal quasi-concave. We then have $\mathfrak{U}_{f,a} = \mathfrak{U}_{f',a}$ for all $a \in \Phi$, and we set $\mathfrak{G}_f = \mathfrak{G}_{f'}$, etc. One should, however, be careful that the root system $\Phi_f = \Phi_{f'}$ is not necessarily the set of $a \in \Phi$ such that $f(a) + f(-a) = 0$. If f is concave, Φ_f is the set of $a \in \Phi$ such that $f(a) \in \Gamma'_a$ and $f(a) + f(-a) = 0$.

Proposition 0.48 (4.6.24)**.** *Let $g : \Phi \to \mathbf{R}$ be another optimal quasi-concave function, satisfying (Sch) when ω is dense and such that $f \leq g$.*

(i) The identity of G extends to a morphism denoted $i_{f,g}^0$ (resp. $i_{f,g}$) from $\mathfrak{G}_g^0$ (resp. $\mathfrak{G}_g$) to $\mathfrak{G}_f^0$ (resp. $\mathfrak{G}_f$).

(ii) Let moreover Y (resp. Y') be a subgroup of $\mathbf{N}_f^\dagger/\mathbf{N}_f^0$ (resp. $\mathbf{N}_g^\dagger/\mathbf{N}_g^0$). If $Y'\mathbf{N}_g^0 \subset Y\mathbf{N}_f^0$, the identity of G extends to a morphism denoted $i_{f,g}^{Y,Y'}$ from $Y'\mathfrak{G}_g^0$ to $Y\mathfrak{G}_f^0$.

For all $a \in \Phi$, the identity of U_a extends to a morphism from $\mathfrak{U}_{g,a}$ to $\mathfrak{U}_{f,a}$ (4.3.10), whence by product an extension of the identity of $\mathfrak{C}_K$ to a morphism of schemes from $\mathfrak{C}_g^\circ$ to $\mathfrak{C}_f^\circ$. It then suffices to apply 1.2.13. Let us remark on the one hand that $\mathbf{N}_g^0 \subset \mathbf{N}_f^0$, since $U_g \subset U_f$, and on the other hand that $\Phi_g \subset \Phi_f$: more precisely, we have $\Phi_g = \{a \in \Phi_f \mid f(\pm a) = g(\pm a)\}$.

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4.6.25. Proposition. — *Keep the notations of 4.6.24 and suppose further that $\Phi_f = \Phi_g$.*

(i) The image under $\bar{i}_{f,g}^0$ of the split unipotent radical of $\overline{\mathfrak{G}}_f^0$ is contained in the split unipotent radical of $\overline{\mathfrak{G}}_g^0$.

(ii) $\bar{i}_{f,g}^0$, defines by passing to the quotients an isomorphism ${}^q\bar{i}_{f,g}^0$ of the quasi-reductive group ${}^q\overline{\mathfrak{G}}_f^0$ (cf. 1.1.12) onto ${}^q\overline{\mathfrak{G}}_g^0$.

We can suppose $\bar{K}$ infinite. Since $\Phi_f = \Phi_g$, we have $f^* \leq g^*$ and (i) follows immediately from 4.6.10. Moreover, $\bar{i}_{f,g}^0$ is the "identity" on $\bar{\mathfrak{T}}^0$ and defines by passing to the quotients a morphism j_a of $q(\bar{\mathfrak{U}}_{f,a})$ into $q(\bar{\mathfrak{U}}_{g,a})$ (for $a \in \Phi$); it is easily verified that j_a is an isomorphism (cf. 4.6.14), hence (ii).

4.6.26. — Let Ω be a non-empty bounded subset of the apartment A. We associate to it a function $f_\Omega : \Phi \to \mathbf{R}$ by setting

$$f_\Omega(a) = \inf\{k \in \mathbf{R} \mid a(x - \phi) + k \geq 0 \text{ for all } x \in \Omega\} \quad (a \in \Phi)$$

(I, 7.1.1). This is obviously a concave function. But when ω is dense, f_Ω does not always satisfy (Sch) (unless $\omega(K^*) = \mathbf{R}$!): if for example $\Omega = \{x\}$, then f_Ω only satisfies (Sch) if $a(x - \phi) \in \Gamma'_a$ for all $a \in \Phi$, which implies that x is a special point (and even hyperspecial: cf. [40], I.10.2, p. 36). Until 4.6.31, all subsets of A denoted Ω (possibly with indices or primes) are assumed such that, when ω is dense, f_Ω satisfies (Sch). According to I, 7.4.9, this property only depends on Ω as a subset of $\mathcal{I}$ and not on the chosen apartment $A \supset \Omega$. The optimized function f'_Ω of f_Ω therefore satisfies the conditions imposed on f in 4.6.1 and we can consider the schemes $\mathfrak{G}_{f'_\Omega}, \ldots$ and the subgroups $U_{f'_\Omega}, \ldots$ which we will denote more simply $\mathfrak{G}_\Omega, \ldots$ and $U_\Omega, \ldots$ (in accordance with I, 7.1, with one exception, due to typographical reasons: the group we denote henceforth $\hat{N}_\Omega^1$ is not the one denoted as such in I, 4.1.4 and 7.3.9, but the intersection of the latter with G^1). In I, 7.1.8, we denoted $\hat{N}_\Omega$ the fixer of Ω in N. The group $\hat{N}_\Omega^1 = \hat{N}_\Omega \cap G^1$ contains N_Ω^1 and is contained in $\hat{N}_\Omega$. One sets

$$\hat{\mathfrak{G}}_\Omega = Y\mathfrak{G}_\Omega \quad \text{with } Y = \hat{N}_\Omega^1/N_\Omega^1.$$

4.6.27. — The function f'_Ω (and all that is associated with it) only depends on the enclosure $cl(\Omega)$ of Ω (I, 7.1.2), which is the intersection of the affine roots $\{x \in A \mid a(x - \phi) + f'_\Omega(a) \geq 0\}$ for $a \in \Phi$. If $\Omega \subset \Omega'$, then $f'_\Omega \leq f'_{\Omega'}$, and one can apply 4.6.24 and 25. One sets $i_{\Omega,\Omega'} = i_{f'_\Omega, f'_{\Omega'}}$, etc.

4.6.28. Proposition. — *(i) $\hat{P}_\Omega^1 = \hat{\mathfrak{G}}_\Omega(\mathcal{O})$ is the fixer of Ω in G^1 and $P_\Omega^0 = \mathfrak{G}_\Omega^0(\mathcal{O})$ is a subgroup, which is called the connected fixer of Ω.*

(ii) If Ω is closed (i.e. equal to its enclosure $cl(\Omega)$), then $\hat{\mathfrak{G}}_\Omega^\dagger(\mathcal{O}) = N_\Omega^\dagger \cdot P_\Omega^0$ is the stabilizer of Ω in G^1.

(iii) $\mathfrak{G}^0_\Omega(\mathcal{O})$ *permutes transitively the apartments of* $\mathcal{I}$ *containing* Ω.
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In view of I, 7.4.9 and the relation $\hat{P}_\Omega = U_\Omega \hat{N}_\Omega$ of I, 7.1.8, the subgroup U_Ω permutes transitively the apartments containing Ω. Since $U_\Omega \subset \mathfrak{G}_\Omega(\mathcal{O})$, assertion (iii) follows from this. On the other hand, we have $\hat{\mathfrak{G}}_\Omega(\mathcal{O}) = \hat{N}^1_\Omega P^0_\Omega = \hat{P}_\Omega \cap G^1$, so (i) results from I, 7.4.4. Finally, if Ω is closed, $\hat{N}^1_\Omega$ (which is by definition the stabilizer of f'_Ω in G^1) stabilizes Ω; conversely, if $g \in G^1$ stabilizes Ω, there exists $u \in U_\Omega$ such that gu stabilizes A and $gu \in \hat{N}^1_\Omega$, whence (ii).

Corollary 0.49 (4.6.29). — *(i) If* Ω *is closed, then* Ω *is exactly the set of fixed points of* $\mathfrak{G}^0_\Omega(\mathcal{O})$ *in the apartment* A [13].
(ii) If $cl(\Omega) \neq \Omega'$, *then* $\mathfrak{G}^0_\Omega(\mathcal{O}) \neq \mathfrak{G}^0_{\Omega'}(\mathcal{O})$.
(iii) If $cl(\Omega) \neq cl(\Omega')$, *then* $\mathfrak{G}^0_\Omega \neq \mathfrak{G}^0_{\Omega'}$.

Let $x \in A, x \notin cl(\Omega)$; according to I, 7.4.5, there exists $u \in U_\Omega \subset \mathfrak{G}^0_\Omega(\mathcal{O})$ not fixing x, whence (i). Assertions (ii) and (iii) result from (i).

4.6.30. — Let us resume the notation of 4.2.12, set $\Omega' = j(\Omega)$, let S'' be a maximal K'-split torus of G', let A'' be the apartment of $\mathscr{I}'$ associated to S'' and suppose $\Omega' \subset A''$. Then Ω' satisfies the desired conditions so that one can define the schemes $\mathfrak{G}'_{\Omega'}$, etc. from the triple (K', G', S''), and j extends to a σ-isomorphism of the $\mathcal{O}$-group scheme $\mathfrak{G}_\Omega$ (resp. $\mathfrak{G}^0_\Omega, \hat{\mathfrak{G}}_\Omega, \mathfrak{G}^\dagger_\Omega$) onto the $\mathcal{O}'$-group scheme $\mathfrak{G}'_{\Omega'}$ (resp. $\mathfrak{G}'^0_{\Omega'}, \hat{\mathfrak{G}}'_{\Omega'}, \mathfrak{G}'^\dagger_{\Omega'}$): up to multiplying j on the right by the inner automorphism of G defined by an element of $\mathfrak{G}^0_\Omega(\mathcal{O})$, one can, in view of (4.6.28) (iii), suppose that $A'' = j(A)$, or even $S'' = j(S)$, and the assertion is evident by transport of structure. In particular, the schemes $\mathfrak{G}^0_\Omega, \mathfrak{G}_\Omega, \hat{\mathfrak{G}}_\Omega$ and $\mathfrak{G}^\dagger_\Omega$ depend only on Ω as a subset of $\mathscr{I}$ and not on the choice of the apartment A containing Ω.

The same holds for the morphisms $i^0_{\Omega,\Omega'}$, etc.

It follows that one can define the schemes $\mathfrak{G}^0_\Omega$, etc. for any bounded subset Ω of the building $\mathscr{I}$ such that there exists an apartment containing Ω, and notably the schemes $\mathfrak{G}^0_x, \ldots$ for every $x \in \mathscr{I}$ (under condition (Sch) in the dense case).

Proposition 0.50 (4.6.31). — *Suppose* $\overline{k}$ *is separably closed and let* $\mathfrak{H}$ *be a smooth* $\mathcal{O}$*-group scheme with generic fiber* G. *The following conditions are equivalent:*
a) The closed fiber $\overline{\mathfrak{H}}$ *is reductive;*
b) The closed fiber $\overline{\mathfrak{H}}$ *is connected reductive;*
c) G *is split over* K *and there exists a special point* [14] $x \in \mathscr{I}$ *such that* $\mathfrak{H} \simeq \mathfrak{G}^0_x$;
d) G *is split over* K *and* $\mathfrak{H}$ *is a Chevalley scheme.*

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We have seen (cf. the proof of 4.6.15) that *c)* and *d)* are equivalent (even if $\overline{k}$ is not separably closed) and imply *b)* hence *a)*. It suffices to show that *b)* implies *d)*, because we will then know that *a)* implies that $\mathfrak{H}^o$ is a Chevalley scheme, whence $\mathfrak{H} = \mathfrak{H}^0$ (4.6.22). Now, let $\mathcal{O}'$ be the henselization of $\mathcal{O}$, which is strictly henselian. According to [SGAD], Exp. XIX, 6.1, $\mathfrak{H}_{\mathcal{O}'}$ possesses a subscheme whose fibers are split maximal tori. Thus, G is split over $K' = {}^hK$, and, as the base change $K \to K'$ is conservative, G is split over K. Moreover, Demazure's uniqueness theorem ([SGAD], Exp. XXIII, cor. 5.2) implies that $\mathfrak{H}_{\mathcal{O}}$ is a Chevalley scheme. There exists therefore a special point x of the building $\mathscr{I}'$ of $G_{K'}$ such that the identity of G extends to an $\mathcal{O}'$-morphism from $\mathfrak{H}_{\mathcal{O}'}$ onto $\mathfrak{G}'^0_x$. But we have seen that the canonical injection of $\mathscr{I}$ into $\mathscr{I}'$ is surjective (4.2.25). Thus, x is a special point of $\mathscr{I}$ and $\mathfrak{G}'^0_x = (\mathfrak{G}^0_x)_{\mathcal{O}'}$. It only remains to invoke 1.2.4 to conclude that $\mathfrak{H} \simeq \mathfrak{G}^0_x$.

Proposition 0.51 (4.6.32). *Suppose* ω *is discrete and* G *is simply connected. If* Ω *is contained in a facet* F *of* A, *then* $\mathfrak{G}^\dagger_\Omega = \mathfrak{G}^0_\Omega = \mathfrak{G}_\Omega = \hat{\mathfrak{G}}_\Omega$ *and* $G_\Omega(\mathcal{O})$ *is both the fixer and the stabilizer of* Ω *in* G.

Proof. Indeed, on the one hand $\mathfrak{T}$ is connected and $G = G^\dagger$, whence $H^0 = H^1 = H$; on the other hand, $W = \nu(N)$ is the affine Weyl group of A (4.2.22) and it follows from well-known properties of affine Weyl groups (cf. [NB] Lie V, § 3, prop. 1) that $W^\dagger_\Omega = \hat{W}_\Omega = W_\Omega$, whence $N^\dagger_\Omega = \hat{N}_\Omega = N_\Omega$. □

Theorem 0.52 (4.6.33). *Suppose* ω *is discrete. Let* F *be a facet of* $\mathscr{I}$ *and let* $\mathscr{I}(F)$ *be the star of* F *in the polysimplicial complex* $\mathscr{I}$, *that is to say the set of facets* F' *of* $\mathscr{I}$ *such that* $F \subset \overline{F}'$, *ordered by the relation* $F' \subset \overline{F}''$.

[13] For a better result, see 5.1.39.
[14] I.e., a point of $\mathscr{I}$ which is special in an apartment containing it, and is then special in every apartment containing it.

(i) The map $F' \mapsto \mathfrak{p}(F') = Im\ i^0_{F,F'}$ is an isomorphism of ordered sets from $\mathscr{I}(F)$ onto the set of $\overline{k}$-pseudo-parabolic subgroups of $\overline{\mathfrak{G}}^0_F$ ordered by the relation opposite to inclusion[15].

(ii) $\mathfrak{p}(F')(\overline{k})$ is the product of the canonical image of $\mathfrak{G}^0_{F'}(\mathcal{O})$ and $\overline{\mathfrak{T}}^0(\overline{k})$.

(iii) The preimage of $\mathfrak{p}(F')(\overline{k})$ in $\mathfrak{G}^0_F(\mathcal{O})$ is equal to $\mathfrak{G}^0_{F'}(\mathcal{O})$.

We can suppose $F \subset A$. Since the walls of A containing F are the hyperplanes $\{a(x-\varphi) + f_F(a) = 0\}$ for $a \in \Phi_F$, there exists a unique chamber C of A such that $F \subset \overline{C}$ and $C \subset \{a(x-\varphi) + f_F(a) > 0\}$ for all $a \in \Phi^+_F = \Phi^+ \cap \Phi_F$. Let B be the basis of Φ_F corresponding to the positive root system Φ^+_F. For $J \subset B$, let us denote by Φ_J the set of $a \in \Phi_F$ which are linear combinations of elements of J and by F_J the facet of A characterized by the relations $F \subset \overline{F}_J \subset \overline{C}$ and $J = \{a \in B \mid f_{F_J}(a) = f_F(a)\}$.

The map $J \mapsto F_J$ is a bijection from $\mathscr{P}(B)$ onto the set of facets F' such that $F \subset \overline{F}' \subset \overline{C}$, and we have $\Phi_{F_J} = \Phi_J$.

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We have $f_F \leq f_{F_J}$, and f_{F_J} is less than or equal to the optimized function of f^*_F. It then follows from 4.6.10 that the image $p(F_J)$ of i^o_{F,F_J} contains the split unipotent radical R of $\mathfrak{G}^o_F$. It also contains $\mathfrak{T}^o$ and, by 4.6.5, its image in the quasi-reductive quotient $\mathfrak{G}^o_F/R$ is generated by the image of $\mathfrak{T}^o$ and the root subgroups of $\mathfrak{G}^o_F/R$ corresponding to $\Phi^+_F \cup \Phi_J$. It is then immediate that $p(F_J)$ is the standard pseudo-parabolic $\overline{K}$-subgroup associated with J (1.1.14).

Furthermore, on the one hand, every facet $F' \in \mathscr{I}(F)$ is the transform of a facet F_J by an element of $\mathfrak{G}^o_F(\mathcal{O})$ (since $\mathfrak{G}^o_F(\mathcal{O})$ permutes transitively the apartments containing F (4.6.28) and W^o_F permutes transitively the chambers of A containing F in their closure), on the other hand, every pseudo-parabolic $\overline{K}$-subgroup of $\mathfrak{G}^o_F$ is conjugate to one $p(F_J)$ by an element of $\mathfrak{G}^o_F(\overline{K})$ which is the image of an element of $\mathfrak{G}^o_F(\mathcal{O})$ (this follows from 1.1.14, taking 4.6.7 (ii) into account). Thus p is a surjective map from $\mathscr{I}(F)$ onto the set of pseudo-parabolic $\overline{K}$-subgroups of $\mathfrak{G}^o_F$.

Let us prove (ii). Suppose first $F' = C$. We have $p(C) = \mathfrak{U}^+_F \cdot \mathfrak{T}^o \cdot R$ and the product map is an isomorphism of varieties from $\mathfrak{U}^+_F \times (\mathfrak{U}^-_F \cap R) \times \mathfrak{T}^o$ onto $p(C)$. Now $\mathfrak{U}^+_F = \mathfrak{U}^o_C$, $\mathfrak{U}^-_F \cap R = \overline{\mathfrak{U}}^o_{\mathscr{I}(F)}$ and $f_C \leq (f^*_C)'$ whence (iii) when $F' = C$. Now maintaining $F' = F_J$, we have $p(F_J)(\overline{K}) = p(C)(\overline{K}) \cdot N_J \cdot p(C)(\overline{K})$, where N_J is the subgroup of the normalizer of $\mathfrak{S}$ in $\mathfrak{G}^o_F(\overline{K})$ which is the preimage of the subgroup of W_F generated by reflections with respect to elements of J. But N_J is equal to the product of $\mathfrak{T}^o(\overline{K})$ by the image of $N^o_{F_J}$, hence (ii) when $F' = F_J$. The general case follows by conjugation.

Let us prove (iii). It follows from 4.6.8 that the kernel of the canonical map from $\mathfrak{G}^o_F(\mathcal{O})$ into $\mathfrak{G}^o_F(\overline{K})$ is contained in $\mathfrak{G}^o_{F'}(\mathcal{O})$. Taking (ii) into account, it thus suffices to show that if the canonical image $\bar{x}$ in $\mathfrak{G}^o_F(\overline{K})$ of an element x of $\mathfrak{G}^o_F(\mathcal{O})$ belongs to $\mathfrak{T}^o(\overline{K})$, then $\bar{x}$ is also the canonical image of an element of $\mathfrak{G}^o_{F'}(\mathcal{O})$. Now, we have $x = uvn$, with $u \in U^+_F$, $v \in U^-_F$, and $n \in N^o_F$ (4.6.7 (i)). Consider the canonical map q from $\mathfrak{G}^o_F$ onto its quasi-reductive quotient $\mathfrak{G}^o_F/R$. We have $q(\bar{v})q(\bar{n}) = q(\bar{u})^{-1}q(\bar{x})$ and it follows then from the Bruhat decomposition of $(\mathfrak{G}^o_F/R)(\overline{K})$ that $q(\bar{n}) = q(\bar{x})$ (and that $q(\bar{u}) = q(\bar{v}) = 1$). Consequently, $q(\bar{n})$ centralizes $q(\mathfrak{S})$. Since the Weyl group of $\mathfrak{G}^o_F/R$ is $W_F = N^o_F/H^o$ (1.1.13 and 4.6.13), we deduce that $n \in H^o = \mathfrak{T}^o(\mathcal{O})$. But the product map from $\mathfrak{U}^+_F \times \mathfrak{U}^-_F \times \mathfrak{T}^o$ into $\mathfrak{G}^o_F$ is injective, thus the relation $\bar{x} = \bar{u} \cdot \bar{v} \cdot \bar{n}$ implies $\bar{x} = \bar{n}$, that is to say that $\bar{x}$ is indeed the image of an element of $\mathfrak{T}^o(\mathcal{O}) \subset \mathfrak{G}^o_{F'}(\mathcal{O})$.

Finally, (iii) implies that p is injective, which completes the proof of (i) and the theorem.

Corollary 0.53 (4.6.34)**.** *Let further Y be a subgroup of N^+_F/N^o_F. For every facet $F' \in \mathscr{I}(F)$, the normalizer of $p(F')$ in $Y\mathfrak{G}^o_F(\overline{K})$ is the product by $\mathfrak{T}^o(\overline{K})$ of the image of $Y'\mathfrak{G}^o_{F'}(\mathcal{O})$, where $Y' = (YN^o_{F'} \cap N^+_F)/N^o_{F'}$, and $Y'\mathfrak{G}^o_{F'}(\mathcal{O})$ is its preimage in $Y\mathfrak{G}^o_F(\mathcal{O})$.*

Proof. Let $x \in Y\mathfrak{G}^o_F(\overline{K})$ normalizing $p(F_J)$. The two maximal $\overline{K}$-split tori $\mathfrak{S}$ and $x\mathfrak{S}x^{-1}$ of $p(F_J)$ are conjugate by an element $y \in p(F_J)(\overline{K})$ ([7]) and in view of assertion (ii) of the theorem, one can take y in the image of $\mathfrak{G}^o_{F_J}(\mathcal{O})$. We are thus reduced to the case where x normalizes $\mathfrak{S}$. But then $x \in YN^o_F \cdot \mathfrak{T}^o(\overline{K})$, from which the corollary easily follows. □

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[15](1) Recall that if $\overline{k}$ is perfect, the pseudo-parabolic subgroups are none other than the parabolic subgroups.

4.6.35. — The map $p \mapsto \overline{p}(\overline{K})$ is a bijection from the set of $\overline{K}$-pseudo-parabolic subgroups of $\overline{\mathfrak{G}}_F^0$ to the set of parabolic subgroups of the Tits system associated with the root datum of $\overline{\mathfrak{G}}_F^0(\overline{K})$ (1.1.14). The preceding theorem shows that, if one endows the star $\mathscr{I}(F)$ of the facet F with an abstract simplicial complex structure by taking as the face relation the relation $F' \subset \overline{F''}$, the map $F' \mapsto p(F')(\overline{K})$ is an isomorphism from $\mathscr{I}(F)$ to the combinatorial building $\mathscr{I}_F$ of this Tits system ([37]), compatible with the actions of $\mathfrak{G}_F^0(\mathfrak{O})$ on $\mathscr{I}(F)$ and of $\overline{\mathfrak{G}}_F^0(\overline{K})$ on $\mathscr{I}_F$. Furthermore, to the set of facets $F' \in \mathscr{I}(F)$ contained in the apartment A corresponds the set of $\overline{K}$-pseudo-parabolic subgroups containing $\overline{S}$, thus forming an apartment of $\mathscr{I}_F$. By conjugation, we see that the inverse images of the apartments of $\mathscr{I}_F$ are the sets of facets of $\mathscr{I}(F)$ contained in the same apartment of $\mathscr{I}$.

4.6.36. — When ω is dense, Theorem 4.6.33 is obviously no longer valid, if only because the facets F are then germs of subsets (I, 7.2) and the schemes $\mathfrak{G}_F$ do not exist in general. However, we shall give a substitute for it.

Until the end of 4.6, we assume ω is dense and the reader who is only interested in the case of discrete valuations may skip directly to the following paragraph.

By hypothesis, the extension $\tilde{K}$ of K is unramified (1.6.1 c)). Consequently, $\Gamma'_a = \omega(K^\times) = \Gamma$ for all $a \in \Phi$, except when $2a \in \Phi$, in which case $\Gamma'_a = \gamma_a + \Gamma$, with γ_a defined as in 4.3.5. We have $2\gamma_a \in \Gamma$. If a and b are multipliable and belong to the same irreducible component of Φ, then $\gamma_a = \gamma_b$ (4.1.17). If a, b and $a + b \in \Phi$, we have

$$\Gamma'_a + \Gamma'_b = \Gamma'_{a+b}. \tag{1}$$

Indeed, if a is not multipliable, then b and $a + b$ are simultaneously multipliable or not multipliable, and in the first case we have $\gamma_{a+b} = \gamma_b$; if a is multipliable and if b is not multipliable (resp. multipliable), then $a + b$ is multipliable (resp. not multipliable) and $\gamma_a = \gamma_{a+b}$ (resp. $\gamma_a = \gamma_b$).

It follows that for any subset Ω of A, the root system Φ_Ω, which is the set of $a \in \Phi$ for which there exists $k \in \Gamma'_a$ such that $\Omega \subset \{x \in A \mid a(x - \phi) + k = 0\}$, is a closed subsystem of Φ.

On the other hand, if F is a facet of A, and $a \in \Phi$, we denote by $f_F(a)$ the infimum in the ordered set $\tilde{\mathbf{R}}$ of $f_\Omega(a)$ for Ω describing the filter F (I, 7.2.1 and 6). We thus obtain a concave function $f_F : \Phi \to \tilde{\mathbf{R}}$. The associated root system $\Phi_F = \Phi_{f_F}$ is the set of $a \in \Phi$ such that $f_F(a) \in \Gamma'_a$ and $f_F(a) + f_F(-a) = 0$.

Lemma 0.54 (4.6.37)**.** *Let X be an (arbitrary) bounded subset of A, or a facet of A, and let Ψ_X be the set of $a \in \Phi$ such that $f_X(a) \in \Gamma'_a$. Let Ξ be the group of affine automorphisms of A preserving Φ and the affine root system of A and stabilizing X. There exists $\eta > 0$ such that for any ε with $0 < \varepsilon < \eta$, one can find a concave function $f : \Phi \to \mathbf{R}$ satisfying (Sch) and the following relations:*

a) $u \cdot f = f$ for every $u \in \Xi$;

b) $f(a) = f_X(a)$ for every $a \in \Psi_X$;

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c) $f_X(a) + \frac{1}{2}\varepsilon < f(a) < f_X(a) + \varepsilon$ for every $a \in \Phi - \Psi_X$;

d) $\Phi_f = \Phi_X$.

Such a function f is called an ε-thickening of X.

Let $u \in \Xi$ and $a \in \Phi$. Let us set $b = {}^t(u)(a)$. If $k \in \Gamma'_a$, the image under u^{-1} of the affine root $\{x \in A \mid a(x-\varphi)+k \geq 0\}$ is the closed half-space

$$\{x \in A \mid b(x-\varphi)+a(u(\varphi)-\varphi)+k \geq 0\},$$

which is thus also an affine root. It follows that

$$a(u(\varphi)-\varphi) \in \Gamma'_b - \Gamma'_a \quad \text{for } a \in \Phi, u \in \Xi \text{ and } b = {}^t(u)(a). \tag{1}$$

On the other hand, in view of 4.6.17 (1 bis) (generalized to functions with values in $\tilde{\mathbf{R}}$), we have

$$u.f_X(a) = f_X(b) - a(u(\varphi)-\varphi), \tag{2}$$

and by transport of structure or by explicit calculation, we see that

$$u.f_X = f_X \quad \text{for all } u \in \Xi. \tag{3}$$

From (1), (2) and (3) it immediately follows that

$${}^t(u)(\Psi_X) = \Psi_X \quad \text{and} \quad {}^t(u)(\Phi_X) = \Phi_X. \tag{4}$$

Let us then prove the lemma when X is the subset reduced to a point x of A. In this case, the function f_X is the linear function $f_x : a \mapsto -a(x-\varphi)$, so Ψ_x coincides with Φ_x and is a closed subsystem of Φ. Furthermore, let $a \in \Phi, b \in \Psi_x$ be such that $a+b \in \Phi$ and let $h \in \mathbf{R}$. In view of 4.6.36 (1) and the linearity of f_x,

$$f_x(a)+h \in \Gamma'_a \quad \text{is equivalent to} \quad f_x(a+b)+h \in \Gamma'_{a+b}. \tag{5}$$

Similarly, if $a \in \Phi, u \in \Xi$ and $h \in \mathbf{R}$, (1), (2) and (3) imply that

$$f_x(a)+h \in \Gamma'_a \quad \text{is equivalent to} \quad f_x(b)+h \in \Gamma'_b \quad \text{with } b = {}^t(u)(a). \tag{6}$$

Let us then consider the smallest equivalence relation $\sim$ on Φ such that ${}^t(u)(a) \sim a$ for $a \in \Phi, u \in \Xi$ and that $a+b \sim b$ whenever $a \in \Psi_x, b, a+b \in \Phi$. Since Ψ_x is closed and stable under Ξ, it is a union of equivalence classes under $\sim$. We then see, thanks to (5) and (6), that for every $\varepsilon > 0$, one can choose a family $(\varepsilon_a)_{a\in\Phi}$ of real numbers possessing the following properties:

$$\varepsilon_a \text{ is constant on the equivalence classes under } \sim; \tag{7}$$

$$f_x(a)+\varepsilon_a \in \Gamma'_a \quad \text{for all } a \in \Phi; \tag{8}$$

$$\varepsilon_a = 0 \quad \text{for all } a \in \Psi_x; \tag{9}$$

$$\frac{1}{2}\varepsilon < \varepsilon_a < \varepsilon \quad \text{for } a \in \Phi - \Psi_x. \tag{10}$$

The function f defined by $f(a) = f_x(a)+\varepsilon_a$ for $a \in \Phi$ is then an ε-thickening of $\{x\}$. Indeed, it is clear that f satisfies (Sch), that $u.f = f$ for all $u \in \Xi$, that $f(a) = f_x(a)$

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for $a \in \Psi_x$ and $f_x(a)+\frac{1}{2}\epsilon < f(a) < f_x(a)+\epsilon$ for $a \in \Phi - \Psi_x$, whence also $\Phi_f = \Phi_x$. It remains to show that f is concave. We evidently have $f(a)+f(-a) \geq f_x(a)+f_x(-a) \geq 0$ for all $a \in \Phi$. Let $a, b \in \Phi$ be such that $a+b \in \Phi$; we have

$$f(a+b) = f(a)+f(b)+\varepsilon_{a+b}-\varepsilon_a-\varepsilon_b$$

and it suffices to show that $\varepsilon_{a+b} \geq \varepsilon_a+\varepsilon_b$. Now, if $a, b \in \Psi_x$, then $a+b \in \Psi_x$ (which is closed) and $\varepsilon_a = \varepsilon_b = \varepsilon_{a+b} = 0$. If $a \in \Psi_x$ and $b \notin \Psi_x$, then $\varepsilon_a = 0$ and $\varepsilon_b = \varepsilon_{a+b}$ according to (7). Finally, if neither a nor b belongs to Ψ_x, we have $\varepsilon_a+\varepsilon_b > \epsilon > \varepsilon_{a+b}$ according to (10) and (9). Consequently, f is indeed concave and the lemma is proven in the case $X = \{x\}$.

Suppose now that X is a facet F of A. Let x be the center of F, D its direction (I, 7.2.4 and 5) and Φ_D^- the set of roots $a \in \Phi$ strictly negative on D. We have $\Phi_D^- \subset \Phi_x = \Psi_x$ and, according to I, 7.2.6,

$$f_F(a) = \begin{cases} f_x(a) & \text{if } a \notin \Phi_D^-, \\ f_x(a)+ & \text{if } a \in \Phi_D^-, \end{cases}$$

whence $\Psi_F = \Phi_x - \Phi_D^-$. Let $\epsilon > 0$. Let us choose an ϵ-thickening g of x and a real number $\epsilon' \in \Gamma$ such that $\epsilon' < \epsilon$ and that $\frac{1}{2}\epsilon < \epsilon' < \inf\{g(a) - f_x(a) \mid a \in \Phi - \Psi_x\}$. Let us set $h(a) = f_x(a)$ if $a \notin \Phi_D^-$ and $h(a) = f_x(a) + \epsilon'$ if $a \in \Phi_D^-$. Since Φ_D^- is closed and f_x is linear, the function $h : \Phi \to \mathbb{R}$ is concave. Finally, let us set $f = \sup(g, h)$, i.e., $f(a) = g(a)$ if $a \notin \Phi_D^-$ and $f(a) = f_x(a) + \epsilon'$ if $a \in \Phi_D^-$. Then f is concave, satisfies (Sch), is invariant under all $u \in \Xi$ (since $u(D) = D$ because u stabilizes F, whence ${}^t(u)(\Phi_D^-) = \Phi_D^-$), and $f(a) = g(a) = f_x(a) = f_F(a)$ for all $a \in \Psi_F = \Phi_x - \Phi_D^-$. If $a \in \Phi - \Phi_x$, we have

$$f_F(a) + \frac{1}{2}\epsilon = f_x(a) + \frac{1}{2}\epsilon < g(a) = f(a) < f_x(a) + \epsilon = f_F(a) + \epsilon,$$

and, if $a \in \Phi_D^-$,

$$f_F(a) + \frac{1}{2}\epsilon = (f_x(a) + \frac{1}{2}\epsilon) + < f_x(a) + \epsilon' = f(a) < f_x(a) + \epsilon < f_F(a) + \epsilon,$$

which finishes showing that f is an ϵ-thickening of F. Let us remark that in these first two cases one can take η to be arbitrary.

Finally, suppose that X is a bounded subset Ω of A. We can suppose Ω to be closed: it is then a closed convex polyhedron and the set Y of its extreme points is finite. We have $f_\Omega = \sup_{y \in Y} f_y$. For $a \in \Phi$, let us denote by Y_a the set of $y \in Y$ such that $f_\Omega(a) = f_y(a)$ and let us set $\eta = \inf\{f_\Omega(a) - f_y(a) \mid a \in \Phi \text{ and } y \in Y - Y_a\}$. Then $\eta > 0$. Let $\epsilon \in \mathbb{R}$ with $0 < \epsilon < \eta$ and let us choose for each $y \in Y$ an ϵ-thickening g_y of y such that if $u \in \Xi$ one has $u \cdot g_y = g_{u(y)}$, which is obviously possible since $u(Y) = Y$ and since, for $y \in Y$, each ϵ-thickening of $\{y\}$ is invariant under the stabilizer of y in Ξ. Let us set $f = \sup_{y \in Y} g_y$ and let us show that f is an ϵ-thickening of Ω. The function f

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is concave, satisfies (Sch) and is invariant by every $u \in \Xi$. Let $a \in \Phi$; since $\varepsilon < \eta$, we have

$$g_y(a) < f_y(a) + \varepsilon < f_\Omega(a) \quad \text{for all } y \in Y - Y_a. \tag{11}$$

Let $y \in Y_a$. If $a \in \Psi_\Omega$, then $a \in \Psi_y$ and $g_y(a) = f_y(a) = f_\Omega(a)$, whence

$$f(a) = f_\Omega(a) \quad \text{for } a \in \Psi_\Omega. \tag{12}$$

If $a \notin \Psi_\Omega$, then $a \notin \Psi_y$, and we have $f_y(a) + \frac{1}{2}\varepsilon < g_y(a) < f_y(a) + \varepsilon$, which, combined with (11), gives

$$f_\Omega(a) + \frac{1}{2}\varepsilon < f(a) < f_\Omega(a) + \varepsilon \quad \text{for } a \in \Phi - \Psi_\Omega. \tag{13}$$

Finally, (12) and (13) imply $\Phi_f = \Phi_\Omega$, which completes the proof of the lemma.

4.6.38. Remarks. — 1) If Ω is a subset of A and if f_Ω satisfies (Sch), every thickening of Ω is equal to f_Ω.
2) Let Ω and Ω' be two bounded subsets of A such that $\Omega \subset \overline{\Omega'}$, whence $f_\Omega \leq f_{\Omega'}$. Let δ be the smallest of the numbers $f_{\Omega'}(a) - f_\Omega(a)$ which are strictly positive. Then one easily sees that every ε-thickening f of Ω with $\varepsilon < \delta$ is less than every 2ε-thickening f' of Ω'.
Similarly, if F and F' are two facets of A such that $F \subset \overline{F'}$ (which implies that F and F' have the same center and that $D \subset \overline{D'}$, whence $\Phi_{\overline{D}} \subset \Phi_{\overline{D'}}$), then for all $\varepsilon > 0$, every ε-thickening f of F is less than every 2ε-thickening f' of F'.
In each of these two cases, we thus obtain a morphism $i^0_{f,f'}$ from $\mathfrak{G}^0_f$ into $\mathfrak{G}^0_{f'}$ (4.6.24).

4.6.39. — It is clear that the family $E = E(X)$ of thickenings of X is a filtered decreasing system: if $2\varepsilon < \varepsilon'$, an ε-thickening is less than an ε'-thickening. One deduces from this a filtered increasing inductive system of schemes $((\mathfrak{G}^0_f)_{f \in E}, (i^0_{f,g})_{f,g \in E, f \leq g})$, from which we shall refrain from considering the inductive limit.
Let $f \in E$; we have $\Phi_f = \Phi_X$ (4.6.37 d)), whence $W_f = W_X$ and $N^0_f = N^0_X$, and f is invariant by every $u \in N^\dagger_X$ (4.6.37 a)), whence $N^\dagger_f \supset N^\dagger_X$. Let Y be a subgroup of $N^\dagger_X/N^0_X \subset N^\dagger_f/N^0_f$; the schemes $Y\mathfrak{G}^0_f$ also form a filtered increasing inductive system. If X is a closed subset or a facet, one easily shows that $N^\dagger_X = N^\dagger_f$ for every ε-thickening f with ε sufficiently small, whence also a filtered increasing inductive system of schemes $\mathfrak{G}^\dagger_f$.
Let $P^\dagger_X$ be the subgroup generated by U_X (or also by the $U_{a,f_X(a)}$ for $a \in \Phi$) and the inverse image of Y in $N^\dagger_X$.
If $Y = N^\dagger_X/N^0_X$ (resp. $\widehat{N}^\dagger_X/N^0_X$), then $P^\dagger_X$ is the stabilizer $P^\dagger_X$ (resp. the fixer $\widehat{P}^\dagger_X$) of X in $G^\dagger$ (I, 7.4.4); if $Y = \{1\}$, then $P^\dagger_X$ is the subgroup $U_X H^0$, which we denote by P^0_X and call the connected fixer of X (cf. 4.6.28).

Proposition 0.55. ***4.6.40. Proposition.*** — *(i) The group U_X is the filtered increasing union of the U_f for $f \in E(X)$.*

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(ii) For every subgroup Y of $N_{\mathfrak{X}}^{\natural}/N_{\mathfrak{X}}^{0}$, the group $P_{\mathfrak{X}}^{Y}$ is the increasing union of the subgroups $Y\mathfrak{G}_f^0(\mathcal{O})$ for $f \in E(\mathfrak{X})$. In particular, the connected stabilizer $P_{\mathfrak{X}}^{0}$ is the increasing filtered union of the $\mathfrak{G}_f^0(\mathcal{O})$ for $f \in E(\mathfrak{X})$.

It is immediate that (i) implies (ii). To prove (i), it suffices to show that, for all $a \in \Phi$, the group $U_{f_{\mathfrak{X}},a}$ is the union of the $U_{f,a}$ for $f \in E$, which is evident since $f_{\mathfrak{X}}(a)$ (resp. $f_{\mathfrak{X}}(2a)$) is the infimum in $\tilde{\mathbb{R}}$ of the $f(a)$ (resp. $f(2a)$) for $f \in E$.

4.6.41. — Let f (resp. g) be an ϵ-thickening (resp. an ϵ'-thickening) of $\mathfrak{X}$, with $2\epsilon' < \epsilon$, hence $g \leq f$, and let us consider the morphism $h = i_{g,f}^0$ from $\overline{\mathfrak{G}}_f^0$ into $\overline{\mathfrak{G}}_g^0$. One sees easily that h is the identity on $\overline{\mathfrak{T}}^0$ and that the restrictions $h_a : \overline{\mathfrak{U}}_{f,a} \to \overline{\mathfrak{U}}_{g,a}$ for $a \in \Phi$ are given by the following rules:

- — if $a \cap \Phi \subset \Psi_{\mathfrak{X}}$, then h_a is the identity;
- — if $a \cap \Psi_{\mathfrak{X}} = \emptyset$, then h_a is the trivial morphism;
- — there remains the case where a is plural, $a \notin \Psi_{\mathfrak{X}}$ and $2a \in \Psi_{\mathfrak{X}}$ (note that if $a \in \Psi_{\mathfrak{X}}$, then $2a \in \Psi_{\mathfrak{X}}$ since the extension $L_a \supset L_{2a}$ is unramified). We then have

$$f(2a) = f_{\mathfrak{X}}(2a) = 2f_{\mathfrak{X}}(a) < 2f(a) - \epsilon$$

and it is verified without difficulty that the group $\overline{\mathfrak{U}}_{f,a}$ is the direct product of $\overline{\mathfrak{U}}_{2a,f(2a)} \simeq \Pi_{\overline{L}_{2a}/\overline{k}}\mathrm{Add}$ by the canonical image of $\overline{\mathfrak{U}}_{a,(f(a),2f(a))}$, an image which is isomorphic to $\Pi_{\overline{L}_a/\overline{k}}\mathrm{Add}$ and on which $\overline{\mathfrak{T}}$ acts by the root a. Furthermore, h_a is the identity on $\overline{\mathfrak{U}}_{2a,f(2a)}$ and is trivial on the image of $\overline{\mathfrak{U}}_{a,(f(a),2f(a))}$.

By considering the "large cells" $\overline{\mathfrak{C}}_f^0$ and $\overline{\mathfrak{C}}_g^0$, we see that h is separable and that the product map is an isomorphism of varieties from the product of the $\ker h_a$ for $a \in \Phi$ onto the kernel $Q_{g,f}$ of h. It follows that $\Phi - \Psi_{\mathfrak{X}}$ is closed in $\overline{\mathfrak{G}}_f^0$ (in the sense of [7], or quasi-closed in the sense of [4] generalized to the non-reductive case) and that $Q_{g,f}$ is the split unipotent subgroup $(\overline{\mathfrak{G}}_f^0)_{(\Phi-\Psi_{\mathfrak{X}})}$ associated to $\Phi - \Psi_{\mathfrak{X}}$ (loc. cit. or 1.1.6). It is therefore independent of the choice of g; we can denote it by Q_f. The determination of the h_a given above shows that $h(Q_f) \subset Q_g$ and that one obtains by passing to quotients an isomorphism from $\overline{\mathfrak{G}}_f^0/Q_f$ onto $\overline{\mathfrak{G}}_g^0/Q_g$ (recall that f is an ϵ-thickening of $\mathfrak{X}$ and g an ϵ'-thickening of $\mathfrak{X}$, with $2\epsilon' < \epsilon$).

Otherwise put, from the inductive system of the $\overline{\mathfrak{G}}_f^0$ for $f \in E$, we deduce by passing to closed fibers, by restriction to a suitable cofinal system and by passing to the quotients by the kernels of the transition morphisms, a trivial inductive system of algebraic groups $\overline{\mathfrak{G}}_f^0/Q_f$; this system clearly admits an inductive limit that we denote by $\overline{\mathfrak{G}}_{\mathfrak{X}}^0$, isomorphic to all the $\overline{\mathfrak{G}}_f^0/Q_f$, and which we call the connected algebraic $\overline{k}$-group attached to $\mathfrak{X}$. In view of 4.6.39, the canonical maps from $\mathfrak{G}_f^0$ into $\overline{\mathfrak{G}}_{\mathfrak{X}}^0$ define a so-called canonical map from $P_{\mathfrak{X}}^0$ into $\overline{\mathfrak{G}}_{\mathfrak{X}}^0(\overline{k})$ and one has $\overline{\mathfrak{G}}_{\mathfrak{X}}^0(\overline{k}) = \mathrm{Im}\, P_{\mathfrak{X}}^0 \cdot \overline{\mathfrak{T}}^0(\overline{k})$ (4.6.7).

Note that it also follows directly from 4.6.25 that the quasi-reductive quotients

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$\mathfrak{G}_f^0$ are isomorphic and are naturally identified with $\overline{\mathfrak{G}}_X^0$. Their root system is Φ_X and 4.6.12 generalizes immediately.

The same considerations applied to the inductive system of the $\hat{\mathfrak{G}}_f$ (resp. of the $\mathfrak{G}_f^\dagger$) allow for the definition of the algebraic $\overline{K}$-group $\overline{\hat{\mathfrak{G}}}_X$ (resp. $\overline{\mathfrak{G}}_X^\dagger$).

Proposition 0.56 (4.6.42)**.** *Let σ be an automorphism of K preserving ω and let j be a σ-automorphism of G. Let j also denote the corresponding automorphism of the building $\mathscr{I}$ (4.2.12). Let X be as in 4.6.37 and suppose that $j(X) = X$. There exists $\delta > 0$ such that for every ε-thickening f of X with $\varepsilon < \delta$, the automorphism j of G extends to a σ-automorphism of the $\mathcal{O}$-scheme $\mathfrak{G}_f^0$ (resp. $\hat{\mathfrak{G}}_f, \mathfrak{G}_f^\dagger$).*

Proof. According to I, 7.4.9, there exists $g \in U_X$ such that $j(A) = g.A$. The map $x \mapsto g^{-1}.j(x)$ preserves A and induces an element $u \in \Xi$ (4.6.37) which preserves every thickening f of X. By transport of structure, it follows that the σ-automorphism $y \mapsto g^{-1}j(y)g$ of G extends to a σ-automorphism of $\mathfrak{G}_f^0$ (resp. $\hat{\mathfrak{G}}_f, \mathfrak{G}_f^\dagger$). On the other hand, 4.6.40 implies that there exists $\delta > 0$ and a 2δ-thickening f_0 of X such that $g \in U_{f_0}$. But then $g \in U_f \subset \mathfrak{G}_f^0(\mathcal{O})$ for every ε-thickening f of X with $\varepsilon < \delta$, hence the proposition. □

Corollary 0.57 (4.6.43)**.** *The algebraic $\overline{K}$-group $\overline{\mathfrak{G}}_X^0$ (resp. $\overline{\hat{\mathfrak{G}}}_X, \overline{\mathfrak{G}}_X^\dagger$) attached to X depends only on X as a subset of the building $\mathscr{I}$ and not on the apartment A chosen to construct it.*

More precisely, if A' is another apartment containing X, the inductive systems of schemes associated with the thickenings of X in A and in A' respectively are cofinal.

4.6.44. — If $X \subset \overline{X}'$, remark 4.6.38 (2) provides a canonical $\overline{K}$-homomorphism denoted $i^0_{X,X'}$ from $\overline{\mathfrak{G}}^0_{X'}$ into $\overline{\mathfrak{G}}^0_X$, which is compatible with the inclusion map $P^0_{X'} \to P^0_X$.

Proposition 0.58 (4.6.45)**.** *Up to replacing $\mathfrak{G}^0_F(\mathcal{O})$ by P^0_F, Theorem 4.6.33 and the assertions of 4.6.35 remain valid when ω is dense.*

We leave to the reader the task of adapting the proofs given above for these results, as well as the statement and proof of 4.6.34.

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§5. ETALE DESCENT

5.1. Descent of the building.

We maintain the previous hypotheses and notation. However, we denote by φ an arbitrary point of the apartment A and no longer necessarily a Chevalley-Steinberg valuation.

Furthermore, we assume that K is strictly Henselian (that is, Henselian with a separably closed residue field) and we are given a Henselian subfield $K^\natural$ of K, of which K is an étale extension (in other words, K is the strict Henselization of $K^\natural$). Then K (resp. $\bar{K}$) is an algebraic Galois extension of $K^\natural$ (resp. of the residue field $\bar{K}^\natural$ of $K^\natural$). We denote by $\mathcal{O}^\natural$ the valuation ring of $K^\natural$ and by Σ the Galois group $\mathrm{Gal}(K/K^\natural)$: it identifies canonically with $\mathrm{Gal}(\bar{K}/\bar{K}^\natural)$.

We further assume that the connected reductive group G is defined over $K^\natural$. We denote by $S^\natural$ a maximal $K^\natural$-split torus of G and we assume that the maximal K-split torus S of G contains $S^\natural$, which is obviously permissible (of course, S is not a priori defined over $K^\natural$: see however 5.1.12). We denote by Z the centralizer of $S^\natural$ in G. For $a \in \Phi$, we denote by $a^\natural$ the restriction of a to $S^\natural$. We denote by Φ_0 the set of $a \in \Phi$ such that $a^\natural = 0$ and by $\Phi^\natural$ the set of $a^\natural$ for $a \in \Phi$ and $a^\natural \neq 0$. It is known that $\Phi^\natural$ is the root system of G relative to $S^\natural$. For $b \in \Phi^\natural$, we denote by Φ^b the set of $a \in \Phi$ such that $a^\natural = b$ or $2b$ (care should be taken that this notation is not that of I, 9.1.1). It is a positively closed subset of Φ; we denote by U_b the unipotent subgroup associated with Φ^b. It is defined over $K^\natural$ and is the root subgroup associated with the root b of G relative to $S^\natural$.

5.1.1. — Conversely, let us start with a Henselian valued field $K^\natural$ and let K be its strict Henselization. Let H be a connected reductive algebraic group defined over $K^\natural$ and set $G = H_K$. Then the previous hypotheses are satisfied if and only if

a) H is quasi-split over K;

b) H is split over an admissible extension of K;

c) the canonical scheme with generic fiber T has a smoothing.

If ω is discrete, condition a) implies b) (1.6.1 f)) and c) (4.6) (let us also recall that, when ω is dense, c) follows from a) and b) as soon as H is split over K, or if H is simply connected or adjoint). If moreover $K^\natural$ is perfect, then a) is satisfied according to a well-known result of Steinberg ([32] and [3, 8.6]). We will be led later to make an additional hypothesis (DE) (5.1.5), but we will see that it is auto-

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matically satisfied when ω is discrete. Consequently, all the results of this paragraph are applicable to any connected reductive group defined over a Henselian field for a discrete valuation, with perfect residue field.

5.1.2. — Let $b \in \Phi^\natural$; set $U^\natural_b = U_b(K^\natural)$. We know that $(Z(K^\natural), (U^\natural_b)_{b\in\Phi^\natural})$ is a generating root datum of type $\Phi^\natural$ in $G^\natural = G(K^\natural)$ (I, 6.1.3 c)), which is compatible with the root datum $(T(K), (U_a(K))_{a\in\Phi})$ of $G(K)$ in the sense of I, § 9. Indeed, the conditions (DDR 1) of 9.1.9 (and even (DDR 1 bis) of 9.1.13), (DDR 2) of 9.1.10 and (DDR 3) of 9.2.9 are satisfied by taking for $V^\natural$ the vector subspace of V defined as follows: the space V is the dual of the vector subspace V^* generated by Φ in $X^*(S) \otimes \mathbf{R}$ (4.1.2), the map which to a character of S associates its restriction to $S^\natural$ defines a surjective linear map u from V^* onto the vector subspace $(V^\natural)^*$ generated by $\Phi^\natural$ in $X^*(S^\natural) \otimes \mathbf{R}$ and $V^\natural$ is the image of the dual of $(V^\natural)^*$ by the injective linear transpose map of u. We will identify in the following each of the two spaces $V^\natural$ and $(V^\natural)^*$ with the dual of the other. It is easily verified that $V^\natural = \nu(S^\natural(K)) \otimes \mathbf{R}$.

The main goal of this paragraph is to demonstrate that, provided an additional condition always satisfied when ω is discrete, one can choose S and the valuation $\varphi \in A$ of $G(K)$ in such a way that φ descends to $G^\natural$ in the sense of I, § 9, and thus defines a valuation of the root datum of $G^\natural$, hence a building, etc.

5.1.3. — Let S_1 be a $K^\natural$-split torus of G contained in S and let Z_1 be the centralizer of S_1. It is the subgroup of G generated by T and the U_a for a ranging over the root system Φ_1 consisting of the $a \in \Phi$ vanishing on S_1. It is defined over $K^\natural$, quasi-split over K, split over $\widetilde{K}$ and admits T as a maximal torus. We can therefore substitute it for G in all the results of the preceding paragraphs and consider in particular its building $\mathcal{I}(Z_1)$. Moreover, we can apply to it the results of I, 7.6 (see also 4.2.17 and 18). Let $\mathcal{I}_{S_1} = Z_1(K).A$: this is the union of the apartments of $\mathcal{I}$ associated with the maximal K-split tori of G containing S_1, since these tori are the maximal K-split tori of Z_1, or even the conjugates of S by the elements of $Z_1(K)$. It was shown (I, 7.6.4) that there exists a map π (denoted $\tilde{\pi}$ in I, 7.6) and only one from $\mathcal{I}_{S_1}$ onto $\mathcal{I}(Z_1)$ commuting with the actions of $Z(K)$ and such that, for any valuation $\varphi \in A$, the image $\pi(\varphi)$ is the valuation of the root datum $(T, (U_a)_{a\in\Phi_1})$ of Z_1 obtained by restriction of φ. Moreover (loc. cit.) the inverse image under π of an apartment (resp. affine root) of $\mathcal{I}(Z_1)$ is an apartment (resp. affine root) of $\mathcal{I}$ and for every $x \in A$, the inverse image $L_x = \pi^{-1}(\pi(x))$ is an affine subspace of A, whose direction is the intersection of the kernels of the $a \in \Phi_1$. In particular, L_x is stable by $\nu(S_1(K))$. If F is a facet of $\mathcal{I}$ contained in $\mathcal{I}_{S_1}$, there exists a facet F' of $\mathcal{I}(Z_1)$ such that $F \subset \pi^{-1}(F')$. Consequently, if F_1 and F_2 are two facets of $\mathcal{I}$ contained in $\mathcal{I}_{S_1}$, there exists an apartment $A' = z.A$ (with $z \in Z_1(K)$) containing $S_1(K).F_1$ and $S_1(K).F_2$.

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5.1.4. — The Galois group $\Sigma = \mathrm{Gal}(K/K^\natural)$ acts on the building $\mathscr{I}$ of G (4.2.12). We denote by $\mathscr{I}_\natural$ the set of fixed points of Σ in $\mathscr{I}$. It is a convex subset of $\mathscr{I}$, stable under $G^\natural$: in other words, the condition (DI 1) of I, 9.2.1 is satisfied.

For every apartment A' of $\mathscr{I}$, there exists a normal subgroup Σ' of finite index in Σ fixing all the points of A'. Indeed, the maximal K-split torus corresponding to A' is defined and split over a finite Galois extension K_1 of $K^\natural$ contained in K, the group G is quasi-split over K_1 and it suffices to take $\Sigma' = \mathrm{Gal}(K/K_1)$ (cf. 4.2.24). In particular, the orbit under Σ of any point of I is finite and the image of Σ in Isom $\mathscr{I}$ is bounded. It follows (I, 3.2, I, 8.2) that if M is a convex subset of $\mathscr{I}$ stable under Σ, then Σ has a fixed point belonging to the closure of M in the completion $\widehat{\mathscr{I}}$ of $\mathscr{I}$. If ω is discrete, $\mathscr{I}$ is complete and this point belongs to $\mathscr{I}_\natural$; taking $M = \mathscr{I}$, we see that $\mathscr{I}_\natural$ is non-empty.

5.1.5. — Of course, Σ also acts on the building $\mathscr{I}(Z)$ of Z and the canonical map from $\mathscr{I}_{S^\natural}$ (which is obviously invariant under Σ as the union of apartments associated with maximal K-split tori containing $S^\natural$) to $\mathscr{I}(Z)$ is compatible with the actions of Σ. From what precedes, it follows that if ω is discrete, the following condition is satisfied:

(DE) *The Galois group $\Sigma = Gal(K/K^\natural)$ has a fixed point in the building $\mathscr{I}(Z)$ of the centralizer Z of $S^\natural$.*

5.1.6. — When ω is dense, we do not know if (DE) is always satisfied or not[16]. However:

Proposition 0.59. *Suppose one of the two following conditions is satisfied:*

a) The completion of $K^\natural$ is maximally complete;

b) There exists a Galois sub-extension K_1 of K, of degree 2^n, such that G is quasi-split over K_1.

Then, the condition (DE) is satisfied.

In case a), let us denote by K_1 a Galois sub-extension of finite degree of K such that G is quasi-split over K_1. In both cases, the smallest extension $\widetilde{K}_1$ of K_1 splitting G is contained in $\widetilde{K}$ and is thus a univalent extension of K_1. Similarly, Z is quasi-split over K_1 and split over $\widetilde{K}_1$, which allows us to speak of the building $\mathscr{I}_1(Z)$ of Z_{K_1}. We have seen (4.2.24) that $\mathscr{I}_1(Z)$ identifies canonically with a subset of $\mathscr{I}(Z)$ formed of points invariant under the Galois group $\mathrm{Gal}(K/K_1)$ and on which the action of Σ on $\mathscr{I}(Z)$ induces the natural action of $\Sigma_1 = \mathrm{Gal}(K_1/K^\natural)$.

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In case a), the completion of K_1 is maximally complete, as a finite degree extension of a maximally complete field, and the building $\mathscr{I}_1(Z)$ is complete (I, 7.5.5), hence (DE) (cf. 5.1.4).

[16] The proof of the existence of fixed points of the Galois group in the building given in [27] is insufficient, as G. Rousseau himself indicated to us: corollary 1.2.9, p. I. 17, on which it relies, is unproven and probably incorrect.

In case b), let us show by induction on n that Σ_1 has a fixed point in $\mathscr{I}_1(Z)$. This is obvious if $n = 0$. If $n > 0$, Σ_1 possesses a normal subgroup Σ_1' of index 2. By replacing K^h with the field of invariants of Σ_1' and applying the induction hypothesis, we see that the set M of points of $\mathscr{I}_1(Z)$ invariant under Σ_1' is non-empty. But Σ_1 operates on M via its quotient Σ_1/Σ_1' of order 2 and if $x \in M$, the midpoint of the segment joining x to its transform by the non-trivial element of Σ_1/Σ_1' is fixed by Σ_1, q.e.d.

5.1.7. — *From now on, we assume the condition (DE) is satisfied.* It immediately follows that Σ has a *fixed point in* $\mathscr{I}_{K^h}$: if x is a fixed point of Σ in $\mathscr{I}(Z)$, the preimage $\pi^{-1}(x)$ is an affine subspace of an apartment of $\mathscr{I}$, it is invariant under Σ and Σ operates on it by affine automorphisms via a finite quotient (5.1.4), hence $\mathscr{I}$ possesses a fixed point.

5.1.8. — Let $\mathfrak{G}$ be a smooth $\mathcal{O}$-group scheme with generic fiber G_K, of affine algebra $\mathcal{O}[\mathfrak{G}] \subset K[\mathfrak{G}] = K \otimes K^h[G]$. Let the Galois group Σ operate in the natural way on $K[\mathfrak{G}]$ and on $G(K)$. The following conditions are equivalent:

a) $\mathcal{O}[\mathfrak{G}]$ is invariant under Σ;

b) $\mathfrak{G}$ comes by base change $\mathcal{O}^h \to \mathcal{O}$ from a smooth $\mathcal{O}^h$-group scheme $\mathfrak{G}^h$ with generic fiber G.

This is obvious by étale descent: the affine algebra $\mathcal{O}^h[\mathfrak{G}^h]$ is none other than the set of fixed points of Σ in $\mathcal{O}[\mathfrak{G}]$. Note that if ω is discrete or more generally if $\mathfrak{G}$ satisfies condition (ET 1) of 1.7.2, then $\mathcal{O}[\mathfrak{G}] = \{f \in K[G] \mid f(G(\mathcal{O})) \subset \mathcal{O}\}$ (since $\mathcal{O}$ is strictly Henselian (1.7.6)) and a) is equivalent to

c) $G(\mathcal{O})$ is invariant under Σ.

5.1.9. — This applies in particular to the schemes $\mathfrak{G}_\Omega$ of 4.6. Notably, let Ω be a bounded subset of an apartment of $\mathscr{I}$, satisfying, when ω is dense, the condition of 4.6.26. If Ω is invariant under Σ, it follows from 4.6.30 that $\mathcal{O}[\mathfrak{G}_\Omega]$ is invariant under Σ, hence that $\mathfrak{G}_\Omega$ comes by base change from an $\mathcal{O}^h$-scheme. The same holds for the schemes $\mathfrak{G}_\Omega, \mathfrak{G}_\Omega^\circ$ and $\mathfrak{G}_\Omega^\natural$.

In particular, *if ω is discrete, to every facet F of $\mathscr{I}$ invariant under Σ is canonically associated a connected smooth $\mathcal{O}^h$-group scheme $\mathfrak{G}_F^h$ of generic fiber G such that $\mathfrak{G}_F = (\mathfrak{G}_F^h)_\mathcal{O}$.* If ω is dense, the schemes $\mathfrak{G}_F^h$ no longer exist in general, but it follows from 4.6.42 (taking 5.1.4 into account) that for every facet F of $\mathscr{I}$ invariant under Σ, contained in an apartment A', there exists $\delta > 0$ tel que, for every ϵ-thickening f of F in A' with $\epsilon < \delta$, the $\mathcal{O}$-scheme $\mathfrak{G}_f$ is invariant under Σ and thus comes from an $\mathcal{O}^h$-scheme $\mathfrak{G}_f^h$. One sees similarly that the algebraic group $\overline{\mathfrak{G}}_F^\circ$ attached to F (4.6.40) is "defined over $\overline{k}^h$".

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Note also that if S' is a torus defined over $K^\natural$ and split over K and if $\mathfrak{S}'$ is the canonical $\mathcal{O}$-scheme with generic fiber S', then $\mathfrak{S}'$ arises by base change from a smooth $\mathcal{O}^\natural$-group scheme whose generic fiber is canonically $K^\natural$-isomorphic to S'. This $\mathcal{O}^\natural$-scheme is called the canonical $\mathcal{O}^\natural$-scheme with generic fiber S' and is denoted $\mathfrak{S}'^\natural$. It is moreover defined in 4.4.

One will sometimes allow oneself to suppress the exponents $\natural$ in these notations.

Proposition 0.60 (5.1.10)**.** *Let F be a facet of $\mathscr{I}$ invariant under Σ, let A' be an apartment containing F and let S' be the maximal K-split torus of G associated with A'. We are also given one of the two following objects:*

a) A sub-torus S_1 of S', defined over $K^\natural$ and $K^\natural$-split;

b) A $\overline{K}^\natural$-split sub-torus $\overline{S}_1^\natural$ of $\overline{G}_F^\natural$.

Then, there exists a maximal K-split torus S'' of G, defined over $K^\natural$, such that F is contained in the apartment A'' associated with S'' and that

- *in case a), S'' contains S_1;*
- *in case b), the canonical $\mathcal{O}^\natural$-scheme $\mathfrak{S}''$ with generic fiber S'' possesses a closed sub-scheme of split tori whose canonical image in $\overline{G}_F^\natural$ is $\overline{S}_1^\natural$.*

Let $\mathfrak{G}^\natural$ be the smooth $\mathcal{O}^\natural$-group scheme defined as follows: if ω is discrete, we take $\mathfrak{G}^\natural = \mathfrak{G}_F^\natural$; if ω is dense, we take $\mathfrak{G}^\natural = \mathfrak{G}_f^\natural$, where f is an ϵ-thickening of F in A' such that $\mathfrak{G}_f^\natural$ is invariant under Σ. Let $\mathfrak{S}'$ be the canonical $\mathcal{O}$-scheme with generic fiber S'. It is a closed sub-scheme of $(\mathfrak{G}^\natural)_\mathcal{O}$.

In case a), let $\mathfrak{S}_1^\natural$ be the canonical $\mathcal{O}^\natural$-scheme with generic fiber S_1 and let u be the closed immersion of $(\mathfrak{S}_1^\natural)_\mathcal{O}$ in $\mathfrak{S}'$ extending the inclusion of S_1 in S'.

In case b), let us define a $\overline{K}$-isomorphism $\bar{u}$ from $\overline{S}_1^\natural$ onto a $\overline{K}^\natural$-split sub-torus of $\overline{\mathfrak{G}}^\natural$ in the following manner: if ω is discrete, $\bar{u}$ is the identity; if ω is dense, the canonical isomorphism from $\overline{\mathfrak{G}}^\natural/Q_f$ onto $\overline{G}_F^\natural$ (4.6.41) allows,

since Q_f is unipotent, to lift $\overline{S}_1^\natural$ to a $\overline{K}^\natural$-split torus of $\overline{\mathfrak{G}}^\natural$, in other words to choose a $\overline{K}^\natural$-isomorphism $\bar{u}$ from $\overline{S}_1^\natural$ onto a $\overline{K}^\natural$-split torus of $\overline{\mathfrak{G}}^\natural$ such that the composite of $\bar{u}$ and the canonical homomorphism from $\overline{\mathfrak{G}}^\natural$ to $\overline{G}_F^\natural$ is the identity.

The closed fiber $\overline{\mathfrak{S}}'$ is a maximal $\overline{K}$-split torus of $(\overline{\mathfrak{G}}^\natural)_{\overline{K}}$ (4.6.4), therefore a maximal torus since $\overline{K}$ is separably closed. Let $\overline{\mathfrak{S}}''$ be a maximal torus defined over $\overline{K}^\natural$ of $\overline{\mathfrak{G}}^\natural$ containing $\bar{u}(\overline{S}_1^\natural)$, and let $\bar{g} \in \overline{\mathfrak{G}}^\natural(\overline{K})$ such that $\overline{\mathfrak{S}}'' = \bar{g} \cdot \overline{\mathfrak{S}}' \cdot \bar{g}^{-1}$ and belonging in case a) to the centralizer of $\overline{\mathfrak{S}}_1^\natural$. From the results of Grothendieck on the existence and smoothness of the scheme of homomorphisms from a group scheme of multiplicative type and of finite type into a smooth (affine) group scheme ([SGAD], Exp. XI, cor. 4.2), we deduce on the one hand that there exists a closed $\mathcal{O}^\natural$-group sub-scheme $\mathfrak{S}''$ of $\mathfrak{G}^\natural$, isomorphic to Mult^r with $r = \dim \overline{\mathfrak{S}}'' = \dim \overline{\mathfrak{S}}' = \dim S$, with closed fiber $\overline{\mathfrak{S}}''$, and on the other hand that $\bar{g}$ lifts to an element $g \in \mathfrak{G}^\natural(\mathcal{O}) \subset P_F^\circ$ (4.6.7 and 4.6.40) such that $\mathfrak{S}'' = g \cdot \mathfrak{S}' \cdot g^{-1}$ and which belongs to the centralizer of $\mathfrak{S}_1^\natural$ in case a) (recall...

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that this centralizer is smooth). Then, the generic fiber $S'' = g.S'.g^{-1}$ is a maximal K-split torus of G, defined over $K^\natural$, and $F = g.F$ belongs to the apartment $g.A'$ associated to S''. Furthermore, $S'' \supset S_1$ in case a). In case b), the same theorem of Grothendieck allows us to lift an $\overline{K^\natural}$-isomorphism $\bar{v} : \mathfrak{Mult}' \to \bar{u}(\overline{\mathfrak{S}}_1)$ into an $\mathcal{O}^\natural$-morphism $v : \mathfrak{Mult}' \to \mathfrak{S}''$ which is a closed immersion since $\bar{v}$ is one ([SGAD], Exp. IX, 2.5 and 6.6). The $\mathcal{O}^\natural$-sub-scheme of $\mathfrak{S}''$ image of v then possesses all the required properties, which completes the proof of the proposition.

Corollary 0.61 (5.1.11)**.** *Let F be a facet of A invariant under Σ. The canonical image of $\mathfrak{S}_F^\natural$ in $\overline{\mathfrak{G}}_F^0$ is a maximal $\overline{K^\natural}$-split torus of $\overline{\mathfrak{G}}_F^0$.*

Corollary 0.62 (5.1.12)**.** *There exists a maximal K-split torus of G, defined over $K^\natural$ and containing the maximal $K^\natural$-split torus $S^\natural$.*

Let us take for F a facet invariant under Σ contained in $\mathscr{I}_{S^\natural}$ (5.1.7); it is then contained in an apartment A' associated to a maximal K-split torus S' containing $S^\natural$ (5.1.3) and it suffices to apply 5.1.10 a), with $S_1 = S^\natural$.

Lemma 0.63 (5.1.13)**.** *Let S' be a maximal K-split torus of G, defined over $K^\natural$, with associated apartment A', and let S_1 be the maximal $K^\natural$-split sub-torus of S'.*

(i) The set $A'_\natural = A' \cap \mathscr{I}_\natural$ of points of A' invariant under Σ is a non-empty affine subspace.

(ii) $A'_\natural$ is the convex hull of $S_1(K).x$ for any $x \in A'_\natural$ and $\dim A'_\natural = \dim S_1/S_1 \cap C$, where C is the center of G.

Proof. Up to replacing G by G/C, we can assume that $C = \{1\}$. The apartment A' is invariant under Σ, which acts on it via a finite group of affine automorphisms (5.1.4), whence (i). Let Φ' be the root system of G relative to S' and V' the vector space of translations of A'. Since Φ' generates a sub-module of finite index of $X^*(S')$, the torus S_1 is the neutral component of the intersection of the kernels (in S) of the characters $a^\sigma - a$ for $a \in \Phi'$ and $\sigma \in \Sigma$. But the subspace $V'_\natural$ of invariants of Σ in V' is also the intersection of the kernels of the $a^\sigma - a$ (this time in V') and (ii) follows from it (cf. 4.2.7). □

Proposition 0.64 (5.1.14)**.** *Let S' be a maximal K-split torus of G, defined over $K^\natural$ and let S_1 be the maximal $K^\natural$-split sub-torus of S'.*
Let F be a facet of the apartment A' associated to S', meeting $A'_\natural = A' \cap \mathscr{I}_\natural$. We assume that the image of $\overline{\mathfrak{S}}_1$ in $\overline{\mathfrak{G}}_F^0$ is a maximal $\overline{K^\natural}$-split torus and that there exists no facet $F' \neq F$ meeting $A'_\natural$ such that $F \subset \overline{F'}$ (e.g. that $\dim F$ is the largest possible).
Then:

(i) $F \cap \mathscr{I}_\natural = F \cap A'_\natural$ is open (cf. I, 7.2.1) in $A'_\natural$ and $A'_\natural$ is the smallest affine subspace of A' containing $F \cap A'_\natural$;

(ii) F is not contained in the closure of any facet $F' \neq F$ meeting $\mathscr{I}_\natural$;

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(iii) For every apartment A'' containing F, the intersection $F \cap \mathscr{I}_{A_\natural}$ is open in $A'' \cap A'_\natural$;

(iv) For every apartment A'' containing $A'_\natural$, one has $A'' \cap \mathscr{I}_{A_\natural} = A'_\natural$.

Assertion (i) is immediate. To prove (ii), let us take up the notation of 4.6.33 and 4.6.45. Since the extension $\overline{K}$ of $K^\natural$ is separable, the $\overline{K}^\natural$-pseudo-parabolic subgroups of $\overline{\mathfrak{G}}_F$ are the $\overline{K}$-pseudo-parabolic subgroups invariant under Σ, i.e., the images under p of the facets $F' \in \mathscr{I}(F)$ invariant under Σ. However, as $\overline{\mathfrak{S}}_1$ is a maximal $\overline{K}^\natural$-split torus of $\overline{\mathfrak{G}}_F$, every $\overline{K}^\natural$-pseudo-parabolic subgroup of $\overline{\mathfrak{G}}_F$ is conjugate to a $\overline{K}^\natural$-pseudo-parabolic subgroup containing $\overline{\mathfrak{S}}_1$, hence containing the centralizer of $\overline{\mathfrak{S}}_1$ and a fortiori containing $\overline{\mathfrak{S}}'$, thus the image of a facet $F' \in \mathscr{I}(F)$ invariant under Σ and contained in A'. But it follows from (i) that such a facet is equal to F. One concludes that $p(F) = \overline{\mathfrak{G}}_F$ is the unique $\overline{K}^\natural$-pseudo-parabolic subgroup of $\overline{\mathfrak{G}}_F$ and that F is the unique element of $\mathscr{I}(F)$ meeting $\mathscr{I}_{A_\natural}$, which was to be shown.

Assertion (iii) then follows from (ii) and from I, 9.2.5 (iii), by taking $M = A'' \cap \mathscr{I}_{A_\natural}$, and (iv) follows from I, 9.2.6 (ii).

5.1.15. — Henceforth, we assume, which is permissible by 5.1.12, that the maximal K-split torus S containing $S^\natural$ is defined over $K^\natural$. It is then the same for T and the group schemes $\mathfrak{S}$, $\mathfrak{T}$ and $\mathfrak{T}^\circ$ arise by base change from $\mathcal{O}^\natural$-group schemes. Furthermore, $A_\natural = \mathscr{I}_{A_\natural} \cap A$ is then a non-empty affine subspace of A (5.1.13): in other words, condition (DI 2) of I, 9.2.1 is satisfied. The direction of $A_\natural$ is the space of invariants of Σ in V, i.e., the subspace $V^\natural$ introduced in 5.1.2, which is consistent with the notation of I, § 9 (9.2.8).

Moreover, we assume henceforth that the valuation φ belongs to $A_\natural$ (one should beware that one cannot in general assume simultaneously that $\varphi \in A_\natural$ and that φ is a Chevalley-Steinberg valuation: it is easy to give examples where $A_\natural$ does not even contain any special point of A). The family of sets of values $(\Gamma'_a)_{a\in\Phi}$ is then invariant under Σ in the sense that $\Gamma'_{\sigma(a)} = \Gamma'_a$ for all $a \in \Phi$ and $\sigma \in \Sigma$. Indeed, if $k \in \Gamma'_a$, the image under σ of the affine root $\alpha_{a,k} = \{x \in A \mid a(x-\varphi) + k \geq 0\}$ is again an affine root, which is $\{x \in A \mid \sigma(a)(x-\varphi) + k \geq 0\}$, whence $k \in \Gamma'_{\sigma(a)}$. Furthermore, one has

$$\mathfrak{U}_{\sigma(a),k} = \sigma(\mathfrak{U}_{a,k}). \tag{1}$$

5.1.16. — Recall (4.2.2 and I, 9.1.6) that the family $\psi^\natural = (\Gamma^\natural_b)_{b\in\Phi^\natural}$ is defined as follows. Let $b \in \Phi^\natural$; we have set

$$\Phi^b = \{a \in \Phi \mid a^\natural = b \text{ or } a^\natural = 2b\}.$$

It is a positively closed subset of Φ, invariant under Σ. For every $k \in \mathbf{R}$, the set

$$U_{b,k} = \prod_{a\in\Phi^b, a^\natural = b} U_{a,k} \cdot \prod_{a\in\Phi^b_{\mathrm{nd}}, a^\natural = 2b} U_{a,2k}$$

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is a subgroup of $U_b(K)$ (I, 9.1.5). We set

$$\begin{aligned}
\varphi_b(u) &= \sup\{k \in \mathbf{R} \mid u \in U_{b,k}\} \quad (u \in U_b(K)),\\
U^\natural_{b,k} &= U_{b,k} \cap U_b(K^\natural) = \{u \in U_b(K^\natural) \mid \varphi_b(u) \geq k\},\\
\Gamma^\natural_b &= \varphi_b(U_b(K^\natural) - \{1\}),\\
\Gamma^{\natural\prime}_b &= \{\varphi_b(u) \mid u \in U_b(K^\natural) - \{1\} \text{ and } \varphi_b(u) = \sup \varphi_b(uU_{2b}(K^\natural))\},
\end{aligned}$$

(cf. I, 6.2.2).

Lemma 0.65 (5.1.17)**.** *Let L be a finite Galois extension of $K^\natural$, contained in K, with valuation ring $\mathcal{O}_L$, Galois group Σ_L, and let M be a free $\mathcal{O}_L$-module of finite type.*
(i) For every operation law of Σ_L on M such that, for $\sigma \in \Sigma_L$, the map $m \mapsto m^\sigma$ is a σ-automorphism of M, we have $H^q(\Sigma_L, M) = \{0\}$ for all $q \geq 1$.
(ii) Under the same hypotheses, we have $H^1(\Sigma_L, GL(M)) = \{1\}$.
(iii) Let $\mathfrak{X}$ be the $\mathcal{O}_L$-group scheme canonically associated to M, equipped with the canonical operation law of $\mathfrak{Mult}$ on M. Let us assume given an operation law $(\sigma, x) \mapsto x^\sigma$ of Σ_L on $\mathfrak{X}$ such that, for all σ, the map $x \mapsto x^\sigma$ is a σ-automorphism of $\mathfrak{X}$ and $(t.x)^\sigma = t^\sigma.x^\sigma$ for $t \in \mathfrak{Mult}$ and $x \in \mathfrak{X}$. Then, $\mathfrak{X}$, equipped with these two operation laws, is isomorphic to the $\mathcal{O}_L$-group scheme obtained by base change $\mathcal{O}^\natural \to \mathcal{O}_L$ from the $\mathcal{O}^\natural$-group scheme $\mathfrak{X}^\natural$ associated to a free $\mathcal{O}^\natural$-module of finite type $M^\natural$ equipped with the canonical operation law of $\mathfrak{Mult}$ and the canonical operation law of Σ_L on $(\mathfrak{X}^\natural)_{\mathcal{O}_L}$.

Let $e \in \mathcal{O}_L$, with image $\bar{e} \in \bar{L}$ such that $(\bar{e}^\sigma)_{\sigma\in\Sigma_L}$ is a normal basis of $\bar{L}$ over $\bar{K}^\natural$. Then, $(e^\sigma)_{\sigma\in\Sigma_L}$ is a basis of $\mathcal{O}_L$ over $\mathcal{O}^\natural$ and $\det(e^{\sigma\tau})_{\sigma,\tau\in\Sigma_L} \in \mathcal{O}_L^\times$. It follows immediately that the set $M^\natural$ of fixed points of Σ_L in M is a free $\mathcal{O}^\natural$-module of finite type and that the canonical map from $\mathcal{O}_L \otimes M^\natural$ into M is an isomorphism of $\mathcal{O}_L$-modules transforming the natural action of Σ_L into the given action on M.

To prove (i), it then suffices to show that $H^q(\Sigma_L, \mathcal{O}_L) = \{0\}$ for $q \geq 1$, which follows from the fact that the Σ_L-module $\mathcal{O}_L$ is an induced module ([28], p. 158) since $(e^\sigma)_{\sigma\in\Sigma_L}$ is a basis of it.

Let us prove (ii). Here too, it suffices to show that $H^1(\Sigma_L, GL_n(\mathcal{O}_L)) = \{1\}$ and for that to reproduce the proof by Cartier (see [28], p. 159) of the analogous assertion in the case of fields, by noting that vectors $y_1, \ldots, y_n$ form a basis of $\mathcal{O}_L^n$ if and only if their images in $\bar{L}^n$ form a basis of it.

Finally, (iii) follows from (ii) by Galois descent, since the group of automorphisms of $\mathfrak{X}$ equipped with the operation law of $\mathfrak{Mult}$ is canonically identified with $GL(M)$.

Proposition 0.66 (5.1.18)**.** *Let $a \in \Phi$. Let Σ_a be the stabilizer of a in Σ, K_a the field of invariants of Σ_a, and $\mathcal{O}_a$ the valuation ring of K_a. Let $k \in \Gamma'_a$.*
(i) The $\mathcal{O}$-group schemes $\mathfrak{U}_{a,k}$, $\mathfrak{U}_{2a,2k}$ and $\mathfrak{U}_{a,k}/\mathfrak{U}_{2a,2k}$ arise by base change from $\mathcal{O}_a$-group schemes whose underlying scheme is associated with a free $\mathcal{O}_a$-module of finite type.
(ii) The canonical map $\mathfrak{U}_{a,k}(\mathcal{O}_a) \to (\mathfrak{U}_{a,k}/\mathfrak{U}_{2a,2k})(K_a)$ is surjective.

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Assertion (i) results from Lemma 5.1.17 (iii), taking into account 5.1.15 (1) and 4.4.19 (1) and (2). Since $H^1(\Sigma_a, \mathfrak{U}_{2a,2k}(\mathcal{O})) = \{0\}$ (5.1.17 (i)) and the canonical map $\mathfrak{U}_{a,k}(\mathcal{O}) \to (\mathfrak{U}_{a,k}/\mathfrak{U}_{2a,2k})(\mathcal{O})$ is surjective (4.3.7), the canonical map $\mathfrak{U}_{a,k}(\mathcal{O}_a) \to (\mathfrak{U}_{a,k}/\mathfrak{U}_{2a,2k})(\mathcal{O}_a)$ is surjective, whence (ii) since $\mathfrak{U}_{a,k}/\mathfrak{U}_{2a,2k}$ is smooth and $\mathcal{O}_a$ is Henselian.

Proposition 0.67 (5.1.19)**.** *Let $b \in \Phi^\natural$ and $k \in \mathbf{R}$. The following conditions are equivalent:*

(i) $k \in \Gamma_b^{\natural\prime}$;

(ii) there exists $a \in \Phi$ such that $a^\natural = b$ and $k \in \Gamma'_a$.

To say that $k \in \Gamma_b^{\natural\prime}$ means that there exists $u \in U_{b,k}^\natural$ such that

$$u \notin \prod_{a\in\Phi^b, a^\natural=b} (U_{a,k+} \cdot U_{2a,2k}) \cdot \prod_{a\in\Phi^b_{nd}, a^\natural=2b} U_{a,2k}. \tag{1}$$

This is impossible if (ii) is not satisfied, since one then has $U_{a,k} = U_{a,k+} \cdot U_{2a,2k}$ for all $a \in \Phi^b$ such that $a^\natural = b$. Thus (i) implies (ii).

Conversely, let $a \in \Phi$ such that $a^\natural = b$ and $k \in \Gamma'_a$. Let us identify $\mathfrak{U}_{a,k}/\mathfrak{U}_{2a,2k}$ with the group scheme associated to a free $\mathcal{O}_a$-module of finite type M. It results from the very construction of the scheme $\mathfrak{U}_{a,k}$ (4.3.2 and 4.3.5) that the canonical image I of $U_{a,k+} \cdot U_{2a,2k} \subset \mathfrak{U}_{a,k}(\mathcal{O})$ in $(\mathfrak{U}_{a,k}/\mathfrak{U}_{2a,2k})(\bar{K}) = M \otimes \bar{K}$ is a $\bar{K}$-sub-vector space of $M \otimes \bar{K}$ distinct from $M \otimes \bar{K}$, obviously invariant under Σ_a. Consequently, $(\mathfrak{U}_{a,k}/\mathfrak{U}_{2a,2k})(\bar{K}_a) \not\subset I$ and it follows from 5.1.18 (ii) that there exists $v \in \mathfrak{U}_{a,k}(\mathcal{O}_a) = U_{a,k} \cap G(K_a)$ such that $v \notin U_{a,k+} \cdot U_{2a,2k}$.

Let $x = \prod_{\sigma\in\Sigma/\Sigma_a} v^\sigma$ (taking into account that $v^\sigma = v$ for $\sigma \in \Sigma_a$). Let us consider the set Ψ of elements of Φ of the form $\sigma(a) + \sigma'(a)$ with $\sigma, \sigma' \in \Sigma$. This is a positively closed subset of Φ and the function g on Ψ constantly equal to $2k$ is concave and satisfies (Sch) when ω is dense (4.6.36 (1)). According to 4.6.2, we can therefore consider the scheme $\mathfrak{X} = \mathfrak{U}_{g,\Psi}$, which is associated to a free $\mathcal{O}$-module of finite type. The scheme $\mathfrak{X}$ is invariant under Σ and the torus $S^\natural$ acts on $\mathfrak{X}$ through the non-zero root $2b$; it is immediately deduced using 5.1.17 (iii) that $\mathfrak{X}$ comes from a scheme $\mathfrak{X}^\natural$ associated to a free $\mathcal{O}^\natural$-module of finite type and 5.1.17 (i) implies that $H^1(\Sigma, \mathfrak{X}(\mathcal{O})) = \{0\}$. However, the group of commutators of the subgroup generated by the $U_{\sigma(a),k}$ for $\sigma \in \Sigma$ is contained in the product of the $U_{\sigma(a)+\sigma'(a),2k}$, that is to say in $\mathfrak{X}(\mathcal{O})$. It follows that the map $\sigma \mapsto x^{-1}x^\sigma$ is a 1-cocycle with values in $\mathfrak{X}(\mathcal{O})$ and there exists therefore $y \in \mathfrak{X}(\mathcal{O})$ such that $u = xy \in G(K^\natural)$. Then, $u \in U_{b,k}^\natural$ and u satisfies (1). Consequently, (ii) implies (i).

Theorem 0.68 (5.1.20)**.** *(i) Every valuation $\varphi \in A_\mathfrak{a}$ descends to $G(K^\natural)$; in other words, the family $\varphi^\natural$ defined in 5.1.16 is a valuation of the root data $(Z(K^\natural), (U_b^\natural)_{b\in\Phi^\natural})$ of $G(K^\natural)$ (cf. I, 9.1.11).*
(ii) Let $\varphi \in A_\mathfrak{a}$ and let $\mathcal{I}^\natural$ be the building of $G(K^\natural)$ defined by $\varphi^\natural$. The map $\psi \mapsto \psi^\natural$ is a bijection from $A^\mathfrak{a}$ onto the apartment $A^\natural = \varphi^\natural + V^\natural$ of $\mathcal{I}^\natural$.

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(iii) Let us thus identify $A_\natural$ with the apartment $A^\natural$. The affine roots of $A^\natural$ are the intersections with $A^\natural$ of the affine roots of A associated with roots $a \in \Phi - \Phi_0$, i.e., non-zero on $V^\natural$.

Let us prove (i). It suffices to verify that the hypotheses of Theorem 9.2.10 of Chapter I are satisfied. Now, we have seen successively that this is the case for conditions (DDR 1), (DDR 2), and (DDR 3) (in 5.1.2), (DI 1) (in 5.1.4), (DI 2) (in 5.1.15), and (DI 3) (in 5.1.14 (ii) by taking $S = S'$ and $S^\natural = S_1$). Finally, (DV 1) is the hypothesis $\varphi \in A_\natural$ and (DV 2) follows from 5.1.19.

Assertion (ii) is obvious, since $A_\natural = \varphi + V^\natural$ (5.1.13 and 15). As for (iii), it is another formulation of 5.1.19.

Remarks. — 1) We have assumed since the beginning of 4.6 the hypothesis that the canonical scheme with generic fiber T admits a smoothing (cf. 5.1.1 c)), a hypothesis which is always satisfied when G is simply connected or adjoint, or when ω is discrete. In reality, the preceding theorem remains true even if this hypothesis is removed: it suffices to consider the canonical morphism from G to the adjoint group ad G (cf. 4.2.14 and 15).

The same holds for the results below concerning the building $\mathscr{I}^\natural$ (5.1.21 to 28), which is the same for G and ad G.

2) Suppose G is quasi-split over $K^\natural$, and let $\tilde{K}^\natural$ be the smallest extension of $K^\natural$ contained in $\tilde{K}$ and splitting G. We have $\tilde{K} = K(\tilde{K}^\natural)$ and one verifies immediately that $\tilde{K}^\natural$ is an admissible extension of $K^\natural$, so that one can apply 4.2. The valuations φ^a of the root datum of $G(K^\natural)$ (for $\varphi \in A_\natural$) are the same as those introduced in 4.2 (i.e., the valuations equipollent to a Chevalley-Steinberg valuation), and the building $\mathscr{I}^\natural$ is the same as that of 4.2. This follows from 4.2.9 and Proposition 5.1.23 below, but can also be seen directly. Consider a Chevalley-Steinberg system $C^\natural$ of G over $K^\natural$ relative to S. It canonically defines, by extension of scalars, a Chevalley-Steinberg system C of G over K. Let $\tilde{\varphi}$ be the Chevalley valuation of $G(\tilde{K})$ corresponding to C. Then, on the one hand $\tilde{\varphi}$ descends to a Chevalley-Steinberg valuation φ of $G(K)$ invariant by Σ, hence to a valuation $\varphi^\natural$ of $G(K^\natural)$ in the sense of the preceding theorem, and on the other hand to the Chevalley valuation $\tilde{\varphi}^\natural$ of $G(\tilde{K}^\natural)$ corresponding to $C^\natural$. By transitivity of descent (cf. I, 9.1.12 b)), it follows that $\varphi^\natural$ is obtained by descent from $\tilde{\varphi}^\natural$ and is the Chevalley-Steinberg valuation of $G(K^\natural)$ corresponding to $C^\natural$.

Theorem 0.69 (5.1.21. Corollary)**.** *The facets of the apartment $A^\natural$ of $\mathscr{I}^\natural$ are the non-empty intersections of $A^\natural$ with the facets of A, or also the intersections with $A^\natural$ of the facets of A invariant by Σ.*

This follows from 5.1.20 (iii), taking into account that a facet of A is invariant by Σ if and only if it meets $A^\natural$.

5.1.22. — Let us apply the results of 4.2.18 by taking $\Phi_1 = \Phi_0$, whence $G_1 = Z$ and $L_1 = V^\natural$. We have seen that the subspace $Z(K).(A \times V^1)$ of the enlarged building $\mathscr{I}^1$

of G (4.2.16) is identified with the extended building $\mathscr{I}(Z) \times V_1^\natural$ of Z (with $V_1^\natural = V^\natural \times V^1$) in the following manner: one chooses a point of A — and we naturally choose the point ϕ — and to $z.(x, v)$, with $z \in Z(K)$, $x \in A$ and $v \in V^1$, one associates $z.(\pi(x), p(x - \phi) + v)$, where p is the orthogonal projection of V onto $V^\natural$ and π is the canonical projection of $Z(K).A$ onto $\mathscr{I}(Z)$ (denoted $\tilde{\pi}$ in I, 7.6.4). In view of 4.2.16 (1) and (3), we have

$$z.(\pi(x), p(x - \phi) + v) = (\pi(z.x), p(x - \phi) + v + \theta_Z(z)),$$

where θ_Z is the homomorphism from $Z(K)$ into $V_1^\natural$ defined by

$$\langle \theta_Z(z), c \rangle = -(\omega \circ c)(z) \quad \text{for } z \in Z(K) \text{ and } c \in X_K^*(Z) \otimes \mathbb{Q}.$$

Let us take in particular $z \in Z(K^\natural)$; then z acting on $\mathscr{I}$ fixes $A^\natural$ and induces there the translation $v^\natural(z) \in V^\natural$. Taking into account that the subspace generated by $\Phi^\natural$ in $X_K^*(Z) \otimes \mathbb{R}$ is the orthogonal of the subspace V^1 of $V_1^\natural$, we see that the translation $v^\natural(z)$ is given by

$$\langle v^\natural(z), b \rangle = -(\omega \circ b)(z) \quad \text{for } b \in \Phi^\natural \tag{1}$$

(with the notations of 4.2.5, identifying $X^*(S^\natural) \otimes \mathbb{Q}$ and $X_{K^\natural}^*(Z) \otimes \mathbb{Q}$), in other words, that we have

$$\phi_b^\natural(zuz^{-1}) = \phi_b^\natural(u) + (\omega \circ b)(z) \quad \text{for } z \in Z(K^\natural) \text{ and } b \in \Phi^\natural. \tag{2}$$

Generally speaking, we say that a valuation ψ of the root datum of $G(K^\natural)$ is *compatible with* ω if it satisfies (2) (cf. 4.2.8). One then shows exactly as in 4.2.9:

Proposition 0.70 (5.1.23)**.** *The valuations of the root datum $(Z(K^\natural), (U_b)_{b\in\Phi^\natural})$ of $G(K^\natural)$ compatible with ω are the valuations ψ^ϕ for $\phi \in A_\natural$.*

5.1.24. — In view of I, 9.2.14 and 15, there exists a unique isometry j of the building $\mathscr{I}^\natural$ of $G(K^\natural)$ for $\psi^\natural$ into the building $\mathscr{I}$ of $G(K)$ such that $j(g.x) = g.j(x)$ for all $x \in \mathscr{I}^\natural$ and $g \in G(K^\natural)$. We have $j(\psi^\phi) = \phi$ and the restriction of j to the apartment $A^\natural$ is the inverse of the map $\psi \mapsto \psi^\natural$ which allowed us to identify $A_\natural$ and $A^\natural$ (5.1.20 (ii)). We say that j is the *canonical injection* of $\mathscr{I}^\natural$ into $\mathscr{I}$. We obviously have $j(\mathscr{I}^\natural) \subset \mathscr{I}_\natural$ but we have seen in Chapter I that in the general situation of theorem I, 9.2.10, we do not always have $j(\mathscr{I}^\natural) = \mathscr{I}_\natural$ (e.g., this equality can be incorrect in the situation studied in 4.2 when $\tilde{K}$ is a ramified extension of K). This is however true in the case of "étale descent" that we are studying at this moment:

Theorem 0.71 (5.1.25)**.** *The canonical injection j of the building $\mathscr{I}^\natural$ of $G(K^\natural)$ into the building $\mathscr{I}$ of $G(K)$ is a bijection from $\mathscr{I}^\natural$ onto the set of points of $\mathscr{I}$ invariant by Σ: in other words, we have*

$$\mathscr{I}_\natural = j(\mathscr{I}^\natural) = G(K^\natural).A_\natural.$$

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Let $x \in \mathcal{I}_\natural$. We need to show that $x \in G(K^\natural).A_\natural$. Let F be a facet of $\mathcal{I}$ such that $x \in \bar{F}$, that $F \cap \mathcal{I}_\natural \neq \varnothing$ and that the dimension of F is as large as possible. By applying 5.1.10 b), starting from a maximal $\bar{K}^\natural$-split torus of $\overline{\mathfrak{G}}^0_\phi$, we construct a $K^\natural$-split torus $S_1^\natural$ of G, contained in a maximal K-split torus S'' defined over $K^\natural$, such that F is contained in the apartment A'' associated to S'' and that $\bar{S}_1^\natural$ is a maximal $\bar{K}^\natural$-split torus of $\overline{\mathfrak{G}}''$ as well as of $\overline{\mathfrak{G}}^0_\phi$.

There exists $g \in G(K^\natural)$ such that $gS_1^\natural g^{-1} \subset S^\natural$ (conjugacy of maximal $K^\natural$-split tori!). Up to replacing x by $g.x$, we can therefore suppose that $S_1^\natural \subset S^\natural$. Since the apartments A and A'' are both associated to K-split tori containing $S_1^\natural$, it follows from 5.1.3 that there exists an apartment A' of $\mathcal{I}$ containing ϕ and $S_1(K^\natural).x$, thus ϕ and $A''_\natural$ (5.1.13). But this implies that $\phi \in A''_\natural$ according to 5.1.14 (iv), whence $S_1^\natural(K^\natural).\phi \subset A''_\natural$, and even that $A''_\natural$ is the convex hull of $S_1^\natural(K^\natural).\phi$ since $\dim A''_\natural = \dim S_1^\natural/S_1^\natural \cap C$ (5.1.13) is equal to the dimension of this convex hull. Consequently $A''_\natural \subset A_\natural$ and $x \in A_\natural$, q.e.d.

Corollary 0.72 (5.1.26)**.** *The Galois group Σ has a unique fixed point in the building $\mathcal{I}(Z)$ of the centralizer Z of $S^\natural$.*

By replacing G with Z in all that precedes, we see that the restriction of ϕ to the root datum $(T(K), (U_a(K))_{a\in\Phi_0})$ of $Z(K)$ descends to a valuation of the root datum $(Z(K^\natural), \varnothing)$ of type $\varnothing$ of $Z(K^\natural)$. The building of the latter is on one hand reduced to a point, and on the other hand is identified, in view of the theorem, with the set of fixed points of Σ in $\mathcal{I}(Z)$, whence the corollary.

Recall nevertheless that to prove this corollary (and the following one), we had to, when ω is dense, postulate the existence of fixed points (5.1.5 to 7).

Corollary 0.73 (5.1.27)**.** *Suppose furthermore that G is anisotropic over $K^\natural$.*

(i) The Galois group Σ has a unique fixed point in the enlarged building $\mathcal{I}^1$ of G.

(ii) The group $G(K^\natural)$ of points of G rational over $K^\natural$ is bounded for the natural bornology $\mathcal{B}(\omega)$.

Proof. Indeed, we have $G = Z$, so Σ has a unique fixed point in $\mathcal{I}$; but Σ also has a unique fixed point in V^1 (the origin), since G has no non-trivial rational character over $K^\natural$. Whence (i).

On the other hand, the unique fixed point of Σ in $\mathcal{I}^1$ is invariant by $G(K^\natural)$. Consequently, the image of $G(K^\natural)$ in Isom $\mathcal{I}^1$ is bounded. In view of 4.2.19, this means that $G(K^\natural)$ is bounded for $\mathcal{B}(\omega)$. □

Remark. — In fact, as shown by G. Rousseau ([27]), the group of rational points of any reductive group anisotropic over a Henselian field is bounded. In other words, assertion (ii) above remains true without the additional hypotheses of this para-

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graph and in particular without imposing *a priori* condition (DE). An elementary proof of this result was recently given by G. Prasad ([23]).

5.1.28. — From now on we will identify the building $\mathscr{I}^\natural$ of $G(K^\natural)$ with the set $\mathscr{I}_\natural$ of fixed points of Σ in $\mathscr{I}$ thanks to the canonical injection j.

Corollary 0.74. *The facets of $\mathscr{I}^\natural$ are the intersections with $\mathscr{I}^\natural$ of the facets of $\mathscr{I}$ invariant under Σ (i.e., meeting $\mathscr{I}^\natural$).*

This results from the theorem and from 5.1.21. On the other hand, while it is true that every apartment of $\mathscr{I}^\natural$ is the intersection with $\mathscr{I}$ of an apartment of A invariant under Σ, the converse is incorrect.

5.1.29. — One can also consider the enlarged building $\mathscr{I}^1$ of $G(K)$ and the stabilizer G^1 of $\mathscr{I} \subset \mathscr{I}^1$ in $G(K)$ operating on $\mathscr{I}^1$ (4.2.16). The Galois group Σ operates on $\mathscr{I}^1$ and the enlarged building of $G(K^\natural)$ (defined as in 4.2.16) identifies canonically with the set $\mathscr{I}_\natural^1$ of fixed points of Σ in $\mathscr{I}^1$: it suffices to note that $\mathscr{I}_\natural^1 = \mathscr{I}_\natural \times V_\natural^1$, where $V_\natural^1$ is the set of fixed points of Σ in V^1 and identifies naturally with the dual of $X_{K^\natural}(G) \otimes \mathbf{R}$. Note that the stabilizer $G(K^\natural)^1$ of $\mathscr{I}_\natural$ considered as a subset of $\mathscr{I}_\natural^1$, is the intersection $G^1 \cap G(K^\natural)$. Indeed, if $g \in G(K^\natural)^1$ and if $c \in X_K(G)$, we have $\omega(c(g)) = \frac{1}{n}\omega(c_0(g))$ where c_0 is the sum of the distinct transforms of c by Σ and where n is the number of these transforms, whence $g \in G^1$. It follows immediately from 4.2.19 that the natural bornology on $G(K^\natural)$ (resp. $G(K^\natural)^1$) is equal to the inverse image bornology of that of Isom $\mathscr{I}^\natural$ (resp. Isom $\mathscr{I}_\natural^1$).

5.1.30. — Let Ω be a non-empty bounded subset of $A^\natural$. It is a subset of A invariant under Σ. If ω is discrete or if, when ω is dense, Ω satisfies *as a subset of* $\mathscr{I}$ the conditions of 4.6.26 (which can only be realized if for every $a \in \Phi$ vanishing on $S^\natural$ there exists $k \in \Gamma'_a$ such that $A^\natural \subset \{x \in A \mid a(x-\phi)+k=0\}$, a condition which is not always satisfied), we have associated to it $\mathcal{O}$-schemes $\mathfrak{G}^0_\Omega, \mathfrak{G}_\Omega, \hat{\mathfrak{G}}_\Omega$ and $\mathfrak{G}^\natural_\Omega$. In view of 4.6.30, each of these $\mathcal{O}$-schemes is invariant under Σ and arises by base change from a smooth $\mathcal{O}^\natural$-group scheme, which we will denote by $\mathfrak{G}^{0\natural}_\Omega$, etc. When G is quasi-split over $K^\natural$, one verifies without difficulty that $\mathfrak{G}^{0\natural}_\Omega$ (resp. $\mathfrak{G}^\natural_\Omega$) identifies canonically with the scheme denoted $\mathfrak{G}^0_\Omega$ (resp. $\mathfrak{G}_\Omega$) in 4.6. This allows us, without any resulting ambiguity, to denote more simply by $\mathfrak{G}^0_\Omega$ the $\mathcal{O}^\natural$-scheme $\mathfrak{G}^{0\natural}_\Omega$, which we will do henceforth. On the other hand, the scheme $\hat{\mathfrak{G}}_\Omega$ (resp. $\mathfrak{G}^\natural_\Omega$) of 4.6 identifies only with the open subscheme of $\hat{\mathfrak{G}}^\natural_\Omega$ (resp. $\mathfrak{G}^{\natural\natural}_\Omega$) whose closed fiber consists of the cosets of $\overline{\mathfrak{G}}_\Omega$ (that is, connected components when $\overline{\mathfrak{T}}$ is connected) that have a rational point over $\overline{K}^\natural$. If ω is dense and if X is any non-empty bounded subset of $A^\natural$ (resp. a...

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facet of $A^\natural$), we see as in 5.1.9 that there exists $\delta > 0$ such that for every ϵ-thickening f of X (resp. of the facet X' of A containing X) with $\epsilon < \delta$, the schemes $\mathfrak{G}^0_f$, etc., are invariant under Σ and descend from $\mathcal{O}^\natural$-schemes $\mathfrak{G}^{0\natural}_f$, etc. By passing to the inductive limit, we see that the $\overline{K}$-algebraic groups $\overline{\mathfrak{G}}^0_X$, etc. (resp. $\overline{\mathfrak{G}}^0_{X'}$, etc.) are canonically equipped with a structure of $\overline{K}^\natural$-algebraic group. Finally, one defines in an obvious way the groups $P^\natural_X$.

5.1.31. — It is then easily seen, and we leave the proof to the reader, that most of the results of 4.6 "descend" to $\mathcal{O}^\natural$ or to $\overline{K}^\natural$. Let us mention in particular 4.6.10 (using 5.1.18 and a reasoning analogous to that of 5.1.19), 4.6.12 (the relative root system of the quasi-reductive quotient $\overline{\mathfrak{G}}^0_X$ with respect to the maximal $\overline{K}^\natural$-split torus image of $\mathfrak{S}^\natural$ is the set of $b \in \Phi^\natural$ such that there exists $k \in \Gamma^\natural_b$ with $X \subset \{x \in A \mid b(x-\phi^\natural)+k=0\}$, 4.6.28, 4.6.29 (one will note that the subgroup $U^\natural_\Omega$ acts transitively on the set of apartments containing Ω and is contained in $\mathfrak{G}^0_\Omega(\mathcal{O}^\natural)$), 4.6.30 to 32.

Proposition 0.75 (5.1.32)**.** *Let F be a facet of $\mathcal{I}^\natural$ and let $\mathcal{I}^\natural(F)$ be the star of F in $\mathcal{I}^\natural$.*

(i) The map p which, to a facet $F' \in \mathcal{I}^\natural(F)$, associates the canonical image of $\overline{\mathfrak{G}}^0_{F'}$ in $\overline{\mathfrak{G}}^0_F$ is an isomorphism of ordered sets from $\mathcal{I}^\natural(F)$ onto the set of pseudo-parabolic $\overline{K}^\natural$-subgroups of $\overline{\mathfrak{G}}^0_F$ ordered by the relation opposite to inclusion.

Let $F' \in \mathcal{I}^\natural(F)$.

(ii) The preimage of $p(F')(\overline{K}^\natural)$ in the connected stabilizer $P^{0\natural}_F$ of F is the connected stabilizer $P^{0\natural}_{F'}$ of F', and the homomorphism $P^{0\natural}_F \to p(F')(\overline{K}^\natural)$ is surjective.

(iii) Let Y be a subgroup of $N^\natural_F/N^{0\natural}_F$; the normalizer of $p(F')$ in $Y\overline{\mathfrak{G}}^0_F(\overline{K}^\natural)$ is the canonical image of $Y'\mathfrak{G}^0_{F'}(\mathcal{O}^\natural)$, where $Y' = (YN^{0\natural}_F \cap N^\natural_{F'})/N^{0\natural}_{F'}$, and $Y'\mathfrak{G}^0_{F'}(\mathcal{O}^\natural)$ is its preimage in $Y\mathfrak{G}^0_F(\mathcal{O}^\natural)$.

(iv) The map $F' \mapsto p(F')(\overline{K}^\natural)$ is an isomorphism from $\mathcal{I}^\natural(F)$ onto the combinatorial building $\mathcal{I}_{\overline{F}}$ of the Tits system of $\overline{\mathfrak{G}}^0_F(\overline{K}^\natural)$.

(v) The preimages of the apartments of $\mathcal{I}_{\overline{F}}$ are the sets of facets of $\mathcal{I}^\natural(F)$ contained in the same apartment of $\mathcal{I}^\natural$.

The proof by descent from 4.6.33, 34, 35 and 45 is easy and left to the reader. For (ii) and (iv), one will take into account that the canonical map from $\mathfrak{T}^0(\mathcal{O}^\natural)$ to $\overline{\mathfrak{T}}^0(\overline{K}^\natural)$ is surjective since $\mathcal{O}^\natural$ is Henselian.

Proposition 0.76 (5.1.33)**.** *Let Ω be a non-empty bounded subset of $A^\natural$ and let $S^\natural$ be a maximal $K^\natural$-split torus of G. Suppose that the canonical scheme $\mathfrak{S}^\natural$ with generic fiber $S^\natural$ is contained in $\mathfrak{G}_f^0$ (that is, the injection from $S^\natural$ into G extends to a closed immersion from $\mathfrak{S}^\natural$ into $\mathfrak{G}_f^0$), where f is equal to f_Ω when ω is discrete and is an ϵ-thickening of Ω when ω is dense. Then, Ω is contained in the apartment $A^\natural$ associated with $S^\natural$.*

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The tori $\mathfrak{S}^h$ and $\mathfrak{S}^{h'}$ are $\overline{K}^h$-split sub-tori of $\overline{\mathfrak{G}}_f^0$ and are isomorphic. Consequently, $\mathfrak{S}^{h'}$ is also a maximal $\overline{K}^h$-split torus of $\overline{\mathfrak{G}}_f^0$ and is conjugate to $\mathfrak{S}^h$ by an element of $\overline{\mathfrak{G}}_f^0(\overline{K}^h)$. The smoothness of the closed sub-scheme $Transp(S, S')$ of $\mathfrak{G}_f^0$ ([SGAD], Exp. XI, cor. 5.2) implies that there exists $g \in \mathfrak{G}_f^0(\mathcal{O}^h)$ such that $\mathfrak{S}^{h'} = g\mathfrak{S}^h g^{-1}$, whence the proposition.

5.1.34. Remark. — The same proof shows more generally that if f is a quasi-concave function on Φ, invariant under Σ, such that $\mathfrak{S}^h \subset \mathfrak{G}_f^0$, then the scheme $\mathfrak{G}_f^0$ is also a scheme $\mathfrak{G}_{f'}^0$ obtained starting from $\mathfrak{S}^{h'}$, a maximal K-split torus S' defined over K^h containing $\mathfrak{S}^{h'}$, the apartment A' of $\mathcal{I}$ associated to S', a valuation $\varphi' \in A' \cap \mathcal{I}^h$ and a quasi-concave function f' on the root system Φ' of G relative to S' invariant under Σ.

Lemma 0.77 (5.1.35)**.** *Let k be a finite field, of cardinality ≥ 4, and let h be a reductive group defined over k. Let s be a maximal k-split torus of h. Every parabolic subgroup p of h containing $s(k)$ contains s.*

Proof. Since h is quasi-split over k, one can choose a Borel subgroup b of h, defined over k and containing s. Since $p \cap b$ is solvable connected ([4], 4.5) and contains a maximal torus of h ([4], 4.6), there exists a maximal torus t of h such that $s(k) \subset t \subset p \cap b$ ([4], 11.2). It then suffices to show that the centralizer z of $s(k)$ in b is reduced to t: since $s \subset z$, this will imply $s \subset t \subset p$. Now, z is connected ([4], 11.2) and, since it contains t, is generated by t and the root subgroups u_a for a describing the set of positive roots of h relative to t that are trivial on $s(k)$ ([4], 3.4). It is therefore sufficient to prove that a root a of h relative to t does not vanish on $s(k)$. Now, either the restriction $a|_s$ of a to s is a primitive element of $X^*(s)$ and $a(s(k)) = k^\times \neq \{1\}$ as soon as $Card\ k \geq 3$, or there exists a primitive element a' in $X^*(s)$ such that $a|_s = 2a'$ and $a(s(k)) = (k^\times)^2 \neq \{1\}$ as soon as $Card\ k \geq 4$, whence the lemma. □

Let us note that if the root system of h relative to s has no component of type C_n or BC_n, then every root is primitive and one can replace 4 by 3 in the statement of the lemma.

5.1.36. — *In this section exclusively, we abandon the conventions in force and we resume the notation of I, § 7. In particular, $\mathcal{I}$ denotes the building of a group G endowed with an arbitrary valued root datum, $\hat{\mathcal{I}}$ is its completion, A the standard apartment of $\mathcal{I}$, etc.*

Proposition. — *Let $x \in \hat{\mathcal{I}}$, $z \in A$, C a chamber of A containing z in its closure and ρ the retraction of $\hat{\mathcal{I}}$ onto A with center C (I, 7.5.7). Let $y = \rho(x)$ and let Δ be the closure of the union of the vector chambers D such that $z \in y + \bar{D}$. Then, there exists a sequence $(z_p)_{p\geq 1}$...*

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of points of the segment $[zy]$, converging monotonically towards y, with $z_1 = z$, and a sequence $(u_p)_{p\geq 1}$ of elements of $U_{z+\Delta}$ such that

$$u_{p+1} \in u_p U_{z_p+\Delta} \quad \text{for all } p \geq 1, \tag{1}$$

and

$$x = \lim_{p\to\infty} u_p \cdot y. \tag{2}$$

This is a variant of the first assertion of I, 7.5.2; as there, there exists a converse, the formulation and proof of which we leave to the reader.

Let us set $B = P_C$. One has

$$B = (B \cap U_{z+\Delta}) \cdot (B \cap P_y). \tag{3}$$

Indeed, let Ψ be the set of roots which take strictly positive values on $y - z$ and Ψ_1 a system of positive roots in the system of roots vanishing on $y - z$. Thus, $\Phi^+ = \Psi_1 \cup (-\Psi)$ is a system of positive roots in Φ. As usual, let us denote by U_Ψ, etc., the product of the U_a for $a \in \Psi$, etc. By I, 6.4.9, one has

$$B = (B \cap U_\Psi) \cdot (B \cap U_{-\Psi_1}) \cdot (B \cap U_{\Phi^+}) \cdot H,$$

whence (3), since $B \cap U_\Psi \subset U_{z+\Delta}$ and $(B \cap U_{-\Psi_1}) \cdot (B \cap U_{\Phi^+}) \cdot H \subset P_y$.

Let $(x_p)_{p\geq 1}$ be a sequence of points of $\mathscr{I}$ tending towards x. Let us set $y_p = \rho(x_p)$ and let $g_p \in B$ and $u_p \in B \cap U_{z+\Delta}$ be such that $x_p = g_p \cdot y_p$ and $g_p \in u_p \cdot P_y$. For all p, one has

$$d(x_p, u_p \cdot y) = d(x_p, g_p \cdot y) = d(y_p, y),$$

where, let us recall, d represents the distance. It is thus seen that the $u_p \cdot y$ tend towards x and, by replacing the sequence (x_p) with a subsequence of $(u_p \cdot y)$, if necessary, we can suppose that $x_p = u_p \cdot y$ for all p, whence (2), and that the sequence $(d(x, x_p))$ is decreasing. Let us arbitrarily choose a vector chamber D contained in Δ and a vector $v \in D$. It is then shown, exactly as in I, 7.5.2, that there exists a decreasing sequence $(\lambda_p)_{p\geq 1}$ of real numbers tending towards zero such that $u_p^{-1} u_q \in U_{y+\lambda_p v+D}$ for $q \geq p$. Since the convex hull of $\{y + \lambda_p v\} \cup (z + \Delta)$ contains $y + \lambda_p v + \Delta$, one has

$$u_p^{-1} u_q \in U_{y+\lambda_p v+D} \cap U_{z+\Delta} \subset U_{y+\lambda_p v+\Delta}.$$

But then, if one defines the point z_p of $[zy]$ by the relation

$$[zz_p] = [zy] \cap (y + \lambda_p v + \Delta)$$

as soon as $p \geq 2$ and the intersection on the right-hand side is not empty, and by $z_p = z$ in the other cases, one sees that condition (1) of the statement is satisfied.

Corollary 0.78. — *Let $x \in \hat{\mathscr{I}}$ and $z \in \mathscr{I}$. All points of the geodesic segment $[xz]$, except perhaps x itself, belong to the building $\mathscr{I}$.*

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We can assume $z \in A$ and resume the notation of the proposition, and we then have

$$[zx[= \bigcup_{p\geq 1} u_p \cdot [z_p\ z_{p+1}] \subset \mathcal{I}.$$

Theorem 0.79 (5.1.37. Proposition)**.** *Assume Card $\overline{K}^\natural \geq 4$. The apartment $A^\natural$ is then the set of fixed points of $\mathfrak{S}^\natural(\mathcal{O}^\natural)$ in the completion $\widehat{\mathcal{I}}^\natural$ of the building $\mathcal{I}^\natural$, and a fortiori in $\mathcal{I}^\natural$.*

Proof. Let $x \in \widehat{\mathcal{I}}^\natural$ fixed by $\mathfrak{S}^\natural(\mathcal{O}^\natural)$. Suppose $x \notin A^\natural$ and let $z \in A^\natural$. There exists $z' \in [zx[$ such that $[zx] \cap A = [zz']$. Replacing z by z', we can therefore assume $]zx[\cap A = \varnothing$. Given the preceding corollary, we have $[zx[\subset \mathcal{I}^\natural$ and there exists a unique facet F' of $\mathcal{I}$ meeting $]zx[$ and whose closure contains z and therefore the facet F of A containing z. Then F', like all points of $[zx]$, is invariant under Σ and under $\mathfrak{S}^\natural(\mathcal{O}^\natural)$. The pseudo-parabolic $\overline{K}^\natural$-subgroup $\mathfrak{p}(F')$ of $\overline{G}_F^\circ$ is normalized by $\overline{\mathfrak{S}}^\natural(\overline{K}^\natural)$, thus contains $\overline{\mathfrak{S}}^\natural(\overline{K}^\natural)$. If $\overline{K}^\natural$ is infinite, this results from the fact that $\overline{\mathfrak{S}}^\natural(\overline{K}^\natural)$ is dense in the split torus $\overline{\mathfrak{S}}^\natural$; if $\overline{K}^\natural$ is finite, then $\overline{K}^\natural$ is perfect and it suffices to apply lemma 5.1.35 after passing to the quotient by the unipotent radical of $\overline{G}_F^\circ$. Consequently, the image of $\mathfrak{p}(F')$ in the combinatorial building $\mathcal{I}_F$ of $\overline{G}_F^\circ(\overline{K}^\natural)$ is contained in the apartment associated with $\overline{\mathfrak{S}}^\natural$ and $F' \cap \mathcal{I}^\natural$ is contained in $A^\natural$ (5.1.32 (v)). Hence a contradiction, since $F' \cap]zx[\neq \varnothing$ and $A^\natural \cap]zx[= \varnothing$. □

Theorem 0.80 (5.1.38. Corollary)**.** *Let us return to the notation of 5.1.33, but assume only that $\mathfrak{S}^{\natural'}(\mathcal{O}^\natural) \subset \mathfrak{S}_\Omega^\circ(\mathcal{O}^\natural)$. Assume furthermore Card $\overline{K}^\natural \geq 4$. Then Ω is contained in the apartment of $\mathcal{I}^\natural$ associated with $\mathfrak{S}^{\natural'}$.*

Indeed, the points of Ω are fixed by $\mathfrak{S}^{\natural'}(\mathcal{O}^\natural)$.

Theorem 0.81 (5.1.39. Corollary)**.** *If Card $\overline{K}^\natural \geq 4$, and if Ω is closed, then Ω is the set of fixed points of $P_\Omega^\natural$ in $\mathcal{I}^\natural$.*

This results from 4.6.29 and 5.1.37 when ω is discrete. If ω is dense, we use 5.1.38 and reason as in 4.6.29 using 4.6.40 (i).

Theorem 0.82 (5.1.40. Proposition)**.** *Let $\mathfrak{H}$ be a smooth $\mathcal{O}^\natural$-group scheme with generic fiber G. For the closed fiber of $\mathfrak{H}$ to be reductive, it is necessary and sufficient that G be split over K and that there exist a point $x \in \mathcal{I}^\natural$ which is special as a point of $\mathcal{I}$, such that $\mathfrak{H} \simeq \mathfrak{G}_x^{\circ\natural}$.*

From 4.6.31, we immediately deduce that the condition is sufficient and that, if $\overline{\mathfrak{H}}$ is reductive, G is split over K and there exists a special point $x \in \mathcal{I}$ tel que $\mathfrak{H}$ is $\mathcal{O}$-isomorphic to $\mathfrak{G}_x^\circ$. But $\mathfrak{H}(\mathcal{O})$ is invariant under Σ and x is the only fixed point of $\mathfrak{G}_x^\circ$ in $\mathcal{I}$ (4.6.29). Consequently, x is invariant under Σ, $x \in \mathcal{I}^\natural$ and $\mathfrak{H}$ is $\mathcal{O}^\natural$-isomorphic to $\mathfrak{G}_x^{\circ\natural}$ (1.2.4).

5.1.41. — The building $\mathcal{I}^\natural$ of G over $K^\natural$ "depends functorially" on $K^\natural$. More precisely, let $K^{\natural\prime}$ be an extension of $K^\natural$, equipped with a valuation ω' extending ω;

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suppose $K'^\natural$ is henselian and let K' be a strict henselization of $K'^\natural$, which can be assumed to contain K; finally, suppose that the smallest extension $\widehat{K}'$ of K' splitting G is an admissible extension of K' (which is always true if ω' is discrete (1.6.1)). Consider the canonical isometry j of the building of G over K into the building $\mathscr{I}'$ of G over K' (4.2.24). By transport of structure, the Galois group $\Sigma' = \mathrm{Gal}(K'/K'^\natural)$ acts on $\mathscr{I}'$, and acts on $\mathscr{I}$ via the restriction homomorphism $\mathrm{Gal}(K'/K'^\natural) \to \mathrm{Gal}(K/K^\natural)$, and these operations are compatible with j. It follows that the points of $j(\mathscr{I}^\natural)$ are fixed by Σ', which immediately implies that condition (DE) of 5.1.5 is satisfied for G considered as an algebraic group over K'. One can therefore consider the building $\mathscr{I}'^\natural$ of G over $K'^\natural$, which is none other than the set of fixed points of Σ' in $\mathscr{I}'$ (5.1.25), and the restriction of j to $\mathscr{I}^\natural$ is an isometry, called *canonical*, from $\mathscr{I}^\natural$ into $\mathscr{I}'^\natural$, compatible with the operations of $G(K^\natural)$. Notice that $\varphi' = j(\varphi)$ is a valuation of the root datum of $G(K')$ associated with the maximal K'-split sub-torus S'' of T (4.2.24) and that φ' descends to $G(K)$ according to φ, thus to $G(K^\natural)$ according to $\varphi^\natural$. On the other hand, φ' is invariant under Σ', so it descends to $G(K'^\natural)$ according to a valuation $\varphi'^\natural$ (5.1.20 and 5.1.25). From I, 9.1.12 b), we deduce that $\varphi'^\natural$ descends to $G(K^\natural)$ according to $\varphi^\natural$, which allows us to apply the results of I, § 9. When $K'^\natural$ is an étale extension of $K^\natural$, we have $K' = K$ and the canonical isometry from $\mathscr{I}^\natural$ into $\mathscr{I}'^\natural$ is simply the canonical injection of the set of points of $\mathscr{I}$ invariant under $\mathrm{Gal}(K/K^\natural)$ into the set of points of $\mathscr{I}'$ invariant by $\mathrm{Gal}(K'/K'^\natural)$. If in addition $K'^\natural$ is a Galois extension of $K^\natural$, then $\mathscr{I}^\natural$ is the set of points of $\mathscr{I}'^\natural$ invariant under $\mathrm{Gal}(K'^\natural/K^\natural)$.

5.2. Parahoric subgroups.

From now on, we assume that the valuation ω of K is discrete. We further maintain the previous hypotheses and notation. As a general rule, the objects attached to the valuation $\varphi^\natural$ of $G^\natural = G(K^\natural)$ will be designated by the letter assigned to them by the conventions of Chapter I and the preceding paragraphs, with the superscript $\natural$, which we will sometimes allow ourselves to omit. For example, $N^\natural$ is the stabilizer of $A^\natural$ in $G^\natural$, that is to say the group of rational points over $K^\natural$ of the normalizer of $S^\natural$ in G, $H^\natural$ is the fixer of $A^\natural$ in $G^\natural$, that is to say the subgroup formed by $z \in Z(K^\natural)$ such that $\nu^\natural(z) = 0$, etc.

5.2.1. — ***The scheme*** $\mathfrak{Z}$. We have already remarked that the centralizer Z of $S^\natural$ satisfies all the conditions placed on G, and that one can therefore apply to it everything that precedes. Furthermore, we have seen that the Galois group Σ has a unique fixed point x in the building $\mathscr{I}(Z)$ (5.1.26). Applying then to Z and to $\Omega = \{x\}$ the results of 5.1.30, we find a smooth $\mathcal{O}^\natural$-group scheme $\mathfrak{Z}_{\{x\}}$ with generic fiber Z, which we will call *canonical* and which we will denote more simply by $\mathfrak{Z}$.

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On the other hand, let us denote by Z^1, as in 4.2.16, the set of $z \in Z(K)$ such that $\omega(\chi(z)) = 0$ for all $\chi \in X_K^*(Z)$. According to 4.6.28 (i), we have

$$\mathfrak{Z}(\mathcal{O}) = \{z \in Z^1 \mid z.x = x\}.$$

But, $Z^1 \cap Z(K^\natural)$ is the set of $z \in Z(K^\natural)$ such that $\omega(\chi(z)) = 0$ for all $\chi \in X_{K^\natural}^*(Z)$ and $X_{K^\natural}^*(Z) \otimes \mathbf{Q}$ is identified with the direct sum of

$$X_{K^\natural}^*(G) \otimes \mathbf{Q} = X_{K^\natural}^*(CG) \otimes \mathbf{Q}$$

and of the vector subspace generated by $\Phi^\natural$ in $X^*(S^\natural)\otimes\mathbf{Q}$. Furthermore, $Z(K^\natural)$ fixes the unique point $x\in\mathcal{I}(Z)$ invariant under Σ. We deduce, in view of 5.1.22 (1), that

$$\mathfrak{Z}(\mathcal{O}^\natural)=\{z\in Z(K^\natural)\cap G^1\mid \nu^\natural(z)=0\}.$$

If the connected center of G is $K^\natural$-anisotropic, for example if G is semisimple, then $G^\natural\subset G^1$ and

$$\mathfrak{Z}(\mathcal{O}^\natural)=\{z\in Z(K^\natural)\mid \nu^\natural(z)=0\}=H^\natural.$$

5.2.2. — *The schemes* $\mathfrak{U}_{b,(k,l)}$. Let $b\in\Phi^\natural$ and let $k,l\in\mathbf{R}$ with $l\leq 2k$. Let us set, as in 5.1.16, $\Phi^b=\{a\in\Phi\mid a^\natural=b \text{ or } 2b\}$. The function $g:\Phi^b\to\mathbf{R}$ defined by $g(a)=k$ if $a^\natural=b$ and $g(a)=l$ if $a^\natural=2b$ is concave. We can therefore, according to 4.6.2, consider the group scheme $\mathfrak{U}_{g,\Phi^b}$, which we will also denote by $\mathfrak{U}_{b,(k,l)}$ (or simply $\mathfrak{U}_{b,k}$ when b is non-multipliable). As the product map

$$\prod_{a\in\Phi^b,a^\natural=b}\mathfrak{U}_{a,k}\times\prod_{a\in\Phi^b,a^\natural=2b}\mathfrak{U}_{a,l}\to\mathfrak{U}_{b,(k,l)}$$

is an isomorphism of schemes (4.2.6), we have, with the notation of 5.1.16:

$$\mathfrak{U}_{b,(k,l)}(\mathcal{O})=U_{b,k}\cdot U_{2b,l},$$

a relation that characterizes $\mathfrak{U}_{b,(k,l)}$ among the smooth schemes with generic fiber U_b. Furthermore, it is clear that $\mathfrak{U}_{b,(k,l)}$ is invariant under Σ (5.1.15), thus it comes from a smooth $\mathcal{O}^\natural$-group scheme also denoted $\mathfrak{U}_{b,(k,l)}$. One easily verifies (by reasoning as in 5.1.19) that

$$\mathfrak{U}_{b,(k,l)}(\mathcal{O}^\natural)=U^\natural_{b,k}\cdot U^\natural_{2b,l}.$$

The group scheme $\mathfrak{U}_{2b,l}$ is identified with a closed normal group sub-scheme of $\mathfrak{U}_{b,(k,l)}$ and one easily deduces from 5.1.18 that the schemes $\mathfrak{U}_{b,(k,l)}$, $\mathfrak{U}_{2b,l}$, and $\mathfrak{U}_{b,(k,l)}/\mathfrak{U}_{2b,l}$ are schemes associated with free $\mathcal{O}^\natural$-modules of finite type.

5.2.3. — More generally, let Ψ be a positively closed subset of $\Phi^\natural$ and let $g:\Psi\to\mathbf{R}$ be a concave function. Let us consider the set $\Psi^\#$ of $a\in\Phi$ such that $a^\natural\in\Psi$ and the concave function $g^\#$ on $\Psi^\#$ defined by $g^\#(a)=g(a^\natural)$. From 4.6.2 we deduce a smooth group scheme $\mathfrak{U}_{g^\#,\Psi^\#}$, which is invariant under Σ (5.1.15 (1)) and thus comes from a smooth $\mathcal{O}^\natural$-group scheme that we will denote $\mathfrak{U}_{g,\Psi}$. The product map

$$\prod_{b\in\Psi_{nd}}\mathfrak{U}_{b,(g(b),g(2b))}\to\mathfrak{U}_{g,\Psi}$$

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is an isomorphism of $\mathcal{O}^\natural$-schemes. If one is given a system of positive roots $\Phi^{\natural+}$ in $\Phi^\natural$, and a concave function g on $\Phi^\natural$, we set $\mathfrak{U}^\pm_g=\mathfrak{U}_{g,\Phi^{\natural\pm}}$. When $g=f_\Omega$, where Ω is a non-empty bounded subset of $\mathbb{A}^\natural$, we set $\mathfrak{U}^\pm_\Omega=\mathfrak{U}^\pm_g$, etc.: these are closed subgroup schemes of $\hat{\mathfrak{G}}^\natural_\Omega$ and $\mathfrak{U}^{\pm\natural}_\Omega(\mathcal{O}^\natural)=\mathbf{U}^{\pm\natural}_\Omega$ (according to our conventions, $\mathbf{U}^{\pm\natural}_\Omega$ is the subgroup generated by the $U^\natural_{b,f_\Omega(b)}$ for $b\in\Phi^{\natural\pm}$ (I, 7.1.1)).

5.2.4. — Let Ω be a non-empty bounded subset of $\mathbb{A}^\natural$. As the image of $\mathbb{A}^\natural$ in $\mathscr{I}(Z)$ is the unique point x invariant by Σ, one sees easily that the schematic closure of Z in $\hat{\mathfrak{G}}^\natural$ is identified with $\mathfrak{Z}$. From Theorem 2.2.3 and its Corollary 2.2.8, it then follows that the product morphism

$$\mathfrak{U}^{-\natural}_\Omega\times\mathfrak{Z}\times\mathfrak{U}^{+\natural}_\Omega\to\hat{\mathfrak{G}}^\natural_\Omega\quad(\text{resp. } \mathfrak{U}^{-\natural}_\Omega\times\mathfrak{Z}^0\times\mathfrak{U}^{+\natural}_\Omega\to\mathfrak{G}^0_\Omega)$$

is an isomorphism of $\mathcal{O}^\natural$-schemes onto an open set $\hat{\mathfrak{C}}^\natural_\Omega$ (resp. $\mathfrak{C}^0_\Omega$) of $\hat{\mathfrak{G}}^\natural_\Omega$ (resp. $\mathfrak{G}^0_\Omega$). One then shows as in 4.6.6 and 7 that

$$\mathfrak{G}^0_\Omega=\mathfrak{U}^{+\natural}_\Omega(\mathcal{O}^\natural)\cdot\mathfrak{C}^0_\Omega,\quad \hat{\mathfrak{G}}^\natural_\Omega=\mathfrak{U}^{+\natural}_\Omega(\mathcal{O}^\natural)\cdot\hat{\mathfrak{C}}^\natural_\Omega=\mathfrak{G}^0_\Omega\cdot\mathfrak{Z},$$
$$\mathfrak{G}^0_\Omega(\mathcal{O}^\natural)=\mathbf{U}^{+\natural}_\Omega(\mathcal{O}^\natural)\mathbf{U}^{-\natural}_\Omega(\mathcal{O}^\natural)\mathbf{U}^{+\natural}_\Omega(\mathcal{O}^\natural)\mathfrak{Z}^0(\mathcal{O}^\natural)=\mathbf{U}^{+\natural}_\Omega\mathbf{U}^{-\natural}_\Omega\mathbf{N}^0_\Omega,$$
$$\hat{\mathfrak{G}}^\natural_\Omega(\mathcal{O}^\natural)=\mathfrak{G}^0_\Omega(\mathcal{O}^\natural)\mathfrak{Z}(\mathcal{O}^\natural),$$

where $\mathbf{N}^0_\Omega=\mathbf{N}^\natural\cap\mathfrak{G}^0_\Omega(\mathcal{O}^\natural)$. Moreover $\mathfrak{G}^0_\Omega(\mathcal{O}^\natural)$ is the subgroup P^0_Ω of $G^\natural$ generated by $\mathbf{U}^\natural_\Omega$ and $\mathfrak{Z}^0(\mathcal{O}^\natural)$.

Proposition 0.83 (5.2.5)**.** *If G is simply connected, then the canonical $\mathcal{O}^\natural$-group scheme $\mathfrak{Z}$ with generic fiber Z is connected, i.e., we have $\mathfrak{Z}^0=\mathfrak{Z}$.*

Let us take for Ω a facet F of $\mathbb{A}^\natural$. Since F is a subset of a facet of $\mathbb{A}$, we have $\mathfrak{G}_F=\mathfrak{G}^0_F$ (4.6.32). Consequently, the open set $\mathfrak{C}^0_F$ from 5.2.4 has connected fibers and $\mathfrak{Z}$ is necessarily connected.

Definition 0.6 (5.2.6)**.** The parahoric subgroups (resp. the Iwahori subgroups) of G are the connected stabilizers in $G(K)$ (4.6.28) of the facets (resp. chambers) of the building $\mathscr{I}$.

In other words, the parahoric subgroups of G are the groups $P_F^0 = \mathfrak{G}_F^0(\mathcal{O})$ of integral points of the connected components of the schemes associated with the facets of $\mathscr{I}$. Recall that the map $F \mapsto P_F^0$ is injective (4.6.29). A $K^\natural$-parahoric subgroup of G is by definition a parahoric subgroup invariant by the Galois group Σ, or, which is equivalent, given 4.6.29, the connected stabilizer in $G(K)$ of a facet of $\mathscr{I}$ invariant by Σ or also of a facet of $\mathscr{I}^\natural$ (cf. 5.1.28). From 5.1.27, it then follows:

Proposition 0.84 (5.2.7)**.** *If G is anisotropic over $K^\natural$, then G possesses a unique $K^\natural$-parahoric subgroup.*

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5.2.8. Proposition. — *Two $K^\natural$-parahoric subgroups are equal if and only if their intersections with $G(K^\natural)$ are equal.*

Proof. Let F and F' be two facets of $\mathscr{I}^\natural$. Suppose that $P_F^\natural \cap G(K^\natural) = P_{F'}^\natural \cap G(K^\natural)$ and let us show that $F = F'$. Since F and F' are contained in the same apartment of $\mathscr{I}^\natural$, we can, possibly by conjugating by an element of $G(K^\natural)$, assume them contained in $A^\natural$. If then $F' \neq F$, there exists an element $u \in \mathbf{U}_F^\natural$ not fixing F' (I, 7.4.5); but $\mathbf{U}_F^\natural \subset P_F^\natural \cap G(K^\natural)$, hence a contradiction. Consequently $\overline{F} = \overline{F}'$ and $F = F'$. □

This proposition justifies that one sometimes calls, by abuse of language, parahoric subgroups of $G(K^\natural)$ the intersections with $G(K^\natural)$ of the $K^\natural$-parahoric subgroups of G.

5.2.9. — If G is simply connected, we have seen (4.6.32) that the connected fixer of a facet F of $\mathscr{I}$ is equal to its fixer or to its stabilizer in $G(K)$ and consequently is also the fixer or the stabilizer of any point of F. The groups of rational points over $K^\natural$ of the $K^\natural$-parahoric subgroups of G are then the fixers in $G(K^\natural)$ of facets of $\mathscr{I}^\natural$.

5.2.10. Proposition. — *Suppose G is simply connected. Let C be a chamber of $A^\natural$, B the fixer of C in $G(K^\natural)$, S the set of reflections with respect to the walls of C in the apartment $A^\natural$.*

(i) The quadruplet $(G(K^\natural), B, N^\natural, S)$ is a double Tits system (I, 5.1.1) whose Weyl group is the affine Weyl group of the system of affine roots of $A^\natural$ and whose parahoric subgroups (I, 1.5.1) are the groups of rational points over $K^\natural$ of the $K^\natural$-parahoric subgroups of G.

(ii) Two $K^\natural$-parahoric subgroups of G are conjugate in $G(K)$ if and only if they are conjugate by an element of $G(K^\natural)$ and if and only if their intersections with $G(K^\natural)$ are conjugate in $G(K^\natural)$.

Proof. In view of I, 6.5, it suffices to show for the proof of (i) that $\nu^\natural(N^\natural)$ reduces to the affine Weyl group of the system of affine roots of $A^\natural$, or even that if $n \in N^\natural$ stabilizes the chamber C, then $\nu^\natural(n) = 1$. This is evident, since as we recalled above, the stabilizer and fixer of C in $G(K)$ are identical.
Let us prove (ii). According to 5.2.8, the two conditions of (ii) are equivalent. It thus suffices for us to show that if F and F' are two facets of $\mathscr{I}^\natural$, if P and P' are the corresponding $K^\natural$-parahoric subgroups and if $g \in G(K)$ is such that $P' = gPg^{-1}$, then P and P' are conjugate by an element of $G(K^\natural)$. Since every facet of $\mathscr{I}^\natural$ is transformed by an element of $G(K^\natural)$ to a facet contained in the closure of the chamber C, we can assume $F, F' \subset \overline{C}$. Let $\tilde{F}$ (resp. $\tilde{F}', \tilde{C}$) be the facet of $\mathscr{I}$ whose intersection with $\mathscr{I}^\natural$ is F (resp. F', C). We have $\tilde{F}, \tilde{F}' \subset \overline{\tilde{C}}$ and $\tilde{F}' = g.\tilde{F}$. But two facets contained in the closure of the same facet cannot be conjugate by an element of $G(K)$ unless they are equal (I, 2.1.6, taking into account 4.2.22), hence $P = P'$. □

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5.2.11. — If G is not simply connected, the two assertions of 5.2.10 are in general incorrect. Assertion (i) is already false for $G = \mathrm{PGL}_2$; we will discuss (ii) further on.
One can say that the transition from the simply connected case to the general case of any connected reductive group brings to light phenomena analogous to those that appear in the classical study of reductive groups over

any field from the point of view of parabolic subgroups and Tits systems with a finite Weyl group, when one passes from the case of connected groups to that of not necessarily connected groups.

Let us introduce a subgroup of $G^\natural$ which will play, in a sense, the role of the "identity component" of $G^\natural$, namely the subgroup G' of $G^\natural$ generated by the groups of rational points over $K^\natural$ of the parahoric $K^\natural$-subgroups of G. In view of 5.2.4, it is also the subgroup generated by $\mathfrak{Z}^0(\mathcal{O}^\natural)$ and the $U_b^\natural$ for $b \in \Phi^\natural$. If G is simply connected, 5.2.10 (i) implies that $G' = G^\natural$. In the general case, let $\varphi : \widetilde{G} \to G$ be a simply connected $K^\natural$-covering of the derived group of G. The restriction of φ to the centralizer $\widetilde{Z}$ of the maximal $K^\natural$-split torus $\widetilde{S}^\natural = \varphi^{-1}(S^\natural)$ of $\widetilde{G}$ extends to a morphism from the canonical $\mathcal{O}^\natural$-scheme $\widetilde{\mathfrak{Z}}$ with generic fiber $\widetilde{Z}$ into $\mathfrak{Z}$ (since $\varphi(\widetilde{\mathfrak{Z}}(\mathcal{O})) \subset \mathfrak{Z}(\mathcal{O})$: see 4.2.15) and, as $\widetilde{\mathfrak{Z}}$ is connected (5.2.5), we have $\varphi(\widetilde{\mathfrak{Z}}(\mathcal{O}^\natural)) \subset \mathfrak{Z}^0(\mathcal{O}^\natural)$, whence $\varphi(\widetilde{G}(K^\natural)) \subset G'$ and $G' = \mathfrak{Z}^0(\mathcal{O}^\natural) \cdot \varphi(\widetilde{G}(K^\natural))$.

5.2.12. Proposition. — *Let C be a chamber of $A^\natural$, B' the stabilizer of C in G', S the set of reflections with respect to the walls of C, and $N' = N^\natural \cap G'$.*

(i) The quadruple (G', B', N', S) is a double Tits system whose Weyl group is the affine Weyl group of the affine root system of $A^\natural$ and whose parahoric subgroups are the groups of rational points over $K^\natural$ of the parahoric $K^\natural$-subgroups of G.

(ii) The injection of G' into $G^\natural$ is a B'-N'-adapted homomorphism, of connected type (I, 4.1.3).

This follows easily from I, 6.5.

Let us note that if H is a non-connected reductive group, defined over $K^\natural$, with identity component $H^0 = G$, the injection of G' into $H(K^\natural)$ is still a B'-N'-adapted homomorphism, but which is no longer necessarily of connected type.

5.2.13. — Let us give a counter-example to assertion (ii) of 5.2.10 in the non-simply connected case. Let L be a quadratic extension of $K^\natural$ contained in K and take for G the adjoint group of the group ${}^L\mathrm{SU}_4$ of the Hermitian form $x_{-2}^\sigma x_2 + x_{-1}^\sigma x_1 + x_1^\sigma x_{-1} + x_2^\sigma x_{-2}$ on L^4 (cf. 4.1.4). The group G is quasi-split over $K^\natural$, split over K, and isomorphic over K to PGL_4; the valuation $\mathscr{E}$ of G_K is of type A_3 (4.2.23) and the image Ξ of $G(K)$ in the group of automorphisms of the Dynkin graph Δ of $\mathscr{E}$ is the group of circular permutations of the vertices $0, 1, 2, 3$:

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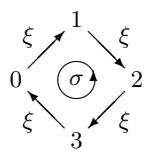

ξ = generator of Ξ,
σ = generator of $\mathrm{Gal}(L/K^\natural)$.

As for the Galois group Σ, it acts on Δ via $\mathrm{Gal}(L/K^\natural)$ and the generator σ of $\mathrm{Gal}(L/K^\natural)$ acts by exchanging the vertices 1 and 3. Let C be a chamber of $\mathscr{I}$ invariant under Σ (cf. 4.2.3). The two parahoric subgroups of G associated with the facets of type $\{0, 2\}$ and $\{1, 3\}$ of C are invariant under Σ and are conjugate by an element of the stabilizer of C in $G(K)$ with image ξ. But one of these two $K^\natural$-parahoric subgroups is maximal and the other is not. They therefore cannot be conjugate by an element of $G(K^\natural)$.

One can also give examples of two maximal parahoric subgroups invariant under Σ which are conjugate in $G(K)$ and not conjugate in $G(K^\natural)$.

However, to answer a question by Y. A. Nisnevich, let us give a positive result.

5.2.14. Proposition. — *Suppose G is semi-simple and split over K and let x, y be two points of $\mathscr{I}^\natural$ which are special points of $\mathscr{I}$. If the two maximal $K^\natural$-parahoric subgroups P_x^0 and P_y^0 are conjugate by an element of $G(K)$, they are so by an element of $G(K^\natural)$.*

Let us first note that the hypotheses made amount to saying that we are given two $\mathcal{O}^\natural$-group schemes (i.e., whose fibers are semi-simple groups) which are smooth with the same generic fiber H. Indeed, as K is strictly Henselian, the group H is necessarily split over K (cf. 4.6.31) and automatically satisfies all the conditions we have imposed on G; finally, 5.1.40 shows that these schemes are those associated with two points of $\mathscr{I}^\natural$ special in $\mathscr{I}$.

We can obviously assume that x and y belong to $A^\natural$. If P_x^0 and P_y^0 are conjugate in $G(K)$, the Bruhat decomposition $G(K) = P_y^0 N(K) P_x^0$ (I, 7.3.4) implies the existence of an element $n \in N(K)$ such that $y = n \cdot x$.

Since x is a special point of A, $P_x^0 \cap N(K)$ contains a system of representatives of the Weyl group ${}^vW = N(K)/T(K)$ and we can take $n \in T(K)$; $\nu(n)$ is then a translation invariant under the Galois group Σ, since its vector $y - x$ is.

Let π be a uniformizer of $K^\natural$, which is also a uniformizer of K, and let T_π be the subgroup of T formed by elements $t \in T(K)$ such that $\chi(t)$ is a power of π for all $\chi \in X^*(T)$. Since T is split over K, we have $T(K) = T_\pi \cdot T_b(K)$ (cf. 4.4.1); as G is semi-simple, the kernel of the restriction of ν to $T(K)$ is equal to $T_b(K)$ and ν defines an isomorphism from T_π onto $\nu(T(K))$; finally, T_π is invariant under Σ and ν is compatible

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with the actions of Σ. Consequently, there exists a unique element $t \in T_\pi \cap T(K^\natural)$ such that $\nu(t) = \nu(n)$. We then have $y = t.x$ and $P_y^0 = tP_x^0t^{-1}$, q.e.d.

Remark 3 (5.2.15). The preceding proposition becomes false if one no longer assumes that G is semi-simple. A counter-example is provided by the group U_2 of an isotropic Hermitian form in two variables relative to an étale quadratic extension of the base field $K^\natural$.

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APPENDIX. — Commutation relations

In this Appendix, K denotes a field (possibly finite) and G a connected reductive group, defined and quasi-split over K. We resume the notations of 1.1 and 4.1. One chooses a coherent system of pinnings $(x_a)_{a\in\Phi}$ of the U_a, deduced from a Chevalley-Steinberg system given in (4.1.16) and one denotes by V_a the algebraic K-group source of x_a.

A.1. — Let $a, b \in \Phi$ with $b \neq -a$, and let us choose a total order on $(a+b) \cap \Phi$ [17]. According to 1.1.8, there exists one and only one family $(\gamma_{a,b}^c)_{c\in(a+b)\cap\Phi}$ of morphisms of K-varieties

$$\gamma_{a,b}^c : V_a \times V_b \to V_c$$

such that the commutator morphism $\gamma_{a,b}$ is the composition

$$U_a \times U_b \xrightarrow{(x_a,x_b)^{-1}} V_a \times V_b \xrightarrow{\prod \gamma_{a,b}^c} \prod_{c\in(a+b)\cap\Phi} V_c \xrightarrow{\text{prod}} G.$$

The goal of this Appendix is to give almost entirely explicit formulas for the $\gamma_{a,b}^c$. Let us say right away that there is nothing truly new here: the results that follow are at least implicitly contained in [14] and [31], but have never been, to our knowledge, completely made explicit and it was perhaps not useless for these formulas to be effectively written somewhere. However, we will leave to the reader the task of performing the calculations that allow them to be established from [14] and [31] (see also I, § 10, [35] and [41]).

A.2. — It is clear that it suffices to study the case where G is simply connected and K-simple, hence of the form $\Pi_{L/K}G'$, where L is a finite separable extension of K and where G' is an algebraic group defined over L, simply connected, absolutely simple and quasi-split over L. The root systems of G and G' identify naturally and the $\gamma_{a,b}^c$ are the $\Pi_{L/K}$ of the analogous morphisms for G'. Consequently, it suffices to consider the case where G is absolutely simple, which we assume from now on.

A.3. — It then follows from 4.1.16 that, for $a \in \Phi$, the source V_a of the pinning x_a is of one of the three following types:

(I) The line $\mathfrak{D}_K$;

(II) The algebraic group $\Pi_{L/K}\mathfrak{D}_L$, where L is a quadratic or cubic separable extension of K;

(III) The algebraic group $H(L, K)$, where L is a quadratic separable extension of K (4.1.9).

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[17] (1) In reality, the U_c for $c \in (a+b) \cap \Phi$ commute with each other, and the choice of this order is without consequence, except in the case where $\{a, b\}$ is a basis of a root subsystem of type G_2 of Φ.

Moreover, the fields L involved for the different $a \in \Phi$ of type (II) or (III) are all equal, provided that, when G is of type ${}^6D_4$, the coherent system of pinnings is chosen appropriately, which we assume to be done.

A.4. — In the case where G is split, the $\gamma_{a,b}$ are given by Chevalley's formulas ([14]; cf. 3.2.3), on which ultimately everything that follows rests. But it is known that these formulas only give the $\gamma_{a,b}$ up to a sign and that there does not exist in general a natural way to determine these signs (cf. [35]). In the non-split case, to this uncertainty of signs is added the impossibility of eliminating that due to the Galois group of L, when there are roots of type (II) or (III) with L quadratic (cf. 4.1.17).

This is why we will give the $\gamma_{a,b}$ only up to "similarity": two morphisms from $V_a \times V_b$ into V_c are said to be similar (symbol: $\sim$) if they differ only by operations of the following types performed successively on one or several of the groups V:

- passing to the inverse;
- automorphism induced by the unique non-trivial K-automorphism σ of L, when V is of type (II) or (III) and $[L : K] = 2$.

For example, the permitted operations on V in case (III) are the following:

$$(u, v) \mapsto (-u, -v + u^\sigma u),$$
$$(u, v) \mapsto (u^\sigma, v^\sigma),$$
$$(u, v) \mapsto (-u^\sigma, -v^\sigma + u^\sigma u).$$

A.5. — Let L be a separable extension of K. We will denote by Nr and Tr the morphisms of K-varieties from $\Pi_{L/K}\mathfrak{D}_L$ into $\mathfrak{D}_K$ defined as follows. For every K-algebra R and every $u \in \Pi_{L/K}\mathfrak{D}_L(R) = L \otimes R$, we set

$$\mathrm{Nr}(u) = \mathrm{Norm}_{L\otimes R/R}\, u \in R = \mathfrak{D}_K(R),$$
$$\mathrm{Tr}(u) = \mathrm{Tr}_{L\otimes R/R}\, u \in R.$$

If, in addition, L is a cubic extension of K, there exists a unique endomorphism θ of $\Pi_{L/K}\mathfrak{D}_L$ such that $\mathrm{Nr}(u) = u\theta(u)$ for every K-algebra R and every $u \in L \otimes R$. Finally, we denote by $*$ the morphism from $(\Pi_{L/K}\mathfrak{D}_L)^2$ into $\Pi_{L/K}\mathfrak{D}_L$ defined by

$$u * v = \theta(u + v) - \theta(u) - \theta(v) \quad (u, v \in L \otimes R).$$

A.6. — Let us now give the $\gamma_{a,b}$ in the different possible cases (restricting ourselves, of course, to those where $(a + b) \cap \Phi \neq \varnothing$!). Let us take up again the notation and the classification of 4.1.16, with the simplifications due to the hypothesis G absolutely simple, which implies $L_2 = K$ and $\Sigma = \mathrm{Gal}(\widetilde{K}/K)$. When Φ is reduced, we identify Φ and $\overline{\Phi}$ via the bijection $a \mapsto \mathbf{R}_+^\times \cdot a$. Finally for $a, b \in \Phi$ or $\overline{\Phi}$, we denote by $\widehat{ab}$ the angle between a and b.

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a) Card $\Sigma = 1$. In other words, $\widetilde{K} = K$ and G is split over K. Then $V_a = \mathfrak{D}_K$ for every $a \in \Phi$ and Chevalley's formulas ([14], 3.2.3) state that there exist integers $C_{a,b}$ (equal to 1, 2 or 3) such that

$$\gamma_{a,b}(u, v) \sim C_{a,b} u^p v^q \quad \text{for } c = pa + qb \in \Phi, \quad p, q \in \mathbf{N}^*.$$

b) Card $\Sigma = 2$ and Φ is reduced. In other words, G is of type ${}^2A_{2n-1}$ (i.e., SU_{2n}) for $n \geq 2$, ${}^2D_n$ (i.e., non-split Spin_{2n}) for $n \geq 4$ or ${}^2E_6$. Then, $L = \widetilde{K}$ and $[L : K] = 2$.

— If $\widehat{ab} = 90°$, which implies that a and b are short, that $c = a + b$ is long, that $V_a = V_b = \Pi_{L/K}\mathfrak{D}_L$ and that $V_c = \mathfrak{D}_K$, we have

$$\gamma_{a,b}^{a+b}(u, v) \sim \mathrm{Tr}(uv).$$

— If $\widehat{ab} = 120°$, which implies that $c = a + b$ and that $V_a = V_b = V_c$ is equal to $\mathfrak{D}_K$ (resp. $\Pi_{L/K}\mathfrak{D}_L$) when a, b, and c are long (resp. short), we have

$$\gamma_{a,b}^{a+b}(u, v) \sim uv.$$

— If $\widehat{ab} = 135°$ and if a is short and b long, we have $V_a = \Pi_{L/K}\mathfrak{D}_L$ and $V_b = \mathfrak{D}_K$; moreover, either $c = a + b$, c is short and $V_c = \Pi_{L/K}\mathfrak{D}_L$, or else $c = 2a + b$, c is long and $V_c = \mathfrak{D}_K$. We have

$$\gamma_{a,b}^{a+b}(u, v) \sim uv,$$
$$\gamma_{a,b}^{2a+b}(u, v) \sim v\,\mathrm{Nr}(u).$$

c) Card $\Sigma = 2$ and Φ is not reduced: in other words, G is of type ${}^2A_{2n}$ (i.e., SU_{2n+1}) with $n \geq 1$. Then, $L = \widetilde{K}$ and $[L : K] = 2$.

— If $\widehat{ab} = 90°$ and if a and b are non-plural, then c is plural and contains the two roots $a + b$ and $\frac{1}{2}(a + b)$. We have $V_a = V_b = \Pi_{L/K}\mathfrak{D}_L$, $V_c = H(L, K)$ and

$$\gamma_{a,b}(u, v) \sim (0, v^\sigma u - uv).$$

— If $\widehat{ab} = 90°$ and if a and b are plural, then c is non-plural and contains the single root $a + b$. We have $V_a = V_b = H(L, K)$, $V_c = \Pi_{L/K}\mathfrak{D}_L$ and

$$\gamma_{a,b}((u, v), (u', v')) \sim uu'.$$

— If $\widehat{ab} = 120°$, then a, b and c are non-plural, $c = a + b$,

$$V_a = V_b = V_c = \Pi_{L/K}\mathfrak{D}_L \quad \text{and} \quad \gamma_{a,b}(u, v) \sim uv.$$

— If $\widehat{ab} = 135°$ and if a is non-plural, then b is plural, $V_a = \Pi_{L/K}\mathfrak{D}_L$, $V_b = H(L, K)$ or else $\widehat{ac} = 45°$, c is plural, contains the two roots $a + b$ and $2a + 2b$, $V_c = H(L, K)$ and

$$\gamma_{a,b}(u, (u', v')) \sim \gamma_{b,a}((u', v'), u) \sim (uu', uu^\sigma v'^\sigma),$$

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or else $\widehat{ac} = 90°$, c is non-plural and contains the single root $a + 2b$, $V_c = \Pi_{L/K}\mathfrak{D}_L$ and

$$\gamma_{a,b}(u, (u', v')) \sim \gamma_{b,a}((u', v'), u) \sim uv'.$$

d) and *e*) Card $\Sigma = 3$ or 6. In other words, G is a trialitarian form of D_4. Then, L is a cubic extension of K and one can choose the coherent system of pinnings in such a way that:

— If $\widehat{ab} = 60°$, then $c = a + b$, $V_a = V_b = \Pi_{L/K}\mathfrak{D}_L$, $V_c = \mathfrak{D}_K$ and

$$\gamma_{a,b}(u, v) \sim \mathrm{Tr}(uv);$$

— If $\widehat{ab} = 120°$ and if a and b are short, then $V_a = V_b = \Pi_{L/K}\mathfrak{D}_L$ and

$$\gamma_{a,b}(u, v) \sim \begin{cases} u * v & \text{when } c = a + b, \quad \text{whence } V_c = \Pi_{L/K}\mathfrak{D}_L, \\ \mathrm{Tr}(\theta(u)v) & \text{when } c = 2a + b, \quad \text{whence } V_c = \mathfrak{D}_K, \\ \mathrm{Tr}(u\theta(v)) & \text{when } c = 2b + a, \quad \text{whence } V_c = \mathfrak{D}_K; \end{cases}$$

— If $\widehat{ab} = 120°$ and if a and b are long, then $c = a + b$ is long, $V_a = V_b = V_c = \mathfrak{D}_K$ and we have

$$\gamma_{a,b}^{a+b}(u, v) \sim uv;$$

— If $\widehat{ab} = 150°$ and if a is short, whence $V_a = \Pi_{L/K}\mathfrak{D}_L$ and $V_b = \mathfrak{D}_K$, we have:

$$\gamma_{a,b}(u, v) \text{ and } \gamma_{b,a}(v, u) \sim \begin{cases} uv & \text{if } c = a + b, \quad \text{whence } V_c = \Pi_{L/K}\mathfrak{D}_L, \\ \theta(u)v & \text{if } c = 2a + b, \quad \text{whence } V_c = \Pi_{L/K}\mathfrak{D}_L, \\ \mathrm{Nr}(u)v & \text{if } c = 3a + b, \quad \text{whence } V_c = \mathfrak{D}_K, \\ C.\mathrm{Nr}(u)v^2 & \text{if } c = 3a + 2b, \quad \text{whence } V_c = \mathfrak{D}_K, \end{cases}$$

where C is an integer equal to 1 or 2 according to the total order chosen on $(a + b) \cap \Phi$ (and which is not the same for $\gamma_{a,b}(u, v)$ and $\gamma_{b,a}(v, u)$).

Remark. — In the form in which they have been stated, the results of this Appendix are valid whether one sets $(x, y) = xyx^{-1}y^{-1}$, as in I, or $(x, y) = x^{-1}y^{-1}xy$, as in [NB] A, chap. I.

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Errata and Addenda to Chapter I

E.1. — P. 29, in the "Table of irreducible valuations", add in eighth position the valuation

$$\text{B-BC}_2 \quad 2 \quad \text{B}_2 \quad \text{BC}_2 \quad \underset{s}{\circ} \equiv \underset{s}{\circ} \overset{x}{\leftarrow} \circ$$

E.2. — P. 136, no. 6.4.8, it is preferable to slightly strengthen the definition of quasi-concave functions by requiring that they satisfy, in addition to (QC 1) and (QC 2), the condition

(QC o) *For every* $a \in \Phi$ *such that* $2a \in \Phi$, *we have* $f(2a) \leq \inf \Gamma_{2a} \cap [2f(a), +\infty[$.

It is clear that a concave function satisfies (QC o), as does every function f_Ω associated with a subset of an apartment (I, 7.1.1). Let us also note that (QC o) implies that the derived group of $U_{a,f(a)}$ is contained in $U_{2a,f(2a)}$. Of course, all results proven in I for quasi-concave functions remain *a fortiori* valid with the new definition.

In fact, this modification does not change much. Indeed, let $f : \Phi \to \tilde{\mathbf{R}}$; let us set $g(a) = f(a)$ for $a \in \Phi, a/2 \notin \Phi$ and $g(2a) = \inf(f(2a), 2f(a))$ for $a, 2a \in \Phi$. It is clear that (with the notation of I, 6.4.9)

$$X_a = U_{a,f(a)}U_{2a,f(2a)} = U_{a,g(a)}U_{2a,g(2a)} \tag{1}$$

for all $a \in \Phi^{\text{red}}$ (on the other hand, one can have $U_{2a,g(2a)} \neq U_{2a,f(2a)}$ for $2a \in \Phi - \Phi^{\text{red}}$). It follows that the groups U_f and U_g are equal, thus also their intersections $U_{f,a}$ and $U_{g,a}$ with U_a for $a \in \Phi$. Let us show that if f satisfies (QC 1) and (QC 2), then g satisfies (QC o), (QC 1) and (QC 2). It is obvious for (QC o). Let us show that g satisfies (QC 1): this is immediate from (1) if $a \in \Phi^{\text{red}}$. If $a, 2a \in \Phi$, let us set

$$h_\pm = \inf\{k \in \Gamma_{\pm 2a} \mid k \geq g(\pm 2a)\} \in \tilde{\mathbf{R}}$$

(cf. I, 6.4.10), so that $U_{\pm 2a,g(\pm 2a)} = U_{\pm 2a,h_\pm}$. One shows as in I, 6.4.11 that $h_+ + h_- \geq 0$, that $h_+ + h_- > 0$ implies $N_g^{(2a)} \subset H$ and that $h_+ + h_- = 0$ implies $h_\pm \in \Gamma_{\pm 2a}$ and $\nu(N_g^{(2a)}) = \{1, r_{2a,h_+}\}$. From I, 6.3.2, 6.3.3 and 6.3.4 it then follows that $U_{-2a,g(-2a)}U_{2a,g(2a)}N_g^{(2a)}$ is a group, q.e.d.

Finally, let us show that g satisfies (QC 2). Since $U_{a,g(a)} \supset U_{a,f(a)}$ for all $a \in \Phi$, with equality except perhaps if $a \in \Phi - \Phi^{\text{red}}$, one sees immediately that the only case to examine is that of the commutator $Y = (U_{2a,g(2a)}, U_{b,g(b)})$ with $a, 2a, b \in \Phi$, which is trivial unless $\{a, b\}$ forms a base of a subsystem of type BC_2 of Φ, composed of...

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roots $a, 2a, 2a+b, a+b, 2(a+b), b$ and their opposites. Let V_1 be the subgroup generated by U_{a+b} and U_{2a+b}, V_2 the subgroup generated by $U_{2(a+b)}$ and U_{2a+b}, and X the one generated by X_{a+b} and X_{2a+b}. Since $U_{2(a+b)}$ and U_{2a+b} are central in V_1, the product map is a bijection of $U_{a+b} \times U_{2a+b}$ onto V_1 (resp. of $U_{2(a+b)} \times U_{2a+b}$ onto V_2, resp. of $X_{a+b} \times X_{2a+b}$ onto X). However, Y is contained in V_2 according to (DR 2) and in X since f satisfies (QC 2) and the injectivity of the product map $U_{a+b} \times U_{2a+b} \to V_1$ implies that $V_2 \cap X = (U_{2(a+b)} \cap X_{a+b}) \cdot (U_{2a+b} \cap X_{2a+b})$, which completes the proof since $U_{2(a+b)} \cap X_{a+b} = U_{2(a+b),g(2(a+b))}$ and $U_{2a+b} \cap X_{2a+b} = X_{2a+b} = U_{2a+b,g(2a+b)}$.

E.3. — P. 136, line 9, read "quasi-concave with values in $\omega(K^\times)$" instead of "quasi-concave".

E.4. — P. 136, add at the end of no. 6.4.8: More generally, if all the roots of Φ are of the same length, it follows easily from 6.1.8 (6) and 6.2.7 that every quasi-concave function f such that $f(a) \in \Gamma_a$ for all $a \in \Phi$ is necessarily concave. If Φ is of type A_n, one can even show that f is of the form f_Ω, where Ω is a non-empty bounded subset of A.

E.5. — P. 136, no. 6.4.9, proposition 6.4.9 and its proof are only correct if one assumes f quasi-concave in the reinforced sense of erratum E.2 above. If one keeps the old definition of quasi-concavity, Φ must be replaced by Φ^{red} in the statement of 6.4.9 (i).

E.6. — P. 143, line 6 of 6.4.24, read (6.4.23) instead of (6.4.22).

E.7. — P. 178, line 1 of 7.4.27, read $v \in D$ instead of $v \in \overline{D}$.

E.8. — P. 179, line 9 from the bottom, read $h = 1/3$ instead of $h = 1/2$. P. 180, lines 1 and 4, read $kr/(1 + cr(1-k))$ instead of kr and $c = r = 1$ instead of $r = 1$.

E.9. — Some readers have pointed out to us that the case of the symplectic group over a field K of characteristic 2 does not seem to be covered by § 10, Valuations of classical groups: if one adheres strictly to the conventions of § 10, one has in this case $\sigma = \text{id}$, $\varepsilon = 1$, hence $K_{\sigma,\varepsilon} = \{0\}$, and if f is a non-degenerate alternating form of rank $2r$ on the vector space X and $q : X \to K/K_{\sigma,\varepsilon} = K$ a pseudo-quadratic form such that $q(x+y) = q(x) + q(y) + f(x,y)$, the group of isometries of the pair (f, q), which is the one considered throughout § 10, is the orthogonal group of q and not the symplectic group of f.

However, it would be easy to see that the results of § 10 remain valid for the symplectic group of f provided that one takes $X_0 = \{0\}$, $q = 0$, replaces $K_{\sigma,\varepsilon}$ by K everywhere, and denotes by Z the set of pairs $(0, k)$ for $k \in K$. The set Φ of roots is, with the notations of 10.1.2, the set $\{2a_i, a_{ij}\}$, of type (C).

In reality, the case of the symplectic group over a field K of characteristic 2 is indeed treated in our § 10, but in an indirect way. It is known indeed (see J. Dieudonné, *Sur les groupes classiques*, Paris, Hermann, 1948, p. 54) that the group $Sp_{2r}(K)$ is isomorphic to the orthogonal group of a (usual) defective quadratic form: if $(e_i)_{1\leq \pm i\leq r}$ is a symplectic basis of K^{2r} and $(x_j)_{j\in J}$ a basis of K over K^2, it suffices to consider the vector space X direct sum of $X_1 = K^{2r}$ and $X_0 = K^{(J)}$ and to set

$$q(\sum \xi_i e_i + \sum \eta_j f_j) = \sum_{1\leq i\leq r} \xi_i \xi_{-i} + \sum_{j\in J} x_j \eta_j^2$$

(where $(f_j)_{j\in J}$ is the canonical basis of $K^{(J)}$). Every element $u \in O(q)$ has for restriction to X_0 the identity (*loc. cit.*, p. 53) and defines by passing to the quotient an automorphism $\bar{u}$ of $X_1 \simeq X/X_0$, and the map $u \mapsto \bar{u}$ is an isomorphism from $O(q)$ onto $Sp_{2r}(K)$ (*loc. cit.*, p. 54).

One can make analogous remarks for the other forms of the symplectic group. Let K be a field of characteristic 2, equipped with an involution σ. We have $K_{\sigma,\epsilon} = TrK$. Let f be a non-degenerate trace-valued σ-sesquilinear form on a vector space X over K. One can always choose a pseudo-quadratic form $q : X \to K/TrK$ which is associated with f: it suffices to choose a basis (e_i) of X and $a_{ij} \in K$ such that $a_{ij} + a_{ji}^{\sigma} = f(e_i, e_j)$ for all i, j, and to set

$$q(\sum \xi_i e_i) = \sum \xi_i^{\sigma} a_{ij} \xi_j + TrK.$$

If the restriction of σ to the center C of K is not the identity, there is a canonical way to make this choice: in this case, there exists indeed an element $\lambda \in C$ such that $\lambda + \lambda^{\sigma} = 1$ and one can take $q(x) = \lambda f(x, x) + TrK$ (cf. I, p. 212); the unitary group of f is then also the group $Is(q, f)$ treated in § 10 of I.

But if the restriction of σ to the center C of K is the identity (involution "of the first kind"), such a $\lambda \in C$ does not exist, there is in general no canonical choice for q and the unitary group $U(f)$ of f is in general strictly larger than $Is(q, f)$ and seems therefore not covered by § 10 of I.

As above, two answers are possible. First of all, one can check that the results of § 10 remain valid in this case, provided $K_{\sigma,\epsilon} = TrK$ is replaced by the group $K^{(\sigma)}$ of elements of K invariant under σ, and by denoting by q the map $x \mapsto \{k \in K \mid f(x, x) = Trk\}$ from X into $K/K^{(\sigma)}$. The set Z of I, 10.1.1 then becomes $\{(z, k) \in X_0 \times K \mid f(z, z) = Trk\}$ and one sets

$$\omega_q(z) = \frac{1}{2} \sup\{\omega(k) \mid k \in K, Trk = f(z, z)\}.$$

One can also, as above, reduce to the case explicitly treated in § 10. We consider the quotient S of $K^{(\sigma)}$ by the subgroup TrK. The law of operation $(k, x) \mapsto k^{\sigma} x k$

from K into $K^{(\sigma)}$ passes to the quotient and defines on S a structure, called canonical, of a right vector space over K. We then set $\tilde{X} = X \oplus S$,

$$\tilde{f}((x, s), (x', s')) = f(x, x')$$
$$\tilde{q}((x, s)) = q(x) + s + \mathrm{Tr}\ K$$

where q is a pseudo-quadratic form associated to f chosen arbitrarily. We then see that S is the kernel of $\tilde{f}$, that the pair $(\tilde{q}, \tilde{f})$ is non-degenerate, that every element $g \in \mathrm{Is}(\tilde{q}, \tilde{f})$ induces the identity on S and defines by passing to the quotient an automorphism $\bar{g}$ of $X = \tilde{X}/S$ which belongs to $U(f)$. Finally, it follows from Prop. 8.2.9 of [37], p. 126, that the map $g \mapsto \bar{g}$ is an isomorphism from $\mathrm{Is}(\tilde{q}, \tilde{f})$ onto $U(f)$. This is moreover easily seen by showing that the set of pseudo-quadratic forms associated to f is in bijective correspondence with the set of complements of S in $\tilde{X}$, or again with the set of linear maps h from X to S: to such a map h corresponds the form $q_h : x \mapsto \tilde{q}((x, h(x)))$.

Let X_+ and X_- be two maximal totally isotropic subspaces for f of X, with intersection $\{0\}$, and let X_0 be the orthogonal of $X_+ + X_-$ in X. One can take for q the form

$$q(x_+ + x_- + x_0) = f(x_+, x_-) + q_0(x_0) \quad (x_+ \in X_+, x_- \in X_-, x_0 \in X_0),$$

where q_0 is a pseudo-quadratic form on X_0 associated to the restriction of f to X_0. For every $x_0 \neq 0$ in X_0, we have $q_0(x_0) + q_0(x_0)^{\sigma} = f(x_0, x_0) \neq 0$ (I, 10.1.1 (5)), hence $\tilde{q}(x_0, s) \neq 0$ for all $s \in S$. It follows that if one chooses a basis $(e_i)_{1\leq i\leq r}$ of X_+ and if one denotes by (e_{-i}) the dual basis of X_-, the hypotheses of I, 10.1.1 are satisfied for the decomposition $\tilde{X} = \coprod e_i K + \tilde{X}_0$, with $\tilde{X}_0 = X_0 + S$. Furthermore, one sees immediately that if $\tilde{Z}$ is the set of $((x_0, s), k) \in \tilde{X}_0 \times K$ such that $k \in \tilde{q}((x_0, s))$ (cf. I, 10.1.1 at the end), then $((x_0, s), k) \in \tilde{Z}$ if and only if $k + k^{\sigma} = f(x_0, x_0)$. Consequently, the map $g \mapsto \bar{g}$ sends the linear transformations of $\tilde{X}$ defined by the formulas of I, 10.1.2 (replacing X_0 with $\tilde{X}_0$ and Z with $\tilde{Z}$) to those of X defined by the same formulas, but taking $Z = \{(x_0, k) \in X_0 \times K \mid k + k^{\sigma} = f(x_0, x_0)\}$. This demonstrates the validity of the "first answer" proposed at the

beginning of the preceding paragraph. Note that for $\sigma = \mathrm{id}$, we recover the case of the ordinary symplectic group and the assertion of the second paragraph of this erratum E.9.

Finally, let us consider the family ϕ^a of functions defined in I, 10.1.13. By translation of I, 10.1.15, we see that it is a valuation of the root datum of $U(f)$ if and only if one of the following two equivalent conditions is satisfied:

(i) $\omega(f(z,z')) \geq \omega_q(z) + \omega_q(z')$ for $z, z' \in X_0$;

(ii) $\omega(f(z,z')) \geq 2\inf(\omega_q(z), \omega_q(z'))$ for $z, z' \in X_0$.

Reasoning as in I, 10.1.21, and taking into account that $\omega_q(z) \leq \frac{1}{2}\omega(f(z,z)) < +\infty$ for $z \in X_0$, $z \neq 0$, one shows that (i) and (ii) are equivalent to

(iii) ω_q is a norm on X_0.

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Let then $\hat{f}$ and $\hat{q}$ be the canonical extensions of f and q to $\hat{X}_0 = X_0 \otimes \hat{K}$. If $\hat{f}$ is anisotropic, then φ^ω is a valuation of the root datum of $U(f)$: otherwise, according to I, 10.1.15 (iii), there would exist $x \in \hat{X}_0 - \{0\}$, $s \in S \otimes \hat{K}$ and $k \in \mathrm{Tr}\,\hat{K} \subset \hat{K}^{(\alpha)}$ such that $k \in \tilde{q}(x,s)$, from which $\hat{f}(x,x) = k + \bar{k} = 0$. Conversely, if $\hat{K}^{(\alpha)} = K^{(\alpha)}$ (a condition which is satisfied when K is of finite dimension over its center) and if φ^ω is a valuation of the root datum of $U(f)$, then $\hat{f}$ is anisotropic and $\omega_{\hat{q}}$ is a norm on $\hat{X}_0$ whose restriction to X_0 is ω_q. Indeed, one easily sees that ω_q is the restriction of $\omega_{\hat{q}}$ to X_0 and that $\omega_{\hat{q}}$ is less than or equal to the extension by continuity of ω_q to a semi-norm on $\hat{X}_0$. By continuity one deduces from (i) that

$$(\hat{\mathrm{i}}) \qquad \omega_{\hat{q}}(\hat{f}(z,z')) \geq \omega_{\hat{q}}(z) + \omega_{\hat{q}}(z') \quad \text{for } z, z' \in \hat{X}_0.$$

Since $\hat{f}$ is non-degenerate, $(\hat{\mathrm{i}})$ implies $\omega_{\hat{q}}(z) < +\infty$ for $z \neq 0$ and $\hat{f}$ is anisotropic. Finally, the equivalence of (i) and (iii) shows that $\omega_{\hat{q}}$ is a norm on $\hat{X}_0$.

Let us note finally that if f is a non-degenerate sesquilinear form, not necessarily trace-like, then the unitary group $U(f)$ is an extension of the unitary group of a non-degenerate trace-like form f' by a nilpotent subgroup (J. Dieudonné, On the structure of the unitary group II, *Amer. J. Math.*, 75 (1953), 665-678). More precisely there exists a decomposition of X into a direct sum $X_1 \oplus X_2 \oplus X_3$ such that the restriction f' of f to X_2 is non-degenerate trace-like and such that every element $g \in U(f)$ is represented in this decomposition by a matrix

$$g = \begin{pmatrix} I & a & b \\ 0 & \bar{g} & c \\ 0 & 0 & I \end{pmatrix},$$

the map $g \mapsto \bar{g}$ being a surjective homomorphism from $U(f)$ onto $U(f')$. In particular, $U(f)$ can be "reductive" only if it is isomorphic to $U(f')$.

E. 10. — The "Note added in proof", pp. 238 and 239, is full of misprints. We will return to the results of this Note and explain the proofs in a forthcoming article. In the meantime, let us point out some corrections on page 239:

— line 1: add "decomposable" after "additive";

— line 2: add "with $u_i = e_i$" after "(1)";

— formula (4): replace "$\sum'_{i=1}$" by "$\sum_{i \in I}$";

— line 6 from the bottom: replace "$u_{-i} = u'_{-i} - \varepsilon u_i \lambda_i$" by "$u_{-i} = u'_{-i} - \sum_{1 \leq j \leq r} u_j a_{ij}$, with $a_{ii} = \varepsilon\lambda_i$, $a_{ij} = \varepsilon f(u'_{-i}, u'_{-j})$ for $i < j$ and $a_{ij} = 0$ for $i > j$";

— line 5 from the bottom: replace "u_i" by "u_{-i}";

— line 2 from the bottom: replace "ω" by "α".

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Special treatment has been reserved for four frequently used references:

— The treatise of Nicolas BOURBAKI (*Elements of Mathematics*, Paris), to which reference is made under the abbreviation [NB], followed by the canonical reference to the various books, notably A (*Algebra*), AC (*Commutative Algebra*), LIE (*Groups and Lie Algebras*) and to the chapters of these books; these references refer to the following editions:

A: *Algebra*, chap. I to III, Hermann, 1970; chap. IV to VII, Masson, 1981;
AC: *Commutative Algebra*, chap. I and II, Hermann, 1961; chap. III and IV, Hermann, 1961; chap. V and VI, Hermann, 1964; chap. VII, Hermann, 1965;
LIE: *Groups and Lie Algebras*, chap. I, Hermann, 1971; chap. II and III, Hermann, 1972; chap. IV to VI, Hermann, 1968; chap. VII and VIII, Hermann, 1975;

— The book by M. DEMAZURE and P. GABRIEL (*Algebraic Groups*, vol. I, Paris, Masson, 1970), noted by the abbreviation [DG];

— The Seminar on Algebraic Geometry of A. GROTHENDIECK, from 1962-1964 (*Lecture Notes in Math.*, 151, 152, 153, Springer, 1970) noted by the abbreviation [SGAD];

— The *Elements of Algebraic Geometry* by A. GROTHENDIECK (*Publ. Math. I.H.E.S.*, 4, 8, 11, 17, 20, 24, 28 and 32) noted by the abbreviation [EGA] followed by the number of the Part.

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